

Md Faizan Ali

Interview
FAQs

14TH
EDITION

Let Us



Yashavant Kanethkar



BPB PUBLICATIONS

Let Us C

Fourteenth Edition

Yashavant Kanetkar



BPB PUBLICATIONS

20 Ansari Road, Daryaganj, New Delhi-110002

Dedicated to baba
Who couldn't be here to see this day...

About the Author

Through his books and Quest Video Courseware DVDs on C, C++, Data Structures, VC++, .NET, Embedded Systems, etc. Yashavant Kanetkar has created, moulded and groomed lacs of IT careers in the last two decades. Yashavant's books and Quest DVDs have made a significant contribution in creating top-notch IT manpower in India and abroad.

Yashavant's books are globally recognized and millions of students / professionals have benefitted from them. Yashavant's books have been translated into Hindi, Gujarati, Japanese, Korean and Chinese languages. Many of his books are published in India, USA, Japan, Singapore, Korea and China.

Yashavant is a much sought after speaker in the IT field and has conducted seminars/workshops at TedEx, IITs, RECs and global software companies.

Yashavant has recently been honored with the prestigious "Distinguished Alumnus Award" by IIT Kanpur for his entrepreneurial, professional and academic excellence. This award was given to top 50 alumni of IIT Kanpur who have made significant contribution towards their profession and betterment of society in the last 50 years.

In recognition of his immense contribution to IT education in India, he has been awarded the "Best .NET Technical Contributor" and "Most Valuable Professional" awards by Microsoft for 5 successive years.

Yashavant holds a BE from VJTI Mumbai and M.Tech. from IIT Kanpur.

Acknowledgments

Let Us C has become an important part of my life. I have created and nurtured it for last decade and half. While doing so, I have received, in addition to the compliments, a lot of suggestions from students, developers, professors, publishers and authors. So much have their inputs helped me in taking this book up to its fourteenth edition that ideally I should put their names too on the cover page.

In particular, I am indebted to Manish Jain who had a faith in this book idea, believed in my writing ability, whispered the words of encouragement and made helpful suggestions from time to time. I hope every author gets a publisher who is as cooperative, knowledgeable and supportive as Manish.

The Fourteen editions of this book saw several changes and facelifts. During this course many people helped in executing programs and spotting bugs. I trust that with their collective acumen, all the programs in this book would run correctly. I value the work that they did a lot. Any errors, omissions or inconsistencies that remain are, alas, my responsibility.

I thank Dada, Ammi—my parents, Seema—my wife, Aditya, Anuj—my sons for enduring the late nights, the clicking keyboard, and mostly for putting up with yet another marathon book effort.

Thinking of a book cover idea is one thing, putting it into action is a different cup of tea. This edition's cover idea has been implemented by Jayant. Many thanks to him.

And finally my heartfelt gratitude to the countless students who made me look into every nook and cranny of C. I want to remain in their debt. It is only because of them that Let Us C is now published from India, Singapore, USA, Japan, Korea and China in multiple languages.

Preface

In this I have reorganized the contents of the book in a major way. After going through the thirteenth edition several times I decided to realign all the chapters in such a manner that if a C programming course is taught using Let Us C, it can roughly be finished in 23 lectures of one hour each, with one chapter's contents devoted to one lecture. I hope this would make the learning path trouble-free. Some end-of-chapter exercises in the book needed a second look to make them more practical. That also stands done now.

Many readers told me that they have immensely benefitted from the inclusion of the chapter on Interview FAQs. I have improved this chapter further. The rationale behind this chapter is simple—ultimately all the readers of Let Us C sooner or later end up in an interview room where they are required to take questions on C programming. I now have a proof that this chapter has helped to make that journey smooth and fruitful.

All the programs present in the book are available in source code form at www.kicit.com/books/letusc/sourcecode. You are free to download them, improve them, change them, do whatever with them. If you wish to get solutions for the Exercises in the book they are available in another book titled 'Let Us C Solutions'. If you want some more problems for practice they are available in the book titled 'Let Us C Workbook'. As usual, new editions of these two books have also been launched along with 14th edition of Let Us C.

If you like 'Let Us C' and want to hear the complete video-recorded lectures created by me on C language (and other subjects like C++, VC++, C#, Java, .NET, Embedded Systems, etc.), then you can visit <http://quest.ksetindia.com> for more details.

'Let Us C' is as much your book as it is mine. So if you feel that I could have done certain job better than what I have, or you have any suggestions about what you would like to see in the next edition, please drop a line to **bpbpublications@gmail.com**.

Countless Indians have relentlessly worked for close to two decades to successfully establish "India" as a software brand. At times, I take secret pleasure in seeing that a Let Us C has contributed in its own little way in shaping so many careers that have made the "India" brand acceptable.

Recently I was presented with “Distinguished Alumnus Award” by IIT Kanpur. It was great to figure in a list that contained Narayan Murthy, Chief Mentor, Infosys, Dr. D. Subbarao, former Governor, Reserve Bank of India, Dr. Rajeev Motwani of Stanford University, Prof. H. C. Verma, Mr. Som Mittal President of NASSCOM, Prof. Minwalla of Harvard University, Dr. Sanjay Dhande former Director of IIT Kanpur, Prof. Arvind and Prof. Sur of MIT USA and Prof. Ashok Jhunjhunwala of IIT Chennai.

I think Let Us C amongst my other books has been primarily responsible for helping me get the “Distinguished Alumnus” award. What was a bit surprising was that almost all who were present knew about the book already and wanted to know from me what it takes to write a book that sells in millions of copies. My reply was—make an honest effort to make the reader understand what you have to say and keep it simple. I don’t know how convincing was this answer, but well, that is what I have been doing with this book in all its previous thirteen editions. I have followed the same principle with this edition too.

All the best and happy programming!

Yashavant Kanetkar

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1

Getting Started

- **What is C?**
- **Getting started with C**
 - The C Character Set
 - Constants, Variables and Keywords
 - Types of C Constants
 - Rules for Constructing Integer Constants
 - Rules for Constructing Real Constants
 - Rules for Constructing Character Constants
 - Types of C Variables
 - Rules for Constructing Variable Names
 - C Keywords
- **The First C Program**
 - Form of a C Program
 - Comments in a C Program
 - What is *main()*?
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 - printf()* and its Purpose
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Before we can begin to write serious programs in C, it would be interesting to find out what really is C, how it came into existence and how does it compare with other programming languages. In this chapter, we would briefly outline these issues.

Four important aspects of any language are the way it stores data, the way it operates upon this data, how it accomplishes input and output, and how it lets you control the sequence of execution of instructions in a program. We would discuss the first three of these building blocks in this chapter.

What is C?

C is a programming language developed at AT & T's Bell Laboratories of USA in 1972. It was designed and written by a man named Dennis Ritchie. In the late seventies C began to replace the more familiar languages of that time like PL/I, ALGOL, etc. No one pushed C. It wasn't made the 'official' Bell Labs language. Thus, without any advertisement, C's reputation spread and its pool of users grew. Ritchie seems to have been rather surprised that so many programmers preferred C to older languages like FORTRAN or PL/I, or the newer ones like Pascal and APL. But, that's what happened.

Possibly why C seems so popular is because it is reliable, simple and easy to use. Moreover, in an industry where newer languages, tools and technologies emerge and vanish day in and day out, a language that has survived for more than three decades has to be *really* good.

An opinion that is often heard today is—"C has been already superseded by languages like C++, C# and Java, so why bother to learn C today". I seriously beg to differ with this opinion. There are several reasons for this. These are as follows:

- (a) C++, C# or Java make use of a principle called Object Oriented Programming (OOP) to organize the program. This organizing principle has lots of advantages to offer. But even while using this organizing principle you would still need a good hold over the language elements of C and the basic programming skills. So it makes more sense to first learn C and then migrate to C++, C# and Java. Though this two-step learning process may take more time, but at the end of it you will definitely find it worth the trouble.
- (b) Major parts of popular operating systems like Windows, UNIX, Linux and Android are written in C. This is because even today when it

comes to performance (speed of execution) nothing beats C. Moreover, if one is to extend the operating system to work with new devices one needs to write device driver programs. These programs are exclusively written in C.

- (c) Mobile devices like Smartphones and Tablets have become rage of today. Also, common consumer devices like microwave ovens, washing machines and digital cameras are getting smarter by the day. This smartness comes from a microprocessor, an operating system and a program embedded in these devices. These programs not only have to run fast but also have to work in limited amount of memory. No wonder that such programs are written in C. With these constraints on time and space, C is the language of choice while building such operating systems and programs.
- (d) You must have seen several professional 3D computer games where the user navigates some object, like say a spaceship and fires bullets at the invaders. The essence of all such games is speed. Needless to say, such games won't become popular if they take a long time to move the spaceship or to fire a bullet. To match the expectations of the player the game has to react fast to the user inputs. This is where C language scores over other languages. Many popular gaming frameworks (like DirectX) have been built using C language.
- (e) At times one is required to very closely interact with the hardware devices. Since C provides several language elements that make this interaction feasible without compromising the performance, it is the preferred choice of the programmer.

I hope that these are very convincing reasons why you should adopt C as the first, and a very important step, in your quest for learning programming.

Getting Started with C

Communicating with a computer involves speaking the language the computer understands, which immediately rules out English as the language of communication with computer. However, there is a close analogy between learning English language and learning C language. The classical method of learning English is to first learn the alphabets used in the language, then learn to combine these alphabets to form words, which, in turn, are combined to form sentences and sentences are combined to form paragraphs.

Learning C is similar and easier. Instead of straight-away learning how to write programs, we must first know what alphabets, numbers and special symbols are used in C, then how using them, constants, variables and keywords are constructed, and finally, how are these combined to form an instruction. A group of instructions would be combined later on to form a program. This is illustrated in the Figure 1.1.

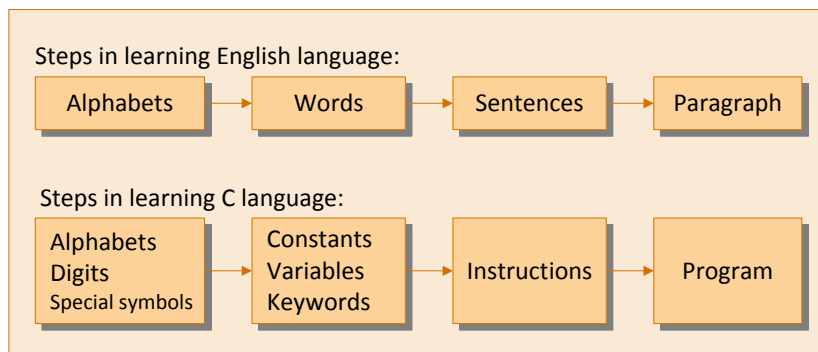


Figure 1.1

The C Character Set

A character denotes any alphabet, digit or special symbol used to represent information. Figure 1.2 shows the valid alphabets, numbers and special symbols allowed in C.

Alphabets	A, B,, Y, Z a, b,, y, z
Digits	0, 1, 2, 3, 4, 5, 6, 7, 8, 9
Special symbols	~ ' ! @ # % ^ & * () _ - + = \ { } [] : ; " ' < > , . ? / \$

Figure 1.2

Constants, Variables and Keywords

The alphabets, digits and special symbols when properly combined form constants, variables and keywords. Let us now understand the meaning of each of them. A constant is an entity that doesn't change, whereas, a variable is an entity that may change. A keyword is a word that carries special meaning.

In any C program we typically do lots of calculations. The results of these calculations are stored in computer's memory. Like human memory, the computer's memory also consists of millions of cells. The calculated values are stored in these memory cells. To make the retrieval and usage of these values easy, these memory cells (also called memory locations) are given names. Since the value stored in each location may change, the names given to these locations are called variable names. Let us understand this with the help of an example.

Consider the memory locations shown in Figure 1.3. Here 3 is stored in a memory location and a name **x** is given to it. Then we have assigned a new value 5 to the same memory location **x**. This would overwrite the earlier value 3, since a memory location can hold only one value at a time.

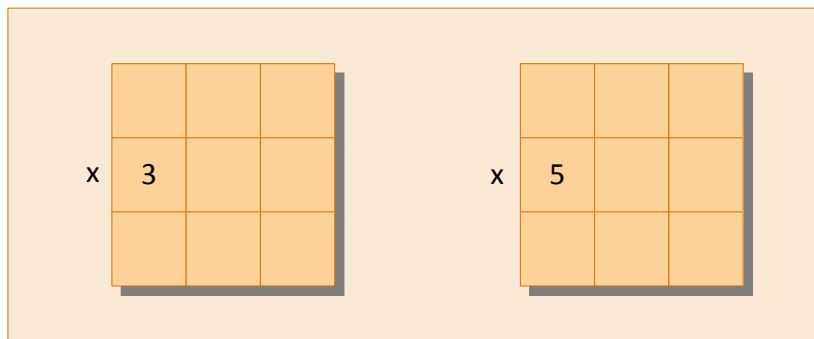


Figure 1.3

Since the location whose name is **x** can hold different values at different times **x** is known as a variable (or a variable name). As against this, 3 or 5 do not change, hence are known as constants.

In programming languages, constants are often called literals, whereas, variables are called identifiers.

Now that we understand the constants and the variables, let us see what different types of constants and variables exist in C.

Types of C Constants

C constants can be divided into two major categories:

- (a) Primary Constants
- (b) Secondary Constants

These constants are further categorized as shown in Figure 1.4.

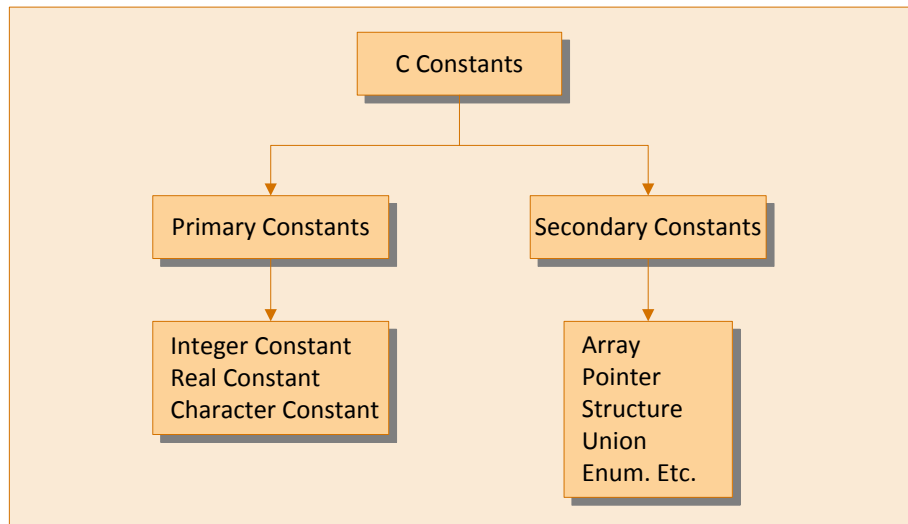


Figure 1.4

At this stage, we would restrict our discussion to only Primary constants, namely, Integer, Real and Character constants. Let us see the details of each of these constants. For constructing these different types of constants, certain rules have been laid down. These rules are as under:

Rules for Constructing Integer Constants

- (a) An integer constant must have at least one digit.
- (b) It must not have a decimal point.
- (c) It can be either positive or negative.
- (d) If no sign precedes an integer constant, it is assumed to be positive.
- (e) No commas or blanks are allowed within an integer constant.
- (f) The allowable range for integer constants is -2147483648 to +2147483647.

Truly speaking, the range of an Integer constant depends upon the compiler. For compilers like Visual Studio, gcc, it is -2147483648 to +214748364, whereas for compilers like Turbo C or Turbo C++ the range is -32768 to +32767.

Ex.: 426
+782
-8000
-7605

Rules for Constructing Real Constants

Real constants are often called Floating Point constants. The real constants could be written in two forms—Fractional form and Exponential form.

Following rules must be observed while constructing real constants expressed in fractional form:

- (a) A real constant must have at least one digit.
- (b) It must have a decimal point.
- (c) It could be either positive or negative.
- (d) Default sign is positive.
- (e) No commas or blanks are allowed within a real constant.

Ex.: +325.34
426.0
-32.76
-48.5792

The exponential form is usually used if the value of the constant is either too small or too large. It, however, doesn't restrict us in any way from using exponential form for other real constants.

In exponential form the real constant is represented in two parts. The part appearing before 'e' is called mantissa, whereas the part following 'e' is called exponent. Thus 0.000342 can be written in exponential form as 3.42×10^{-4} (which in normal arithmetic means 3.42×10^{-4}).

Following rules must be observed while constructing real constants expressed in exponential form:

- (a) The mantissa part and the exponential part should be separated by a letter e or E.
- (b) The mantissa part may have a positive or negative sign.
- (c) Default sign of mantissa part is positive.
- (d) The exponent must have at least one digit, which must be a positive or negative integer. Default sign is positive.
- (e) Range of real constants expressed in exponential form is -3.4×10^{38} to 3.4×10^{38} .

Ex.: +3.2e-5

```
4.1e8  
-0.2E+3  
-3.2e-5
```

Rules for Constructing Character Constants

- (a) A character constant is a *single* alphabet, a single digit or a single special symbol enclosed within single inverted commas.
- (b) Both the inverted commas should point to the left. For example, 'A' is a valid character constant whereas 'A' is not.

```
Ex.: 'A'  
      'I'  
      '5'  
      '=_'
```

Types of C Variables

A particular type of variable can hold only the same type of constant. For example, an integer variable can hold only an integer constant, a real variable can hold only a real constant and a character variable can hold only a character constant. The rules for constructing different types of constants are different. However, for constructing variable names of all types, the same set of rules applies. These rules are given below.

Rules for Constructing Variable Names

- (a) A variable name is any combination of 1 to 31 alphabets, digits or underscores. Some compilers allow variable names whose length could be up to 247 characters. Still, it would be safer to stick to the rule of 31 characters. Do not create unnecessarily long variable names as it adds to your typing effort.
- (b) The first character in the variable name must be an alphabet or underscore (_).
- (c) No commas or blanks are allowed within a variable name.
- (d) No special symbol other than an underscore (as in **gross_sal**) can be used in a variable name.

```
Ex.: si_int  
      m_hra  
      pop_e_89
```

Since, the maximum allowable length of a variable name is 31 characters, an enormous number of variable names can be constructed using the above-mentioned rules. It is a good practice to exploit this abundant choice in naming variables by using meaningful variable names.

Thus, if we want to calculate simple interest, it is always advisable to construct meaningful variable names like **prin**, **roi**, **noy** to represent Principle, Rate of interest and Number of years rather than using the variables **a**, **b**, **c**.

The rules for creating variable names remain same for all the types of primary and secondary variables. Naturally, the question follows... how is C able to differentiate between these variables? This is a rather simple matter. C compiler is able to distinguish between the variable names by making it compulsory for you to declare the type of any variable name that you wish to use in a program. This type declaration is done at the beginning of the program. Examples of type declaration statements are given below.

```
Ex.: int si, m_hra ;  
      float bassal ;  
      char code ;
```

C Keywords

Keywords are the words whose meaning has already been explained to the C compiler (or in a broad sense to the computer). There are only 32 keywords available in C. Figure 1.5 gives a list of these keywords for your ready reference. A detailed discussion of each of these keywords would be taken up in later chapters wherever their use is relevant.

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

Figure 1.5

The keywords **cannot** be used as variable names because if we do so, we are trying to assign a new meaning to the keyword, which is not allowed. Some C compilers allow you to construct variable names that exactly resemble the keywords. However, it would be safer not to mix up the variable names and the keywords.

Note that compiler vendors (like Microsoft, Borland, etc.) provide their own keywords apart from the ones mentioned in Figure 1.5. These include extended keywords like **near**, **far**, **asm**, etc. Though it has been suggested by the ANSI committee that every such compiler-specific keyword should be preceded by two underscores (as in **__asm**), not every vendor follows this rule.

The First C Program

Once armed with the knowledge of variables, constants & keywords, the next logical step would be to combine them to form instructions. However, instead of this, we would write our first C program now. Once we have done that we would see in detail the instructions that it made use of. The first program is very simple. It calculates simple interest for a set of values representing principal, number of years and rate of interest.

```
/* Calculation of simple interest */
/* Author: gekay Date: 25/06/2016 */
#include <stdio.h>

int main( )
{
    int  p, n ;
    float r, si ;

    p = 1000 ;
    n = 3 ;
    r = 8.5 ;

    /* formula for simple interest */
    si = p * n * r / 100 ;

    printf ( "%f\n" , si ) ;
    return 0 ;
}
```

Let us now understand this program in detail.

Form of a C Program

Form of a C program indicates how it has to be written/typed. There are certain rules about the form of a C program that are applicable to all C programs. These are as under:

- (a) Each instruction in a C program is written as a separate statement.
- (b) The statements in a program must appear in the same order in which we wish them to be executed.
- (c) Blank spaces may be inserted between two words to improve the readability of the statement.
- (d) All statements should be in lower case letters.
- (e) C has no specific rules for the position at which a statement is to be written in a given line. That's why it is often called a free-form language.
- (f) Every C statement must end with a semicolon (;). Thus ; acts as a statement terminator.

Comments in a C Program

Comments are used in a C program to clarify either the purpose of the program or the purpose of some statement in the program. It is a good practice to begin a program with a comment indicating the purpose of the program, its author and the date on which the program was written.

Here are a few things that you must remember while writing comments in a C program:

- (a) Comment about the program should be enclosed within `/* */`. Thus, the first two statements in our program are comments.
- (b) Sometimes it is not very obvious as to what a particular statement in a program accomplishes. At such times it is worthwhile mentioning the purpose of the statement (or a set of statements) using a comment. For example:

```
/* formula for simple interest */  
si = p * n * r / 100 ;
```

- (c) Any number of comments can be written at any place in the program. For example, a comment can be written before the

statement, after the statement or within the statement as shown below.

```
/* formula */ si = p * n * r / 100 ;  
si = p * n * r / 100 ; /* formula */  
si = p * n * r /* formula */ 100 ;
```

- (d) The normal language rules do not apply to text written within `/* .. */`. Thus we can type this text in small case, capital or a combination. This is because the comments are solely given for the understanding of the programmer or the fellow programmers and are completely ignored by the compiler.
- (e) Comments cannot be nested. This means one comment cannot be written inside another comment. For example,

```
/* Cal of SI /* Author: gekay date: 25/06/2016 */ */
```

is invalid.

- (f) A comment can be split over more than one line, as in,

```
/* This comment has  
three lines  
in it */
```

Such a comment is often called a multi-line comment.

- (g) ANSI C permits comments to be written in the following way:

```
// Calculation of simple interest  
// Formula
```

What is *main()*?

main() forms a crucial part of any C program. Let us understand its purpose as well as its intricacies.

- (a) **main()** is a function. A function is nothing but a container for a set of statements. In a C program there can be multiple functions. To begin with, we would concentrate only on those programs which have only one function. The name of this function has to be **main()**, it cannot be anything else. All statements that belong to **main()** are enclosed within a pair of braces `{ }` as shown below.

```
int main( )  
{  
    statement 1 ;  
    statement 2 ;  
    statement 3 ;  
}
```

- (b) The way functions in a calculator return a value, similarly, functions in C also return a value. **main()** function always returns an integer value, hence there is an **int** before **main()**. The integer value that we are returning is 0. 0 indicates success. If for any reason the statements in **main()** fail to do their intended work we can return a non-zero number from **main()**. This would indicate failure.
- (c) Some compilers like Turbo C/C++ even permit us to return nothing from **main()**. In such a case we should precede it with the keyword **void**. But this is non-standard way of writing the **main()** function. We would discuss functions and their working in great detail in Chapter 8.

Variables and their Usage

We have learnt constants and variables in isolation. Let us understand their significance with reference to our first C program.

- (a) Any variable used in the program must be declared before using it. For example,

```
int  p, n ;           /* declaration */  
float r, si ;         /* declaration */  
si = p * n * r / 100 ; /* usage */
```

- (b) In the statement,

```
si = p * n * r / 100 ;
```

***** and **/** are the arithmetic operators. The arithmetic operators available in C are **+**, **-**, ***** and **/**. C is very rich in operators. There are as many as 45 operators available in C.

Surprisingly there is no operator for exponentiation... a slip, which can be forgiven considering the fact that C has been developed by an individual, not by a committee.

printf() and its Purpose

C does not contain any instruction to display output on the screen. All output to screen is achieved using readymade library functions. One such function is **printf()**. Let us understand this function with respect to our program.

- (a) Once the value of **si** is calculated it needs to be displayed on the screen. We have used **printf()** to do so.
- (b) For us to be able to use the **printf()** function, it is necessary to use **#include <stdio.h>** at the beginning of the program. **#include** is a preprocessor directive. Its purpose will be clarified in Chapter 8. For now, use it whenever you use **printf()**.
- (c) The general form of **printf()** function is,

```
printf ( "<format string>", <list of variables> );
```

<format string> can contain,

```
%f for printing real values
%d for printing integer values
%c for printing character values
```

In addition to format specifiers like **%f**, **%d** and **%c**, the format string may also contain any other characters. These characters are printed as they are when **printf()** is executed.

- (d) Given below are some more examples of usage of **printf()** function:

```
printf ( "%f", si );
printf ( "%d %d %f %f", p, n, r, si );
printf ( "Simple interest = Rs. %f", si );
printf ( "Principal = %d \nRate = %f", p, r );
```

The output of the last statement would look like this...

```
Principal = 1000
Rate = 8.500000
```

What is **'\n'** doing in this statement? It is called newline and it takes the cursor to the next line. Therefore, you get the output split over two lines. **'\n'** is one of the several Escape Sequences available in C. These are discussed in detail in Chapter 18. Right now, all that we

can say is '\n' comes in handy when we want to format the output properly on separate lines.

- (e) **printf()** can not only print values of variables, it can also print the result of an expression. An expression is nothing but a valid combination of constants, variables and operators. Thus, 3, 3 + 2, c and a + b * c – d all are valid expressions. The results of these expressions can be printed as shown below.

```
printf ( "%d %d %d %d", 3, 3 + 2, c, a + b * c – d );
```

Note that **3** and **c** also represent valid expressions.

Compilation and Execution

Once you have written the program you need to type it and instruct the machine to execute it. To type your C program you need another program called Editor. Once the program has been typed it needs to be converted to machine language instructions before the machine can execute it. To carry out this conversion we need another program called Compiler. Compiler vendors provide an Integrated Development Environment (IDE) which consists of an Editor as well as the Compiler.

There are several IDEs available in the market targeted towards different operating systems and microprocessors. Details of which IDE to use, how to procure, install and use it are given in Appendix A. Please go through this appendix and install the right IDE on your machine before you try rest of the programs in this book.

Receiving Input

In the program discussed above we assumed the values of **p**, **n** and **r** to be 1000, 3 and 8.5. Every time we run the program we would get the same value for simple interest. If we want to calculate simple interest for some other set of values then we are required to make the relevant changes in the program, and again compile and execute it. Thus the program is not general enough to calculate simple interest for any set of values without being required to make a change in the program. Moreover, if you distribute the EXE file of this program to somebody he would not even be able to make changes in the program. Hence it is a good practice to create a program that is general enough to work for any set of values.

To make the program general, the program itself should ask the user to supply the values of **p**, **n** and **r** through the keyboard during execution. This can be achieved using a function called **scanf()**. This function is a counter-part of the **printf()** function. **printf()** outputs the values to the screen whereas **scanf()** receives them from the keyboard. This is illustrated in the program given below.

```
/* Calculation of simple interest */
/* Author gekay Date 25/06/2016 */
#include <stdio.h>
int main( )
{
    int p, n ;
    float r, si ;

    printf ( "Enter values of p, n, r" ) ;
    scanf ( "%d %d %f", &p, &n, &r ) ;

    si = p * n * r / 100 ;
    printf ( "%f\n", si ) ;
    return 0 ;
}
```

The first **printf()** outputs the message ‘Enter values of p, n, r’ on the screen. Here we have not used any expression in **printf()** which means that using expressions in **printf()** is optional.

Note the use of ampersand (&) before the variables in the **scanf()** function is a must. **&** is an ‘Address of’ operator. It gives the location number (address) used by the variable in memory. When we say **&a**, we are telling **scanf()** at which memory location should it store the value supplied by the user from the keyboard. The detailed working of the **&** operator would be taken up in Chapter 6.

Note that a blank, a tab or a new line must separate the values supplied to **scanf()**. A blank is created using a spacebar, tab using the Tab key and new line using the Enter key. This is shown below.

Ex.: The three values separated by blank:

```
1000 5 15.5
```

Ex.: The three values separated by tab:

```
1000 5 15.5
```

Ex.: The three values separated by newline:

```
1000
5
15.5
```

So much for the tips. How about writing another program to give you a feel of things. Here it is...

```
/* Just for fun. Author: Bozo */
#include <stdio.h>
int main( )
{
    int num ;

    printf ( "Enter a number" ) ;
    scanf ( "%d", &num ) ;

    printf ( "Now I am letting you on a secret...\n" ) ;
    printf ( "You have just entered the number %d\n", num ) ;
    return 0 ;
}
```

Summary

- (a) Constant is an entity whose value remains fixed.
- (b) Variable is an entity whose value can change during course of execution of the program.
- (c) Keywords are special words whose meaning is known to the Compiler.
- (d) There are certain rules that must be followed while building constants or variables.
- (e) The three primary constants and variable types in C are integer, float and character.
- (f) We should not use a keyword as a variable name.
- (g) Comments should be used to indicate the purpose of the program or statements in a program.
- (h) Comments can be single line or multi-line.

- (i) Input/output in C can be achieved using `scanf( )` and `printf( )` functions.

Exercise

[A] Which of the following are invalid C constants and why?

'3.15'	35,550	3.25e2
2e-3	'eLearning'	"show"
'Quest'	2 ³	4 6 5 2

[B] Which of the following are invalid variable names and why?

B'day	int	\$hello
#HASH	dot.	number
totalArea	_main()	temp_in_Deg
total%	1st	stack-queue
variable name	%name%	salary

[C] State whether the following statements are True or False:

- C language has been developed by Dennis Ritchie.
- Operating systems like Windows, UNIX, Linux and Android are written in C.
- C language programs can easily interact with hardware of a PC / Laptop.
- A real constant in C can be expressed in both Fractional and Exponential forms.
- A character variable can at a time store only one character.
- The maximum value that an integer constant can have varies from one compiler to another.
- Usually all C statements are written in small case letters.
- Spaces may be inserted between two words in a C statement.
- Spaces cannot be present within a variable name.
- C programs are converted into machine language with the help of a program called Editor.
- Most development environments provide an Editor to type a C program and a Compiler to convert it into machine language.
- `int`, `char`, `float`, `real`, `integer`, `character`, `char`, `main`, `printf` and `scanf` all are keywords.

[D] Match the following:

(a) \n	Literal
(b) 3.145	Statement terminator
(c) -6513	Character constant
(d) 'D'	Escape sequence
(e) 4.25e-3	Input function
(f) main()	Function
(g) %f, %d, %c	Integer constant
(h) ;	Address of operator
(i) Constant	Output function
(j) Variable	Format specifier
(k) &	Exponential form
(l) printf()	Real constant
(m) scanf()	Identifier

[E] Point out the errors, if any, in the following programs:

- (a)

```
int main( )
{
    int a, float b, int c ;
    a = 25 ; b = 3.24 ; c = a + b * b - 35 ;
}
```
- (b)

```
/* Calculation of average
/* Author: Sanjay */
/* Place – Whispering Bytes */
*/

#include <stdio.h>
int main( )
{
    int a = 35 ; float b = 3.24 ;
    printf ( "%d %f %d", a, b + 1.5, 235 ) ;
}
```
- (c)

```
#include <stdio.h>
int main( )
{
    int a, b, c ;
    scanf ( "%d %d %d", a, b, c ) ;
}
```
- (d)

```
#include <stdio.h>
```



```
int main( )
{
    int m1, m2, m3
    printf ( "Enter values of marks in 3 subjects" )
    scanf ( "%d %d %d", &m1, &m2, &m3 )
    printf ( "You entered %d %d %d", m1, m2, m3 ) ;
}
```

[F] Attempt the following:

- (a) Ramesh's basic salary is input through the keyboard. His dearness allowance is 40% of basic salary, and house rent allowance is 20% of basic salary. Write a program to calculate his gross salary.
- (b) The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches and centimeters.
- (c) If the marks obtained by a student in five different subjects are input through the keyboard, write a program to find out the aggregate marks and percentage marks obtained by the student. Assume that the maximum marks that can be obtained by a student in each subject is 100.
- (d) Temperature of a city in Fahrenheit degrees is input through the keyboard. Write a program to convert this temperature into Centigrade degrees.
- (e) The length and breadth of a rectangle and radius of a circle are input through the keyboard. Write a program to calculate the area and perimeter of the rectangle, and the area and circumference of the circle.
- (f) Paper of size A0 has dimensions 1189 mm x 841 mm. Each subsequent size A(n) is defined as A(n-1) cut in half parallel to its shorter sides. Thus paper of size A1 would have dimensions 841 mm x 594 mm. Write a program to calculate and print paper sizes A0, A1, A2, ... A8.

2

C Instructions

- **Types of Instructions**
- **Type Declaration Instruction**
- **Arithmetic Instruction**
- **Integer and Float Conversions**
- **Type Conversion in Assignments**
- **Hierarchy of Operations**
- **Associativity of Operators**
- **Control Instructions**
- **Summary**
- **Exercise**



2 ***C Instructions***

- Types of Instructions
- Type Declaration Instruction
- Arithmetic Instruction
- Integer and Float Conversions
- Type Conversion in Assignments
- Hierarchy of Operations
- Associativity of Operators
- Control Instruction in C
- Summary
- Exercise

A program is nothing but a set of instructions. The program behaves as per the instructions that we give in it. Different instructions help us achieve different tasks in a program. In the last chapter we saw how to write simple C programs. In these programs knowingly or unknowingly we have used instructions to achieve the intended purpose of the program. In this chapter we would explore the instructions that we have used in these programs.

Types of Instructions

There are basically three types of instructions in C:

- (a) Type Declaration Instruction – This instruction is used to declare the type of variables used in a C program.
- (b) Arithmetic Instruction – This instruction is used to perform arithmetic operations on constants and variables.
- (c) Control Instruction – This instruction is used to control the sequence of execution of various statements in a C program.

Since, the elementary C programs would usually contain only the type declaration and the arithmetic instructions; we would discuss only these two instructions at this stage. The control instruction would be discussed in detail in the subsequent chapters.

Type Declaration Instruction

This instruction is used to declare the type of variables being used in the program. Any variable used in the program must be declared before using it in any statement. The type declaration statement is written at the beginning of **main()** function.

```
Ex.: int  bas ;  
      float  rs, grosssal ;  
      char  name, code ;
```

There are several subtle variations of the type declaration instruction. These are discussed below:

- (a) While declaring the type of variable we can also initialize it as shown below.

```
int  i = 10, j = 25 ;  
float a = 1.5, b = 1.99 + 2.4 * 1.44 ;
```

- (b) The order in which we define the variables is sometimes important sometimes not. For example,

```
int i = 10, j = 25 ;
```

is same as

```
int j = 25, i = 10 ;
```

However,

```
float a = 1.5, b = a + 3.1 ;
```

is alright, but

```
float b = a + 3.1, a = 1.5 ;
```

is not. This is because here we are trying to use **a** before defining it.

- (c) The following statements would work

```
int a, b, c, d ;  
a = b = c = 10 ;
```

However, the following statement would not work

```
int a = b = c = d = 10 ;
```

Once again we are trying to use **b** (to assign to **a**) before defining it.

Arithmetic Instruction

A C arithmetic instruction consists of a variable name on the left hand side of = and variable names and constants on the right hand side of =. The variables and constants appearing on the right hand side of = are connected by arithmetic operators like +, -, *, and /.

```
Ex.: int ad ;  
float kot, deta, alpha, beta, gamma ;  
ad = 3200 ;  
kot = 0.0056 ;  
deta = alpha * beta / gamma + 3.2 * 2 / 5 ;
```

Here,

*****, **/**, **-**, **+** are the arithmetic operators.

= is the assignment operator.

2, 5 and 3200 are integer constants.

3.2 and 0.0056 are real constants.

ad is an integer variable.

kot, deta, alpha, beta, gamma are real variables.

The variables and constants together are called 'operands'. While executing an arithmetic statement the operands on right hand side are operated upon by the 'arithmetic operators' and the result is then assigned, using the assignment operator, to the variable on left-hand side.

A C arithmetic statement could be of three types. These are as follows:

- (a) Integer mode arithmetic statement – In this statement all operands are either integer variables or integer constants.

```
Ex.: int i, king, issac, noteit ;  
      i = i + 1 ;  
      king = issac * 234 + noteit - 7689 ;
```

- (b) Real mode arithmetic statement – In this statement all operands are either real constants or real variables.

```
Ex.: float qbee, antink, si, prin, anoy, roi ;  
      qbee = antink + 23.123 / 4.5 * 0.3442 ;  
      si = prin * anoy * roi / 100.0 ;
```

- (c) Mixed mode arithmetic statement – In this statement some operands are integers and some operands are real.

```
Ex.: float si, prin, anoy, roi, avg ;  
      int a, b, c, num ;  
      si = prin * anoy * roi / 100.0 ;  
      avg = ( a + b + c + num ) / 4 ;
```

Though Arithmetic instructions look simple to use, one often commits mistakes in writing them. Let us take a closer look at these statements. Note the following points carefully:

- (a) C allows only one variable on left-hand side of `=`. That is, `z = k * l` is legal, whereas `k * l = z` is illegal.
- (b) In addition to the division operator C also provides a modular division operator. This operator returns the remainder on dividing one integer with another. Thus the expression `10 / 2` yields 5, whereas, `10 % 2` yields 0. Note that the modulus operator (`%`) cannot be applied on a float. Also note that on using `%` the sign of the remainder is always same as the sign of the numerator. Thus `-5 % 2` yields `-1`, whereas, `5 % -2` yields `1`.
- (c) An arithmetic instruction is at times used for storing character constants in character variables.

```
char a, b, d;  
a = 'F';  
b = 'G';  
d = '+';
```

When we do this, the ASCII values of the characters are stored in the variables. ASCII codes are used to represent any character in memory. For example, ASCII codes of 'F' and 'G' are 01000110 and 01000111. ASCII values are nothing but the decimal equivalent of ASCII codes. Thus ASCII values of 'F' and 'G' are 70 and 71.

- (d) Arithmetic operations can be performed on **ints**, **floats** and **chars**.
Thus the statements,

```
char x, y;  
int z;  
x = 'a';  
y = 'b';  
z = x + y;
```

are perfectly valid, since the addition is performed on the ASCII values of the characters and not on characters themselves. The ASCII values of 'a' and 'b' are 97 and 98, and hence can definitely be added.

- (e) No operator is assumed to be present. It must be written explicitly. In the following example, the multiplication operator after `b` must be explicitly written.

<code>a = c.d.b(xy)</code>	usual arithmetic statement
<code>a = c * d * b * (x * y)</code>	C statement

- (f) There is no operator in C to perform exponentiation operation. Exponentiation has to be carried out as shown below:

```
#include <math.h>
#include <stdio.h>
int main( )
{
    float a ;
    a = pow ( 3.0, 2.0 ) ;
    printf ( "%f", a ) ;
}
```

Here **pow()** function is a standard library function. It is being used to raise 3.0 to the power of 2.0. The **pow()** function works only with real numbers, hence we have used 3.0 and 2.0 instead of 3 and 2.

#include <math.h> is a preprocessor directive. It is being used here to ensure that the **pow()** function works correctly. We would learn more about standard library functions in Chapter 6 and about preprocessor in Chapter 8.

You can explore other mathematical functions like **abs()**, **sqrt()**, **sin()**, **cos()**, **tan()**, etc., declared in **math.h** on your own.

Integer and Float Conversions

In order to effectively develop C programs, it will be necessary to understand the rules that are used for the implicit conversion of floating point and integer values in C. These are mentioned below. Note them carefully.

- (a) An arithmetic operation between an integer and integer always yields an integer result.
- (b) An operation between a real and real always yields a real result.

- (c) An operation between an integer and real always yields a real result. In this operation the integer is first promoted to a real and then the operation is performed. Hence the result is real.

I think a few practical examples shown in Figure 2.1 would put the issue beyond doubt.

Operation	Result	Operation	Result
5 / 2	2	2 / 5	0
5.0 / 2	2.5	2.0 / 5	0.4
5 / 2.0	2.5	2 / 5.0	0.4
5.0 / 2.0	2.5	2.0 / 5.0	0.4

Figure 2.1

Type Conversion in Assignments

It may so happen that the type of the expression on right hand side and the type of the variable on the left-hand side of an assignment operator may not be same. In such a case, the value of the expression is promoted or demoted depending on the type of the variable on left-hand side of =.

For example, consider the following assignment statements.

```
int i;
float b;
i = 3.5;
b = 30;
```

Here in the first assignment statement, though the expression's value is a **float** (3.5), it cannot be stored in **i** since it is an **int**. In such a case, the **float** is demoted to an **int** and then its value is stored. Hence what gets stored in **i** is 3. Exactly opposite happens in the next statement. Here, 30 is promoted to 30.0 and then stored in **b**, since **b** being a **float** variable cannot hold anything except a **float** value.

Instead of a simple expression used in the above examples, if a complex expression occurs, still the same rules apply. For example, consider the following program fragment.

```
float a, b, c; int s;
```

```
s = a * b * c / 100 + 32 / 4 - 3 * 1.1 ;
```

Here, in the assignment statement, some operands are **ints** whereas others are **floats**. As we know, during evaluation of the expression, the **ints** would be promoted to **floats** and the result of the expression would be a **float**. But when this **float** value is assigned to **s** it is again demoted to an **int** and then stored in **s**.

Observe the results of the arithmetic statements shown in Figure 2.2. It has been assumed that **k** is an integer variable and **a** is a real variable.

Arithmetic Instruction	Result	Arithmetic Instruction	Result
k = 2 / 9	0	a = 2 / 9	0.0
k = 2.0 / 9	0	a = 2.0 / 9	0.222222
k = 2 / 9.0	0	a = 2 / 9.0	0.222222
k = 2.0 / 9.0	0	a = 2.0 / 9.0	0.222222
k = 9 / 2	4	a = 9 / 2	4.0
k = 9.0 / 2	4	a = 9.0 / 2	4.5
k = 9 / 2.0	4	a = 9 / 2.0	4.5
k = 9.0 / 2.0	4	a = 9.0 / 2.0	4.5

Figure 2.2

Note that though the following statements give the same result, 0, the results are obtained differently.

```
k = 2 / 9 ;
k = 2.0 / 9 ;
```

In the first statement, since both 2 and 9 are integers, the result is an integer, i.e. 0. This 0 is then assigned to **k**. In the second statement 9 is promoted to 9.0 and then the division is performed. Division yields 0.222222. However, this cannot be stored in **k**, **k** being an **int**. Hence it gets demoted to 0 and then stored in **k**.

Hierarchy of Operations

While executing an arithmetic statement that has multiple operators, there might be some issues about their evaluation. For example, does the expression $2 * x - 3 * y$ correspond to $(2x)-(3y)$ or to $2(x-3y)$? Similarly, does $A / B * C$ correspond to $A / (B * C)$ or to $(A / B) * C$? To answer these questions satisfactorily, one has to understand the 'hierarchy' of operations. The priority or precedence in which the

operations in an arithmetic statement are performed is called the hierarchy of operations. The hierarchy of commonly used operators is shown in Figure 2.3.

Priority	Operators	Description
1 st	* / %	Multiplication, Division, Modular division
2 nd	+ -	Addition, Subtraction
3 rd	=	Assignment

Figure 2.3

Now a few tips about usage of operators in general.

- Within parentheses the same hierarchy as mentioned in Figure 2.3 is operative. Also, if there are more than one set of parentheses, the operations within the innermost parentheses would be performed first, followed by the operations within the second innermost pair and so on.
- We must always remember to use pairs of parentheses. A careless imbalance of the right and left parentheses is a common error. Best way to avoid this error is to type () and then type an expression inside it.

A few examples would clarify the issue further.

Example 2.1: Determine the hierarchy of operations and evaluate the following expression, assuming that *i* is an integer variable:

$$i = 2 * 3 / 4 + 4 / 4 + 8 - 2 + 5 / 8$$

Stepwise evaluation of this expression is shown below:

$i = 2 * 3 / 4 + 4 / 4 + 8 - 2 + 5 / 8$	
$i = 6 / 4 + 4 / 4 + 8 - 2 + 5 / 8$	operation: *
$i = 1 + 4 / 4 + 8 - 2 + 5 / 8$	operation: /
$i = 1 + 1 + 8 - 2 + 5 / 8$	operation: /
$i = 1 + 1 + 8 - 2 + 0$	operation: /
$i = 2 + 8 - 2 + 0$	operation: +
$i = 10 - 2 + 0$	operation: +
$i = 8 + 0$	operation: -
$i = 8$	operation: +

Note that $6 / 4$ gives 1 and not 1.5. This so happens because 6 and 4 both are integers and therefore $6 / 4$ must evaluate to an integer. Similarly $5 / 8$ evaluates to zero, since 5 and 8 are integers and hence $5 / 8$ must return an integer value.

Example 2.2: Determine the hierarchy of operations and evaluate the following expression, assuming that **kk** is a float variable:

```
kk = 3 / 2 * 4 + 3 / 8
```

Stepwise evaluation of this expression is shown below:

kk = 3 / 2 * 4 + 3 / 8	
kk = 1 * 4 + 3 / 8	operation: /
kk = 4 + 3 / 8	operation: *
kk = 4 + 0	operation: /
kk = 4	operation: +

Note that $3 / 8$ gives zero, again for the same reason mentioned in the previous example.

All operators in C are ranked according to their precedence. And mind you, there are as many as 45 odd operators in C, and these can affect the evaluation of an expression in subtle and unexpected ways if we aren't careful. Unfortunately, there are no simple rules that one can follow, such as "BODMAS" that tells algebra students in which order does an expression evaluate. We have not encountered many out of these 45 operators, so we won't pursue the subject of precedence any further here. However, it can be realized at this stage that it would be almost impossible to remember the precedence of all these operators. So a full-fledged list of all operators and their precedence is given in Appendix A. This may sound daunting, but when its contents are absorbed in small bites, it becomes more palatable.

So far we have seen how arithmetic statements written in C are evaluated. But our knowledge would be incomplete unless we know how to convert a general algebraic expression to a C statement. C can handle any complex algebraic expressions with ease. Some examples of algebraic expressions and their equivalent C expressions are shown in Figure 2.4.

Algebraic Expression	C Expression
$a \times b - c \times d$ $(m + n)(a + b)$ $3x^2 + 2x + 5$ $\frac{a + b + c}{d + e}$ $\left[\frac{2BY}{d + 1} - \frac{x}{3(z + y)} \right]$	$a * b - c * d$ $(m + n) * (a + b)$ $3 * x * x + 2 * x + 5$ $(a + b + c) / (d + e)$ $2 * b * y / (d + 1) - x / 3 * (z + y)$

Figure 2.4

Associativity of Operators

When an expression contains two operators of equal priority the tie between them is settled using the associativity of the operators. All operators in C have either Left to Right associativity or Right to Left associativity. Let us understand this with the help of a few examples.

Consider the expression $a = 3 / 2 * 5$;

Here there is a tie between operators of same priority, that is between / and *. This tie is settled using the associativity of / and *. Both enjoy Left to Right associativity. Therefore firstly / operation is done followed by *.

Consider one more expression.

$a = b = 3$;

Here both assignment operators have the same priority. So order of operations is decided using associativity of = operator. = associates from Right to Left. Therefore, second = is performed earlier than first =.

Consider yet another expression.

$z = a * b + c / d$;

Here * and / enjoys same priority and same associativity (Left to Right). Compiler is free to perform * or / operation as per its convenience, since no matter which is performed earlier, the result would be same.

Appendix B gives the associativity of all the operators available in C. Note that the precedence and associativity of all operators is predetermined and we cannot change it.

Control Instructions

As the name suggests, the 'Control Instructions' enable us to specify the order in which the various instructions in a program are to be executed by the computer. In other words, the control instructions determine the 'flow of control' in a program. There are four types of control instructions in C. They are:

- (a) Sequence Control Instruction
- (b) Selection or Decision Control Instruction
- (c) Repetition or Loop Control Instruction
- (d) Case Control Instruction

The Sequence control instruction ensures that the instructions are executed in the same order in which they appear in the program. Decision and Case control instructions allow the computer to take a decision as to which instruction is to be executed next. The Loop control instruction helps computer to execute a group of statements repeatedly. In the following chapters we are going to discuss these instructions in detail.

Summary

- (a) Instructions in a program control the behavior/working of the program.
- (b) A C program can contain three types of instructions—Type declaration instruction, Arithmetic instruction, Control instruction.
- (c) An expression may contain any sequence of constants, variables and operators.
- (d) An expression is evaluated based on the hierarchy or precedence of operators.
- (e) Operators having equal precedence are evaluated using associativity of operators.
- (f) Associativity of all operators is either left to right or right to left.

Exercise

[A] Point out the errors, if any, in the following C statements:

- (a) $x = (y + 3);$
- (b) $\text{cir} = 2 * 3.141593 * r;$
- (c) $\text{char} = '3';$
- (d) $4 / 3 * 3.14 * r * r * r = \text{vol_of_sphere};$
- (e) $\text{volume} = a^3;$
- (f) $\text{area} = 1 / 2 * \text{base} * \text{height};$
- (g) $\text{si} = p * r * n / 100;$
- (h) $\text{area of circle} = 3.14 * r * r;$
- (i) $\text{peri_of_tri} = a + b + c;$
- (j) $\text{slope} = (y2 - y1) \div (x2 - x1);$
- (k) $3 = b = 4 = a;$
- (l) $\text{count} = \text{count} + 1;$
- (m) $\text{char ch} = '25 \text{ Apr } 12';$

[B] Evaluate the following expressions and show their hierarchy.

- (a) $\text{ans} = 5 * b * b * x - 3 * a * y * y - 8 * b * b * x + 10 * a * y;$
($a = 3, b = 2, x = 5, y = 4$ assume **ans** to be an int)
- (b) $\text{res} = 4 * a * y / c - a * y / c;$
($a = 4, y = 1, c = 3$, assume **res** to be an int)
- (c) $s = c + a * y * y / b;$
($a = 2.2, b = 0.0, c = 4.1, y = 3.0$, assume **s** to be a float)
- (d) $R = x * x + 2 * x + 1 / 2 * x * x + x + 1;$
($x = 3.5$, assume **R** to be a float)

[C] Indicate the order in which the following expressions would be evaluated:

- (a) $g = 10 / 5 / 2 / 1;$
- (b) $b = 3 / 2 + 5 * 4 / 3;$
- (c) $a = b = c = 3 + 4;$
- (d) $x = 2 - 3 + 5 * 2 / 8 \% 3;$
- (e) $z = 5 \% 3 / 8 * 3 + 4$

(f) $y = z = -3 \% -8 / 2 + 7 ;$

[D] Convert the following algebraic expressions into equivalent C statements:

$$(a) \quad Z = \frac{(x+3)x^3}{(y-4)(y+5)}$$

$$(b) \quad R = \frac{2v + 6.22 (c + d)}{g + v}$$

$$(c) \quad A = \frac{7.7b (xy + a) / c - 0.8 + 2b}{(x + a) (1 / y)}$$

$$(d) \quad X = \frac{12x^3}{4x} + \frac{8x^2}{4x} + \frac{x}{8x} + \frac{8}{8x}$$

[E] What will be the output of the following programs:

```
(a) #include <stdio.h>
int main( )
{
    int i = 2, j = 3, k, l ;
    float a, b ;
    k = i / j * j ;
    l = j / i * i ;
    a = i / j * j ;
    b = j / i * i ;
    printf ( "%d %d %f %f\n", k, l, a, b ) ;
    return 0 ;
}
```

```
(b) #include <stdio.h>
int main( )
{
    int a, b, c, d ;
    a = 2 % 5 ;
    b = -2 % 5 ;
    c = 2 % -5 ;
    d = -2 % -5 ;
    printf ( "a = %d b = %d c = %d d = %d\n", a, b, c, d ) ;
}
```



```
    return 0 ;  
}
```

```
(c) # include <stdio.h>  
int main( )  
{  
    float a = 5, b = 2 ;  
    int c, d ;  
    c = a % b ;  
    d = a / 2 ;  
    printf ( "%d\n", d ) ;  
    return 0 ;  
}
```

```
(d) # include <stdio.h>  
int main( )  
{  
    printf ( "nn \n\n nn\n" ) ;  
    printf ( "nn /n/n nn/n" ) ;  
    return 0 ;  
}
```

```
(e) # include <stdio.h>  
int main( )  
{  
    int a, b ;  
    printf ( "Enter values of a and b" ) ;  
    scanf ( " %d %d ", &a, &b ) ;  
    printf ( "a = %d b = %d", a, b ) ;  
    return 0 ;  
}
```

[F] State whether the following statements are True or False:

- (a) * or /, + or – represents the correct hierarchy of arithmetic operators in C.
- (b) [] and { } can be used in Arithmetic instructions.
- (c) Hierarchy decides which operator is used first.
- (d) In C, Arithmetic instruction cannot contain constants on left side of =.
- (e) In C ** operator is used for exponentiation operation.

- (f) % operator cannot be used with floats.

[G] Fill in the blanks:

- (a) In $y = 10 * x / 2 + z$; ___ operation will be performed first.
 (b) If **a** is an integer variable, $a = 11 / 2$; will store ___ in **a**.
 (c) The expression, $a = 22 / 7 * 5 / 3$; would evaluate to _____.
 (d) The expression $x = -7 \% 2 - 8$ would evaluate to _____.
 (e) If **d** is a **float** the operation $d = 2 / 7.0$ would store ___ in **d**.

[H] Attempt the following:

- (a) If a five-digit number is input through the keyboard, write a program to calculate the sum of its digits. (Hint: Use the modulus operator '%')
 (b) If a five-digit number is input through the keyboard, write a program to reverse the number.
 (c) If lengths of three sides of a triangle are input through the keyboard, write a program to find the area of the triangle.
 (d) Write a program to receive Cartesian co-ordinates (x, y) of a point and convert them into polar co-ordinates (r, ϕ).

Hint: $r = \text{sqrt}(x^2 + y^2)$ and $\phi = \tan^{-1}(y/x)$

- (e) Write a program to receive values of latitude (L1, L2) and longitude (G1, G2), in degrees, of two places on the earth and output the distance (D) between them in nautical miles. The formula for distance in nautical miles is:

$$D = 3963 \cos^{-1}(\sin L1 \sin L2 + \cos L1 \cos L2 * \cos(G2 - G1))$$

- (f) Wind chill factor is the felt air temperature on exposed skin due to wind. The wind chill temperature is always lower than the air temperature, and is calculated as per the following formula:

$$wcf = 35.74 + 0.6215t + (0.4275t - 35.75) * v^{0.16}$$

where t is the temperature and v is the wind velocity. Write a program to receive values of t and v and calculate wind chill factor (wcf).

- (g) If value of an angle is input through the keyboard, write a program to print all its Trigonometric ratios.

-
- (h) Two numbers are input through the keyboard into two locations C and D. Write a program to interchange the contents of C and D.
 - (i) Consider a currency system in which there are notes of seven denominations, namely, Re. 1, Rs. 2, Rs. 5, Rs. 10, Rs. 50, Rs. 100. If a sum of Rs. N is entered through the keyboard, write a program to compute the smallest number of notes that will combine to give Rs. N.

3

Decision Control Instruction

- **Decisions! Decisions!**
- **The *if* Statement**
 - The Real Thing
 - Multiple Statements within *if*
- **The *if-else* Statement**
 - Nested if-elses
 - Forms of if
- **Summary**
- **Exercise**



3 *Decision*

Control Instruction

- Decisions! Decisions!
- The *if* Statement
 - The Real Thing
 - Multiple Statements within *if*
- The *if-else* Statement
 - Nested *if-elses*
 - Forms of *if*
- Summary
- Exercise

We all need to alter our actions in the face of changing circumstances. If the weather is fine, then I will go for a stroll. If the highway is busy, I would take a diversion. If the pitch takes spin, we would win the match. If you join our WhatsApp group, I would send you interesting videos. If you like this book, I would write the next edition. You can notice that all these decisions depend on some condition being met.

C language too must be able to perform different sets of actions depending on the circumstances. C has three major decision making instructions—the **if** statement, the **if-else** statement, and the **switch** statement. A fourth, somewhat less important structure is the one that uses conditional operators. In this chapter, we will explore **if** and the **if-else** statement. Conditional operators would be dealt with in Chapter 4 and **switch** statement in Chapter 7.

Decisions! Decisions!

In the programs written in Chapters 1 and 2, we have used sequence control instruction in which the various statements are executed sequentially, i.e., in the same order in which they appear in the program. In fact, to execute the instructions sequentially, we don't have to do anything at all. That is, by default, the instructions in a program are executed sequentially. However, in serious programming situations, seldom do we want the instructions to be executed sequentially. Many a time, we want a set of instructions to be executed in one situation, and an entirely different set of instructions to be executed in another situation. This kind of situation is dealt with in C programs using a decision control instruction. As mentioned earlier, a decision control instruction can be implemented in C using:

- (a) The **if** statement
- (b) The **if-else** statement
- (c) The conditional operators

Now let us learn each of these and their variations in turn.

The *if* Statement

C uses the keyword **if** to implement the decision control instruction. The general form of **if** statement looks like this:

```
if ( this condition is true )  
    execute this statement ;
```

The keyword **if** tells the compiler that what follows is a decision control instruction. The condition following the keyword **if** is always enclosed within a pair of parentheses. If the condition, whatever it is, is true, then the statement is executed. If the condition is not true, then the statement is not executed; instead the program skips past it. But how do we express the condition itself in C? And how do we evaluate its truth or falsity? As a general rule, we express a condition using C's 'relational' operators. The relational operators allow us to compare two values to see whether they are equal to each other, unequal, or whether one is greater than the other. Figure 3.1 shows how they look and how they are evaluated in C.

this expression	is true if
<code>x == y</code>	x is equal to y
<code>x != y</code>	x is not equal to y
<code>x < y</code>	x is less than y
<code>x > y</code>	x is greater than y
<code>x <= y</code>	x is less than or equal to y
<code>x >= y</code>	x is greater than or equal to y

Figure 3.1

The relational operators should be familiar to you except for the equality operator `==` and the inequality operator `!=`. Note that `=` is used for assignment, whereas, `==` is used for comparison of two quantities. Here is a simple program that demonstrates the use of **if** and the relational operators.

```
/* Demonstration of if statement */
#include <stdio.h>
int main( )
{
    int num ;

    printf ( "Enter a number less than 10 " );
    scanf ( "%d", &num );

    if ( num < 10 )
        printf ( "What an obedient servant you are !\n" );
}
```



```
    return 0 ;  
}
```

On execution of this program, if you type a number less than 10, you get a message on the screen through **printf()**. If you type some other number the program doesn't do anything. The flowchart given in Figure 3.2 would help you understand the flow of control in the program.

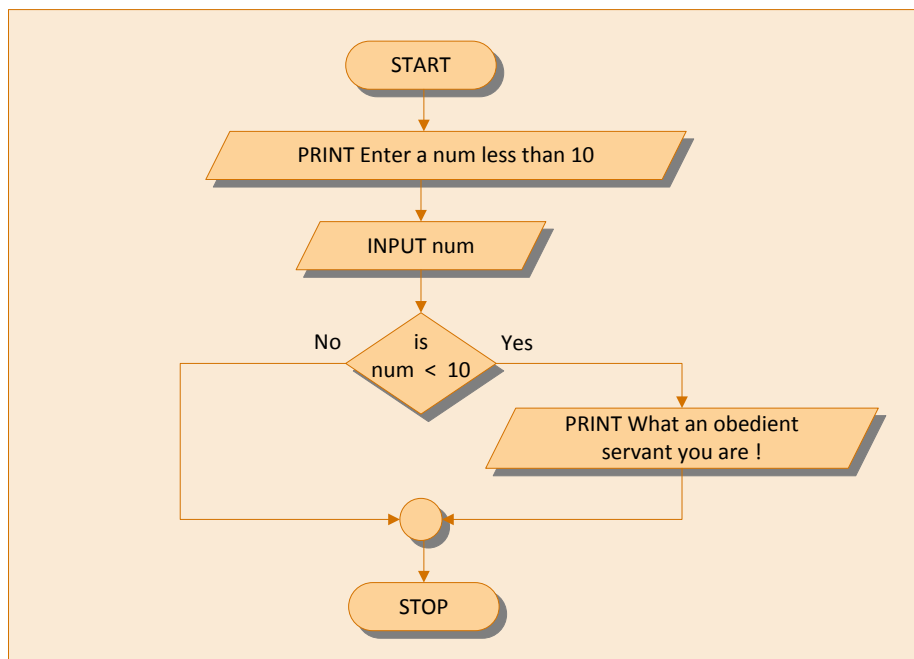


Figure 3.2

To make you comfortable with the decision control instruction, one more example has been given below. Study it carefully before reading further. To help you understand it easily, the program is accompanied by an appropriate flowchart shown in Figure 3.3.

Example 3.1: While purchasing certain items, a discount of 10% is offered if the quantity purchased is more than 1000. If quantity and price per item are input through the keyboard, write a program to calculate the total expenses.

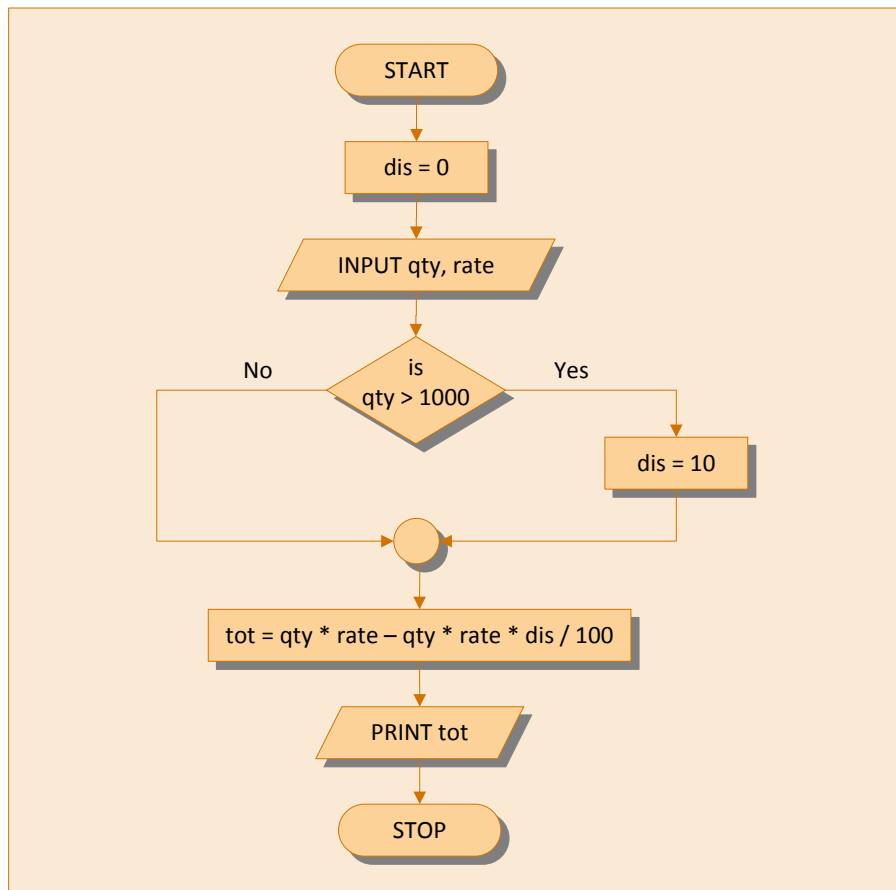


Figure 3.3

```
/* Calculation of total expenses */
#include <stdio.h>
int main( )
{
    int  qty, dis = 0 ;
    float  rate, tot ;
    printf ( "Enter quantity and rate " ) ;
    scanf ( "%d %f", &qty, &rate) ;

    if ( qty > 1000 )
        dis = 10 ;

    tot = ( qty * rate ) - ( qty * rate * dis / 100 ) ;
    printf ( "Total expenses = Rs. %f\n", tot ) ;
}
```

```
    return 0 ;  
}
```

Here is some sample interaction with the program.

```
Enter quantity and rate 1200 15.50  
Total expenses = Rs. 16740.000000
```

```
Enter quantity and rate 200 15.50  
Total expenses = Rs. 3100.000000
```

In the first run of the program, the condition evaluates to true, as 1200 (value of **qty**) is greater than 1000. Therefore, the variable **dis**, which was earlier set to 0, now gets a new value 10. Using this new value, total expenses are calculated and printed.

In the second run, the condition evaluates to false, as 200 (the value of **qty**) isn't greater than 1000. Thus, **dis**, which is earlier set to 0, remains 0, and hence the expression after the minus sign evaluates to zero, thereby offering no discount.

Is the statement **dis = 0** necessary? The answer is yes, since in C, a variable, if not specifically initialized, contains some unpredictable value (garbage value).

The Real Thing

We mentioned earlier that the general form of the **if** statement is as follows:

```
if ( condition )  
    statement ;
```

Here the expression can be any valid expression including a relational expression. We can even use arithmetic expressions in the **if** statement. For example all the following **if** statements are valid.

```
if ( 3 + 2 % 5 )  
    printf ( "This works" ) ;  
  
if ( a = 10 )  
    printf ( "Even this works" ) ;  
  
if ( -5 )
```

```
printf ( "Surprisingly even this works" ) ;
```

Note that in C a non-zero value is considered to be true, whereas a 0 is considered to be false. In the first **if**, the expression evaluates to **5** and since **5** is non-zero it is considered to be true. Hence the **printf()** gets executed.

In the second **if**, 10 gets assigned to **a** so the **if** is now reduced to **if (a)** or **if (10)**. Since 10 is non-zero, it is true hence again **printf()** goes to work.

In the third **if**, -5 is a non-zero number, hence true. So again **printf()** goes to work. In place of -5 even if a float like 3.14 were used it would be considered to be true. So the issue is not whether the number is integer or float, or whether it is positive or negative. Issue is whether it is zero or non-zero.

Multiple Statements within *if*

It may so happen that in a program we want more than one statement to be executed if the expression following **if** is satisfied. If such multiple statements are to be executed, then they must be placed within a pair of braces, as illustrated in the following example:

Example 3.2: The current year and the year in which the employee joined the organization are entered through the keyboard. If the number of years for which the employee has served the organization is greater than 3, then a bonus of Rs. 2500/- is given to the employee. If the years of service are not greater than 3, then the program should do nothing.

```
/* Calculation of bonus */
#include <stdio.h>
int main( )
{
    int  bonus, cy, yoj, yos ;

    printf ( "Enter current year and year of joining " ) ;
    scanf ( "%d %d", &cy, &yoy ) ;

    yos = cy - yoj ;

    if ( yos > 3 )
    {
```

```
        bonus = 2500 ;  
        printf ( "Bonus = Rs. %d\n", bonus ) ;  
    }  
  
    return 0 ;  
}
```

Observe that here the two statements to be executed on satisfaction of the condition have been enclosed within a pair of braces. If a pair of braces is not used, then the C compiler assumes that the programmer wants only the immediately next statement after the **if** to be executed on satisfaction of the condition. In other words, we can say that the default scope of the **if** statement is the immediately next statement after it. Figure 3.4 would help you understand the flow of control in the program.

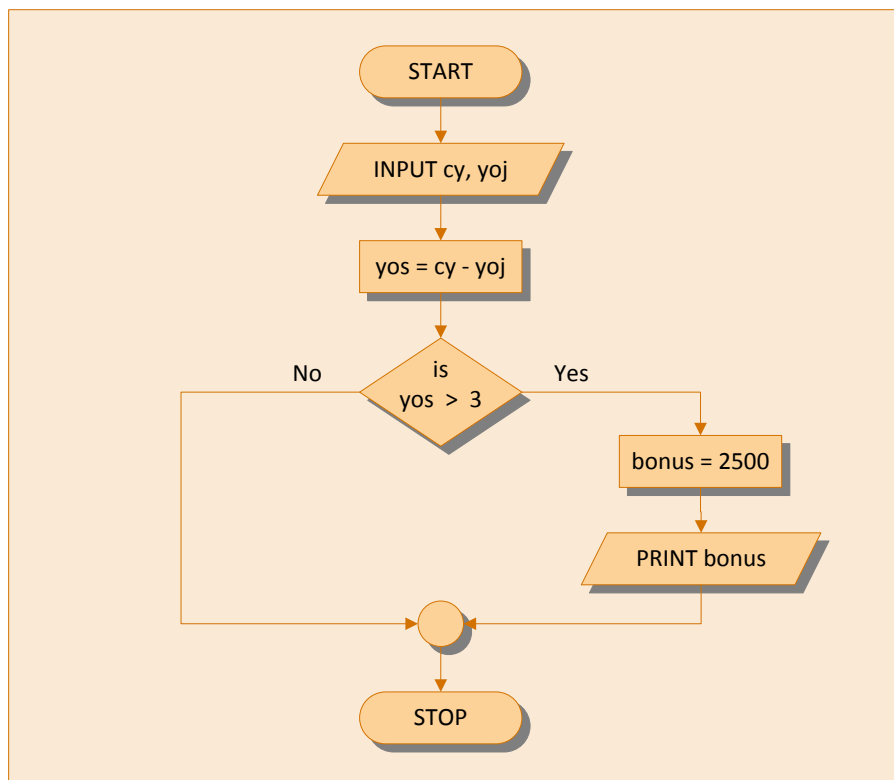


Figure 3.4

The *if-else* Statement

The **if** statement by itself will execute a single statement, or a group of statements, when the expression following **if** evaluates to true. It does nothing when the expression evaluates to false. Can we execute one group of statements if the expression evaluates to true and another group of statements if the expression evaluates to false? Of course! This is what is the purpose of the **else** statement that is demonstrated in the following example:

Example 3.3: In a company an employee is paid as under:

If his basic salary is less than Rs. 1500, then HRA = 10% of basic salary and DA = 90% of basic salary. If his salary is either equal to or above Rs. 1500, then HRA = Rs. 500 and DA = 98% of basic salary. If the employee's salary is input through the keyboard write a program to find his gross salary.

```
/* Calculation of gross salary */
#include <stdio.h>
int main( )
{
    float bs, gs, da, hra ;

    printf ( "Enter basic salary " ) ;
    scanf ( "%f", &bs ) ;

    if ( bs < 1500 )
    {
        hra = bs * 10 / 100 ;
        da = bs * 90 / 100 ;
    }
    else
    {
        hra = 500 ;
        da = bs * 98 / 100 ;
    }
    gs = bs + hra + da ;
    printf ( "gross salary = Rs. %f\n", gs ) ;

    return 0 ;
}
```

Figure 3.5 would help you understand the flow of control in the program.

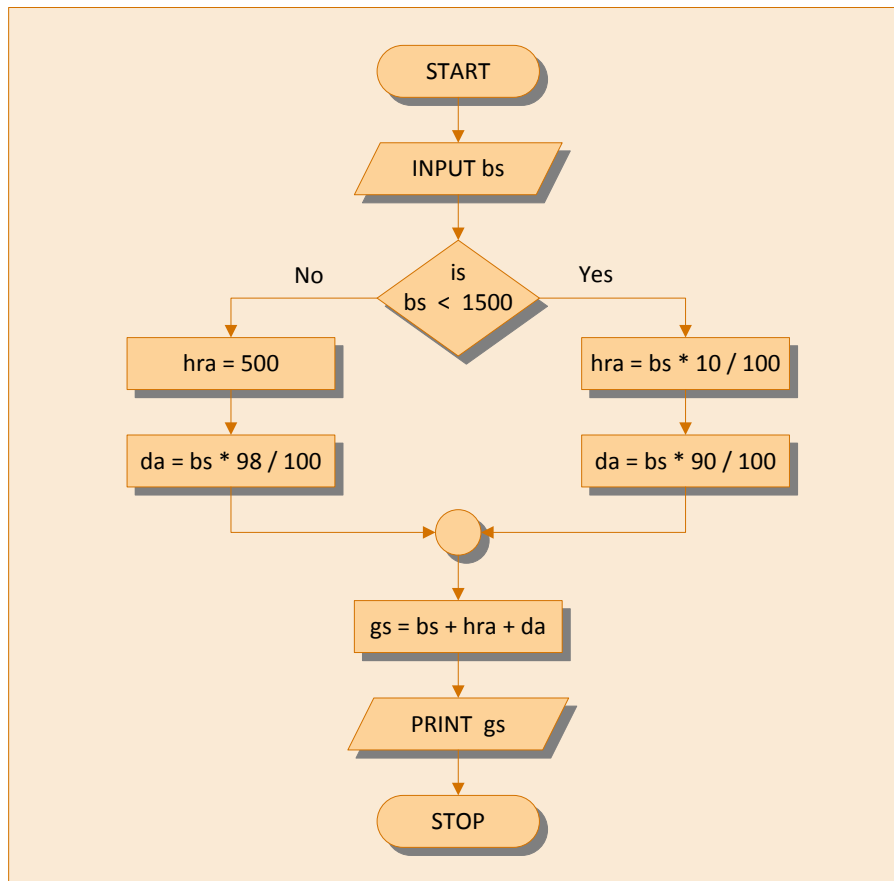


Figure 3.5

A few points worth noting about the program...

- (a) The group of statements after the **if** upto and not including the **else** is called an 'if block'. Similarly, the statements after the **else** form the 'else block'.
- (b) Notice that the **else** is written exactly below the **if**. The statements in the if block and those in the else block have been indented to the right. This formatting convention is followed throughout the book to enable you to understand the working of the program better.
- (c) Had there been only one statement to be executed in the if block and only one statement in the else block we could have dropped the pair of braces.

- (d) As with the **if** statement, the default scope of **else** is also the statement immediately after the **else**. To override this default scope, a pair of braces, as shown in the above example, must be used.

Nested *if-elses*

It is perfectly alright if we write an entire **if-else** construct within either the body of the **if** statement or the body of an **else** statement. This is called 'nesting' of **ifs**. This is shown in the following program:

```
/* A quick demo of nested if-else */
#include <stdio.h>
int main( )
{
    int i;

    printf ( "Enter either 1 or 2 " );
    scanf ( "%d", &i );

    if ( i == 1 )
        printf ( "You would go to heaven !\n" );
    else
    {
        if ( i == 2 )
            printf ( "Hell was created with you in mind\n" );
        else
            printf ( "How about mother earth !\n" );
    }

    return 0 ;
}
```

Note that the second **if-else** construct is nested in the first **else** statement. If the condition in the first **if** statement is false, then the condition in the second **if** statement is checked. If it is false as well, then the final **else** statement is executed.

You can see in the program how each time a **if-else** construct is nested within another **if-else** construct, it is also indented to add clarity to the program. Inculcate this habit of indentation; otherwise, you would end up writing programs which nobody (you included) can understand easily at a later date. Note that whether we indent or do not indent the

program, it doesn't alter the flow of execution of instructions in the program.

In the above program, an **if-else** occurs within the '**else** block' of the first **if** statement. Similarly, in some other program, an **if-else** may occur in the '**if** block' as well. There is no limit on how deeply the **ifs** and the **elses** can be nested.

Forms of *if*

The **if** statement can take any of the following forms:

- (a) `if ( condition )`
 `do this ;`
- (b) `if ( condition )`
 `{`
 `do this ;`
 `and this ;`
 `}`
- (c) `if ( condition )`
 `do this ;`
 `else`
 `do this ;`
- (d) `if ( condition )`
 `{`
 `do this ;`
 `and this ;`
 `}`
 `else`
 `{`
 `do this ;`
 `and this ;`
 `}`
- (e) `if ( condition )`
 `{`
 `if ( condition )`
 `do this ;`
 `else`
 `{`
 `do this ;`

```
        and this ;
    }
}
else
    do this ;

(f) if ( condition )
    do this ;
else
{
    if ( condition )
        do this ;
    else
    {
        do this ;
        and this ;
    }
}
```

Summary

- (a) There are three ways for taking decisions in a program. First way is to use the **if-else** statement, second way is to use the conditional operators and third way is to use the **switch** statement.
- (b) The condition associated with an if statement is built using relational operators <, >, <=, >=, == and !=.
- (c) The default scope of **if** statement is only the next statement. So, to execute more than one statement they must be written in a pair of braces.
- (d) An '**if** block' need not always be associated with an '**else** block'. However, an '**else** block' must always be associated with an **if**.
- (e) An **if-else** statement can be nested inside another **if-else** statement.

Exercise

[A] What will be the output of the following programs:

```
(a) #include <stdio.h>
int main( )
{
```

```
    int a = 300, b, c ;
    if ( a >= 400 )
        b = 300 ;
    c = 200 ;
    printf ( "%d %d\n", b, c ) ;
    return 0 ;
}
```

(b) # include <stdio.h>

```
int main( )
{
    int a = 500, b, c ;
    if ( a >= 400 )
        b = 300 ;
    c = 200 ;
    printf ( "%d %d\n", b, c ) ;
    return 0 ;
}
```

(c) # include <stdio.h>

```
int main( )
{
    int x = 10, y = 20 ;
    if ( x == y ) ;
        printf ( "%d %d\n", x, y ) ;
    return 0 ;
}
```

(d) # include <stdio.h>

```
int main( )
{
    int x = 3 ;
    float y = 3.0 ;
    if ( x == y )
        printf ( "x and y are equal\n" ) ;
    else
        printf ( "x and y are not equal\n" ) ;
    return 0 ;
}
```

(e) # include <stdio.h>

```
int main( )
{
```

```
int x = 3, y, z ;
y = x = 10 ;
z = x < 10 ;
printf ( "x = %d y = %d z = %d\n", x, y, z ) ;
return 0 ;
}
```

```
(f) # include <stdio.h>
int main( )
{
    int i = 65 ;
    char j = 'A' ;
    if ( i == j )
        printf ( "C is WOW\n" ) ;
    else
        printf ( "C is a headache\n" ) ;
    return 0 ;
}
```

[B] Point out the errors, if any, in the following programs:

```
(a) # include <stdio.h>
int main( )
{
    float a = 12.25, b = 12.52 ;
    if ( a = b )
        printf ( "a and b are equal\n" ) ;
    return 0 ;
}
```

```
(b) # include <stdio.h>
int main( )
{
    int j = 10, k = 12 ;
    if ( k >= j )
    {
        {
            k = j ;
            j = k ;
        }
    }
    return 0 ;
}
```

- (c) `# include <stdio.h>`
`int main( )`
`{`
 `if ( 'X' < 'x' )`
 `printf ( "ascii value of X is smaller than that of x\n" );`
`}`
- (d) `# include <stdio.h>`
`int main( )`
`{`
 `int x = 10 ;`
 `if ( x >= 2 ) then`
 `printf ( "%d\n", x );`
 `return 0 ;`
`}`
- (e) `# include <stdio.h>`
`int main( )`
`{`
 `int x = 10, y = 15 ;`
 `if ( x % 2 = y % 3 )`
 `printf ( "Carpathians\n" );`
`}`
- (f) `# include <stdio.h>`
`int main( )`
`{`
 `int x = 30, y = 40 ;`
 `if ( x == y )`
 `printf ( "x is equal to y\n" );`
 `elseif ( x > y )`
 `printf ( "x is greater than y\n" );`
 `elseif ( x < y )`
 `printf ( "x is less than y\n" );`
 `return 0 ;`
`}`
- (g) `# include <stdio.h>`
`int main( )`
`{`
 `int a, b ;`
 `scanf ( "%d %d", a, b );`
 `if ( a > b );`

```
        printf ( "This is a game\n" ) ;  
    else  
        printf ( "You have to play it\n" ) ;  
    return 0 ;  
}
```

[C] Attempt the following:

- (a) If cost price and selling price of an item are input through the keyboard, write a program to determine whether the seller has made profit or incurred loss. Also determine how much profit he made or loss he incurred.
- (b) Any integer is input through the keyboard. Write a program to find out whether it is an odd number or even number.
- (c) Any year is input through the keyboard. Write a program to determine whether the year is a leap year or not.
(Hint: Use the % (modulus) operator)
- (d) According to the Gregorian calendar, it was Monday on the date 01/01/01. If any year is input through the keyboard write a program to find out what is the day on 1st January of this year.
- (e) A five-digit number is entered through the keyboard. Write a program to obtain the reversed number and to determine whether the original and reversed numbers are equal or not.
- (f) If the ages of Ram, Shyam and Ajay are input through the keyboard, write a program to determine the youngest of the three.
- (g) Write a program to check whether a triangle is valid or not, when the three angles of the triangle are entered through the keyboard. A triangle is valid if the sum of all the three angles is equal to 180 degrees.
- (h) Write a program to find the absolute value of a number entered through the keyboard.
- (i) Given the length and breadth of a rectangle, write a program to find whether the area of the rectangle is greater than its perimeter. For example, the area of the rectangle with length = 5 and breadth = 4 is greater than its perimeter.
- (j) Given three points **(x1, y1)**, **(x2, y2)** and **(x3, y3)**, write a program to check if all the three points fall on one straight line.
- (k) Given the coordinates **(x, y)** of center of a circle and its radius, write a program that will determine whether a point lies inside the circle,

on the circle or outside the circle. (Hint: Use **sqrt()** and **pow()** functions)

- (l) Given a point **(x, y)**, write a program to find out if it lies on the X-axis, Y-axis or on the origin.

4

More Complex Decision Making

- **Use of Logical Operators**
 - The *else if* Clause
 - The ! Operator
 - Hierarchy of Operators Revisited
- **A Word of Caution**
- **The Conditional Operators**
- **Summary**
- **Exercise**



4 ***More Complex Decision Making***

- Use of Logical Operators
 - The *else if* Clause
 - The ! Operator
 - Hierarchy of Operators Revisited
- A Word of Caution
- The Conditional Operators
- Summary
- Exercise

We all face situations in real-life where the action that we carry out is based on multiple conditions. For example, I will join a company if the company allocates a metro location, gives me a good pay package and permits a joining period of 4 weeks. In programming too action performed may be based on result of multiple conditions. Such programming situations can be handled elegantly using Logical Operators. This chapter also explores the use of another type of operators called Conditional Operators.

Use of Logical Operators

C allows usage of three logical operators, namely, `&&`, `||` and `!`. These are to be read as 'AND', 'OR' and 'NOT', respectively.

There are several things to note about these logical operators. Most obviously, two of them are composed of double symbols: `||` and `&&`. Don't use the single symbol `|` and `&`. These single symbols also have a meaning. They are bitwise operators, which we would examine in Chapter 21.

The first two operators, `&&` and `||`, allow two or more conditions to be combined in an `if` statement. Let us see how they are used in a program. Consider the following example:

Example 4.1: The marks obtained by a student in 5 different subjects are input through the keyboard. The student gets a division as per the following rules:

Percentage above or equal to 60 - First division
Percentage between 50 and 59 - Second division
Percentage between 40 and 49 - Third division
Percentage less than 40 - Fail

Write a program to calculate the division obtained by the student.

There are two ways in which we can write a program for this example. These methods are given below.

```
/* Method – I */
#include <stdio.h>
int main( )
{
    int  m1, m2, m3, m4, m5, per ;

    printf ( "Enter marks in five subjects " ) ;
```

```
scanf ( "%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5 );
per = ( m1 + m2 + m3 + m4 + m5 ) * 100 / 500 ;

if ( per >= 60 )
    printf ( "First division\n" ) ;
else
{
    if ( per >= 50 )
        printf ( "Second division\n" ) ;
    else
    {
        if ( per >= 40 )
            printf ( "Third division\n" ) ;
        else
            printf ( "Fail\n" ) ;
    }
}

return 0 ;
}
```

This is a straight-forward program. Observe that the program uses nested **if-elses**. Though the program works fine, it has three disadvantages:

- (a) As the number of conditions go on increasing the level of indentation also goes on increasing. As a result, the whole program creeps to the right. So much so that entire program is not visible on the screen. So if something goes wrong with the program locating what is wrong where becomes difficult.
- (b) Care needs to be exercised to match the corresponding **ifs** and **elses**.
- (c) Care needs to be exercised to match the corresponding pair of braces.

All these three problems can be eliminated by usage of 'Logical Operators'. The following program illustrates this:

```
/* Method – II */
#include <stdio.h>
int main( )
{
```

```
int m1, m2, m3, m4, m5, per ;

printf ( "Enter marks in five subjects " ) ;
scanf ( "%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5 ) ;

per = ( m1 + m2 + m3 + m4 + m5 ) / 500 * 100 ;

if ( per >= 60 )
    printf ( "First division\n" ) ;

if ( ( per >= 50 ) && ( per < 60 ) )
    printf ( "Second division\n" ) ;

if ( ( per >= 40 ) && ( per < 50 ) )
    printf ( "Third division\n" ) ;

if ( per < 40 )
    printf ( "Fail\n" ) ;

return 0 ;
}
```

As can be seen from the second **if** statement, the **&&** operator is used to combine two conditions. 'Second division' gets printed if both the conditions evaluate to true. If one of the conditions evaluate to false then the whole thing is treated as false.

Two distinct advantages can be cited in favor of this program:

- (a) The matching (or do I say mismatching) of the **ifs** with their corresponding **elses** gets avoided, since there are no **elses** in this program.
- (b) In spite of using several conditions, the program doesn't creep to the right. In the previous program the statements went on creeping to the right. This effect becomes more pronounced as the number of conditions goes on increasing. This would make the task of matching the **ifs** with their corresponding **elses** and matching of opening and closing braces much more difficult.

There is a negative side to the program too. Even if the first condition turns out to be true, still all other conditions are checked. This will

increase the time of execution of the program. This can be avoided using the **else if** clause discussed in the next section.

The **else if** Clause

There is one more way in which we can write program for Example 4.1. This involves usage of **else if** blocks as shown below.

```
/* else if ladder demo */
#include <stdio.h>
int main( )
{
    int m1, m2, m3, m4, m5, per ;

    per = ( m1+ m2 + m3 + m4+ m5 ) / 500 * 100 ;

    if ( per >= 60 )
        printf ( "First division\n" ) ;
    else if ( per >= 50 )
        printf ( "Second division\n" ) ;
    else if ( per >= 40 )
        printf ( "Third division\n" ) ;
    else
        printf ( "fail\n" ) ;

    return 0 ;
}
```

You can note that this program reduces the indentation of the statements. In this case, every **else** is associated with its previous **if**. The last **else** goes to work only if all the conditions fail. Also, if a condition is satisfied, other conditions below it are not checked. Even in **else if** ladder, the last **else** is optional.

Note that the **else if** clause is nothing different. It is just a way of rearranging the **else** with the **if** that follows it. This would be evident if you look at the following code:

```
/* code using ifs */
if ( i == 2 )
    printf ( "With you..." ) ;
else
{
```

```
    if ( j == 2 )
        printf ( "...All the time" );
}
```

```
/* code using if - else if - else */
if ( i == 2 )
    printf ( "With you..." );
else if ( j == 2 )
    printf ( "...All the time " );
```

Another place where logical operators are useful is when we want to write programs for complicated logics that ultimately boil down to only two answers. For example, consider the following example:

Example 4.2: A company insures its drivers in the following cases:

- If the driver is married.
- If the driver is unmarried, male & above 30 years of age.
- If the driver is unmarried, female & above 25 years of age.

In all other cases, the driver is not insured. If the marital status, sex and age of the driver are the inputs, write a program to determine whether the driver should be insured or not.

Here after checking a complicated set of instructions the final output of the program would be one of the two—either the driver should be insured or the driver should not be insured. As mentioned above, since these are the only two outcomes this problem can be solved using logical operators. But before we do that, let us write a program that does not make use of logical operators.

```
/* Insurance of driver - without using logical operators */
#include <stdio.h>
int main( )
{
    char sex, ms ;
    int age ;

    printf ( "Enter age, sex, marital status " );
    scanf ( "%d %c %c", &age, &sex, &ms );

    if ( ms == 'M' )
        printf ( "Driver should be insured\n" );
```

```
else
{
    if ( sex == 'M' )
    {
        if ( age > 30 )
            printf ( "Driver should be insured\n" );
        else
            printf ( "Driver should not be insured\n" );
    }
    else
    {
        if ( age > 25 )
            printf ( "Driver should be insured\n" );
        else
            printf ( "Driver should not be insured\n" );
    }
}

return 0 ;
}
```

From the program it is evident that we are required to match several **ifs** and **elses** and several pairs of braces. In a more real-life situation there would be more conditions to check leading to the program creeping to the right. Let us now see how to avoid these problems by using logical operators.

As mentioned above, in this example, we expect the answer to be either 'Driver should be insured' or 'Driver should not be insured'. If we list down all those cases in which the driver is insured, then they would be:

- (a) Driver is married.
- (b) Driver is an unmarried male above 30 years of age.
- (c) Driver is an unmarried female above 25 years of age.

Since all these cases lead to the driver being insured, they can be combined together using **&&** and **||** as shown in the program below.

```
/* Insurance of driver - using logical operators */
#include <stdio.h>
int main( )
{
    char sex, ms ;
```

```
int age ;

printf ( "Enter age, sex, marital status " ) ;
scanf ( "%d %c %c", &age, &sex, &ms ) ;

if ( ( ms == 'M') || ( ms == 'U' && sex == 'M' && age > 30 ) ||
    ( ms == 'U' && sex == 'F' && age > 25 ) )
    printf ( "Driver should be insured\n" ) ;
else
    printf ( "Driver should not be insured\n" ) ;

return 0 ;
}
```

In this program, it is important to note that:

- The driver will be insured only if one of the conditions enclosed in parentheses evaluates to true.
- For the second pair of parentheses to evaluate to true, each condition in the parentheses separated by && must evaluate to true.
- Even if one of the conditions in the second parentheses evaluates to false, then the whole of the second parentheses evaluates to false.
- The last two of the above arguments apply to third pair of parentheses as well.

Thus, we can conclude that the **&&** and **||** are useful in the following programming situations:

- (a) When it is to be checked in which range does a value fall.
- (b) When after testing several conditions, the outcome is only one of the two answers. (This problem is often called yes/no problem).

In some programming situations we may combine the usage of **if— else if—else** and logical operators. This is demonstrated in the following program.

Example 4.3: Write a program to calculate the salary as per the following table:

Gender	Years of Service	Qualifications	Salary
Male	>= 10	Post-Graduate	15000
	>= 10	Graduate	10000
	< 10	Post-Graduate	10000
	< 10	Graduate	7000
Female	>= 10	Post-Graduate	12000
	>= 10	Graduate	9000
	< 10	Post-Graduate	10000
	< 10	Graduate	6000

Figure 4.1

```
# include <stdio.h>
int main( )
{
    char g ;
    int  yos, qual, sal = 0 ;

    printf ( "Enter Gender, Years of Service and
              Qualifications ( 0 = G, 1 = PG ):" ) ;
    scanf ( "%c%d%d", &g, &yos, &qual ) ;

    if ( g == 'm' && yos >= 10 && qual == 1 )
        sal = 15000 ;
    else if ( ( g == 'm' && yos >= 10 && qual == 0 ) ||
              ( g == 'm' && yos < 10 && qual == 1 ) )
        sal = 10000 ;
    else if ( g == 'm' && yos < 10 && qual == 0 )
        sal = 7000 ;
    else if ( g == 'f' && yos >= 10 && qual == 1 )
        sal = 12000 ;
    else if ( g == 'f' && yos >= 10 && qual == 0 )
        sal = 9000 ;
    else if ( g == 'f' && yos < 10 && qual == 1 )
        sal = 10000 ;
    else if ( g == 'f' && yos < 10 && qual == 0 )
        sal = 6000 ;
```

```
printf ( "\nSalary of Employee = %d\n", sal ) ;  
return 0 ;  
}
```

I hope you can follow the implementation of this program on your own.

The ! Operator

So far we have used only the logical operators **&&** and **||**. The third logical operator is the NOT operator, written as **!**. This operator reverses the result of the expression it operates on. For example, if the expression evaluates to a non-zero value, then applying **!** operator to it results into a 0. Vice versa, if the expression evaluates to zero then on applying **!** operator to it makes it 1, a non-zero value. The final result (after applying **!**) 0 or 1 is considered to be false or true, respectively. Here is an example of the NOT operator applied to a relational expression.

```
! ( y < 10 )
```

This means ‘not **y** less than 10’. In other words, if **y** is less than 10, the expression will be false, since **(y < 10)** is true. We can express the same condition as **(y >= 10)**.

The NOT operator is often used to reverse the logical value of a single variable, as in the expression

```
if ( ! flag )
```

This is another way of saying:

```
if ( flag == 0 )
```

Does the NOT operator sound confusing? Avoid it if you want, as the same thing can be achieved without using the NOT operator.

Hierarchy of Operators Revisited

Since we have now added the logical operators to the list of operators we know, it is time to review these operators and their priorities. Figure 4.2 summarizes the operators we have seen so far. The higher the position of an operator is in the table, higher is its priority. (A full-fledged precedence table of operators is given in Appendix B.)

Operators	Type
!	Logical NOT
* / %	Arithmetic and modulus
+ -	Arithmetic
< > <= >=	Relational
== !=	Relational
&&	Logical AND
	Logical OR
=	Assignment

Figure 4.2

A Word of Caution

Can you guess what will be the output of the following program?

```
#include <stdio.h>
int main( )
{
    int i;
    printf ( "Enter value of i " );
    scanf ( "%d", &i );
    if ( i = 5 )
        printf ( "You entered 5\n" );
    else
        printf ( "You entered something other than 5\n" );

    return 0 ;
}
```

Well, here is the output of two runs of this program...

```
Enter value of i 200
You entered 5
Enter value of i 9999
You entered 5
```

Surprised? You have entered 200 and 9999, and still you find in either case the output is 'You entered 5'. This is because we have written the condition wrongly. We have used the assignment operator = instead of the relational operator ==. As a result, the condition gets reduced to **if (**

5), irrespective of what you supply as the value of **i**. And remember that in C, 'truth' is always non-zero, whereas, 'falsity' is always zero. Therefore, **if (5)** always evaluates to true and hence the result.

Another common mistake while using the **if** statement is to write a semicolon (;) after the condition, as shown below.

```
#include <stdio.h>
int main( )
{
    int i;

    printf ( "Enter value of i " );
    scanf ( "%d", &i );

    if ( i == 5 );
        printf ( "You entered 5\n" );
    return 0 ;
}
```

The ; makes the compiler to interpret the statement as if you have written it in following manner:

```
if ( i == 5 )
;
printf ( "You entered 5\n" );
```

Here, if the condition evaluates to true, the ; (null statement, which does nothing on execution) gets executed, following which the **printf()** gets executed. If the condition fails, then straightaway the **printf()** gets executed. So, irrespective of whether the condition evaluates to true or false, **printf()** is bound to get executed. Remember that compiler would not point out this as an error, since as far as the syntax is concerned, nothing has gone wrong, but the logic has certainly gone awry. Moral is, beware of such pitfalls.

The following figure summarizes the working of all the three logical operators.

Operands		Results			
x	y	!x	!y	x && y	x y
0	0	1	1	0	0
0	non-zero	1	0	0	1
non-zero	0	0	1	0	1
non-zero	non-zero	0	0	1	1

Figure 4.3

The Conditional Operators

The conditional operators **?** and **:** are sometimes called ternary operators since they take three arguments. In fact, they form a kind of foreshortened if-then-else. Their general form is,

expression 1 ? expression 2 : expression 3

What this expression says is: “if **expression 1** is true (that is, if its value is non-zero), then the value returned will be **expression 2**, otherwise the value returned will be **expression 3**”. Let us understand this with the help of a few examples.

```
(a) int x, y;
    scanf ("%d", &x);
    y = ( x > 5 ? 3 : 4 );
```

This statement will store 3 in **y** if **x** is greater than 5, otherwise it will store 4 in **y**.

The equivalent **if-else** form would be,

```
if ( x > 5 )
    y = 3;
else
    y = 4;
```

```
(a) char a;
    int y;
    scanf ("%c", &a);
    y = ( a >= 65 && a <= 90 ? 1 : 0 );
```

Here 1 would be assigned to **y** if **a >=65 && a <=90** evaluates to true, otherwise 0 would be assigned.

The following points may be noted about the conditional operators:

- (a) It's not necessary that the conditional operators should be used only in arithmetic statements. This is illustrated in the following examples:

```
Ex.: int i ;  
      scanf ( "%d", &i ) ;  
      ( i == 1 ? printf ( "Amit" ) : printf ( "All and sundry" ) ) ;
```

```
Ex.: char a = 'z' ;  
      printf ( "%c", ( a >= 'a' ? a : '!' ) ) ;
```

- (b) The conditional operators can be nested as shown below.

```
int big, a, b, c ;  
big = ( a > b ? ( a > c ? 3: 4 ) : ( b > c ? 6: 8 ) ) ;
```

- (c) Check out the following conditional expression:

```
a > b ? g = a : g = b ;
```

This will give you an error 'Lvalue Required'. The error can be overcome by enclosing the statement in the **:** part within a pair of parentheses. This is shown below.

```
a > b ? g = a : ( g = b ) ;
```

In absence of parentheses, the compiler believes that **b** is being assigned to the result of the expression to the left of second **=**. Hence it reports an error.

The limitation of the conditional operators is that after the **?** or after the **:**, only one C statement can occur. In practice, rarely does this requirement exist. Therefore, in serious C programming, conditional operators aren't as frequently used as the **if-else**.

Summary

- (a) If the outcome of an if-else ladder is only one of two answers then the ladder should be replaced either with an else-if clause or by logical operators.
- (b) **&&** and **||** are binary operators, whereas, **!** is a unary operator.
- (c) In C every test expression is evaluated in terms of zero and non-zero values. A zero value is considered to be false and a non-zero value is considered to be true.
- (d) Conditional operators can be used as an alternative to if-else statement if there is a single statement in the 'if block' and a single statement in the 'else block'.
- (e) Assignment statements used with conditional operators must be enclosed within a pair of parentheses.

Exercise

- [A]** If $a = 10$, $b = 12$, $c = 0$, find the values of the expressions in the following table:

Expression	Value
$a != 6 \ \&\& \ b > 5$	1
$a == 9 \ \ b < 3$	
$! (a < 10)$	
$! (a > 5 \ \&\& \ c)$	
$5 \ \&\& \ c != 8 \ \ !c$	

- [B]** What will be the output of the following programs:

```
(a) #include <stdio.h>
int main( )
{
    int i = 4, z = 12 ;
    if ( i = 5 || z > 50 )
        printf ( "Dean of students affairs\n" ) ;
    else
        printf ( "Dosa\n" ) ;
```

```
    return 0 ;  
}
```

```
(b) #include <stdio.h>  
int main( )  
{  
    int i = 4, j = -1, k = 0, w, x, y, z ;  
    w = i || j || k ;  
    x = i && j && k ;  
    y = i || j && k ;  
    z = i && j || k ;  
    printf ( "w = %d x = %d y = %d z = %d\n", w, x, y, z ) ;  
    return 0 ;  
}
```

```
(c) # include <stdio.h>  
int main( )  
{  
    int x = 20, y = 40, z = 45 ;  
    if ( x > y && x > z )  
        printf ( "biggest = %d\n", x ) ;  
    else if ( y > x && y > z )  
        printf ( "biggest = %d\n", y ) ;  
    else if ( z > x && z > y )  
        printf ( "biggest = %d\n", z ) ;  
    return 0 ;  
}
```

```
(d) # include <stdio.h>  
int main( )  
{  
    int i = -1, j = 1, k, l ;  
    k = !i && j ;  
    l = !i || j ;  
    printf ( "%d %d\n", i, j ) ;  
    return 0 ;  
}
```

```
(e) # include <stdio.h>  
int main( )  
{  
    int i = -4, j, num ;  
    j = ( num < 0 ? 0 : num * num ) ;
```



```
    printf ( "%d\n", j ) ;  
    return 0 ;  
}
```

```
(f) # include <stdio.h>  
int main( )  
{  
    int k, num = 30 ;  
    k = ( num > 5 ? ( num <= 10 ? 100 : 200 ) : 500 ) ;  
    printf ( "%d\n", num ) ;  
    return 0 ;  
}
```

[C] Point out the errors, if any, in the following programs:

```
(a) # include <stdio.h>  
int main( )  
{  
    int code, flag ;  
    if ( code == 1 & flag == 0 )  
        printf ( "The eagle has landed\n" ) ;  
    return 0 ;  
}
```

```
(b) # include <stdio.h>  
int main( )  
{  
    char spy = 'a', password = 'z' ;  
    if ( spy == 'a' or password == 'z' )  
        printf ( "All the birds are safe in the nest\n" ) ;  
    return 0 ;  
}
```

```
(c) # include <stdio.h>  
int main( )  
{  
    int i = 10, j = 20 ;  
    if ( i = 5 ) && if ( j = 10 )  
        printf ( "Have a nice day\n" ) ;  
    return 0 ;  
}
```

```
(d) # include <stdio.h>  
int main( )
```

```
{
    int x = 10, y = 20 ;
    if ( x >= 2 and y <= 50 )
        printf ( "%d\n", x ) ;
    return 0 ;
}
```

(e) # include <stdio.h>

```
int main( )
{
    int x = 2 ;
    if ( x == 2 && x != 0 ) ;
        printf ( "Hello\n" ) ;
    else
        printf ( "Bye\n" ) ;
    return 0 ;
}
```

(f) # include <stdio.h>

```
int main( )
{
    int i = 10, j = 10 ;
    if ( i && j == 10 )
        printf ( "Have a nice day\n" ) ;
    return 0 ;
}
```

(g) # include <stdio.h>

```
int main( )
{
    int j = 65 ;
    printf ( "j >= 65 ? %d : %c\n", j ) ;
    return 0 ;
}
```

(h) # include <stdio.h>

```
int main( )
{
    int i = 10, j ;
    i >= 5 ? j = 10 : j = 15 ;
    printf ( "%d %d\n", i, j ) ;
    return 0 ;
}
```

- (i)

```
#include <stdio.h>
int main( )
{
    int a = 5, b = 6 ;
    ( a == b ? printf ( "%d\n", a ) ) ;
    return 0 ;
}
```
- (j)

```
#include <stdio.h>
int main( )
{
    int n = 9 ;
    ( n == 9 ? printf ( "Correct\n" ) ; : printf ( "Wrong\n" ) ; ) ;
    return 0 ;
}
```

[D] Attempt the following:

- (a) Any year is entered through the keyboard, write a program to determine whether the year is leap or not. Use the logical operators **&&** and **||**.
- (b) Any character is entered through the keyboard, write a program to determine whether the character entered is a capital letter, a small case letter, a digit or a special symbol.

The following table shows the range of ASCII values for various characters:

Characters	ASCII Values
A – Z	65 – 90
a – z	97 – 122
0 – 9	48 – 57
special symbols	0 - 47, 58 - 64, 91 - 96, 123 - 127

- (c) A certain grade of steel is graded according to the following conditions:
- (i) Hardness must be greater than 50
 - (ii) Carbon content must be less than 0.7
 - (iii) Tensile strength must be greater than 5600

The grades are as follows:

Grade is 10 if all three conditions are met

Grade is 9 if conditions (i) and (ii) are met

Grade is 8 if conditions (ii) and (iii) are met

Grade is 7 if conditions (i) and (iii) are met

Grade is 6 if only one condition is met

Grade is 5 if none of the conditions are met

Write a program, which will require the user to give values of hardness, carbon content and tensile strength of the steel under consideration and output the grade of the steel.

- (d) If the three sides of a triangle are entered through the keyboard, write a program to check whether the triangle is valid or not. The triangle is valid if the sum of two sides is greater than the largest of the three sides.
- (e) If the three sides of a triangle are entered through the keyboard, write a program to check whether the triangle is isosceles, equilateral, scalene or right angled triangle.
- (f) In boxing the weight class of a boxer is decided as per the following table. Write a program that receives weight as input and prints out the boxer's weight class.

Boxer Class	Weight in Pounds
Flyweight	< 115
Bantamweight	115 - 121
Featherweight	122 - 153
Middleweight	154 – 189
Heavyweight	>= 190

- (g) In digital world colors are specified in Red-Green-Blue (RGB) format, with values of R, G, B varying on an integer scale from 0 to 255. In print publishing the colors are mentioned in Cyan-Magenta-Yellow-Black (CMYK) format, with values of C, M, Y, and K varying on a real scale from 0.0 to 1.0. Write a program that converts RGB color to CMYK color as per the following formulae:

$$White = \text{Max}(Red / 255, Green / 255, Blue / 255)$$

$$Cyan = \left(\frac{White - Red / 255}{White} \right)$$

$$Magenta = \left(\frac{White - Green / 255}{White} \right)$$

$$Yellow = \left(\frac{White - Blue / 255}{White} \right)$$

$$Black = 1 - White$$

Note that if the RGB values are all 0, then the CMY values are all 0 and the K value is 1.

- (h) Write a program that receives month and date of birth as input and prints the corresponding Zodiac sign based on the following table:

Sun Sign	From - To
Capricorn	December 22 - January 19
Aquarius	January 20 - February 17
Pisces	February 18 - March 19
Aries	March 20 - April 19
Taurus	April 20 - May 20
Gemini	May 21 - June 20
Cancer	June 21 - July 22
Leo	July 23 - August 22
Virgo	August 23 - September 22
Libra	September 23 - October 22
Scorpio	October 23 - November 21
Sagittarius	November 22 - December 21

- (i) The Body Mass Index (BMI) is defined as ratio of the weight of a person (in kilograms) to the square of the height (in meters). Write a program that receives weight and height, calculates the BMI, and reports the BMI category as per the following table:

BMI Category	BMI
Starvation	< 15
Anorexic	15.1 to 17.5
Underweight	17.6 to 18.5
Ideal	18.6 to 24.9
Overweight	25 to 25.9
Obese	30 to 30.9
Morbidly Obese	>= 40

[E] Attempt the following:

- (a) Using conditional operators determine:
 - (1) Whether the character entered through the keyboard is a lower case alphabet or not.
 - (2) Whether a character entered through the keyboard is a special symbol or not.
- (b) Write a program using conditional operators to determine whether a year entered through the keyboard is a leap year or not.
- (c) Write a program to find the greatest of the three numbers entered through the keyboard. Use conditional operators.
- (d) Write a program to receive value of an angle in degrees and check whether sum of squares of sine and cosine of this angle is equal to 1.
- (e) Rewrite the following program using conditional operators.

```
# include <stdio.h>
int main( )
{
    float sal ;

    printf ( "Enter the salary" ) ;
    scanf ( "%f", &sal ) ;
    if ( sal >= 25000 && sal <= 40000 )
        printf ( "Manager\n" ) ;
    else
        if ( sal >= 15000 && sal < 25000 )
            printf ( "Accountant\n" ) ;
```

```
        else  
            printf ( "Clerk\n" );  
    return 0 ;  
}
```


5

Loop Control Instruction

- **Loops**
- **The *while* Loop**
 - Tips and Traps
 - More Operators
- **Summary**
- **Exercise**



5 *Loop Control Instruction*

- Loops
- The *while* Loop
 - Tips and Traps
 - More Operators
- Summary
- Exercise

The programs that we have developed so far used either a sequential or a decision control instruction. In the first one, the calculations were carried out in a fixed order; while in the second, an appropriate set of instructions were executed depending upon the outcome of the condition(s) being tested.

These programs were of limited nature, because when executed, they always performed the same series of actions, in the same way, exactly once. Almost always, if something is worth doing, it's worth doing more than once. You can probably think of several examples of this from real life, such as eating a good dinner or going for a movie. Programming is the same; we frequently need to perform an action over and over, often with variations in the details each time. The mechanism, which meets this need, is the 'Loop Control Instruction', and loops are the subject of this chapter.

Loops

The versatility of the computer lies in its ability to perform a set of instructions repeatedly. This involves repeating some portion of the program either a specified number of times or until a particular condition is being satisfied. This repetitive operation is done through a loop control instruction.

There are three methods by way of which we can repeat a part of a program. They are:

- (a) Using a **for** statement
- (b) Using a **while** statement
- (c) Using a **do-while** statement

Each of these methods is discussed in the following pages.

The *while* Loop

It is often the case in programming that you want to repeat something a fixed number of times. Perhaps you want to calculate gross salaries of ten different persons, or you want to convert temperatures from Centigrade to Fahrenheit for 15 different cities. The **while** loop is ideally suited for this.

Let us look at a simple example that uses a **while** loop to calculate simple interest for 3 sets of values of principal, number of years and rate of interest. The flowchart shown in Figure 5.1 would help you to understand the operation of the **while** loop.

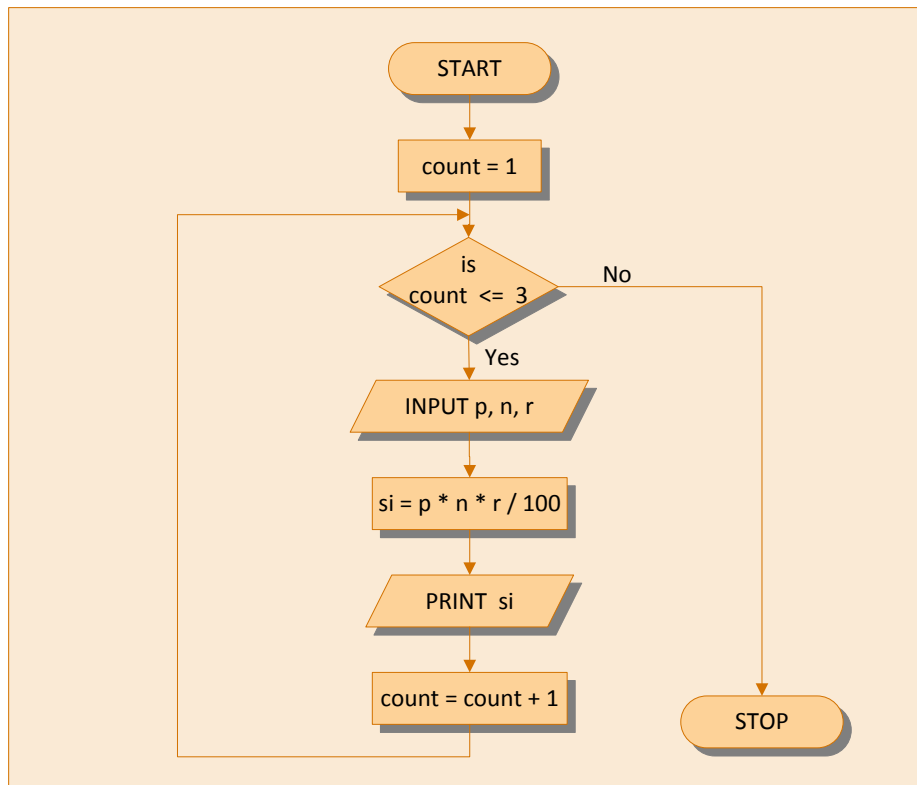


Figure 5.1

Let us now write a program that implements the logic of this flowchart.

```
/* Calculation of simple interest for 3 sets of p, n and r */
#include <stdio.h>
int main( )
{
    int  p, n, count ;
    float  r, si ;

    count = 1 ;
    while ( count <= 3 )
    {
        printf ( "\nEnter values of p, n and r " ) ;
        scanf ( "%d %d %f", &p, &n, &r ) ;
        si = p * n * r / 100 ;
        printf ( "Simple interest = Rs. %\nf", si ) ;

        count = count + 1 ;
    }
}
```

```
}  
    return 0 ;  
}
```

And here are a few sample runs of the program...

```
Enter values of p, n and r 1000 5 13.5  
Simple interest = Rs. 675.000000  
Enter values of p, n and r 2000 5 13.5  
Simple interest = Rs. 1350.000000  
Enter values of p, n and r 3500 5 3.5  
Simple interest = Rs. 612.500000
```

The program executes all statements after the **while** 3 times. The logic for calculating the simple interest is written in these statements and they are enclosed within a pair of braces. These statements form the 'body' of the **while** loop. The parentheses after the **while** contain a condition. So long as this condition remains true the statements in the body of the **while** loop keep getting executed repeatedly. To begin with, the variable **count** is initialized to 1 and every time the simple interest logic is executed, the value of **count** is incremented by one. The variable **count** is often called either a 'loop counter' or an 'index variable'.

The operation of the **while** loop is illustrated in Figure 5.2.

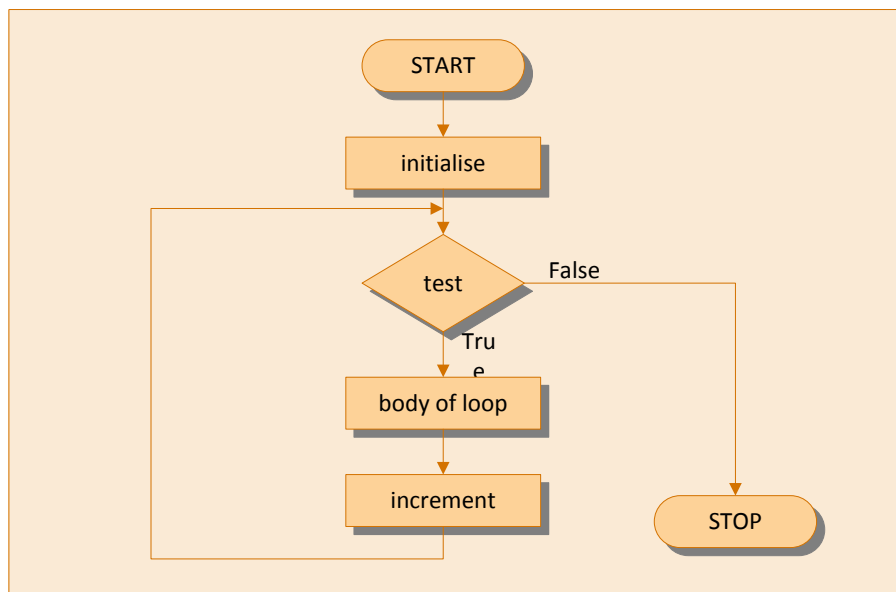


Figure 5.2

Tips and Traps

The general form of **while** is as shown below.

```
initialize loop counter ;  
while ( test loop counter using a condition )  
{  
    do this ;  
    and this ;  
    increment loop counter ;  
}
```

Note the following points about **while**...

- The statements within the **while** loop would keep getting executed till the condition being tested remains true. When the condition becomes false, the control passes to the first statement that follows the body of the **while** loop.
- In place of the condition there can be any other valid expression. So long as the expression evaluates to a non-zero value the statements within the loop would get executed.
- The condition being tested may use relational or logical operators as shown in the following examples:

```
while ( i <= 10 )  
while ( i >= 10 && j <= 15 )  
while ( j > 10 && ( b < 15 || c < 20 ) )
```

- The statements within the loop may be a single line or a block of statements. In the first case, the braces are optional. For example,

```
while ( i <= 10 )  
    i = i + 1 ;
```

is same as

```
while ( i <= 10 )  
{  
    i = i + 1 ;  
}
```

- Almost always, the while must test a condition that will eventually become false, otherwise the loop would be executed forever, indefinitely.

```
# include <stdio.h>
int main( )
{
    int i = 1 ;
    while ( i <= 10 )
        printf ( "%d\n", i ) ;
    return 0 ;
}
```

This is an indefinite loop, since *i* remains equal to 1 forever. The correct form would be as under:

```
# include <stdio.h>
int main( )
{
    int i = 1 ;
    while ( i <= 10 )
    {
        printf ( "%d\n", i ) ;
        i = i + 1 ;
    }
    return 0 ;
}
```

- Instead of incrementing a loop counter, we can decrement it and still manage to get the body of the loop executed repeatedly. This is shown below.

```
# include <stdio.h>
int main( )
{
    int i = 5 ;
    while ( i >= 1 )
    {
        printf ( "Make the computer literate!\n" ) ;
        i = i - 1 ;
    }
}
```

- It is not necessary that a loop counter must only be an **int**. It can even be a **float**.

```
# include <stdio.h>
int main( )
{
    float a = 10.0 ;
    while ( a <= 10.5 )
    {
        printf ( "Raindrops on roses..." );
        printf ( "...and whiskers on kittens\n" );
        a = a + 0.1 ;
    }
    return 0 ;
}
```

- Even floating point loop counters can be decremented. Once again, the increment and decrement could be by any value, not necessarily 1.
- What will be the output of the following program:

```
# include <stdio.h>
int main( )
{
    int i = 1 ;
    while ( i <= 10 ) ;
    {
        printf ( "%d\n", i );
        i = i + 1 ;
    }
    return 0 ;
}
```

This is an indefinite loop, and it doesn't give any output at all. The reason is, we have carelessly given a ; after the **while**. It would make the loop work like this...

```
while ( i <= 10 )
;
{
    printf ( "%d\n", i );
}
```



```
i = i + 1 ;  
}
```

Since the value of **i** is not getting incremented, the control would keep rotating within the loop, eternally. Note that enclosing **printf()** and **i = i +1** within a pair of braces is not an error. In fact we can put a pair of braces around any individual statement or set of statements without affecting the execution of the program.

More Operators

There are several operators that are frequently used with **while**. To illustrate their usage, let us consider a problem wherein numbers from 1 to 10 are to be printed on the screen. The program for performing this task can be written using **while** in following different ways:

```
(a) # include <stdio.h>  
int main( )  
{  
    int i = 1 ;  
    while ( i <= 10 )  
    {  
        printf ( "%d\n", i ) ;  
        i = i + 1 ;  
    }  
    return 0 ;  
}
```

This is the most straight-forward way of printing numbers from 1 to 10.

```
(b) # include <stdio.h>  
int main( )  
{  
    int i = 1 ;  
    while ( i <= 10 )  
    {  
        printf ( "%d\n", i ) ;  
        i++ ;  
    }  
    return 0 ;  
}
```

Note that the increment operator `++` increments the value of `i` by 1, every time the statement `i++` gets executed. Similarly, to reduce the value of a variable by 1, a decrement operator `--` is also available.

However, never use `n+++` to increment the value of `n` by 2, since there doesn't exist an operator `+++` in C.

```
(c) #include <stdio.h>
int main( )
{
    int i = 1 ;
    while ( i <= 10 )
    {
        printf ( "%d\n", i ) ;
        i += 1 ;
    }
    return 0 ;
}
```

Note that `+=` is a compound assignment operator. It increments the value of `i` by 1. Similarly, `j = j + 10` can also be written as `j += 10`. Other compound assignment operators are `-=`, `*=`, `/=` and `%=`.

```
(d) #include <stdio.h>
int main( )
{
    int i = 0 ;
    while ( i++ < 10 )
        printf ( "%d\n", i ) ;
    return 0 ;
}
```

In the statement **`while ( i++ < 10 )`**, first the comparison of value of `i` with 10 is performed, and then the incrementation of `i` takes place. Since the incrementation of `i` happens after the comparison, here the `++` operator is called a post-incrementation operator. When the control reaches **`printf( )`**, `i` has already been incremented, hence `i` must be initialized to 0, not 1.

```
(e) #include <stdio.h>
int main( )
{
```

```
int i = 0 ;
while ( ++i <= 10 )
    printf ( "%d\n", i ) ;
return 0 ;
}
```

In the statement **while (++i <= 10)**, first incrementation of **i** takes place, then the comparison of value of **i** with 10 is performed. Since the incrementation of **i** happens before the comparison, here the **++** operator is called a pre-incrementation operator.

Summary

- (a) The three type of loops available in C are **for**, **while**, and **do-while**.
- (b) In a **while** loop there are three distinct steps of initialization, testing and incrementation of loop counter.
- (c) The loop counter can be incremented or decremented by any suitable step value.
- (d) There are two forms of **++** operator—pre-increment and post-increment.
- (e) There are two forms of **--** operator—pre-decrement and post-decrement.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    int i = 1 ;
    while ( i <= 10 ) ;
    {
        printf ( "%d\n", i ) ;
        i++ ;
    }
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
```

```
{
    int x = 4, y, z ;
    y = --x ;
    z = x-- ;
    printf ( "%d %d %d\n", x, y, z ) ;
    return 0 ;
}
```

```
(c) # include <stdio.h>
int main( )
{
    int x = 4, y = 3, z ;
    z = x-- - y ;
    printf ( "%d %d %d\n", x, y, z ) ;
    return 0 ;
}
```

```
(d) # include <stdio.h>
int main( )
{
    while ( 'a' < 'b' )
        printf ( "malayalam is a palindrome\n" ) ;
    return 0 ;
}
```

```
(e) # include <stdio.h>
int main( )
{
    int i ;
    while ( i = 10 )
    {
        printf ( "%d\n", i ) ;
        i = i + 1 ;
    }
    return 0 ;
}
```

```
(f) # include <stdio.h>
int main( )
{
    float x = 1.1 ;
    while ( x == 1.1 )
    {
```

```
        printf ( "%f\n", x ) ;  
        x = x - 0.1 ;  
    }  
    return 0 ;  
}
```

[B] Attempt the following:

- (a) Write a program to calculate overtime pay of 10 employees. Overtime is paid at the rate of Rs. 12.00 per hour for every hour worked above 40 hours. Assume that employees do not work for fractional part of an hour.
- (b) Write a program to find the factorial value of any number entered through the keyboard.
- (c) Two numbers are entered through the keyboard. Write a program to find the value of one number raised to the power of another.
- (d) Write a program to print all the ASCII values and their equivalent characters using a while loop. The ASCII values vary from 0 to 255.
- (e) Write a program to print out all Armstrong numbers between 1 and 500. If sum of cubes of each digit of the number is equal to the number itself, then the number is called an Armstrong number. For example, $153 = (1 * 1 * 1) + (5 * 5 * 5) + (3 * 3 * 3)$.
- (f) Write a program for a matchstick game being played between the computer and a user. Your program should ensure that the computer always wins. Rules for the game are as follows:
 - There are 21 matchsticks.
 - The computer asks the player to pick 1, 2, 3, or 4 matchsticks.
 - After the person picks, the computer does its picking.
 - Whoever is forced to pick up the last matchstick loses the game.
- (g) Write a program to enter numbers till the user wants. At the end it should display the count of positive, negative and zeros entered.
- (h) Write a program to receive an integer and find its octal equivalent.
- (i) (Hint: To obtain octal equivalent of an integer, divide it continuously by 8 till dividend doesn't become zero, then write the remainders obtained in reverse direction.)

- (j) Write a program to find the range of a set of numbers entered through the keyboard. Range is the difference between the smallest and biggest number in the list.

6

More Complex Repetitions

- The *for* Loop
 - Nesting of Loops
 - Multiple Initializations in the *for* Loop
- The *break* Statement
- The *continue* Statement
- The *do-while* Loop
- The Odd Loop
- Summary
- Exercise



6 *More Complex Repetitions*

- The *for* Loop
 - Nesting of Loops
 - Multiple Initializations in the *for* Loop
- The *break* Statement
 - The *continue* Statement
- The *do-while* Loop
- The Odd Loop
- Summary
- Exercise

The programs that we have developed so far used either a sequential or a decision control instruction. In the first one, the calculations were carried out in a fixed order; while in the second, an appropriate set of instructions were executed depending upon the outcome of the condition(s) being tested.

The **for** Loop

Perhaps one reason why few programmers use **while** is that they are too busy using the **for**, which is probably the most popular looping instruction. The **for** allows us to specify three things about a loop in a single line:

- (a) Setting a loop counter to an initial value.
- (b) Testing the loop counter to determine whether its value has reached the number of repetitions desired.
- (c) Increasing the value of loop counter each time the body of the loop has been executed.

The general form of **for** statement is as under:

```
for ( initialize counter ; test counter ; increment counter )  
{  
    do this ;  
    and this ;  
    and this ;  
}
```

Let us now write down the simple interest program using **for**. Compare this program with the one that we wrote using **while**. The flowchart is also given in Figure 6.1 for a better understanding.

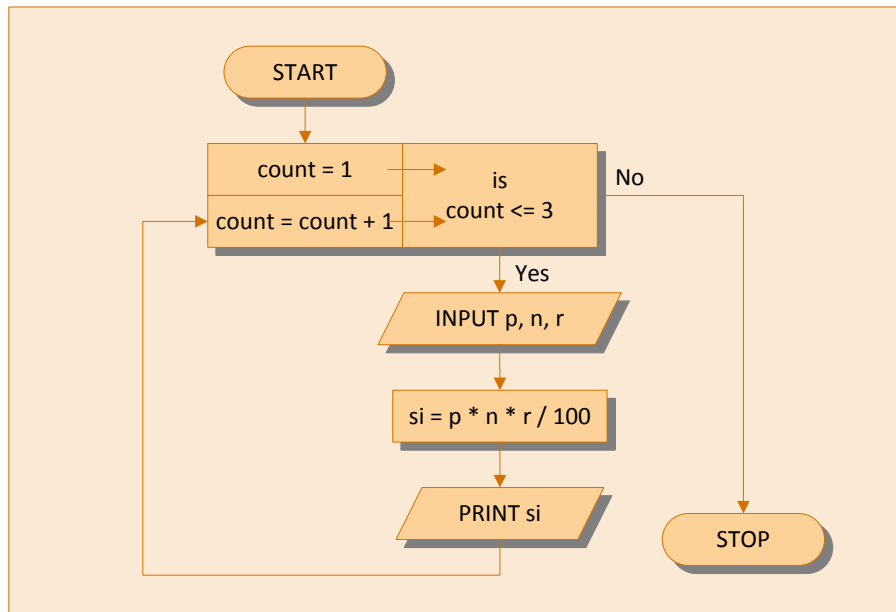


Figure 6.1

```

/* Calculation of simple interest for 3 sets of p, n and r */
#include <stdio.h>
int main( )
{
    int p, n, count ;
    float r, si ;

    for ( count = 1 ; count <= 3 ; count = count + 1 )
    {
        printf ( "Enter values of p, n, and r " ) ;
        scanf ( "%d %d %f", &p, &n, &r ) ;

        si = p * n * r / 100 ;
        printf ( "Simple Interest = Rs.%f\n", si ) ;
    }
    return 0 ;
}

```

If you compare this program with the one written using **while**, you can observe that the three steps—initialization, testing and incrementation—required for the loop construct have now been incorporated in the **for** statement.

Let us now examine how the **for** statement gets executed:

- When the **for** statement is executed for the first time, the value of **count** is set to an initial value 1.
- Next the condition **count** $\leq$ 3 is tested. Since **count** is 1, the condition is satisfied and the body of the loop is executed for the first time.
- Upon reaching the closing brace of **for**, control is sent back to the **for** statement, where the value of **count** gets incremented by 1.
- Again the test is performed to check whether the new value of **count** exceeds 3.
- If the value of **count** is less than or equal to 3, the statements within the braces of **for** are executed again.
- The body of the **for** loop continues to get executed till **count** doesn't exceed the final value 3.
- When **count** reaches the value 4, the control exits from the loop and is transferred to the statement (if any) immediately after the body of **for**.

Figure 6.2 would help in further clarifying the concept of execution of the **for** loop.

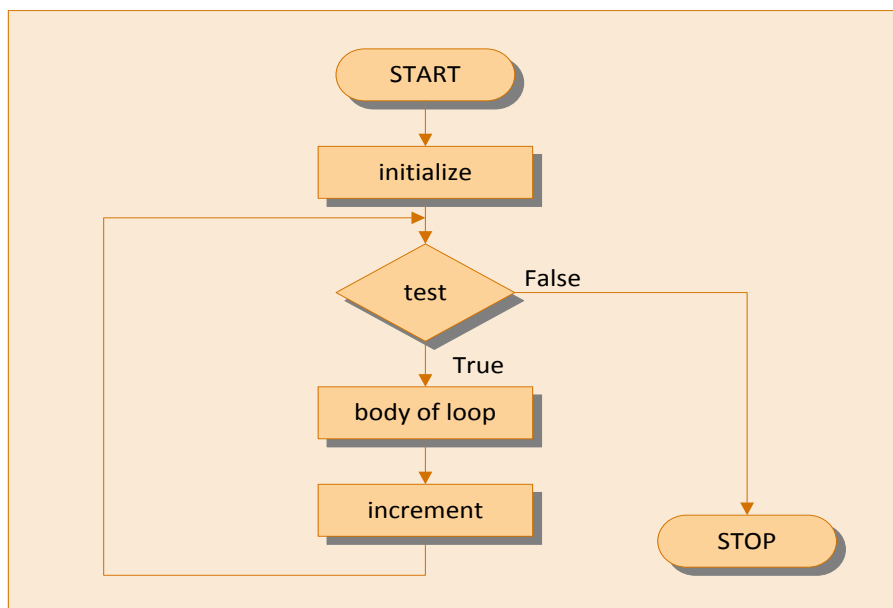


Figure 6.2

It is important to note that the initialization, testing and incrementation part of a **for** loop can be replaced by any valid expression. Thus the following **for** loops are perfectly ok.

```
for ( i = 10 ; i ; i -- )
    printf ( "%d ", i ) ;
for ( i < 4 ; j = 5 ; j = 0 )
    printf ( "%d ", i ) ;
for ( i = 1 ; i <= 10 ; printf ( "%d ", i++ ) )
    ;
for ( scanf ( "%d", &i ) ; i <= 10 ; i++ )
    printf ( "%d", i ) ;
```

Let us now write down the program to print numbers from 1 to 10 in different ways. This time we would use a **for** loop instead of a **while** loop.

```
(a) # include <stdio.h>
    int main( )
    {
        int i ;
        for ( i = 1 ; i <= 10 ; i = i + 1 )
            printf ( "%d\n", i ) ;
        return 0 ;
    }
```

Note that the initialization, testing and incrementation of loop counter is done in the **for** statement itself. Instead of **i = i + 1**, the statements **i++** or **i += 1** can also be used.

Since there is only one statement in the body of the **for** loop, the pair of braces have been dropped. As with the **while**, the default scope of **for** is the immediately next statement after **for**.

```
(b) # include <stdio.h>
    int main( )
    {
        int i ;
        for ( i = 1 ; i <= 10 ; )
        {
            printf ( "%d\n", i ) ;
            i = i + 1 ;
        }
    }
```

```
    return 0 ;  
}
```

Here, the incrementation is done within the body of the **for** loop and not in the **for** statement. Note that, in spite of this, the semicolon (;) after the condition is necessary.

```
(c) #include <stdio.h>  
int main( )  
{  
    int i = 1 ;  
    for ( ; i <= 10 ; i = i + 1 )  
        printf ( "%d\n", i ) ;  
    return 0 ;  
}
```

Here the initialization is done in the declaration statement itself, but still the semicolon before the condition is necessary.

```
(d) #include <stdio.h>  
int main( )  
{  
    int i = 1 ;  
    for ( ; i <= 10 ; )  
    {  
        printf ( "%d\n", i ) ;  
        i = i + 1 ;  
    }  
    return 0 ;  
}
```

Here, neither the initialization, nor the incrementation is done in the **for** statement, but still the two semicolons are necessary.

```
(e) #include <stdio.h>  
int main( )  
{  
    int i ;  
    for ( i = 0 ; i++ < 10 ; )  
        printf ( "%d\n", i ) ;  
    return 0 ;  
}
```

Here, the comparison as well as the incrementation is done through the same expression, **i++ < 10**. Since the **++** operator comes after **i**, first comparison is done, followed by incrementation. Note that it is necessary to initialize **i** to 0.

```
(f) #include <stdio.h>
int main( )
{
    int i;
    for ( i = 0 ; ++i <= 10 ; )
        printf ( "%d\n", i );
    return 0 ;
}
```

Here again, both, the comparison and the incrementation are done through the same expression, **++i <= 10**. Since **++** precedes **i** firstly incrementation is done, followed by comparison. Note that it is necessary to initialize **i** to 0.

Nesting of Loops

The way **if** statements can be nested, similarly **whiles** and **fors** can also be nested. To understand how nested loops work, look at the program given below.

```
/* Demonstration of nested loops */
#include <stdio.h>
int main( )
{
    int r, c, sum ;
    for ( r = 1 ; r <= 3 ; r++ ) /* outer loop */
    {
        for ( c = 1 ; c <= 2 ; c++ ) /* inner loop */
        {
            sum = r + c ;
            printf ( "r = %d c = %d sum = %d\n", r, c, sum ) ;
        }
    }
    return 0 ;
}
```

When you run this program, you will get the following output:

```
r = 1 c = 1 sum = 2
r = 1 c = 2 sum = 3
r = 2 c = 1 sum = 3
r = 2 c = 2 sum = 4
r = 3 c = 1 sum = 4
r = 3 c = 2 sum = 5
```

Here, for each value of **r**, the inner loop is cycled through twice, with the variable **c** taking values from 1 to 2. The inner loop terminates when the value of **c** exceeds 2, and the outer loop terminates when the value of **r** exceeds 3.

As you can see, the body of the outer **for** loop is indented, and the body of the inner **for** loop is further indented. These multiple indentations make the program easier to understand.

Instead of using two statements, one to calculate **sum** and another to print it out, we can compact them into one single statement by saying:

```
printf ( "r = %d c = %d sum = %d\n", r, c, r + c );
```

The way **for** loops have been nested here, similarly, two **while** loops can also be nested. Not only this, a **for** loop can occur within a **while** loop, or a **while** within a **for**.

Multiple Initializations in the **for** Loop

The initialization expression in the **for** loop can contain more than one statement separated by a comma. For example,

```
for ( i = 1, j = 2 ; j <= 10 ; j++ )
```

Multiple statements can also be used in the incrementation expression of **for** loop; i.e., you can increment (or decrement) two or more variables at the same time. Similarly multiple conditions are allowed in the test expression. These conditions must be linked together using logical operators **&&** and/or **||**.

The **break** Statement

We often come across situations where we want to jump out of a loop instantly, without waiting to get back to the condition. The keyword **break** allows us to do this. When **break** is encountered inside any loop, control automatically passes to the first statement after the loop. A

break is usually associated with an **if**. Let's consider the following example:

Example 6.1: Write a program to determine whether a number is prime or not. A prime number is said to be prime if it is divisible only by 1 or itself.

All we have to do to test whether a number is prime or not, is to divide it successively by all numbers from 2 to one less than itself. If remainder of any of these divisions is zero, the number is not a prime. If no division yields a zero then the number is a prime number. Following program implements this logic:

```
#include <stdio.h>
int main( )
{
    int num, i ;

    printf ( "Enter a number " ) ;
    scanf ( "%d", &num ) ;

    i = 2 ;
    while ( i <= num - 1 )
    {
        if ( num % i == 0 )
        {
            printf ( "Not a prime number\n" ) ;
            break ;
        }
        i++ ;
    }

    if ( i == num )
        printf ( "Prime number\n" ) ;
}
```

In this program, the moment **num % i** turns out to be zero, (i.e., **num** is exactly divisible by **i**), the message “Not a prime number” is printed and the control breaks out of the **while** loop. Why does the program require the **if** statement after the **while** loop at all? Well, there are two possibilities the control could have reached outside the **while** loop:

- (a) It jumped out because the number proved to be not a prime.

- (b) The loop came to an end because the value of **i** became equal to **num**.

When the loop terminates in the second case, it means that there was no number between 2 to **num - 1** that could exactly divide **num**. That is, **num** is indeed a prime. If this is true, the program should print out the message "Prime number".

The keyword **break**, breaks the control only from the **while** in which it is placed. Consider the following program, which illustrates this fact:

```
# include <stdio.h>
int main( )
{
    int i = 1 , j = 1 ;

    while ( i++ <= 100 )
    {
        while ( j++ <= 200 )
        {
            if ( j == 150 )
                break ;
            else
                printf ( "%d %d\n", i, j ) ;
        }
    }
    return 0 ;
}
```

In this program when **j** equals 150, **break** takes the control outside the inner **while** only, since it is placed inside the inner **while**.

The *continue* Statement

In some programming situations, we want to take the control to the beginning of the loop, bypassing the statements inside the loop, which have not yet been executed. The keyword **continue** allows us to do this. When **continue** is encountered inside any loop, control automatically passes to the beginning of the loop.

A **continue** is usually associated with an **if**. As an example, let's consider the following program:

```
# include <stdio.h>
```

```
int main( )
{
    int i, j;

    for ( i = 1 ; i <= 2 ; i++ )
    {
        for ( j = 1 ; j <= 2 ; j++ )
        {
            if ( i == j )
                continue ;

            printf ( "%d %d\n", i, j ) ;
        }
    }
    return 0 ;
}
```

The output of the above program would be...

```
1 2
2 1
```

Note that when the value of **i** equals that of **j**, the **continue** statement takes the control to the **for** loop (inner) bypassing the rest of the statements pending execution in the **for** loop (inner).

The *do-while* Loop

The **do-while** loop looks like this:

```
do
{
    this ;
    and this ;
    and this ;
    and this ;
} while ( this condition is true ) ;
```

There is a minor difference between the working of **while** and **do-while** loops. This difference is the place where the condition is tested. The **while** tests the condition before executing any of the statements within the **while** loop. As against this, the **do-while** tests the condition after

having executed the statements within the loop. Figure 6.3 would clarify the execution of **do-while** loop still further.

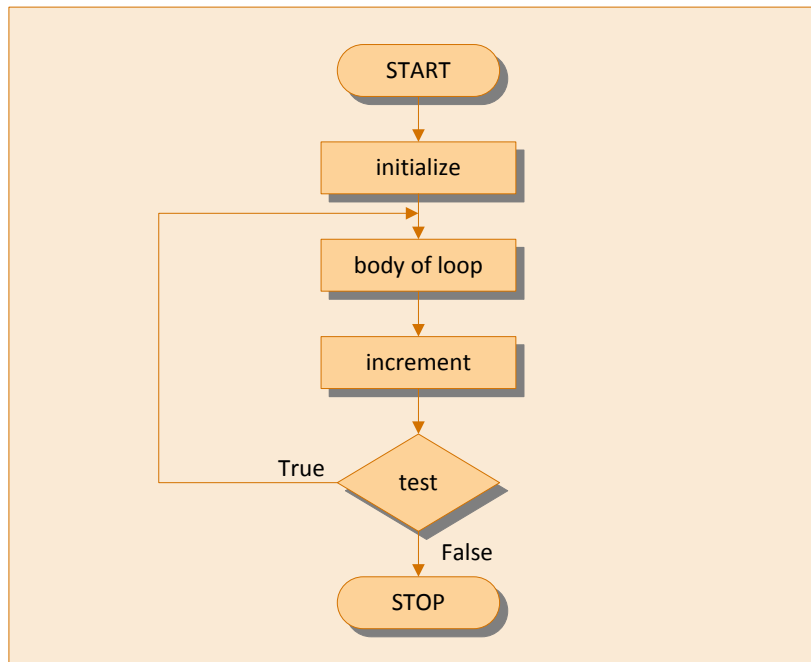


Figure 6.3

This means that **do-while** would execute its statements at least once, even if the condition fails for the first time. The **while**, on the other hand will not execute its statements if the condition fails for the first time. This difference is brought about more clearly by the following program:

```
# include <stdio.h>
int main( )
{
    while ( 4 < 1 )
        printf ( "Hello there \n" ) ;
    return 0 ;
}
```

Here, since the condition fails the first time itself, the **printf()** will not get executed at all. Let's now write the same program using a **do-while** loop.

```
# include <stdio.h>
int main( )
{
```

```
do
{
    printf ( "Hello there \n" );
} while ( 4 < 1 );

return 0 ;
}
```

In this program, the **printf()** would be executed once, since first the body of the loop is executed and then the condition is tested.

The Odd Loop

The loops that we have used so far executed the statements within them a finite number of times. However, in real life programming, one comes across a situation when it is not known beforehand how many times the statements in the loop are to be executed. This situation can be programmed as shown below.

```
/* Execution of a loop an unknown number of times */
# include <stdio.h>
int main( )
{
    char another ;
    int num ;
    do
    {
        printf ( "Enter a number " ) ;
        scanf ( "%d", &num ) ;
        printf ( "square of %d is %d\n", num, num * num ) ;
        printf ( "Want to enter another number y/n " ) ;
        fflush ( stdin ) ;
        scanf ( "%c", &another ) ;
    } while ( another == 'y' ) ;

    return 0 ;
}
```

And here is the sample output...

```
Enter a number 5
square of 5 is 25
Want to enter another number y/n y
```

```
Enter a number 7
square of 7 is 49
Want to enter another number y/n n
```

In this program, the **do-while** loop would keep getting executed till the user continues to answer **y**. The moment user answers **n**, the loop terminates, since the condition (**another == 'y'**) fails. Note that this loop ensures that statements within it are executed at least once even if **n** is supplied first time itself.

Perhaps you are wondering what for have we used the function **fflush()**. The reason is to get rid of a peculiarity of **scanf()**. After supplying a number when we hit the Enter key, **scanf()** assigns the number to variable **num** and keeps the Enter key unread in the keyboard buffer. So when it's time to supply Y or N for the question 'Want to enter another number (y/n)', **scanf()** will read the Enter key from the buffer thinking that user has entered the Enter key. To avoid this problem, we use the function **fflush()**. It is designed to remove or 'flush out' any data remaining in the buffer. The argument to **fflush()** must be the buffer which we want to flush out. Here we have used 'stdin', which means buffer related with standard input device, i.e., keyboard.

Though it is simpler to program such a requirement using a **do-while** loop, the same functionality if required, can also be accomplished using **for** and **while** loops as shown below.

```
/* odd loop using a for loop */
#include <stdio.h>
int main( )
{
    char another = 'y';
    int num;
    for ( ; another == 'y'; )
    {
        printf ( "Enter a number " );
        scanf ( "%d", &num );
        printf ( "square of %d is %d\n", num, num * num );
        printf ( "Want to enter another number y/n " );
        fflush ( stdin );
        scanf ( "%c", &another );
    }
    return 0;
}
```

```
/* odd loop using a while loop */
#include <stdio.h>
int main( )
{
    char another = 'y';
    int num;

    while ( another == 'y' )
    {
        printf ( "Enter a number " );
        scanf ( "%d", &num );
        printf ( "square of %d is %d\n", num, num * num );
        printf ( "Want to enter another number y/n " );
        fflush ( stdin );
        scanf ( " %c", &another );
    }
    return 0;
}
```

break and **continue** are used with **do-while** just as they would be in a **while** or a **for** loop. A **break** takes you out of the **do-while** bypassing the conditional test. A **continue** sends you straight to the test at the end of the loop.

Summary

- (a) Unlike a **while** loop, in a **for** loop the initialization, test and incrementation are written in a single line.
- (b) A **break** statement takes the execution control out of the loop.
- (c) A **continue** statement skips the execution of the statements after it and takes the control through the next cycle of the loop.
- (d) A **do-while** loop is used to ensure that the statements within the loop are executed at least once.
- (e) The operators **+=**, **-=**, ***=**, **/=**, **%=** are compound assignment operators. They modify the value of the operand to the left of them.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    int i = 0 ;
    for ( ; i ; )
        printf ( "Here is some mail for you\n" ) ;
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
{
    int i ;
    for ( i = 1 ; i <= 5 ; printf ( "%d\n", i ) ) ;
        i++ ;
    return 0 ;
}
```
- (c)

```
#include <stdio.h>
int main( )
{
    int i = 1, j = 1 ;
    for ( ; ; )
    {
        if ( i > 5 )
            break ;
        else
            j += i ;
        printf ( "%d\n", j ) ;
        i += j ;
        return 0 ;
    }
}
```

[B] Answer the following:

- (a) The three parts of the loop expression in the **for** loop are:

the i _____ expression
the t _____ expression

the `i` _____ expression

(b) The **break** statement is used to exit from:

1. An **if** statement
2. A **for** loop
3. A program
4. The **main()** function

(c) A **do-while** loop is useful when we want that the statements within the loop must be executed:

1. Only once
2. At least once
3. More than once
4. None of the above

(d) In what sequence the initialization, testing and execution of body is done in a **do-while** loop

1. Initialization, execution of body, testing
2. Execution of body, initialization, testing
3. Initialization, testing, execution of body
4. None of the above

(e) Which of the following is not an infinite loop?

- | | |
|--|--|
| 1. <pre>int i = 1 ; while (1) { i++ ; }</pre> | 2. <pre>for (; ;)</pre> |
| 3. <pre>int t = 0, f ; while (t) { f = 1 ; }</pre> | 4. <pre>int y, x = 0 ; do { y = x ; } while (x == 0</pre> |

(f) Which keyword is used to take the control to the beginning of the loop?

(g) How many times the **while** loop in the following C code will get executed?

```
# include <stdio.h>
int main( )
{
    int j = 1 ;
    while ( j <= 255 ) ;
    {
        printf ( "%c %d ", j, j ) ;
        j++;
    }
    return 0 ;
}
```

- (h) Which of the following statements is true for the following program?

```
# include <stdio.h>
int main( )
{
    int x=10, y = 100%90 ;
    for ( i = 1 ; i <= 10 ; i++ ) ;
    if ( x != y ) ;
        printf ( "x = %d y = %d\n", x, y ) ;
    return 0 ;
}
```

1. The **printf()** function is called 10 times.
 2. The program will produce the output x=10 y=10.
 3. The ; after the **if (x!=y)** would not produce an error.
 4. The program will not produce any output.
 5. The **printf()** function is called infinite times.
- (i) Which of the following statement is true about a **for** loop used in a C program?
1. **for** loop works faster than a **while** loop.
 2. All things that can be done using a **for** loop can also be done using a **while** loop.
 3. **for (; ;)** implements an infinite loop.
 4. **for** loop can be used if we want statements in a loop to get executed at least once.
 5. **for** loop works faster than a **do-while** loop.

[C] Attempt the following:

- (a) Write a program to print all prime numbers from 1 to 300. (Hint: Use nested loops, **break** and **continue**)
- (b) Write a program to fill the entire screen with a smiling face. The smiling face has an ASCII value 1.
- (c) Write a program to add first seven terms of the following series

$$\frac{1}{1!} + \frac{2}{2!} + \frac{3}{3!} + \dots$$

using a **for** loop:

- (d) Write a program to generate all combinations of 1, 2 and 3 using **for** loop.
- (e) A machine is purchased which will produce earning of Rs. 1000 per year while it lasts. The machine costs Rs. 6000 and will have a salvage value of Rs. 2000 when it is condemned. If 9 percent per annum can be earned on alternate investments, write a program to determine what will be the minimum life of the machine to make it a more attractive investment compared to alternative investment?
- (c) Write a program to print the multiplication table of the number entered by the user. The table should get displayed in the following form:

29 * 1 = 29

29 * 2 = 58

...

- (f) According to a study, the approximate level of intelligence of a person can be calculated using the following formula:

$$i = 2 + (y + 0.5x)$$

Write a program that will produce a table of values of **i**, **y** and **x**, where **y** varies from 1 to 6, and, for each value of **y**, **x** varies from 5.5 to 12.5 in steps of 0.5.

- (g) When interest compounds **q** times per year at an annual rate of **r** % for **n** years, the principal **p** compounds to an amount **a** as per the following formula

$$a = p (1 + r / q)^{nq}$$

Write a program to read 10 sets of **p**, **r**, **n** & **q** and calculate the corresponding **a**s.

- (h) The natural logarithm can be approximated by the following series.

$$\frac{x-1}{x} + \frac{1}{2}\left(\frac{x-1}{x}\right)^2 + \frac{1}{2}\left(\frac{x-1}{x}\right)^3 + \frac{1}{2}\left(\frac{x-1}{x}\right)^4 + \dots$$

If x is input through the keyboard, write a program to calculate the sum of first seven terms of this series.

- (i) Write a program to generate all Pythagorean Triplets with side length less than or equal to 30.
- (j) Population of a town today is 100000. The population has increased steadily at the rate of 10 % per year for last 10 years. Write a program to determine the population at the end of each year in the last decade.
- (k) Ramanujan number is the smallest number that can be expressed as sum of two cubes in two different ways. Write a program to print all such numbers up to a reasonable limit.
- (l) Write a program to print 24 hours of day with suitable suffixes like AM, PM, Noon and Midnight.
- (m) Write a program to produce the following output:

```

          1
        2   3
      4   5   6
    7   8   9  10

```

- (n) Write a program to produce the following output:

```

A B C D E F G F E D C B A
A B C D E F   F E D C B A
A B C D E     E D C B A
A B C D       D C B A
A B C         C B A
A B           B A
A             A

```

- (o) Write a program to produce the following output:

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

7

Case Control Instruction

- **Decisions using *switch***
The Tips and Traps
- ***switch* versus *if-else* Ladder**
- **The *goto* Keyword**
- **Summary**
- **Exercise**



7 *Case Control Instruction*

- Decisions Using *switch*
The Tips and Traps
- *switch* Versus *if-else* Ladder
- The *goto* Keyword
- Summary
- Exercise

In real life, we are often faced with situations where we are required to make a choice between a number of alternatives rather than only one or two. For example, which school to join or which hotel to visit, or still harder, which girl to marry. Serious C programming is same; the choice we are asked to make is more complicated than merely selecting between two alternatives. C provides a special control statement that allows us to handle such cases effectively; rather than using a series of **if** statements. This control instruction is, in fact, the topic of this chapter. Towards the end of the chapter, we would also study a keyword called **goto**, and understand why we should avoid its usage in C programming.

Decisions using *switch*

The control statement that allows us to make a decision from the number of choices is called a **switch**, or more correctly a **switch-case-default**, since these three keywords go together to make up the control statement. They most often appear as follows:

```
switch ( integer expression )
{
    case constant 1 :
        do this ;
    case constant 2 :
        do this ;
    case constant 3 :
        do this ;
    default :
        do this ;
}
```

The integer expression following the keyword **switch** is any C expression that will yield an integer value. It could be an integer constant like 1, 2 or 3, or an expression that evaluates to an integer. The keyword **case** is followed by an integer or a character constant. Each constant in each **case** must be different from all the others. The “do this” lines in the above form of **switch** represent any valid C statement.

What happens when we run a program containing a **switch**? First, the integer expression following the keyword **switch** is evaluated. The value it gives is then matched, one-by-one, against the constant values that follow the **case** statements. When a match is found, the program executes the statements following that **case**, and all subsequent **case** and **default** statements as well. If no match is found with any of the **case**

statements, only the statements following the **default** case are executed. A few examples will show how this control instruction works.

Consider the following program:

```
# include <stdio.h>
int main( )
{
    int i = 2 ;

    switch ( i )
    {
        case 1 :
            printf ( "I am in case 1 \n" ) ;
        case 2 :
            printf ( "I am in case 2 \n" ) ;
        case 3 :
            printf ( "I am in case 3 \n" ) ;
        default :
            printf ( "I am in default \n" ) ;
    }
    return 0 ;
}
```

The output of this program would be:

```
I am in case 2
I am in case 3
I am in default
```

The output is definitely not what we expected! We didn't expect the second and third line in the above output. The program prints case 2 and case 3 and the default case. Well, yes. We said the **switch** executes the case where a match is found and all the subsequent **cases** and the **default** as well.

If you want that only case 2 should get executed, it is upto you to get out of the **switch** then and there by using a **break** statement. The following example shows how this is done. Note that there is no need for a **break** statement after the **default**, since on reaching the **default** case, the control comes out of the **switch** anyway.

```
# include <stdio.h>
```

```
int main( )
{
    int i = 2 ;
    switch ( i )
    {
        case 1 :
            printf ( "I am in case 1 \n" ) ;
            break ;
        case 2 :
            printf ( "I am in case 2 \n" ) ;
            break ;
        case 3 :
            printf ( "I am in case 3 \n" ) ;
            break ;
        default :
            printf ( "I am in default \n" ) ;
    }
    return 0 ;
}
```

The output of this program would be:

```
I am in case 2
```

The operation of **switch** is shown in Figure 7.1 in the form of a flowchart for a better understanding.

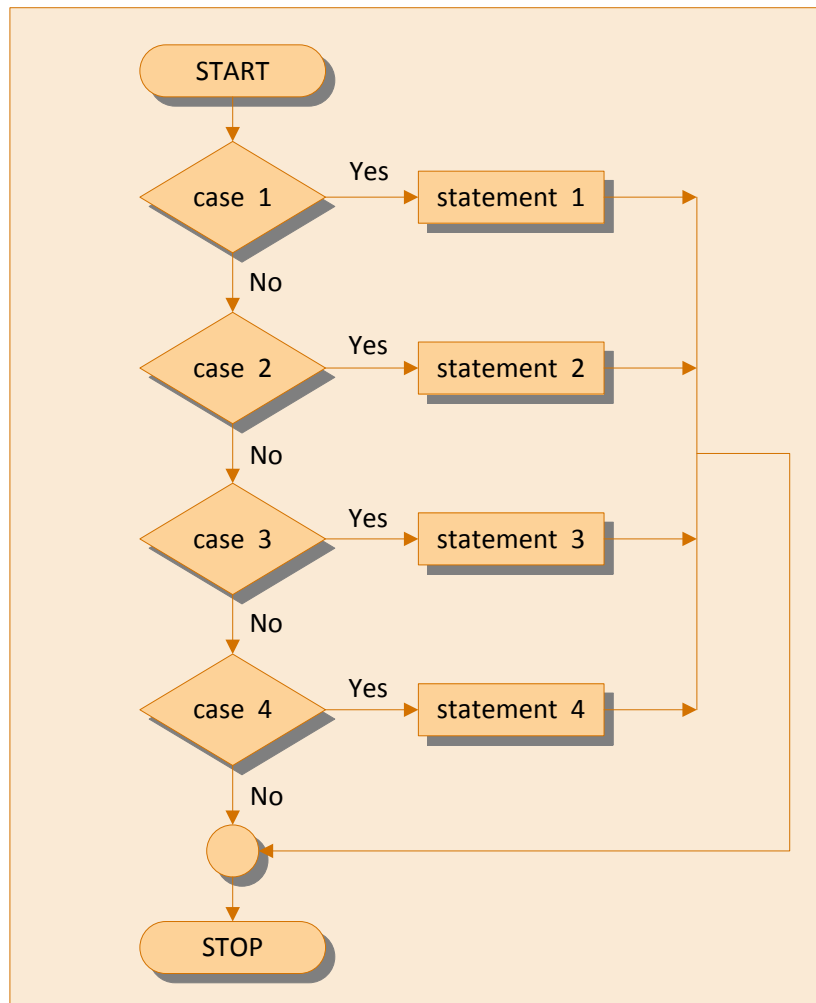


Figure 7.1

The Tips and Traps

A few useful tips about the usage of **switch** and a few pitfalls to be avoided:

- (a) The earlier program that used **switch** may give you the wrong impression that you can use only cases arranged in ascending order, 1, 2, 3 and default. You can, in fact, put the cases in any order you please. Here is an example of scrambled case order:

```
# include <stdio.h>
int main( )
{
    int i = 22 ;
```

```
switch ( i )
{
    case 121 :
        printf ( "I am in case 121 \n" ) ;
        break ;
    case 7 :
        printf ( "I am in case 7 \n" ) ;
        break ;
    case 22 :
        printf ( "I am in case 22 \n" ) ;
        break ;
    default :
        printf ( "I am in default \n" ) ;
}
return 0 ;
}
```

The output of this program would be:

```
I am in case 22
```

- (b) You are also allowed to use **char** values in **case** and **switch** as shown in the following program:

```
# include <stdio.h>
int main( )
{
    char c = 'x' ;

    switch ( c )
    {
        case 'v' :
            printf ( "I am in case v \n" ) ;
            break ;
        case 'a' :
            printf ( "I am in case a \n" ) ;
            break ;
        case 'x' :
            printf ( "I am in case x \n" ) ;
            break ;
        default :
```

```
        printf ( "I am in default \n" );
    }
    return 0 ;
}
```

The output of this program would be:

```
I am in case x
```

In fact here when we use 'v', 'a', 'x' they are actually replaced by the ASCII values (118, 97, 120) of these character constants.

- (c) At times we may want to execute a common set of statements for multiple **cases**. The following example shows how this can be achieved:

```
# include <stdio.h>
int main( )
{
    char ch ;

    printf ( "Enter any one of the alphabets a, b, or c " );
    scanf ( "%c", &ch );

    switch ( ch )
    {
        case 'a' :
        case 'A' :
            printf ( "a as in ashar\n" );
            break ;
        case 'b' :
        case 'B' :
            printf ( "b as in brain\n" );
            break ;
        case 'c' :
        case 'C' :
            printf ( "c as in cookie\n" );
            break ;
        default :
            printf ( "wish you knew what are alphabets\n" );

    }

    return 0 ;
}
```

```
}  
}
```

Here, we are making use of the fact that once a **case** is satisfied; the control simply falls through the **switch** till it doesn't encounter a **break** statement. That is why if an alphabet **a** is entered, the **case 'a'** is satisfied and since there are no statements to be executed in this **case**, the control automatically reaches the next **case**, i.e., **case 'A'** and executes all the statements in this **case**.

- (d) Even if there are multiple statements to be executed in each **case**, there is no need to enclose them within a pair of braces (unlike **if** and **else**).
- (e) Every statement in a **switch** must belong to some **case** or the other. If a statement doesn't belong to any **case**, the compiler won't report an error. However, the statement would never get executed. For example, in the following program, the **printf()** never goes to work:

```
#include <stdio.h>  
int main( )  
{  
    int i, j;  
  
    printf ( "Enter value of i" );  
    scanf ( "%d", &i );  
  
    switch ( i )  
    {  
        printf ( "Hello\n" );  
        case 1 :  
            j = 10 ;  
            break ;  
        case 2 :  
            j = 20 ;  
            break ;  
    }  
    return 0 ;  
}
```

- (f) If we have no **default** case, then the program simply falls through the entire **switch** and continues with the next instruction (if any,) that follows the closing brace of **switch**.
- (g) Is **switch** a replacement for **if**? Yes and no. Yes, because it offers a better way of writing programs as compared to **if**, and no, because, in certain situations, we are left with no choice but to use **if**. The disadvantage of **switch** is that one cannot have a case in a **switch** which looks like:

```
case i <= 20 :
```

All that we can have after the case is an **int** constant or a **char** constant or an expression that evaluates to one of these constants. Even a **float** is not allowed.

The advantage of **switch** over **if** is that it leads to a more structured program and the level of indentation is manageable, more so, if there are multiple statements within each **case** of a **switch**.

- (h) We can check the value of any expression in a **switch**. Thus, the following **switch** statements are legal:

```
switch ( i + j * k )  
switch ( 23 + 45 % 4 * k )  
switch ( a < 4 && b > 7 )
```

Expressions can also be used in cases provided they are constant expressions. Thus, **case 3 + 7** is correct, however, **case a + b** is incorrect.

- (i) The **break** statement when used in a **switch** takes the control outside the **switch**. However, use of **continue** will not take the control to the beginning of **switch** as one is likely to believe. This is because **switch** is not a looping statement unlike **while**, **for** or **do-while**.
- (j) In principle, a **switch** may occur within another, but in practice, this is rarely done. Such statements would be called nested **switch** statements.
- (k) The **switch** statement is very useful while writing menu driven programs. This aspect of **switch** is discussed in the exercise at the end of this chapter.

switch versus if-else Ladder

There are some things that you simply cannot do with a **switch**. These are:

- (a) A float expression cannot be tested using a **switch**.
- (b) Cases can never have variable expressions (for example, it is wrong to say **case a +3 :**).
- (c) Multiple cases cannot use same expressions. Thus the following **switch** is illegal:

```
switch ( a )
{
    case 3 :
        ...
    case 1 + 2 :
        ...
}
```

(a), (b) and (c) above may lead you to believe that these are obvious disadvantages with a **switch**, especially since there weren't any such limitations with **if-else**. Then why use a **switch** at all? For speed—**switch** works faster than an equivalent **if-else** ladder. How come? This is because the compiler generates a jump table for a **switch** during compilation. As a result, during execution it simply refers the jump table to decide which case should be executed, rather than actually checking which case is satisfied. As against this, **if-elses** are slower because the conditions in them are evaluated at execution time. Thus a **switch** with 10 cases would work faster than an equivalent **if-else** ladder. If the 10th **case** is satisfied then jump table would be referred and statements for the 10th **case** would be executed. As against this, in an **if-else** ladder 10 conditions would be evaluated at execution time, which makes it slow. Note that a lookup in the jump table is faster than evaluation of a condition, especially if the condition is complex.

The goto Keyword

Avoid **goto** keyword! It makes a C programmer's life miserable. There is seldom a legitimate reason for using **goto**, and its use is one of the reasons that programs become unreliable, unreadable, and hard to debug. And yet many programmers find **goto** seductive.

In a difficult programming situation, it seems so easy to use a **goto** to take the control where you want. However, almost always, there is a more elegant way of writing the same program using **if**, **for**, **while**, **do-while** and **switch**. These constructs are far more logical and easy to understand.

The big problem with **gotos** is that when we do use them we can never be sure how we got to a certain point in our code. They obscure the flow of control. So as far as possible skip them. You can always get the job done without them. Trust me, with good programming skills **goto** can always be avoided. This is the first and last time that we are going to use **goto** in this book. However, for sake of completeness of the book, the following program shows how to use **goto**:

```
# include <stdio.h>
# include <stdlib.h>

int main( )
{
    int goals ;

    printf ( "Enter the number of goals scored against India" ) ;
    scanf ( "%d", &goals ) ;

    if ( goals <= 5 )
        goto sos ;
    else
    {
        printf ( "About time soccer players learnt C\n" ) ;
        printf ( "and said goodbye! adieu! to soccer\n" ) ;
        exit ( 1 ) ; /* terminates program execution */
    }

    sos :
        printf ( "To err is human!\n" ) ;

    return 0 ;
}
```

And here are two sample runs of the program...

```
Enter the number of goals scored against India 3
```

To err is human!
Enter the number of goals scored against India 7
About time soccer players learnt C
and said goodbye! adieu! to soccer

A few remarks about the program would make the things clearer.

- If the condition is satisfied the **goto** statement transfers control to the label **sos**, causing **printf()** following **sos** to be executed.
- The label can be on a separate line or on the same line as the statement following it, as in,

```

sos : printf ( "To err is human!\n" ) ;

```

- Any number of **gotos** can take the control to the same label.
- The **exit()** function is a standard library function which terminates the execution of the program. It is necessary to use this function since we don't want the statement

```
printf ( "To err is human!\n" );
```

to get executed after execution of the **else** block.

- The only programming situation in favour of using **goto** is when we want to take the control out of the loop that is contained in several other loops. The following program illustrates this:

```
# include <stdio.h>

int main( )
{
    int i,j,k;

    for ( i = 1 ; i <= 3 ; i++ )
    {
        for ( j = 1 ; j <= 3 ; j++ )
        {
            for ( k = 1 ; k <= 3 ; k++ )
            {
                if ( i == 3 && j == 3 && k == 3 )
                    goto out ;
                else
```

```
        printf ( "%d %d %d\n", i, j, k ) ;
    }
}
out :
    printf ( "Out of the loops at last!\n" ) ;

return 0 ;
}
```

Go through the program carefully and find out how it works. Also write a program to implement the same logic without using **goto**.

Summary

- (a) Ramesh's basic salary is input through the keyboard. His dearness allowance is 40% of basic salary, and house rent allowance is 20% of basic salary. Write a program to calculate his gross salary.
- (b) The
- (c) When we need to choose one amongst number of alternatives, a **switch** statement is used.
- (d) The **switch** keyword is followed by an integer or an expression that evaluates to an integer.
- (e) The **case** keyword is followed by an integer or a character constant.
- (f) The control falls through all the cases unless the fall is stopped using a **break** statement.
- (g) The usage of the **goto** keyword should be avoided as it usually disturbs the normal flow of execution.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    char suite = 3 ;
    switch ( suite )
    {
        case 1 :
```

```
        printf ( "Diamond\n" );
    case 2 :
        printf ( "Spade\n" );
    default :
        printf ( "Heart\n" );
    }
    printf ( "I thought one wears a suite\n" );
    return 0 ;
}
```

(b) # include <stdio.h>

```
int main( )
{
    int c = 3 ;
    switch ( c )
    {
        case '3' :
            printf ( "You never win the silver prize\n" );
            break ;
        case 3 :
            printf ( "You always lose the gold prize\n" );
            break ;
        default :
            printf ( "Of course provided you win a prize\n" );
    }
    return 0 ;
}
```

(c) # include <stdio.h>

```
int main( )
{
    int i = 3 ;
    switch ( i )
    {
        case 0 :
            printf ( "Customers are dicey\n" );
        case 1 + 0 :
            printf ( "Markets are pricey\n" );
        case 4 / 2 :
            printf ( "Investors are moody\n" );
        case 8 % 5 :
            printf ( "At least employees are good\n" );
    }
```

```
    }  
    return 0 ;  
}
```

- (d) # include <stdio.h>
int main()
{
 int k ;
 float j = 2.0 ;
 switch (k = j + 1)
 {
 case 3 :
 printf ("Trapped\n") ;
 break ;
 default :
 printf ("Caught!\n") ;
 }
 return 0 ;
}
- (e) # include <stdio.h>
int main()
{
 int ch = 'a' + 'b' ;
 switch (ch)
 {
 case 'a' :
 case 'b' :
 printf ("You entered b\n") ;
 case 'A' :
 printf ("a as in ashar\n") ;
 case 'b' + 'a' :
 printf ("You entered a and b\n") ;
 }
 return 0 ;
}
- (f) # include <stdio.h>
int main()
{
 int i = 1 ;
 switch (i - 2)
 {

```
        case -1 :
            printf ( "Feeding fish\n" ) ;
        case 0 :
            printf ( "Weeding grass\n" ) ;
        case 1 :
            printf ( "Mending roof\n" ) ;
        default :
            printf ( "Just to survive\n" ) ;
    }
    return 0 ;
}
```

[B] Point out the errors, if any, in the following programs:

(a) #include <stdio.h>

```
int main( )
{
    int suite = 1 ;
    switch ( suite ) ;
    {
        case 0 ;
            printf ( "Club\n" ) ;
        case 1 ;
            printf ( "Diamond\n" ) ;
    }
    return 0 ;
}
```

(b) #include <stdio.h>

```
int main( )
{
    int temp ;
    scanf ( "%d", &temp ) ;
    switch ( temp )
    {
        case ( temp <= 20 ) :
            printf ( "Ooooooohhhh! Damn cool!\n" ) ;
        case ( temp > 20 && temp <= 30 ) :
            printf ( "Rain rain here again!\n" ) ;
        case ( temp > 30 && temp <= 40 ) :
            printf ( "Wish I am on Everest\n" ) ;
        default :
            printf ( "Good old nagpur weather\n" ) ;
    }
```

```
    }
    return 0 ;
}

(c) # include <stdio.h>
int main( )
{
    float a = 3.5 ;
    switch ( a )
    {
        case 0.5 :
            printf ( "The art of C\n" ) ; break ;
        case 1.5 :
            printf ( "The spirit of C\n" ) ; break ;
        case 2.5 :
            printf ( "See through C\n" ) ; break ;
        case 3.5 :
            printf ( "Simply c\n" ) ;
    }
    return 0 ;
}

(d) # include <stdio.h>
int main( )
{
    int a = 3, b = 4, c ;
    c = b - a ;
    switch ( c )
    {
        case 1 || 2 :
            printf ( "God give me a chance to change things\n" ) ;
            break ;

        case a || b :
            printf ( "God give me a chance to run my show\n" ) ;
            break ;
    }
    return 0 ;
}
```

[C] Write a menu driven program which has following options:

1. Factorial of a number

2. Prime or not
3. Odd or even
4. Exit

Once a menu item is selected the appropriate action should be taken and once this action is finished, the menu should reappear. Unless the user selects the 'Exit' option the program should continue to run.

Hint: Make use of an infinite **while** and a **switch** statement.

[D] Write a program to find the grace marks for a student using **switch**. The user should enter the class obtained by the student and the number of subjects he has failed in. Use the following logic:

- If the student gets first class and the number of subjects he failed in is greater than 3, then he does not get any grace. Otherwise the grace is of 5 marks per subject.
- If the student gets second class and the number of subjects he failed in is greater than 2, then he does not get any grace. Otherwise the grace is of 4 marks per subject.
- If the student gets third class and the number of subjects he failed in is greater than 1, then he does not get any grace. Otherwise the grace is of 5 marks.

8

Functions

- **What is a Function?**
Why use Functions?
- **Passing Values between Functions**
- **Scope Rule of Functions**
- **Order of Passing Arguments**
- **Using Library Functions**
- **One Dicey Issue**
- **Return Type of Function**
- **Summary**
- **Exercise**



8 *Functions*

- What is a Function
 - Why Use Functions
- Passing Values between Functions
- Scope Rule of Functions
- Order of Passing Arguments
- Using Library Functions
- One Dicey Issue
- Return Type of Function
- Summary
- Exercise

Man is an intelligent species, but still cannot perform all of life's tasks all alone. He has to rely on others. You may call a mechanic to fix up your bike, hire a gardener to mow your lawn, or rely on a store to supply you groceries every month. A computer program (except for the simplest one) finds itself in a similar situation. It cannot handle all the tasks by itself. Instead, it requests other program-like entities—called 'functions' in C—to get its tasks done. In this chapter we will study these functions. We will look at a variety of features of these functions, starting with the simplest one and then working towards those that demonstrate the power of C functions.

What is a Function?

A function is a self-contained block of statements that perform a coherent task of some kind. Every C program can be thought of as a collection of these functions. As we noted earlier, using a function is something like hiring a person to do a specific job for you. Sometimes the interaction with this person is very simple; sometimes it's complex.

Suppose you have a task that is always performed exactly in the same way—say a bimonthly servicing of your motorbike. When you want it to be done, you go to the service station and say, "It's time, do it now". You don't need to give instructions, because the mechanic knows his job. You don't need to be told how the job is done. You assume the bike would be serviced in the usual way, the mechanic does it and that's that.

Let us now look at a simple C function that operates in much the same way as the mechanic. Actually, we will be looking at two things—a function that calls or activates the function and the function itself.

```
#include <stdio.h>
void message( ) ; /* function prototype declaration */
int main( )
{
    message( ) ; /* function call */
    printf ( "Cry, and you stop the monotony!\n" ) ;
    return 0 ;
}
void message( ) /* function definition */
{
    printf ( "Smile, and the world smiles with you...\n" ) ;
}
```

And here's the output...

```
Smile, and the world smiles with you...  
Cry, and you stop the monotony!
```

Here, we have defined two functions—**main()** and **message()**. In fact, we have used the word **message** at three places in the program. Let us understand the meaning of each.

The first is the function prototype declaration and is written as:

```
void message( );
```

This prototype declaration indicates that **message()** is a function which after completing its execution does not return any value. This 'does not return any value' is indicated using the keyword **void**. It is necessary to mention the prototype of every function that we intend to define in the program.

The second usage of **message** is...

```
void message( )  
{  
    printf ( "Smile, and the world smiles with you...\n" );  
}
```

This is the function definition. In this definition right now we are having only **printf()**, but we can also use **if**, **for**, **while**, **switch**, etc., within this function definition.

The third usage is...

```
message( );
```

Here the function **message()** is being called by **main()**. What do we mean when we say that **main()** 'calls' the function **message()**? We mean that the control passes to the function **message()**. The activity of **main()** is temporarily suspended; it falls asleep while the **message()** function wakes up and goes to work. When the **message()** function runs out of statements to execute, the control returns to **main()**, which comes to life again and begins executing its code at the exact point where it left off. Thus, **main()** becomes the 'calling' function, whereas **message()** becomes the 'called' function.

If you have grasped the concept of 'calling' a function you are prepared for a call to more than one function. Consider the following example:

```
# include <stdio.h>
void italy( );
void brazil( );
void argentina( );
int main( )
{
    printf ( "I am in main\n" );
    italy( );
    brazil( );
    argentina( );
    return 0 ;
}
void italy( )
{
    printf ( "I am in italy\n" );
}
void brazil( )
{
    printf ( "I am in brazil\n" );
}
void argentina( )
{
    printf ( "I am in argentina\n" );
}
```

The output of the above program when executed would be as under:

```
I am in main
I am in italy
I am in brazil
I am in argentina
```

A number of conclusions can be drawn from this program:

- A C program is a collection of one or more functions.
- If a C program contains only one function, it must be **main()**.
- If a C program contains more than one function, then one (and only one) of these functions must be **main()**, because program execution always begins with **main()**.

- There is no limit on the number of functions that might be present in a C program.
- Each function in a program is called in the sequence specified by the function calls in **main()**.
- After each function has done its thing, control returns to **main()**. When **main()** runs out of statements and function calls, the program ends.

As we have noted earlier, the program execution always begins with **main()**. Except for this fact, all C functions enjoy a state of perfect equality. No precedence, no priorities, nobody is nobody's boss. One function can call another function it has already called but has in the meantime left temporarily in order to call a third function which will sometime later call the function that has called it, if you understand what I mean. No? Well, let me illustrate with an example.

```
#include <stdio.h>
void italy( );
void brazil( );
void argentina( );
int main( )
{
    printf ( "I am in main\n" );
    italy( );
    printf ( "I am finally back in main\n" );
    return 0 ;
}
void italy( )
{
    printf ( "I am in italy\n" );
    brazil( );
    printf ( "I am back in italy\n" );
}
void brazil( )
{
    printf ( "I am in brazil\n" );
    argentina( );
}
void argentina( )
{
    printf ( "I am in argentina\n" );
```

```
}
```

And the output would look like...

```
I am in main
I am in italy
I am in brazil
I am in argentina
I am back in italy
I am finally back in main
```

Here, **main()** calls other functions, which in turn call still other functions. Trace carefully the way control passes from one function to another. Since the compiler always begins the program execution with **main()**, every function in a program must be called directly or indirectly by **main()**. In other words, the **main()** function drives other functions.

Let us now summarize what we have learnt so far.

- (a) A function gets called when the function name is followed by a semicolon (;). For example,

```
argentina( ) ;
```

- (b) A function is defined when function name is followed by a pair of braces ({ }) in which one or more statements may be present. For example,

```
void argentina( )
{
    statement 1 ;
    statement 2 ;
    statement 3 ;
}
```

- (c) Any function can be called from any other function. Even **main()** can be called from other functions. For example,

```
# include <stdio.h>
void message( ) ;
int main( )
{
```



```
    message( ) ;  
    return 0 ;  
}  
void message( )  
{  
    printf ( "Can't imagine life without C\n" ) ;  
    main( ) ;  
}
```

- (d) A function can be called any number of times. For example,

```
# include <stdio.h>  
void message( ) ;  
int main( )  
{  
    message( ) ;  
    message( ) ;  
    return 0 ;  
}  
void message( )  
{  
    printf ( "Jewel Thief!!\n" ) ;  
}
```

- (e) The order in which the functions are defined in a program and the order in which they get called need not necessarily be same. For example,

```
# include <stdio.h>  
void message1( ) ;  
void message2( ) ;  
int main( )  
{  
    message1( ) ;  
    message2( ) ;  
    return 0 ;  
}  
void message2( )  
{  
    printf ( "But the butter was bitter\n" ) ;  
}
```

```
void message1( )
{
    printf ( "Mary bought some butter\n" );
}
```

Here, even though **message1()** is getting called before **message2()**, still, **message1()** has been defined after **message2()**. However, it is advisable to define the functions in the same order in which they are called. This makes the program easier to understand.

- (f) A function can call itself. Such a process is called 'recursion'. We would discuss this aspect of C functions in Chapter 10.
- (g) A function can be called from another function, but a function cannot be defined in another function. Thus, the following program code would be wrong, since **argentina()** is being defined inside another function, **main()**:

```
int main( )
{
    printf ( "I am in main\n" );
    void argentina( )
    {
        printf ( "I am in argentina\n" );
    }
}
```

- (h) There are basically two types of functions:

Library functions Ex. **printf()**, **scanf()**, etc.

User-defined functions Ex. **argentina()**, **brazil()**, etc.

As the name suggests, library functions are nothing but commonly required functions grouped together and stored in a Library file on the disk. These library of functions come ready-made with development environments like Turbo C, Visual Studio, NetBeans, gcc, etc. The procedure for calling both types of functions is exactly same.

Why use Functions?

Why write separate functions at all? Why not squeeze the entire logic into one function, **main()**? Two reasons:

- (a) Writing functions avoids rewriting the same code over and over. Suppose you have a section of code in your program that calculates area of a triangle. If later in the program you want to calculate the area of a different triangle, you won't like it if you are required to write the same instructions all over again. Instead, you would prefer to jump to a 'section of code' that calculates area and then jump back to the place from where you left off. This section of code is nothing but a function.
- (b) By using functions it becomes easier to write programs and keep track of what they are doing. If the operation of a program can be divided into separate activities, and each activity placed in a different function, then each could be written and checked more or less independently. Separating the code into modular functions also makes the program easier to design and understand.

What is the moral of the story? Don't try to cram the entire logic in one function. It is a very bad style of programming. Instead, break a program into small units and write functions for each of these isolated subdivisions. Don't hesitate to write functions that are called only once. What is important is that these functions perform some logically isolated task.

Passing Values between Functions

The functions that we have used so far haven't been very flexible. We call them and they do what they are designed to do. Like our mechanic who always services the motorbike in exactly the same way, we haven't been able to influence the functions in the way they carry out their tasks. It would be nice to have a little more control over what functions do, in the same way it would be nice to be able to tell the mechanic, 'Also change the engine oil, I am going for an outing'. In short, now we want to communicate between the 'calling' and the 'called' functions.

The mechanism used to convey information to the function is the 'argument'. You have unknowingly used the arguments in the **printf()** and **scanf()** functions; the format string and the list of variables used inside the parentheses in these functions are arguments. The arguments are sometimes also called 'parameters'.

Consider the following program. In this program, in **main()** we receive the values of **a**, **b** and **c** through the keyboard and then output the sum of **a**, **b** and **c**. However, the calculation of sum is done in a different function called **calsum()**. If sum is to be calculated in **calsum()** and

values of **a**, **b** and **c** are received in **main()**, then we must pass on these values to **calsum()**, and once **calsum()** calculates the sum, we must return it from **calsum()** back to **main()**.

```
/* Sending and receiving values between functions */
# include <stdio.h>
int calsum ( int x, int y, int z ) ;
int main( )
{
    int a, b, c, sum ;
    printf ( "Enter any three numbers " ) ;
    scanf ( "%d %d %d", &a, &b, &c ) ;
    sum = calsum ( a, b, c ) ;
    printf ( "Sum = %d\n", sum ) ;
    return 0 ;
}
int calsum ( int x, int y, int z )
{
    int d ;

    d = x + y + z ;
    return ( d ) ;
}
```

And here is the output...

```
Enter any three numbers 10 20 30
Sum = 60
```

There are a number of things to note about this program:

- (a) In this program, from the function **main()**, the values of **a**, **b** and **c** are passed on to the function **calsum()**, by making a call to the function **calsum()** and mentioning **a**, **b** and **c** in the parentheses:

```
sum = calsum ( a, b, c ) ;
```

In the **calsum()** function these values get collected in three variables **x**, **y** and **z**:

```
int calsum ( int x, int y, int z )
```

- (b) The variables **a**, **b** and **c** are called ‘actual arguments’, whereas the variables **x**, **y** and **z** are called ‘formal arguments’. Any number of arguments can be passed to a function being called. However, the type, order and number of the actual and formal arguments must always be same.

Instead of using different variable names **x**, **y** and **z**, we could have used the same variable names **a**, **b** and **c**. But the compiler would still treat them as different variables since they are in different functions.

- (c) Note the function prototype declaration of **calsum()**. Instead of the usual **void**, we are using **int**. This indicates that **calsum()** is going to return a value of the type **int**. It is not compulsory to use variable names in the prototype declaration. Hence we could as well have written the prototype as:

```
int calsum ( int, int, int ) ;
```

In the definition of **calsum** too, **void** has been replaced by **int**.

- (d) In the earlier programs, the moment closing brace (**}**) of the called function was encountered, the control returned to the calling function. No separate **return** statement was necessary to send back the control.

This approach is fine if the called function is not going to return any meaningful value to the calling function. In the above program, however, we want to return the sum of **x**, **y** and **z**. Therefore, it is necessary to use the **return** statement.

The **return** statement serves two purposes:

- (1) On executing the **return** statement, it immediately transfers the control back to the calling function.
 - (2) It returns the value present in the parentheses after **return**, to the calling function. In the above program, the value of sum of three numbers is being returned.
- (e) There is no restriction on the number of **return** statements that may be present in a function. Also, the **return** statement need not always be present at the end of the called function. The following program illustrates these facts:

```
int fun( )
```

```
{  
    int n ;  
    printf ( "Enter any number " ) ;  
    scanf ( "%d", &n ) ;  
    if ( n >= 10 && n <= 90 )  
        return ( n ) ;  
    else  
        return ( n + 32 ) ;  
}
```

In this function, different **return** statements will be executed depending on whether **n** is between 10 and 90 or not.

- (f) Whenever the control returns from a function, the sum being returned is collected in the calling function by assigning the called function to some variable. For example,

```
sum = calsum ( a, b, c ) ;
```

- (g) All the following are valid **return** statements.

```
return ( a ) ;  
return ( 23 ) ;  
return ;
```

In the last statement, a garbage value is returned to the calling function since we are not returning any specific value. Note that, in this case, the parentheses after **return** are dropped. In the other **return** statements too, the parentheses can be dropped.

- (h) A function can return only one value at a time. Thus, the following statements are invalid:

```
return ( a, b ) ;  
return ( x, 12 ) ;
```

There is a way to get around this limitation, which would be discussed later in this chapter, when we learn pointers.

- (i) If the value of a formal argument is changed in the called function, the corresponding change does not take place in the calling function. For example,

```
# include <stdio.h>
void fun ( int ) ;
int main( )
{
    int a = 30 ;
    fun ( a ) ;
    printf ( "%d\n", a ) ;
    return 0 ;
}
void fun ( int b )
{
    b = 60 ;
    printf ( "%d\n", b ) ;
}
```

The output of the above program would be:

```
60
30
```

Thus, even though the value of **b** is changed in **fun()**, the value of **a** in **main()** remains unchanged. This means that when values are passed to a called function, the values present in actual arguments are not physically moved to the formal arguments; just a photocopy of values in actual argument is made into formal arguments.

Scope Rule of Functions

Look at the following program:

```
# include <stdio.h>
void display ( int ) ;
int main( )
{
    int i = 20 ;
    display ( i ) ;
    return 0 ;
}

void display ( int j )
{
    int k = 35 ;
```

```
printf ( "%d\n", j ) ;  
printf ( "%d\n", k ) ;  
}
```

In this program, is it necessary to pass the value of the variable **i** to the function **display()**? Will it not become automatically available to the function **display()**? No. Because, by default, the scope of a variable is local to the function in which it is defined. The presence of **i** is known only to the function **main()** and not to any other function. Similarly, the variable **k** is local to the function **display()** and hence it is not available to **main()**. That is why to make the value of **i** available to **display()**, we have to explicitly pass it to **display()**. Likewise, if we want **k** to be available to **main()**, we will have to return it to **main()** using the **return** statement. In general, we can say that the scope of a variable is local to the function in which it is defined.

Order of Passing Arguments

Consider the following function call:

```
fun (a, b, c, d ) ;
```

In this call, it doesn't matter whether the arguments are passed from left to right or from right to left. However, in some function calls, the order of passing arguments becomes an important consideration. For example:

```
int a = 1 ;  
printf ( "%d %d %d\n", a, ++a, ++a ) ;
```

It appears that this **printf()** would output 1 2 2.

This however is not the case. Surprisingly, it outputs 3 3 1. This is because, in C during a function call the arguments are passed from right to left. That is, firstly 1 is passed through the expression **a++** and then **a** is incremented to 2. Then result of **++a** is passed. That is, **a** is incremented to 3 and then passed. Finally, latest value of **a**, i.e., 3, is passed. Thus in right to left order, 1, 3, 3 get passed. Once **printf()** collects them, it prints them in the order in which we have asked it to get them printed (and not the order in which they were passed). Thus 3 3 1 gets printed.

Using Library Functions

Consider the following program:

```
# include <stdio.h>
# include <math.h>

int main( )
{
    float a = 0.5 ;
    float w, x, y, z ;

    w = sin ( a ) ;
    x = cos ( a ) ;
    y = tan ( a ) ;
    z = pow ( a, 2 ) ;
    printf ( "%f %f %f %f\n", w, x, y, z ) ;
    return 0 ;
}
```

Here we have called four standard library functions—**sin()**, **cos()**, **tan()** and **pow()**. As we know, before calling any function, we must declare its prototype. This helps the compiler in checking whether the values being passed and returned are as per the prototype declaration. But since we didn't define the library functions (we merely called them), we do not know the prototype declarations of library functions. Hence, when the library of functions is provided, a set of '.h' files is also provided. These files contain the prototype declarations of library functions. But why multiple files? Because the library functions are divided into different groups and one file is provided for each group. For example, prototypes of all input/output functions are provided in the file 'stdio.h', prototypes of all mathematical functions are provided in the file 'math.h', etc.

Prototypes of functions **sin()**, **cos()**, **tan()** and **pow()** are declared in the file 'math.h'. To make these prototypes available to our program we need to **#include** the file 'math.h'. You can even open this file and look at the prototypes. They would appear as shown below.

```
double sin ( double ) ;
double cos ( double ) ;
double tan ( double ) ;
double pow ( double, double ) ;
```

Here **double** indicates a real number. We would learn more about **double** in Chapter 11.

Whenever we wish to call any library function we must include the header file which contains its prototype declaration.

One Dicey Issue

Now consider the following program:

```
# include <stdio.h>
int main( )
{
    int i = 10, j = 20 ;
    printf ( "%d %d %d\n", i, j ) ;
    printf ( "%d\n", i, j ) ;
    return 0 ;
}
```

This program gets successfully compiled, even though there is a mismatch in the format specifiers and the variables in the list used in **printf()**. This is because, **printf()** accepts *variable* number of arguments (sometimes 2 arguments, sometimes 3 arguments, etc.), and even with the mismatch above, the call still matches with the prototype of **printf()** present in 'stdio.h'. At run-time, when the first **printf()** is executed, since there is no variable matching with the last specifier **%d**, a garbage integer gets printed. Similarly, in the second **printf()**, since the format specifier for **j** has not been mentioned, its value does not get printed.

Return Type of Function

Suppose we want to find out square of a floating point number using a function. This is how this simple program would look like:

```
# include <stdio.h>
float square ( float ) ;
int main( )
{
    float a, b ;
    printf ( "Enter any number " ) ;
    scanf ( "%f", &a ) ;
    b = square ( a ) ;
    printf ( "Square of %f is %f\n", a, b ) ;
    return 0 ;
}
```

```
}  
float square ( float x )  
{  
    float y ;  
    y = x * x ;  
    return ( y ) ;  
}
```

And here are three sample runs of this program...

```
Enter any number 3  
Square of 3 is 9.000000  
Enter any number 1.5  
Square of 1.5 is 2.250000  
Enter any number 2.5  
Square of 2.5 is 6.250000
```

Since we are returning a **float** value from this function we have indicated the return type of the **square()** function as **float** in the prototype declaration as well as in the function definition. Had we dropped **float** from the prototype and the definition, the compiler would have assumed that **square()** is supposed to return an integer value.

Summary

- (a) To avoid repetition of code and bulky programs functionally related statements are isolated into a function.
- (b) Function declaration specifies the return type of the function and the types of parameters it accepts.
- (c) Function definition defines the body of the function.
- (d) Variables declared in a function are not available to other functions in a program. So, there won't be any clash even if we give same name to the variables declared in different functions.
- (e) In C order of passing arguments to a function is from right to left.

Exercise

[A] What will be the output of the following programs:

- (a)

```
# include <stdio.h>  
void display( ) ;
```

```
int main( )
{
    printf ( "Learn C\n" );
    display( );
    return 0 ;
}
void display( )
{
    printf ( "Followed by C++, C# and Java!\n" );
    main( );
}
```

```
(b) # include <stdio.h>
int check ( int );
int main( )
{
    int i = 45, c ;
    c = check ( i );
    printf ( "%d\n", c );
    return 0 ;
}
int check ( int ch )
{
    if ( ch >= 45 )
        return ( 100 );
    else
        return ( 10 * 10 );
}
```

```
(c) # include <stdio.h>
float circle ( int );
int main( )
{
    float area ;
    int radius = 1 ;
    area = circle ( radius );
    printf ( "%f\n", area );
    return 0 ;
}
float circle ( int r )
{
    float a ;
```

```
    a = 3.14 * r * r ;  
    return ( a ) ;  
}
```

```
(d) #include <stdio.h>  
int main( )  
{  
    void slogan( ) ;  
    int c = 5 ;  
    c = slogan( ) ;  
    printf ( "%d\n", c ) ;  
    return 0 ;  
}  
void slogan( )  
{  
    printf ( "Only He men use C!\n" ) ;  
}
```

[B] Point out the errors, if any, in the following programs:

```
(a) # include <stdio.h>  
int addmult ( int, int )  
int main( )  
{  
    int i = 3, j = 4, k, l ;  
    k = addmult ( i, j ) ;  
    l = addmult ( i, j ) ;  
    printf ( "%d %d\n", k, l ) ;  
    return 0 ;  
}  
int addmult ( int ii, int jj )  
{  
    int kk, ll ;  
    kk = ii + jj ;  
    ll = ii * jj ;  
    return ( kk, ll ) ;  
}
```

```
(b) # include <stdio.h>  
void message( ) ;  
int main( )  
{  
    int a ;
```

```
    a = message( );
    return 0 ;
}
void message( )
{
    printf ( "Viruses are written in C\n" );
    return ;
}
```

(c) # include <stdio.h>

```
int main( )
{
    float a = 15.5 ;
    char ch = 'C' ;
    printit ( a, ch ) ;
    return 0 ;
}
printit ( a, ch )
{
    printf ( "%f %c\n", a, ch ) ;
}
```

(d) # include <stdio.h>

```
void message( ) ;
int main( )
{
    message( ) ;
    message( ) ;
    return 0 ;
}
void message( ) ;
{
    printf ( "Praise worthy and C worthy are synonyms\n" ) ;
}
```

(e) # include <stdio.h>

```
int main( )
{
    let_us_c( )
    {
        printf ( "C is a Cimple minded language !\n" ) ;
        printf ( "Others are of course no match !\n" ) ;
    }
}
```

```
    }  
    return 0 ;  
}
```

```
(f) #include <stdio.h>  
void message( ) ;  
int main( )  
{  
    message ( message( ) ) ;  
    return 0 ;  
}  
void message( )  
{  
    printf ( "It's a small world after all...\n" ) ;  
}
```

[C] Answer the following:

(a) Is this a correctly written function:

```
int sqr ( int a ) ;  
{  
    return ( a * a ) ;  
}
```

(b) State whether the following statements are True or False:

1. The variables commonly used in C functions are available to all the functions in a program.
2. To return the control back to the calling function we must use the keyword **return**.
3. The same variable names can be used in different functions without any conflict.
4. Every called function must contain a **return** statement.
5. A function may contain more than one **return** statements.
6. Each **return** statement in a function may return a different value.
7. A function can still be useful even if you don't pass any arguments to it and the function doesn't return any value back.

8. Same names can be used for different functions without any conflict.
9. A function may be called more than once from any other function.
10. It is necessary for a function to return some value.

[D] Answer the following:

- (a) Write a function to calculate the factorial value of any integer entered through the keyboard.
- (b) Write a function **power (a, b)**, to calculate the value of **a** raised to **b**.
- (c) Write a general-purpose function to convert any given year into its Roman equivalent. Use these Roman equivalents for decimal numbers: 1 – I, 5 – V, 10 – X, 50 – L, 100 – C, 500 – D, 1000 – M.

Example:

Roman equivalent of 1988 is mdcccclxxxviii.

Roman equivalent of 1525 is mdxxv.

- (d) Any year is entered through the keyboard. Write a function to determine whether the year is a leap year or not.
- (e) A positive integer is entered through the keyboard. Write a function to obtain the prime factors of this number.

For example, prime factors of 24 are 2, 2, 2 and 3, whereas prime factors of 35 are 5 and 7.

9

Pointers

- Call by Value and Call by Reference
- An Introduction to Pointers
- Pointer Notation
- Back to Function Calls
- Conclusions
- Summary
- Exercise



9 *Pointers*

- Call by Value and Call by Reference
- An Introduction to Pointers
- Pointer Notation
- Back to Function Calls
- Conclusions
- Summary
- Exercise

Which feature of C do beginners find most difficult to understand? The answer is easy: pointers. Other languages have pointers but few use them so frequently as C does. And why not? It is C's clever use of pointers that makes it the excellent language it is. This chapter is devoted to pointers and their usage in function calls. Let us begin with the function calls.

Call by Value and Call by Reference

By now, we are well familiar with how to call functions. But, if you observe carefully, whenever we called a function and passed something to it we have always passed the 'values' of variables to the called function. Such function calls are called 'calls by value'. By this what we mean is, on calling a function, we are passing values of variables to it. The examples of call by value are shown below:

```
sum = calsum ( a, b, c );  
f = factr ( a );
```

We have also learnt that variables are stored somewhere in memory. So instead of passing the value of a variable, can we not pass the location number (also called address) of the variable to a function? If we were able to do so, it would become a 'call by reference'. What purpose a 'call by reference' serves we would find out a little later. First we must equip ourselves with knowledge of how to make a 'call by reference'. This feature of C functions needs at least an elementary knowledge of a concept called 'pointers'. So let us first acquire the basics of pointers after which we would take up this topic once again.

An Introduction to Pointers

The difficulty beginners have with pointers has much to do with C's pointer terminology than the actual concept. For instance, when a C programmer says that a certain variable is a "pointer", what does that mean? It is hard to see how a variable can point to something, or in a certain direction.

It is hard to get a grip on pointers just by listening to programmer's jargon. In our discussion of C pointers, therefore, we will try to avoid this difficulty by explaining pointers in terms of programming concepts we already understand. The first thing we want to do is to explain the rationale of C's pointer notation.

Pointer Notation

Consider the declaration,

```
int i = 3 ;
```

This declaration tells the C compiler to:

- (a) Reserve space in memory to hold the integer value.
- (b) Associate the name `i` with this memory location.
- (c) Store the value 3 at this location.

We may represent `i`'s location in memory by the memory map shown in Figure 9.1.

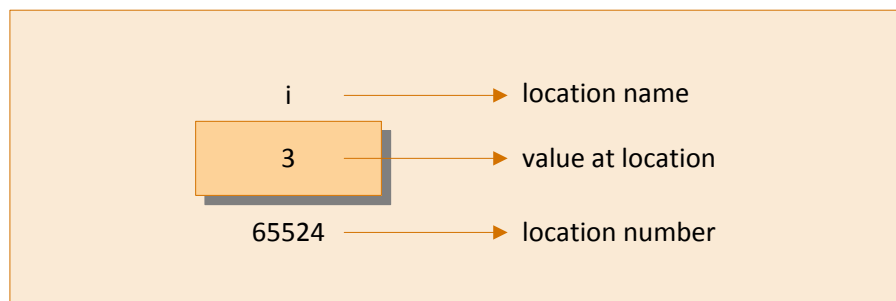


Figure 9.1

We see that the computer has selected memory location 65524 as the place to store the value 3. The location number 65524 is not a number to be relied upon, because some other time the computer may choose a different location for storing the value 3. The important point is, `i`'s address in memory is a number.

We can print this address number through the following program:

```
#include <stdio.h>
int main( )
{
    int i = 3 ;
    printf ( "Address of i = %u\n", &i );
    printf ( "Value of i = %d\n", i );
    return 0 ;
}
```

The output of the above program would be:

Address of i = 65524

Value of i = 3

Look at the first **printf()** statement carefully. ‘&’ used in this statement is C’s ‘address of’ operator. The expression **&i** returns the address of the variable **i**, which in this case happens to be 65524. Since 65524 represents an address, there is no question of a sign being associated with it. Hence it is printed out using **%u**, which is a format specifier for printing an unsigned integer. We have been using the ‘&’ operator all the time in the **scanf()** statement.

The other pointer operator available in C is ‘*’, called ‘value at address’ operator. It gives the value stored at a particular address. The ‘value at address’ operator is also called ‘indirection’ operator.

Observe carefully the output of the following program:

```
#include <stdio.h>
int main( )
{
    int i = 3 ;
    printf ( "Address of i = %u\n", &i ) ;
    printf ( "Value of i = %d\n", i ) ;
    printf ( "Value of i = %d\n", *( &i ) ) ;
    return 0 ;
}
```

The output of the above program would be:

Address of i = 65524

Value of i = 3

Value of i = 3

Note that printing the value of ***(&i)** is same as printing the value of **i**.

The expression **&i** gives the address of the variable **i**. This address can be collected in a variable, by saying,

```
j = &i ;
```

But remember that **j** is not an ordinary variable like any other integer variable. It is a variable that contains the address of other variable (**i** in this case). Since **j** is a variable, the compiler must provide it space in the

memory. Once again, the memory map shown in Figure 9.2 would illustrate the contents of **i** and **j**.

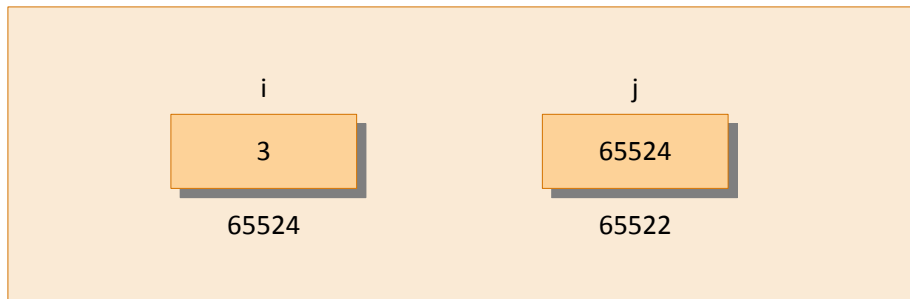


Figure 9.2

As you can see, **i**'s value is 3 and **j**'s value is **i**'s address.

But wait, we can't use **j** in a program without declaring it. And since **j** is a variable that contains the address of **i**, it is declared as,

```
int *j;
```

This declaration tells the compiler that **j** will be used to store the address of an integer value. In other words, **j** points to an integer. How do we justify the usage of ***** in the declaration,

```
int *j;
```

Let us go by the meaning of *****. It stands for 'value at address'. Thus, **int *j** would mean, the value at the address contained in **j** is an **int**.

Here is a program that demonstrates the relationships we have been discussing.

```
#include <stdio.h>
int main( )
{
    int i = 3;
    int *j;

    j = &i;
    printf ( "Address of i = %u\n", &i );
    printf ( "Address of i = %u\n", j );
    printf ( "Address of j = %u\n", &j );
    printf ( "Value of j = %u\n", j );
    printf ( "Value of i = %d\n", i );
```

```
printf ( "Value of i = %d\n", *( &i ) );  
printf ( "Value of i = %d\n", *j );  
return 0 ;  
}
```

The output of the above program would be:

```
Address of i = 65524  
Address of i = 65524  
Address of j = 65522  
Value of j = 65524  
Value of i = 3  
Value of i = 3  
Value of i = 3
```

Work through the above program carefully, taking help of the memory locations of **i** and **j** shown earlier. This program summarizes everything that we have discussed so far. If you don't understand the program's output, or the meanings of **&i**, **&j**, ***j** and ***(&i)**, re-read the last few pages. Everything we say about pointers from here onwards will depend on your understanding these expressions thoroughly.

Look at the following declarations:

```
int *alpha ;  
char *ch ;  
float *s ;
```

Here, **alpha**, **ch** and **s** are declared as pointer variables, i.e., variables capable of holding addresses. Remember that, addresses (location nos.) are always going to be whole numbers, therefore pointers always contain whole numbers. Now we can put these two facts together and say—pointers are variables that contain addresses, and since addresses are always whole numbers, pointers would always contain whole numbers.

The declaration **float *s** does not mean that **s** is going to contain a floating-point value. What it means is, **s** is going to contain the address of a floating-point value. Similarly, **char *ch** means that **ch** is going to contain the address of a char value. Or in other words, the value at address stored in **ch** is going to be a **char**.

The concept of pointers can be further extended. Pointer, we know is a variable that contains address of another variable. Now this variable

itself might be another pointer. Thus, we now have a pointer that contains another pointer's address. The following example should make this point clear:

```
#include <stdio.h>
int main( )
{
    int i = 3, *j, **k;

    j = &i;
    k = &j;
    printf ( "Address of i = %u\n", &i );
    printf ( "Address of i = %u\n", j );
    printf ( "Address of i = %u\n", *k );
    printf ( "Address of j = %u\n", &j );
    printf ( "Address of j = %u\n", k );
    printf ( "Address of k = %u\n", &k );
    printf ( "Value of j = %u\n", j );
    printf ( "Value of k = %u\n", k );
    printf ( "Value of i = %d\n", i );
    printf ( "Value of i = %d\n", * ( &i ) );
    printf ( "Value of i = %d\n", *j );
    printf ( "Value of i = %d\n", **k );
    return 0;
}
```

The output of the above program would be:

```
Address of i = 65524
Address of i = 65524
Address of i = 65524
Address of j = 65522
Address of j = 65522
Address of k = 65520
Value of j = 65524
Value of k = 65522
Value of i = 3
Value of i = 3
Value of i = 3
Value of i = 3
```


Figure 9.3 would help you in tracing out how the program prints the above output.

Remember that when you run this program, the addresses that get printed might turn out to be something different than the ones shown in Figure 9.3. However, with these addresses too, the relationship between **i**, **j** and **k** can be easily established.

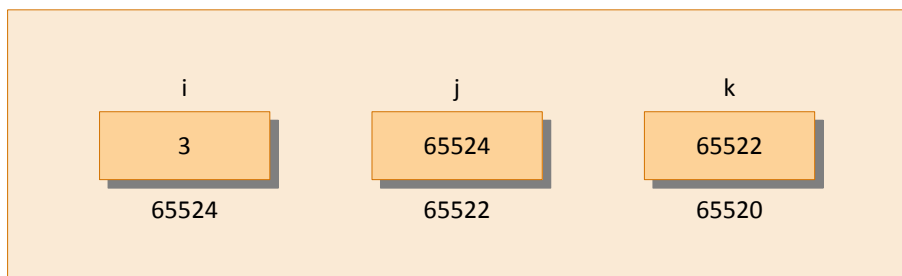


Figure 9.3

Observe how the variables **j** and **k** have been declared,

```
int i, *j, **k;
```

Here, **i** is an ordinary **int**, **j** is a pointer to an **int** (often called an integer pointer), whereas **k** is a pointer to an integer pointer. We can extend the above program still further by creating a pointer to a pointer to an integer pointer. In principle, you would agree that likewise there could exist a pointer to a pointer to a pointer to a pointer to a pointer. There is no limit on how far can we go on extending this definition. Possibly, till the point we can comprehend it. And that point of comprehension is usually a pointer to a pointer. Beyond this, one rarely requires to extend the definition of a pointer. But just in case...

Back to Function Calls

Having had the first tryst with pointers, let us now get back to what we had originally set out to learn—the two types of function calls—call by value and call by reference. Arguments can generally be passed to functions in one of the two ways:

- (a) sending the values of the arguments
- (b) sending the addresses of the arguments

In the first method, the 'value' of each of the actual arguments in the calling function is copied into corresponding formal arguments of the

called function. With this method, the changes made to the formal arguments in the called function have no effect on the values of actual arguments in the calling function. The following program illustrates the 'Call by Value':

```
#include <stdio.h>
void swapv ( int x, int y );
int main( )
{
    int a = 10, b = 20 ;
    swapv ( a, b );
    printf ( "a = %d b = %d\n", a, b );
    return 0 ;
}
void swapv ( int x, int y )
{
    int t ;
    t = x ;
    x = y ;
    y = t ;
    printf ( "x = %d y = %d\n", x, y );
}
```

The output of the above program would be:

```
x = 20 y = 10
a = 10 b = 20
```

Note that values of **a** and **b** remain unchanged even after exchanging the values of **x** and **y**.

In the second method (call by reference), the addresses of actual arguments in the calling function are copied into the formal arguments of the called function. This means that, using these addresses, we would have an access to the actual arguments and hence we would be able to manipulate them. The following program illustrates this fact:

```
#include <stdio.h>
void swapr ( int *, int * );
int main( )
{
    int a = 10, b = 20 ;
    swapr ( &a, &b );
}
```

```
    printf ( "a = %d b = %d\n", a, b );
    return 0 ;
}
void swapr ( int *x, int *y )
{
    int t ;

    t = *x ;
    *x = *y ;
    *y = t ;
}
```

The output of the above program would be:

```
a = 20 b = 10
```

Note that this program manages to exchange the values of **a** and **b** using their addresses stored in **x** and **y**.

Usually, in C programming, we make a call by value. This means that, in general, you cannot alter the actual arguments. But if desired, it can always be achieved through a call by reference.

Using a call by reference intelligently, we can make a function return more than one value at a time, which is not possible ordinarily. This is shown in the program given below.

```
# include <stdio.h>
void areaperi ( int, float *, float * ) ;
int main( )
{
    int radius ;
    float area, perimeter ;

    printf ( "Enter radius of a circle " ) ;
    scanf ( "%d", &radius ) ;
    areaperi ( radius, &area, &perimeter ) ;

    printf ( "Area = %f\n", area ) ;
    printf ( "Perimeter = %f\n", perimeter ) ;
    return 0 ;
}
void areaperi ( int r, float *a, float *p )
```

```
{  
    *a = 3.14 * r * r ;  
    *p = 2 * 3.14 * r ;  
}
```

And here is the output...

```
Enter radius of a circle 5  
Area = 78.500000  
Perimeter = 31.400000
```

Here, we are making a mixed call, in the sense, we are passing the value of **radius** but, addresses of **area** and **perimeter**. And since we are passing the addresses, any change that we make in values stored at addresses contained in the variables **a** and **p**, would make the change effective in **main()**. That is why, when the control returns from the function **areaperi()**, we are able to output the values of **area** and **perimeter**.

Thus, we have been able to indirectly return two values from a called function, and hence, have overcome the limitation of the **return** statement, which can return only one value from a function at a time.

Conclusions

From the programs that we discussed here, we can draw the following conclusions:

- (a) If we want that the value of an actual argument should not get changed in the function being called, pass the actual argument by value.
- (b) If we want that the value of an actual argument should get changed in the function being called, pass the actual argument by reference.
- (c) If a function is to be made to return more than one value at a time, then return these values indirectly by using a call by reference.

Summary

- (a) Pointers are variables which hold addresses of other variables.
- (b) A pointer to a pointer is a variable that holds address of a pointer variable.
- (c) The **&** operator fetches the address of the variable in memory.

- (d) The * operator lets us access the value present at an address in memory with an intension of reading it or modifying it.
- (e) A function can be called either by value or by reference.
- (f) Pointers can be used to make a function return more than one value simultaneously in an indirect manner.

Exercise

[A] What will be the output of the following programs:

```
(a) #include <stdio.h>
void fun ( int, int );
int main( )
{
    int i = 5, j = 2 ;
    fun ( i, j );
    printf ( "%d %d\n", i, j );
    return 0 ;
}
void fun ( int i, int j )
{
    i = i * i ;
    j = j * j ;
}
```

```
(b) #include <stdio.h>
void fun ( int *, int * );
int main( )
{
    int i = 5, j = 2 ;
    fun ( &i, &j );
    printf ( "%d %d\n", i, j );
    return 0 ;
}
void fun ( int *i, int *j )
{
    *i = *i * *i ;
    *j = *j * *j ;
}
```

```
(c) #include <stdio.h>
int main( )
```

```
{
    float a = 13.5 ;
    float *b, *c ;
    b = &a ; /* suppose address of a is 1006 */
    c = b ;
    printf ( "%u %u %u\n", &a, b, c ) ;
    printf ( "%f %f %f %f %f\n", a, *(&a), *&a, *b, *c ) ;
    return 0 ;
}
```

[B] Point out the errors, if any, in the following programs:

- (a) `# include <stdio.h>`
`void pass ( int, int ) ;`
`int main( )`
`{`
 `int i = 135, a = 135, k ;`
 `k = pass ( i, a ) ;`
 `printf ( "%d\n", k ) ;`
 `return 0 ;`
`}`
`void pass ( int j, int b )`
`int c ;`
`{`
 `c = j + b ;`
 `return ( c ) ;`
`}`
- (b) `# include <stdio.h>`
`void jiaayjo ( int , int )`
`int main( )`
`{`
 `int p = 23, f = 24 ;`
 `jiaayjo ( &p, &f ) ;`
 `printf ( "%d %d\n", p, f ) ;`
 `return 0 ;`
`}`
`void jiaayjo ( int q, int g )`
`{`
 `q = q + q ;`
 `g = g + g ;`
`}`

- (c)

```
#include <stdio.h>
void check ( int );
int main( )
{
    int k = 35, z ;
    z = check ( k );
    printf ( "%d\n", z );
    return 0 ;
}
void check ( m )
{
    int m ;
    if ( m > 40 )
        return ( 1 );
    else
        return ( 0 );
}
```
- (d)

```
#include <stdio.h>
void function ( int * );
int main( )
{
    int i = 35, *z ;
    z = function ( &i );
    printf ( "%d\n", z );
    return 0 ;
}
void function ( int *m )
{
    return ( *m + 2 );
}
```

[C] Attempt the following:

- Write a function that receives 5 integers and returns the sum, average and standard deviation of these numbers. Call this function from **main()** and print the results in **main()**.
- Write a function that receives marks received by a student in 3 subjects and returns the average and percentage of these marks. Call this function from **main()** and print the results in **main()**.
- Write a C function to evaluate the series

$$\sin(x) = x - (x^3 / 3!) + (x^5 / 5!) - (x^7 / 7!) + \dots$$

up to 10 terms.

- (d) Given three variables **x**, **y**, **z** write a function to circularly shift their values to right. In other words if $x = 5$, $y = 8$, $z = 10$, after circular shift $y = 5$, $z = 8$, $x = 10$. Call the function with variables **a**, **b**, **c** to circularly shift values.
- (e) If the lengths of the sides of a triangle are denoted by **a**, **b**, and **c**, then area of triangle is given by

$$area = \sqrt{S(S-a)(S-b)(S-c)}$$

where, $S = (a + b + c) / 2$. Write a function to calculate the area of the triangle.

- (f) Write a function to compute the distance between two points and use it to develop another function that will compute the area of the triangle whose vertices are **A(x1, y1)**, **B(x2, y2)**, and **C(x3, y3)**. Use these functions to develop a function which returns a value 1 if the point **(x, y)** lies inside the triangle ABC, otherwise returns a value 0.
- (g) Write a function to compute the greatest common divisor given by Euclid's algorithm, exemplified for $J = 1980$, $K = 1617$ as follows:

$1980 / 1617 = 1$	$1980 - 1 * 1617 = 363$
$1617 / 363 = 4$	$1617 - 4 * 363 = 165$
$363 / 165 = 2$	$363 - 2 * 165 = 33$
$5 / 33 = 5$	$165 - 5 * 33 = 0$

Thus, the greatest common divisor is 33.

10

Recursion

- Recursion
- Recursion and Stack
- Summary
- Exercise



10 Recursion

- Recursion
- Recursion and Stack
- Summary
- Exercise

There is one important feature associated with functions in C. It is known as recursion. Though a bit difficult to understand, it is often the most direct way of programming a complicated logic. This chapter explores recursion in detail.

Recursion

In C, it is possible for the functions to call themselves. A function is called 'recursive' if a statement within the body of a function calls the same function. Sometimes called 'circular definition', recursion is thus the process of defining something in terms of itself.

Let us now see a simple example of recursion. Suppose we want to calculate the factorial value of an integer. As we know, the factorial of a number is the product of all the integers between 1 and that number. For example, 4 factorial is $4 * 3 * 2 * 1$. This can also be expressed as $4! = 4 * 3!$ where '!' stands for factorial. Thus factorial of a number can be expressed in the form of itself. Hence this can be programmed using recursion. However, before we try to write a recursive function for calculating factorial let us take a look at the non-recursive function for calculating the factorial value of an integer.

```
#include <stdio.h>
int factorial ( int );
int main( )
{
    int a, fact ;

    printf ( "Enter any number " );
    scanf ( "%d", &a );

    fact = factorial ( a );
    printf ( "Factorial value = %d\n", fact );
    return 0 ;
}

int factorial ( int x )
{
    int f = 1, i ;

    for ( i = x ; i >= 1 ; i-- )
        f = f * i ;
}
```

```
    return ( f );  
}
```

And here is the output...

```
Enter any number 3  
Factorial value = 6
```

Work through the above program carefully, till you understand the logic of the program properly. Recursive factorial function can be understood only if you are thorough with the above logic.

Following is the recursive version of the function to calculate the factorial value:

```
# include <stdio.h>  
int rec ( int );  
int main( )  
{  
    int a, fact ;  
  
    printf ( "Enter any number " );  
    scanf ( "%d", &a );  
  
    fact = rec ( a );  
    printf ( "Factorial value = %d\n", fact );  
    return 0 ;  
}  
  
int rec ( int x )  
{  
    int f ;  
  
    if ( x == 1 )  
        return ( 1 );  
    else  
        f = x * rec ( x - 1 );  
  
    return ( f );  
}
```

And here is the output for four runs of the program...

```
Enter any number 1
Factorial value = 1
Enter any number 2
Factorial value = 2
Enter any number 3
Factorial value = 6
Enter any number 5
Factorial value = 120
```

Let us understand this recursive factorial function thoroughly. In the first run when the number entered through `scanf()` is 1, let us see what action does `rec()` take. The value of `a` (i.e., 1) is copied into `x`. Since `x` turns out to be 1, the condition `if ( x == 1 )` is satisfied and hence 1 (which indeed is the value of 1 factorial) is returned through the `return` statement.

When the number entered through `scanf()` is 2, the `( x == 1 )` test fails, so we reach the statement,

```
f = x * rec ( x - 1 );
```

And here is where we meet recursion. How do we handle the expression `x * rec ( x - 1 )`? We multiply `x` by `rec ( x - 1 )`. Since the current value of `x` is 2, it is same as saying that we must calculate the value `( 2 * rec ( 1 ) )`. We know that the value returned by `rec ( 1 )` is 1, so the expression reduces to `( 2 * 1 )`, or simply 2. Thus the statement,

```
x * rec ( x - 1 );
```

evaluates to 2, which is stored in the variable `f`, and is returned to `main()`, where it is duly printed as

```
Factorial value = 2
```

Now perhaps you can see what would happen if the value of `a` is 3, 4, 5 and so on.

In case the value of `a` is 5, `main()` would call `rec()` with 5 as its actual argument, and `rec()` will send back the computed value. But before sending the computed value, `rec()` calls `rec()` and waits for a value to be returned. It is possible for the `rec()` that has just been called to call yet another `rec()`, the argument `x` being decreased in value by 1 for each of these recursive calls. We speak of this series of calls to `rec()` as being

different invocations of **rec()**. These successive invocations of the same function are possible because the C compiler keeps track of which invocation calls which. These recursive invocations end finally when the last invocation gets an argument value of 1, which the preceding invocation of **rec()** now uses to calculate its own **f** value and so on up the ladder. So we might say what happens is,

```
rec ( 5 ) returns ( 5 times rec ( 4 ),  
    which returns ( 4 times rec ( 3 ),  
        which returns ( 3 times rec ( 2 ),  
            which returns ( 2 times rec ( 1 ),  
                which returns ( 1 ) ) ) ) )
```

Foxed? Well, that is recursion for you in its simplest garbs. I hope you agree that it's difficult to visualize how the control flows from one function call to another. Possibly Figure 10.1 would make things a bit clearer.

Assume that the number entered through **scanf()** is 3. Using Figure 10.1 let's visualize what exactly happens when the recursive function **rec()** gets called. Go through the figure carefully. The first time when **rec()** is called from **main()**, **x** collects 3. From here, since **x** is not equal to 1, the **if** block is skipped and **rec()** is called again with the argument (**x - 1**), i.e. 2. This is a recursive call. Since **x** is still not equal to 1, **rec()** is called yet another time, with argument (2 - 1). This time as **x** is 1, control goes back to previous **rec()** with the value 1, and **f** is evaluated as 2.

Similarly, each **rec()** evaluates its **f** from the returned value, and finally 6 is returned to **main()**. The sequence would be grasped better by following the arrows shown in Figure 10.1. Let it be clear that while executing the program, there do not exist so many copies of the function **rec()**. These have been shown in the figure just to help you keep track of how the control flows during successive recursive calls.

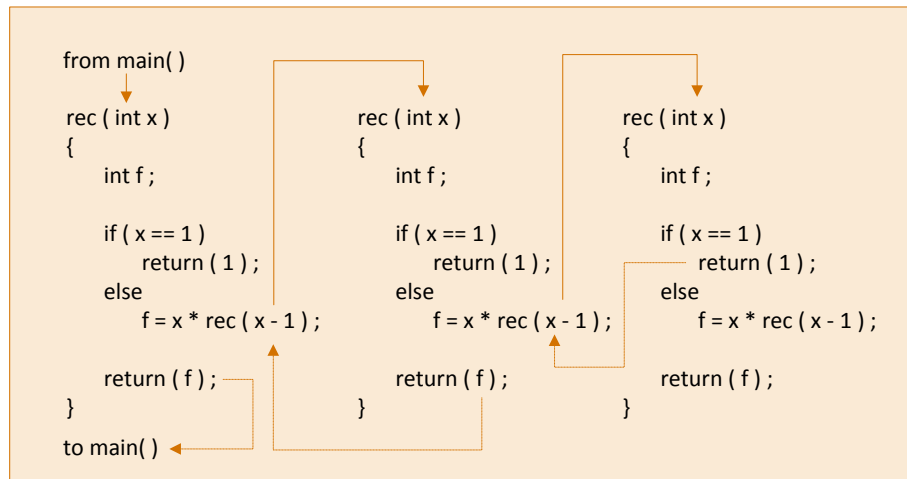


Figure 10.1

Recursion may seem strange and complicated at first glance, but it is often the most direct way to code an algorithm, and once you are familiar with recursion, the clearest way of doing so.

Recursion and Stack

There are different ways in which data can be organized. For example, if you are to store five numbers then we can store them in five different variables, an array, a linked list, a binary tree, etc. All these different ways of organizing the data are known as data structures. The compiler uses one such data structure called stack for implementing normal as well as recursive function calls.

A stack is a Last In First Out (LIFO) data structure. This means that the last item to get stored on the stack (often called Push operation) is the first one to get out of it (often called as Pop operation). You can compare this to the stack of plates in a cafeteria—the last plate that goes on the stack is the first one to get out of it. Now let us see how the stack works in case of the following program:

```

#include <stdio.h>
int add ( int, int ) ;
int main( )
{
    int a = 5, b = 2, c ;
    c = add ( a, b ) ;
    printf ( "sum = %d\n", c ) ;
    return 0 ;
}

```



```
}  
  
int add ( int i, int j )  
{  
    int sum ;  
    sum = i + j ;  
    return sum ;  
}
```

In this program, before transferring the execution control to the function **add()**, the values of parameters **a** and **b** are pushed on the stack. Following this, the address of the statement **printf()** is pushed on the stack and the control is transferred to **add()**. It is necessary to push this address on the stack. In **add()**, the values of **a** and **b** that were pushed on the stack are referred as **i** and **j**. Once inside **add()**, the local variable **sum** gets pushed on the stack. When value of **sum** is returned, **sum** is popped off from the stack. Next, the address of the statement where the control should be returned, is popped off from the stack. Using this address, the control returns to the **printf()** statement in **main()**. Before execution of **printf()** begins, the two integers that were earlier pushed on the stack are now popped off.

How the values are being pushed and popped even though we didn't write any code to do so? Simple—the compiler, on encountering the function call, would generate code to push the parameters and the address. Similarly, it would generate code to clear the stack when the control returns back from **add()**. Figure 10.2 shows the contents of the stack at different stages of execution.

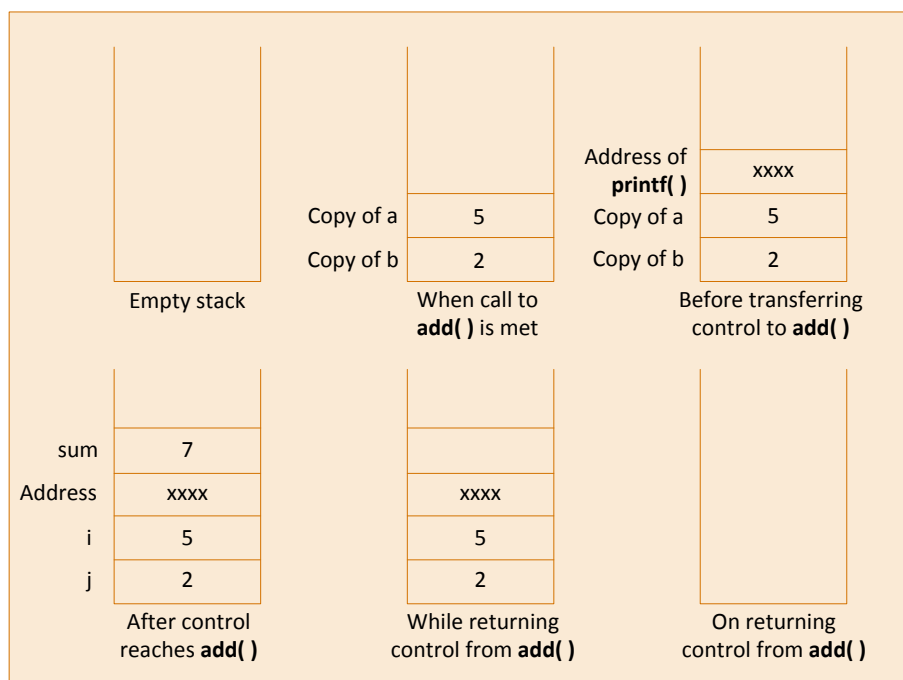


Figure 10.2

Note that in this program, popping of **sum** and address is done by **add()**, whereas, popping of the two integers is done by **main()**. When it is done this way, it is known as 'cdecl Calling Convention'. There are other calling conventions as well, where instead of **main()**, **add()** itself clears the two integers. The calling convention also decides whether the parameters being passed to the function are pushed on the stack in left-to-right or right-to-left order. The standard calling convention always uses the right-to-left order. Thus, during the call to **add()** firstly value of **b** is pushed to the stack, followed by the value of **a**.

The recursive calls are no different. Whenever we make a recursive call the parameters and the return address gets pushed on the stack. The stack gets unwound when the control returns from the called function. Thus during every recursive function call we are working with a fresh set of parameters.

Also, note that while writing recursive functions, you must have an **if** statement somewhere in the recursive function to force the function to return without recursive call being executed. If you don't do this and you call the function, you will fall in an indefinite loop, and the stack will keep on getting filled with parameters and the return address each time there is a call. Soon the stack would become full and you would get a

run-time error indicating that the stack has become full. This is a very common error while writing recursive functions. My advice is to use **printf()** statement liberally during the development of recursive function, so that you can watch what is going on and can abort execution if you see that you have made a mistake.

Summary

- (a) When a function calls itself from within its body it is known as a recursive function.
- (b) A fresh set of variables are created every time a function gets called (normally or recursively).
- (c) Understanding how a recursive function is working becomes easy if you make several copies of the same function on paper and then perform a dry run of the program.
- (d) Adding too many functions and calling them frequently may slow down the program execution.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    printf ( "C to it that C survives\n" );
    main( ) ;
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
{
    int i = 0 ;
    i++ ;
    if ( i <= 5 )
    {
        printf ( "C adds wings to your thoughts\n" );
        exit ( 0 ) ;
        main( ) ;
    }
    return 0 ;
}
```

```
}
```

[B] Attempt the following:

- (a) A 5-digit positive integer is entered through the keyboard, write a recursive and a non-recursive function to calculate sum of digits of the 5-digit number.
- (b) A positive integer is entered through the keyboard, write a program to obtain the prime factors of the number. Modify the function suitably to obtain the prime factors recursively.
- (c) Write a recursive function to obtain the first 25 numbers of a Fibonacci sequence. In a Fibonacci sequence the sum of two successive terms gives the third term. Following are the first few terms of the Fibonacci sequence:
1 1 2 3 5 8 13 21 34 55 89...
- (d) A positive integer is entered through the keyboard, write a function to find the binary equivalent of this number :
 - (1) Without using recursion
 - (2) Using recursion
- (e) Write a recursive function to obtain the running sum of first 25 natural numbers.

11

Data Types Revisited

- **Integers, *long and short***
- **Integers, *signed and unsigned***
- **Chars, *signed and unsigned***
- **Floats and Doubles**
- **A Few More Issues...**
- **Storage Classes in C**
 - Automatic Storage Class
 - Register Storage Class
 - Static Storage Class
 - External Storage Class
 - A Few Subtle Issues
 - Which to Use When
- **Summary**
- **Exercise**



11 ***Data Types*** ***Revisited***

- Integers, *long* and *short*
- Integers, *signed* and *unsigned*
- Chars, *signed* and *unsigned*
- Floats and Doubles
- A Few More Issues...
- Storage Classes in C
 - Automatic Storage Class
 - Register Storage Class
 - Static Storage Class
 - External Storage Class
 - A Few Subtle Issues
 - Which to Use When
- Summary
- Exercise

As seen in the first chapter, the primary data types could be of three varieties—**char**, **int**, and **float**. It may seem odd to many, how C programmers manage with such a tiny set of data types. Fact is, the C programmers aren't really deprived. They can derive many data types from these three types. In fact, the number of data types that can be derived in C, is in principle, unlimited. A C programmer can always invent whatever data type he needs, as we would see later in Chapter 17.

Not only this, the primary data types themselves can be of several types. For example, a **char** can be an **unsigned char** or a **signed char**. Or an **int** can be a **short int** or a **long int**. Sufficiently confusing? Well, let us take a closer look at these variations of primary data types in this chapter.

To fully define a variable, one needs to mention not only its type but also its storage class. In this chapter we would also explore the different storage classes and their relevance in C programming.

Integers, long and short

We had seen earlier that the range of an Integer constant depends upon the compiler. For a 16-bit compiler like Turbo C or Turbo C++ the range is -32768 to 32767 . For a 32-bit compiler like Visual Studio or gcc the range would be -2147483648 to $+2147483647$. Here a 16-bit compiler means that when it compiles a C program it generates machine language code that is targeted towards working on a 16-bit microprocessor like Intel 8086/8088. As against this, a 32-bit compiler like Visual Studio generates machine language code that is targeted towards a 32-bit microprocessor like Intel Pentium. Note that this does not mean that a program compiled using Turbo C would not work on 32-bit processor. It would run successfully but at that time the 32-bit processor would work as if it were a 16-bit processor. This happens because a 32-bit processor provides support for programs compiled using 16-bit compilers. If this backward compatibility support is not provided the 16-bit program would not run on it. This is precisely what happens on the new Intel Itanium processors (64-bit), which have withdrawn support for 16-bit code.

Remember that out of the two/four bytes used to store an integer, the highest bit ($16^{\text{th}}/32^{\text{nd}}$ bit) is used to store the sign of the integer. This bit is 1 if the number is negative and 0 if the number is positive.

C offers a variation of the integer data type that provides what are called **short** and **long** integer values. The intention of providing these variations is to provide integers with different ranges wherever possible. Though not a rule, **short** and **long** integers would usually occupy two and four bytes respectively. Each compiler can decide appropriate sizes depending on the operating system and hardware, for which it is being written, subject to the following rules:

- (a) **shorts** are at least 2 bytes big
- (b) **longs** are at least 4 bytes big
- (c) **shorts** are never bigger than **ints**
- (d) **ints** are never bigger than **longs**

Figure 11.1 shows the sizes of different integers based upon the compiler used.

Compiler	short	int	long
16-bit (Turbo C/C++)	2	2	4
32-bit (Visual Studio, gcc)	2	4	4

Figure 11.1

long variables which hold **long** integers are declared using the keyword **long**, as in,

```
long int i ;  
long int abc ;
```

long integers cause the program to run a bit slower, but the range of values that we can use is expanded tremendously. The value of a **long** integer typically can vary from -2147483648 to +2147483647. More than this you should not need unless you are taking a world census.

If there are such things as **longs**, symmetry requires **shorts** as well—integers that need less space in memory and thus help speed up program execution. **short** integer variables are declared as,


```
short int j ;  
short int height ;
```

C allows the abbreviation of **short int** to **short** and of **long int** to **long**. So the declarations made above can be written as,

```
long i ;  
long abc ;  
short j ;  
short height ;
```

Naturally, most C programmers prefer this short-cut.

Sometimes, we come across situations where the constant is small enough to be an **int**, but still we want to give it as much storage as a **long**. In such cases, we add the suffix 'L' or 'l' at the end of the number, as in 23L.

Integers, signed and unsigned

Sometimes, we know in advance that the value stored in a given integer variable will always be positive—when it is being used to only count things, for example. In such a case we can declare the variable to be **unsigned**, as in,

```
unsigned int num_students ;
```

With such a declaration, the range of permissible integer values (for a 32-bit compiler) will shift from the range -2147483648 to $+2147483647$ to the range 0 to 4294967295. Thus, declaring an integer as **unsigned** almost doubles the size of the largest possible value that it can otherwise take. This so happens because on declaring the integer as **unsigned**, the left-most bit is now free and is not used to store the sign of the number. Note that an **unsigned** integer still occupies two bytes. This is how an **unsigned** integer can be declared:

```
unsigned int i ;  
unsigned i ;
```

Like an **unsigned int**, there also exists a **short unsigned int** and a **long unsigned int**. By default, a **short int** is a **signed short int** and a **long int** is a **signed long int**.

Chars, signed and unsigned

Parallel to **signed** and **unsigned ints** (either **short** or **long**), there also exist **signed** and **unsigned chars**, both occupying one byte each, but having different ranges. To begin with, it might appear strange as to how a **char** can have a sign. Consider the statement

```
char ch = 'A' ;
```

Here what gets stored in **ch** is the binary equivalent of the ASCII / Unicode value of 'A' (i.e. binary of 65). And if 65's binary can be stored, then -54's binary can also be stored (in a **signed char**).

A **signed char** is same as an ordinary **char** and has a range from -128 to +127; whereas, an **unsigned char** has a range from 0 to 255. Let us now see a program that illustrates this range:

```
#include <stdio.h>
int main( )
{
    char ch = 291 ;
    printf ( "\n%d %c\n", ch, ch ) ;
    return 0 ;
}
```

What output do you expect from this program? Possibly, 291 and the character corresponding to it. Well, not really. Surprised? The reason is that **ch** has been defined as a **char**, and a **char** cannot take a value bigger than +127. Hence when value of **ch** exceeds +127, an appropriate value from the other side of the range is picked up and stored in **ch**. This value in our case happens to be 35, hence 35 and its corresponding character #, gets printed out.

Here is another program that would make the concept clearer.

```
#include <stdio.h>
int main( )
{
```

```
char ch ;

for ( ch = 0 ; ch <= 255 ; ch++ )
    printf ( "%d %c\n", ch, ch ) ;
return 0 ;
}
```

This program should output ASCII values and their corresponding characters. Well, No! This is an indefinite loop. The reason is that **ch** has been defined as a **char**, and a **char** cannot take values bigger than +127. Hence when value of **ch** is +127 and we perform **ch++** it becomes -128 instead of +128. -128 is less than 255, hence the condition is still satisfied. Here onwards **ch** would take values like -127, -126, -125, -2, -1, 0, +1, +2, ... +127, -128, -127, etc. Thus the value of **ch** would keep oscillating between -128 to +127, thereby ensuring that the loop never gets terminated. How do you overcome this difficulty? Would declaring **ch** as an **unsigned char** solve the problem? Even this would not serve the purpose since when **ch** reaches a value 255, **ch++** would try to make it 256 which cannot be stored in an **unsigned char**. Thus the only alternative is to declare **ch** as an **int**. However, if we are bent upon writing the program using **unsigned char**, it can be done as shown below. The program is definitely less elegant, but workable all the same.

```
#include <stdio.h>
int main( )
{
    unsigned char ch ;

    for ( ch = 0 ; ch <= 254 ; ch++ )
        printf ( "%d %c\n", ch, ch ) ;

    printf ( "%d %c\n", ch, ch ) ;
    return 0 ;
}
```

Floats and Doubles

A **float** occupies four bytes in memory and can range from -3.4e38 to +3.4e38. If this is insufficient, then C offers a **double** data type that

occupies 8 bytes in memory and has a range from $-1.7e308$ to $+1.7e308$. A variable of type **double** can be declared as,

```
double a, population ;
```

If the situation demands usage of real numbers that lie even beyond the range offered by **double** data type, then there exists a **long double** that can range from $-1.7e4932$ to $+1.7e4932$. A **long double** occupies 10 bytes in memory.

You would see that most of the times in C programming, one is required to use either **chars** or **ints** and cases where **floats**, **doubles** or **long doubles** would be used are indeed rare.

Let us now write a program that puts to use all the data types that we have learnt in this chapter. Go through the following program carefully, which shows how to use these different data types. Note the format specifiers used to input and output these data types.

```
#include <stdio.h>
int main( )
{
    char c ;
    unsigned char d ;
    int i ;
    unsigned int j ;
    short int k ;
    unsigned short int l ;
    long int m ;
    unsigned long int n ;
    float x ;
    double y ;
    long double z ;

    /* char */
    scanf ( "%c %c", &c, &d ) ;
    printf ( "%c %c\n", c, d ) ;

    /* int */
    scanf ( "%d %u", &i, &j ) ;
    printf ( "%d %u\n", i, j ) ;
```

```

/* short int */
scanf ( "%d %u", &k, &l );
printf ( "%d %u\n", k, l );

/* long int */
scanf ( "%ld %lu", &m, &n );
printf ( "%ld %lu\n", m, n );

/* float, double, long double */
scanf ( "%f %lf %Lf", &x, &y, &z );
printf ( "%f %lf %Lf\n", x, y, z );
}

```

The essence of all the data types that we have learnt so far has been captured in Figure 11.2.

Data Type	Range	Bytes	Format
signed char	-128 to +127	1	%c
unsigned char	0 to 255	1	%c
short signed int	-32768 to +32767	2	%d
short unsigned int	0 to 65535	2	%u
signed int	-2147483648 to +2147483647	4	%d
unsigned int	0 to 4294967295	4	%u
long signed int	-2147483648 to +2147483647	4	%ld
long unsigned int	0 to 4294967295	4	%lu
float	-3.4e38 to +3.4e38	4	%f
double	-1.7e308 to +1.7e308	8	%lf
long double	-1.7e4932 to +1.7e4932	10	%Lf
Note: The sizes and ranges of int, short and long are compiler dependent. Sizes in this figure are for 32-bit compiler.			

Figure 11.2

A Few More Issues...

Having seen all the variations of the primary types let us take a look at some more related issues.

- (a) We saw earlier that size of an integer is compiler dependent. This is even true in case of **chars** and **floats**. Also, depending upon the microprocessor for which the compiler targets its code the accuracy of floating point calculations may change. For example, the result of $22.0/7.0$ would be reported more accurately by Visual Studio compiler as compared to TC/TC++ compilers. This is because TC/TC++ targets its compiled code to 8088/8086 (16-bit) microprocessors. Since these microprocessors do not offer floating point support, TC/TC++ performs all float operations using a software piece called Floating Point Emulator. This emulator has limitations and hence produces less accurate results. Also, this emulator becomes part of the EXE file, thereby increasing its size. In addition to this increased size there is a performance penalty since this bigger code would take more time to execute.
- (b) If you look at ranges of **chars** and **ints** there seems to be one extra number on the negative side. This is because a negative number is always stored as 2's complement of its binary. For example, let us see how -128 is stored. Firstly, binary of 128 is calculated (10000000), then its 1's complement is obtained (01111111). A 1's complement is obtained by changing all 0s to 1s and 1s to 0s. Finally, 2's complement of this number, i.e. 10000000, gets stored. A 2's complement is obtained by adding 1 to the 1's complement. Thus, for -128, 10000000 gets stored. This is an 8-bit number and it can be easily accommodated in a **char**. As against this, +128 cannot be stored in a **char** because its binary 010000000 (left-most 0 is for positive sign) is a 9-bit number. However +127 can be stored as its binary 01111111 turns out to be a 8-bit number.
- (c) What happens when we attempt to store +128 in a **char**? The first number on the negative side, i.e. -128 gets stored. This is because from the 9-bit binary of +128, 010000000, only the right-most 8 bits get stored. But when 10000000 is stored the left-most bit is 1 and it is treated as a sign bit. Thus the value of the number becomes -128 since it is indeed the binary of -128, as can be understood from (b) above. Similarly, you can verify that an attempt to store +129 in a

char results in storing -127 in it. In general, if we exceed the range from positive side we end up on the negative side. Vice versa is also true. If we exceed the range from negative side we end up on positive side.

Storage Classes in C

We have already said all that needs to be said about constants, but we are not finished with variables. To fully define a variable, one needs to mention not only its 'type' but also its 'storage class'. In other words, not only do all variables have a data type, they also have a 'storage class'.

We have not mentioned storage classes yet, though we have written several programs in C. We were able to get away with this because storage classes have defaults. If we don't specify the storage class of a variable in its declaration, the compiler will assume a storage class depending on the context in which the variable is used. Thus, variables have certain default storage classes.

From C compiler's point of view, a variable name identifies some physical location within the computer where the string of bits representing the variable's value is stored. There are basically two kinds of locations in a computer where such a value may be kept— Memory and CPU registers. It is the variable's storage class that determines in which of these two types of locations, the value is stored.

Moreover, a variable's storage class tells us:

- (a) Where the variable would be stored.
- (b) What will be the initial value of the variable, if initial value is not specifically assigned.(i.e. the default initial value).
- (c) What is the scope of the variable; i.e. in which functions the value of the variable would be available.
- (d) What is the life of the variable; i.e. how long would the variable exist.

There are four storage classes in C:

- (a) Automatic storage class
- (b) Register storage class
- (c) Static storage class

(d) External storage class

Let us examine these storage classes one by one.

Automatic Storage Class

The features of a variable defined to have an automatic storage class are as under:

Storage:	Memory.
Default value:	An unpredictable value, often called a garbage value.
Scope:	Local to the block in which the variable is defined.
Life:	Till the control remains within the block in which the variable is defined.

Following program shows how an automatic storage class variable is declared, and the fact that if the variable is not initialized, it contains a garbage value.

```
#include <stdio.h>
int main( )
{
    auto int i, j ;
    printf ( "%d %d\n", i, j ) ;
    return 0 ;
}
```

The output of the above program could be...

```
1211 221
```

where, 1211 and 221 are garbage values of **i** and **j**. When you run this program, you may get different values, since garbage values are unpredictable. So always make it a point that you initialize the automatic variables properly, otherwise you are likely to get unexpected results. Note that the keyword for this storage class is **auto**, and not automatic.

Scope and life of an automatic variable is illustrated in the following program.


```
# include <stdio.h>
int main( )
{
    auto int i = 1 ;
    {
        auto int i = 2 ;
        {
            auto int i = 3 ;
            printf ( "%d ", i ) ;
        }
        printf ( "%d ", i ) ;
    }
    printf ( "%d\n", i ) ;
    return 0 ;
}
```

The output of the above program would be:

```
3 2 1
```

Note that the Compiler treats the three **i**'s as totally different variables, since they are defined in different blocks. All three **i**'s are available to the innermost **printf()**. This is because the innermost **printf()** lies in all the three blocks (a block is all statements enclosed within a pair of braces) in which the three **i**'s are defined. This **printf()** prints 3 because when all three **i**'s are available, the one which is most local (nearest to **printf()**) is given a priority.

Once the control comes out of the innermost block, the variable **i** with value 3 is lost, and hence the **i** in the second **printf()** refers to **i** with value 2. Similarly, when the control comes out of the next innermost block, the third **printf()** refers to the **i** with value 1.

Understand the concept of life and scope of an automatic storage class variable thoroughly before proceeding with the next storage class.

Register Storage Class

The features of a variable defined to be of **register** storage class are as under:

Storage:	CPU registers.
Default value:	Garbage value.
Scope:	Local to the block in which the variable is defined.
Life:	Till the control remains within the block in which the variable is defined.

A value stored in a CPU register can always be accessed faster than the one that is stored in memory. Therefore, if a variable is used at many places in a program, it is better to declare its storage class as **register**. A good example of frequently used variables is loop counters. We can name their storage class as **register**.

```
#include <stdio.h>
int main( )
{
    register int i ;

    for ( i = 1 ; i <= 10 ; i++ )
        printf ( "%d\n", i ) ;
    return 0 ;
}
```

Here, even though we have declared the storage class of **i** as **register**, we cannot say for sure that the value of **i** would be stored in a CPU register. Why? Because the number of CPU registers are limited, and they may be busy doing some other task. What happens in such an event... the variable works as if its storage class is **auto**.

Not every type of variable can be stored in a CPU register. For example, if the microprocessor has 16-bit registers then they cannot hold a **float** value or a **double** value, which require 4 and 8 bytes respectively. However, if you use the **register** storage class for a **float** or a **double** variable, you won't get any error messages. All that would happen is the compiler would treat the variables to be of **auto** storage class.

Static Storage Class

The features of a variable defined to have a **static** storage class are as under:

Storage: Memory.
 Default value: Zero.
 Scope: Local to the block in which the variable is defined.
 Life: Value of the variable persists between different function calls.

Compare the two programs and their output given in Figure 11.3 to understand the difference between the **automatic** and **static** storage classes.

The programs in Figure 11.3 consist of two functions **main()** and **increment()**. The function **increment()** gets called from **main()** thrice. Each time it prints the value of **i** and then increments it. The only difference in the two programs is that one uses an **auto** storage class for variable **i**, whereas the other uses **static** storage class.

<pre> #include <stdio.h> void increment(); int main() { increment(); increment(); increment(); return 0 ; } void increment() { auto int i = 1 ; printf ("%d\n", i); i = i + 1 ; } </pre>	<pre> #include <stdio.h> void increment(); int main() { increment(); increment(); increment(); return 0 ; } void increment() { static int i = 1 ; printf ("%d\n", i); i = i + 1 ; } </pre>
The output of the above programs would be:	
1	1
1	2
1	3

Figure 11.3

Like **auto** variables, **static** variables are also local to the block in which they are declared. The difference between them is that **static** variables don't disappear when the function is no longer active. Their values

persist. If the control comes back to the same function again, the **static** variables have the same values they had last time around.

In the above example, when variable **i** is **auto**, each time **increment()** is called, it is re-initialized to one. When the function terminates, **i** vanishes and its new value of 2 is lost. The result: no matter how many times we call **increment()**, **i** is initialized to 1 every time.

On the other hand, if **i** is **static**, it is initialized to 1 only once. It is never initialized again. During the first call to **increment()**, **i** is incremented to 2. Because **i** is static, this value persists. The next time **increment()** is called, **i** is not re-initialized to 1; on the contrary, its old value 2 is still available. This current value of **i** (i.e. 2) gets printed and then **i = i + 1** adds 1 to **i** to get a value of 3. When **increment()** is called the third time, the current value of **i** (i.e. 3) gets printed and once again **i** is incremented. In short, if the storage class is **static**, then the statement **static int i = 1** is executed only once, irrespective of how many times the same function is called.

Consider one more program.

```
#include <stdio.h>

int * fun( );
int main( )
{
    int *j;
    j = fun( );
    printf ( "%d\n", *j );
    return 0;
}

int *fun( )
{
    int k = 35;
    return ( &k );
}
```

Here we are returning an address of **k** from **fun()** and collecting it in **j**. Thus **j** becomes pointer to **k**. Then using this pointer we are printing the value of **k**. This correctly prints out 35. Now try calling any function (even **printf()**) immediately after the call to **fun()**. This time **printf()**

prints a garbage value. Why does this happen? In the first case, when the control returned from **fun()** though **k** went dead it was still left on the stack. We then accessed this value using its address that was collected in **j**. But when we precede the call to **printf()** by a call to any other function, the stack is now changed, hence we get the garbage value. If we want to get the correct value each time then we must declare **k** as **static**. By doing this when the control returns from **fun()**, **k** would not die.

All this having been said, a word of advice—avoid using **static** variables unless you really need them. Because their values are kept in memory when the variables are not active, which means they take up space in memory that could otherwise be used by other variables.

External Storage Class

The features of a variable whose storage class has been defined as external are as follows:

Storage:	Memory.
Default value:	Zero.
Scope:	Global.
Life:	As long as the program's execution doesn't come to an end.

External variables differ from those we have already discussed in that their scope is global, not local. External variables are declared outside all functions, yet are available to all functions that care to use them. Here is an example to illustrate this fact.

```
#include <stdio.h>

int i ;
void increment( ) ;
void decrement( ) ;

int main( )
{
    printf ( "\ni = %d", i ) ;
    increment( ) ;
    increment( ) ;
```

```
    decrement( );
    decrement( );
    return 0 ;
}

void increment( )
{
    i = i + 1 ;
    printf ( "on incrementing i = %d\n", i ) ;
}

void decrement( )
{
    i = i - 1 ;
    printf ( "on decrementing i = %d\n", i ) ;
}
```

The output would be:

```
i = 0
on incrementing i = 1
on incrementing i = 2
on decrementing i = 1
on decrementing i = 0
```

As is obvious from the above output, the value of **i** is available to the functions **increment()** and **decrement()** since **i** has been declared outside all functions.

Look at the following program.

```
# include <stdio.h>
int x = 21 ;
int main( )
{
    extern int y ;
    printf ( "%d %d\n", x, y ) ;
    return 0 ;
}
int y = 31 ;
```

Here, **x** and **y** both are global variables. Since both of them have been defined outside all the functions both enjoy external storage class. Note the difference between the following:

```
extern int y ;  
int y = 31 ;
```

Here the first statement is a declaration, whereas the second is the definition. When we declare a variable no space is reserved for it, whereas, when we define it space gets reserved for it in memory. We had to declare **y** since it is being used in **printf()** before it's definition is encountered. There was no need to declare **x** since its definition is done before its usage. Also remember that a variable can be declared several times but can be defined only once.

Another small issue—what will be the output of the following program?

```
# include <stdio.h>  
int x = 10 ;  
void display( ) ;  
int main( )  
{  
    int x = 20 ;  
    printf ( "%d\n", x ) ;  
    display( ) ;  
    return 0 ;  
}  
  
void display( )  
{  
    printf ( "%d\n", x ) ;  
}
```

Here **x** is defined at two places, once outside **main()** and once inside it. When the control reaches the **printf()** in **main()** which **x** gets printed? Whenever such a conflict arises, it's the local variable that gets preference over the global variable. Hence the **printf()** outputs 20. When **display()** is called and control reaches the **printf()** there is no such conflict. Hence, this time, the value of the global **x**, i.e. 10 gets printed.

One last thing—a **static** variable can also be declared outside all the functions. For all practical purposes it will be treated as an **extern** variable. However, the scope of this variable is limited to the same file in which it is declared. This means that the variable would not be available to any function that is defined in a file other than the file in which the variable is defined.

A Few Subtle Issues

Let us now look at some subtle issues about storage classes.

- (a) All variables that are defined inside a function are normally created on the stack each time the function is called. These variables die as soon as control goes back from the function. However, if the variables inside the function are defined as static then they do not get created on the stack. Instead they are created in a place in memory called 'Data Segment'. Such variables die only when program execution comes to an end.
- (b) If a variable is defined outside all functions, then not only is it available to all other functions in the file in which it is defined, but is also available to functions defined in other files. In the other files the variable should be declared as extern. This is shown in the following program:

```
/* PR1.C */
# include <stdio.h>
# include <functions.c>

int i = 35 ;
int fun1( ) ;
int fun2( ) ;

int main( )
{
    printf ( "%d\n", i ) ;
    fun1( ) ;
    fun2( ) ;
    return 0 ;
}
```



```
/* FUNCTIONS.C */
extern int i ;
int fun1( )
{
    i++ ;
    printf ( "%d\n", i ) ;
    return 0 ;
}
int fun2( )
{
    i-- ;
    printf ( "%d\n", i ) ;
    return 0 ;
}
```

The output of the program would be

```
35
36
35
```

However, if we place the word **static** in front of an external variable (**i** in our case) it makes the variable private and not accessible to use in any other file.

- (c) In the following statements the first three are definitions, whereas, the last one is a declaration.

```
auto int i ;
static int j ;
register int k ;
extern int l ;
```

Which to Use When

Dennis Ritchie has made available to the C programmer a number of storage classes with varying features, believing that the programmer would be in a best position to decide which one of these storage classes should be used when. We can make a few ground rules for usage of

different storage classes in different programming situations with a view to:

- (a) economise the memory space consumed by the variables
- (b) improve the speed of execution of the program

The rules are as under:

- Use **static** storage class only if you want the value of a variable to persist between different function calls.
- Use **register** storage class for only those variables that are being used very often in a program. Reason is, there are very few CPU registers at our disposal and many of them might be busy doing something else. Make careful utilization of the scarce resources. A typical application of **register** storage class is loop counters, which get used a number of times in a program.
- Use **extern** storage class for only those variables that are being used by almost all the functions in the program. This would avoid unnecessary passing of these variables as arguments when making a function call. Declaring all the variables as **extern** would amount to a lot of wastage of memory space because these variables would remain active throughout the life of the program.
- If you don't have any of the express needs mentioned above, then use the **auto** storage class. In fact, most of the times, we end up using the **auto** variables. This is because once we have used the variables in a function and are returning from it, we don't mind losing them.

Summary

- (a) We can use different variations of the primary data types, namely **signed** and **unsigned char**, **long** and **short int**, **float**, **double** and **long double**. There are different format specifications for all these data types when they are used in **scanf()** and **printf()** functions.
- (b) The maximum value a variable can hold depends upon the number of bytes it occupies in memory.
- (c) By default all the variables are **signed**. We can declare a variable as **unsigned** to accommodate bigger value without increasing the bytes occupied.

- (d) We can make use of proper storage classes like **auto**, **register**, **static** and **extern** to control four properties of a variable—storage, default initial value, scope and life.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    int i;
    for ( i = 0 ; i <= 50000 ; i++ )
        printf ( "%d\n", i );
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
{
    float a = 13.5 ;
    double b = 13.5 ;
    printf ( "%f %lf\n", a, b ) ;
    return 0 ;
}
```
- (c)

```
#include <stdio.h>
int i = 0 ;
void val( ) ;
int main( )
{
    printf ( "main's i = %d\n", i ) ;
    i++ ;
    val( ) ;
    printf ( "main's i = %d\n", i ) ;
    val( ) ;
    return 0 ;
}
void val( )
{
```

```
    i = 100 ;  
    printf ( "val's i = %d\n", i ) ;  
    i++ ;  
}
```

```
(d) #include <stdio.h>  
int f ( int ) ;  
int g ( int ) ;  
int main( )  
{  
    int x, y, s = 2 ;  
    s *= 3 ;  
    y = f ( s ) ;  
    x = g ( s ) ;  
    printf ( "%d %d %d\n", s, y, x ) ;  
    return 0 ;  
}  
int t = 8 ;  
  
int f ( int a )  
{  
    a += -5 ;  
    t -= 4 ;  
    return ( a + t ) ;  
}  
  
int g ( int a )  
{  
    a = 1 ;  
    t += a ;  
    return ( a + t ) ;  
}
```

```
(e) #include <stdio.h>  
int main( )  
{  
    static int count = 5 ;  
    printf ( "count = %d\n", count-- ) ;  
    if ( count != 0 )  
        main( ) ;  
}
```

```
    return 0 ;  
}
```

```
(f) #include <stdio.h>  
int g ( int ) ;  
int main( )  
{  
    int i, j ;  
    for ( i = 1 ; i < 5 ; i++ )  
    {  
        j = g ( i ) ;  
        printf ( "%d\n", j ) ;  
    }  
    return 0 ;  
}  
int g ( int x )  
{  
    static int v = 1 ;  
    int b = 3 ;  
    v += x ;  
    return ( v + x + b ) ;  
}
```

```
(g) #include <stdio.h>  
int main( )  
{  
    func( ) ;  
    func( ) ;  
    return 0 ;  
}  
void func( )  
{  
    auto int i = 0 ;  
    register int j = 0 ;  
    static int k = 0 ;  
    i++ ; j++ ; k++ ;  
    printf ( "%d %d %d\n", i, j, k ) ;  
}
```

```
(h) #include <stdio.h>  
int x = 10 ;
```

```
int main( )
{
    int x = 20 ;
    {
        int x = 30 ;
        printf ( "%d\n", x ) ;
    }
    printf ( "%d\n", x ) ;
    return 0 ;
}
```

[B] Point out the errors, if any, in the following programs:

- (a) # include <stdio.h>
int main()
{
 long num ;
 num = 2 ;
 printf ("%d\n", num) ;
 return 0 ;
}
- (b) # include <stdio.h>
int main()
{
 char ch = 200 ;
 printf ("%d\n", ch) ;
 return 0 ;
}
- (c) # include <stdio.h>
int main()
{
 unsigned a = 25 ;
 long unsigned b = 25l ;
 printf ("%lu %u\n", a, b) ;
 return 0 ;
}
- (d) # include <stdio.h>
int main()
{

```
    long float a = 25.345e454 ;  
    unsigned double b = 25 ;  
    printf ( "%lf %d\n", a, b ) ;  
    return 0 ;  
}
```

```
(e) # include <stdio.h>  
    static int y ;  
    int main( )  
    {  
        static int z ;  
        printf ( "%d %d\n", y, z ) ;  
        return 0 ;  
    }
```

[C] State whether the following statements are True or False:

- (a) Storage for a register storage class variable is allocated each time the control reaches the block in which it is present.
- (b) An extern storage class variable is not available to the functions that precede its definition, unless the variable is explicitly declared in these functions.
- (c) The value of an automatic storage class variable persists between various function invocations.
- (d) If the CPU registers are not available, the register storage class variables are treated as static storage class variables.
- (e) The register storage class variables cannot hold float values.
- (f) If we try to use register storage class for a **float** variable the compiler will report an error message.
- (g) If the variable **x** is defined outside all functions and a variable **x** is also defined as a local variable of some function, then the global variable gets preference over the local variable.
- (h) The default value for automatic variable is zero.
- (i) The life of static variable is till the control remains within the block in which it is defined.
- (j) If a global variable is to be defined, then the **extern** keyword is necessary in its declaration.
- (k) The address of register variable is not accessible.

- (l) A variable that is defined outside all functions can also have a static **storage** class.
- (m) One variable can have multiple storage classes.

- **Features of C Preprocessor**
- **Macro Expansion**
 - Macros with Arguments
 - Macros versus Functions
- **File Inclusion**
- **Conditional Compilation**
- ***#if* and *#elif* Directives**
- **Miscellaneous Directives**
 - #undef* Directive
 - #pragma* Directive
- **The Build Process**
 - Preprocessing
 - Compilation
 - Assembling
 - Linking
 - Loading
- **Summary**
- **Exercise**



12 *The C*

Preprocessor

- Features of C Preprocessor
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The C preprocessor is exactly what its name implies. It is a program that processes our source program before it is passed to the compiler. Preprocessor commands (often known as directives) form what can almost be considered a language within C language. We can certainly write C programs without knowing anything about the preprocessor or its facilities. But preprocessor is such a great convenience that virtually all C programmers rely on it. This chapter explores the preprocessor directives, and discusses the pros and cons of using them in programs.

Features of C Preprocessor

There are several steps involved from the stage of writing a C program to the stage of getting it executed. The combination of these steps is known as the 'Build Process'. The detailed build process is discussed in the last section of this chapter. At this stage it would be sufficient to note that before a C program is compiled it is passed through another program called 'Preprocessor'. The C program is often known as 'Source Code'. The Preprocessor works on the source code and creates 'Expanded Source Code'. If the source code is stored in a file PR1.C, then the expanded source code gets stored in a file PR1.I. It is this expanded source code that is sent to the compiler for compilation.

The preprocessor offers several features called preprocessor directives. Each of these preprocessor directives begins with a # symbol. The directives can be placed anywhere in a program but are most often placed at the beginning of a program, before the first function definition. We would learn the following preprocessor directives here:

- (a) Macro expansion
- (b) File inclusion
- (c) Conditional compilation
- (d) Miscellaneous directives

Let us understand these features of preprocessor one-by-one.

Macro Expansion

Have a look at the following program:

```
#include <stdio.h>
#define UPPER 25
int main( )
{
    int i;
```

```
    for ( i = 1 ; i <= UPPER ; i++ )  
        printf ( "%d\n", i ) ;  
    return 0 ;  
}
```

In this program, instead of writing 25 in the **for** loop we are writing it in the form of UPPER, which has already been defined before **main()** through the statement,

```
# define UPPER 25
```

This statement is called ‘macro definition’ or more commonly, just a ‘macro’. What purpose does it serve? During preprocessing, the preprocessor replaces every occurrence of UPPER in the program with 25. Here is another example of macro definition.

```
# include <stdio.h>  
# define PI 3.1415  
int main( )  
{  
    float r = 6.25 ;  
    float area ;  
  
    area = PI * r * r ;  
    printf ( "Area of circle = %f\n", area ) ;  
    return 0 ;  
}
```

UPPER and PI in the above programs are often called ‘macro templates’, whereas, 25 and 3.1415 are called their corresponding ‘macro expansions’.

When we compile the program, before the source code passes to the compiler, it is examined by the C preprocessor for any macro definitions. When it sees the **#define** directive, it goes through the entire program in search of the macro templates; wherever it finds one, it replaces the macro template with the appropriate macro expansion. Only after this procedure has been completed, is the program handed over to the compiler.

In C programming, it is customary to use capital letters for macro template. This makes it easy for programmers to pick out all the macro templates when reading through the program.

Note that a macro template and its macro expansion are separated by blanks or tabs. A space between **#** and **define** is optional. Remember that a macro definition is never to be terminated by a semicolon.

And now a million dollar question... why use **#define** in the above programs? What have we gained by substituting PI for 3.1415 in our program? Probably, we have made the program easier to read.

There is perhaps a more important reason for using macro definition than mere readability. Suppose a constant like 3.1415 appears many times in your program. This value may have to be changed some day to 3.141592. Ordinarily, you would need to go through the program and manually change each occurrence of the constant. However, if you have defined PI in a **#define** directive, you only need to make one change, in the **#define** directive itself:

```
#define PI 3.141592
```

Beyond this, the change will be made automatically to all occurrences of PI before the beginning of compilation.

In short, it is nice to know that you would be able to change values of a constant at all the places in the program by just making a change in the **#define** directive. This convenience may not matter for small programs shown above, but with large programs, macro definitions are almost indispensable.

But the same purpose could have been served had we used a variable **pi** instead of a macro template **PI**. A variable could also have provided a meaningful name for a constant and permitted one change to affect many occurrences of the constant. It's true that a variable can be used in this way. Then, why not use it? For three reasons it's a bad idea.

Firstly, it is inefficient, since the compiler can generate faster and more compact code for constants than it can for variables. Secondly, using a variable for what is really a constant encourages sloppy thinking and makes the program more difficult to understand: if something never changes, it is hard to imagine it as a variable. And thirdly, there is always a danger that the variable may inadvertently get altered somewhere in the program. So it's no longer a constant that you think it is.

Thus, using **#define** can produce more efficient and more easily understandable programs. This directive is used extensively by C programmers, as you will see in many programs in this book.

Following three examples show places where a **#define** directive is popularly used by C programmers.

A **#define** directive is many a time used to define operators as shown below.

```
# include <stdio.h>
# define AND &&
# define OR ||
int main( )
{
    int f = 1, x = 4, y = 90 ;

    if ( ( f < 5 ) AND ( x <= 20 OR y <= 45 ) )
        printf ( "Your PC will always work fine...\n" ) ;
    else
        printf ( "In front of the maintenance man\n" ) ;
    return 0 ;
}
```

A **#define** directive could be used even to replace a condition, as shown below.

```
# include <stdio.h>
# define AND &&
# define ARANGE ( a > 25 AND a < 50 )

int main( )
{
    int a = 30 ;

    if ( ARANGE )
        printf ( "within range\n" ) ;
    else
        printf ( "out of range\n" ) ;
    return 0 ;
}
```

A **#define** directive could be used to replace even an entire C statement. This is shown below.

```
# include <stdio.h>
# define FOUND printf ( "The Yankee Doodle Virus\n" ) ;
```

```
int main( )
{
    char signature ;

    if ( signature == 'Y' )
        FOUND
    else
        printf ( "Safe... as yet !\n" ) ;
    return 0 ;
}
```

Macros with Arguments

The macros that we have used so far are called simple macros. Macros can have arguments, just as functions can. Here is an example that illustrates this fact.

```
# include <stdio.h>
# define AREA(x) ( 3.14 * x * x )

int main( )
{
    float r1 = 6.25, r2 = 2.5, a ;

    a = AREA ( r1 ) ;
    printf ( "Area of circle = %f\n", a ) ;
    a = AREA ( r2 ) ;
    printf ( "Area of circle = %f\n", a ) ;
    return 0 ;
}
```

Here's the output of the program...

```
Area of circle = 122.656250
Area of circle = 19.625000
```

In this program, wherever the preprocessor finds the phrase **AREA(x)** it expands it into the statement (**3.14 * x * x**). However, that's not all that it does. The **x** in the macro template **AREA(x)** is an argument that matches the **x** in the macro expansion (**3.14 * x * x**). The statement **AREA(r1)** in the program causes the variable **r1** to be substituted for **x**. Thus the statement **AREA(r1)** is equivalent to:


```
( 3.14 * r1 * r1 )
```

After the above source code has passed through the preprocessor, what the compiler gets to work on will be this:

```
# include <stdio.h>
int main( )
{
    float r1 = 6.25, r2 = 2.5, a ;

    a = 3.14 * r1 *r1 ;
    printf ( "Area of circle = %f\n", a ) ;
    a = 3.14 *r2 * r2 ;
    printf ( "Area of circle = %f\n", a ) ;
    return 0 ;
}
```

Here is another example of macros with arguments:

```
# include <stdio.h>
# define ISDIGIT(y) ( y >= 48 && y <= 57 )

int main( )
{
    char ch ;

    printf ( "Enter any digit " ) ;
    scanf ( "%c", &ch ) ;

    if ( ISDIGIT ( ch ) )
        printf ( "You entered a digit\n" ) ;
    else
        printf ( "Illegal input\n" ) ;
    return 0 ;
}
```

Here are some important points to remember while writing macros with arguments:

- (a) Be careful not to leave a blank between the macro template and its argument while defining the macro. For example, there should be

no blank between **AREA** and **(x)** in the definition, `#define AREA(x) (3.14 * x * x)`

If we were to write **AREA (x)** instead of **AREA(x)**, the **(x)** would become a part of macro expansion, which we certainly don't want. What would happen is, the template would be expanded to

```
( r1 ) ( 3.14 * r1 * r1 )
```

which won't run. Not at all what we wanted.

- (b) The entire macro expansion should be enclosed within parentheses. Here is an example of what would happen if we fail to enclose the macro expansion within parentheses.

```
# include <stdio.h>
# define SQUARE(n) n * n

int main( )
{
    int j ;

    j = 64 / SQUARE ( 4 ) ;
    printf ( "j = %d\n", j ) ;
    return 0 ;
}
```

The output of the above program would be:

```
j = 64
```

whereas, what we expected was `j = 4`.

What went wrong? The macro was expanded into

```
j = 64 / 4 * 4 ;
```

which yielded 64.

- (c) Macros can be split into multiple lines, with a `'\'` (backslash) present at the end of each line. Following program shows how we can define and use multiple line macros.

```
# include <stdio.h>
```

```
#include <windows.h>
void gotoxy ( short int col, short int row ) ;

# define HLINE  for ( i = 0 ; i < 79 ; i++ ) \
                printf ( "%c", 196 ) ;

# define VLINE( X, Y ) { \
                        gotoxy ( X, Y ) ; \
                        printf ( "%c", 179 ) ; \
                    }

int main( )
{
    int i, y ;

    /* system( ) will work in Visual Studio. In Turbo C / C++ use
       clrscr( ) */
    system ( "cls" ) ;

    /* position cursor in row x and column y. Use this function in
       Visual Studio. If you are using Turbo C / C++ use standard
       library function gotoxy( ) declared in file conio.h */
    gotoxy ( 1, 12 ) ;

    HLINE

    for ( y = 1 ; y < 25 ; y++ )
        VLINE ( 39, y ) ;
    return 0 ;
}

void gotoxy ( short int col, short int row )
{
    HANDLE hStdout = GetStdHandle ( STD_OUTPUT_HANDLE ) ;
    COORD position = { col, row } ;
    SetConsoleCursorPosition ( hStdout, position ) ;
}
```

This program draws a vertical and a horizontal line in the center of the screen.

- (d) If for any reason, you are unable to debug a macro, then you should view the expanded code of the program to see how the macros are getting expanded. If your source code is present in the file PR1.C then the expanded source code would be stored in PR1.I. You need to generate this file at the command prompt by saying:

```
cpp pr1.c
```

Here CPP stands for C PreProcessor. It generates the expanded source code and stores it in a file called PR1.I. You can now open this file and see the expanded source code. Note that the file PR1.I gets generated in C:\TC\BIN directory. The procedure for generating expanded source code for compilers other than Turbo C/C++ might be a little different.

Macros versus Functions

In the above example, a macro was used to calculate the area of the circle. As we know, even a function can be written to calculate the area of the circle. Though macro calls are 'like' function calls, they are not really the same things. Then what is the difference between the two?

In a macro call, the preprocessor replaces the macro template with its macro expansion, in a stupid, unthinking, literal way. As against this, in a function call the control is passed to a function along with certain arguments, some calculations are performed in the function and a useful value is returned back from the function.

This brings us to a question: when is it best to use macros with arguments and when is it better to use a function? Usually macros make the program run faster but increase the program size, whereas functions make the program smaller and compact.

If we use a macro hundred times in a program, the macro expansion goes into our source code at hundred different places, thus increasing the program size. On the other hand, if a function is used, then even if it is called from hundred different places in the program, it would take the same amount of space in the program.

But passing arguments to a function and getting back the returned value does take time and would therefore slow down the program. This gets avoided with macros since they have already been expanded and placed in the source code before compilation. Thus the trade-off is between memory space and time.

Moral of the story is—if the macro is simple and sweet like in our examples, it makes nice shorthand and avoids the overheads associated with function calls. On the other hand, if we have a fairly large macro and it is used fairly often, perhaps we ought to replace it with a function.

If memory space in the machine in which the program is being executed is less (like a mobile phone or a tablet), then it may make sense to use functions instead of macros. By doing so, the program may run slower, but will need less memory space.

File Inclusion

The second preprocessor directive we'll explore in this chapter is file inclusion. This directive causes one file to be included in another. The preprocessor command for file inclusion looks like this:

```
#include "filename"
```

and it simply causes the entire contents of **filename** to be inserted into the source code at that point in the program. Of course, this presumes that the file being included exists. When and why this feature is used? It can be used in two cases:

- (a) If we have a very large program, the code is best divided into several different files, each containing a set of related functions. It is a good programming practice to keep different sections of a large program separate. These files are **#included** at the beginning of main program file.
- (b) There are some functions and some macro definitions that we need almost in all programs that we write. These commonly needed functions and macro definitions can be stored in a file, and that file can be included in every program we write, which would add all the statements in this file to our program as if we have typed them in.
- (c) When creating our own library of functions which we wish to distribute to others. In this situation the functions are defined in a ".c" file and their corresponding prototype declarations and macros are declared in a ".h" file. While distributing the definitions are compiled into a library file (in machine language) and then the library file and the ".h" file. This way the function definitions in the ".c" file remain with you and are not exposed to users of these functions.

It is common for the files that are to be included to have a .h extension. This extension stands for 'header file', because it contains statements which when included go to the head of your program. The prototypes of all the library functions are grouped into different categories and then stored in different header files. For example, prototypes of all mathematics related functions are stored in the header file 'math.h', prototypes of console input/output functions are stored in the header file 'conio.h', and so on.

Actually there exist two ways to write **#include** statement. These are:

```
# include "filename"  
# include <filename>
```

The meaning of each of these forms is given below.

# include "mylib.h"	This command would look for the file mylib.h in the current directory as well as the specified list of directories as mentioned in the include search path that might have been set up.
---------------------	--

# include <mylib.h>	This command would look for the file mylib.h in the specified list of directories only.
---------------------	--

The include search path is nothing but a list of directories that would be searched for the file being included. Different C compilers let you set the search path in different manners. If you are using Turbo C/C++ compiler, then the search path can be set up by selecting 'Directories' from the 'Options' menu. On doing this, a dialog box appears. In this dialog box against 'Include Directories', we can specify the search path. We can also specify multiple include paths separated by ';' (semicolon) as shown below.

```
c:\tc\lib ; c:\mylib ; d:\libfiles
```

The path can contain maximum of 127 characters. Both relative and absolute paths are valid. For example, '..\dir\incfiles' is a valid path.

In Visual Studio the search path for a project can be set by right-clicking the project name in Solution Explorer and selecting "Properties" from the menu that pops up. This brings up a dialog box. You can now set up the search path by going to "Include Directories" in "Configuration Properties" tab.

Conditional Compilation

We can, if we want, have the compiler skip over part of a source code by inserting the preprocessing commands **#ifdef** and **#endif**, which have the general form given below.

```
# ifdef macroname
    statement 1 ;
    statement 2 ;
    statement 3 ;
# endif
```

If **macroname** has been **#defined**, the block of code will be processed as usual; otherwise not.

Where would **#ifdef** be useful? When would you like to compile only a part of your program? In three cases:

- (a) To “comment out” obsolete lines of code. It often happens that a program is changed at the last minute to satisfy a client. This involves rewriting some part of source code to the client’s satisfaction and deleting the old code. But veteran programmers are familiar with the clients who change their mind and want the old code back again just the way it was. Now you would definitely not like to retype the deleted code again.

One solution in such a situation is to put the old code within a pair of `/* - - */` combination. But we might have already written a comment in the code that we are about to “comment out”. This would mean we end up with nested comments. Obviously, this solution won’t work since we can’t nest comments in C.

Therefore, the solution is to use conditional compilation as shown below.

```
int main( )
{
    # ifdef OKAY
        statement 1 ;
        statement 2 ; /* detects virus */
        statement 3 ;
        statement 4 ; /* specific to stone virus */
    # endif
```

```
statement 5 ;  
statement 6 ;  
statement 7 ;  
}
```

Here, statements 1, 2, 3 and 4 would get compiled only if the macro OKAY has been defined, and we have purposefully omitted the definition of the macro OKAY. At a later date, if we want that these statements should also get compiled, all that we are required to do is to delete the **#ifdef** and **#endif** statements.

- (b) A more sophisticated use of **#ifdef** has to do with making the programs portable, i.e., to make them work on two totally different computers. Suppose an organization has two different types of computers and you are expected to write a program that works on both the machines. You can do so by isolating the lines of code that must be different for each machine by marking them off with **#ifdef**. For example:

```
int main( )  
{  
    # ifdef INTEL  
        code suitable for an Intel PC  
    # else  
        code suitable for a Motorola PC  
    # endif  
    code common to both the computers  
}
```

When you compile this program, it would compile only the code suitable for Motorola PC and the common code. This is because the macro INTEL has not been defined. Note that, the working of **#ifdef** - **#else** - **#endif** is similar to the ordinary **if** - **else** control instruction of C.

If you want to run your program on a Intel PC, just add a statement at the top saying,

```
# define INTEL
```

Sometimes, instead of **#ifdef**, the **#ifndef** directive is used. The **#ifndef** (which means 'if not defined') works exactly opposite to

#ifdef. The above example, if written using **#ifdef**, would look like this:

```
int main( )
{
    # ifdef INTEL
        code suitable for a Motorola PC
    # else
        code suitable for a Intel PC
    # endif
    code common to both the computers
}
```

- (c) Suppose a function **myfunc()** is defined in a file 'myfile.h' which is **#included** in a file 'myfile1.h'. Now in your program file, if you **#include** both 'myfile.h' and 'myfile1.h', the compiler flashes an error 'Multiple declaration for **myfunc**'. This is because the same file 'myfile.h' gets included twice. To avoid this, we can write following code in the 'myfile.h' header file:

```
/* myfile.h */
# ifndef __myfile_h
    # define __myfile_h

    myfunc( )
    {
        /* some code */
    }

# endif
```

First time the file 'myfile.h' gets included, the preprocessor checks whether a macro called **__myfile_h** has been defined or not. If it has not been, then it gets defined and the rest of the code gets included. Next time we attempt to include the same file, the inclusion is prevented since **__myfile_h** already stands defined. Note that there is nothing special about **__myfile_h**. In its place, we can use any other macro as well.

#if and #elif Directives

The **#if** directive can be used to test whether an expression evaluates to a non-zero value or not. If the result of the expression is non-zero, then subsequent lines upto a **#else**, **#elif** or **#endif** are compiled, otherwise they are skipped.

A simple example of **#if** directive is shown below:

```
int main( )
{
    # if TEST <= 5
        statement 1 ;
        statement 2 ;
        statement 3 ;
    # else
        statement 4 ;
        statement 5 ;
        statement 6 ;
    # endif
}
```

If the expression, **TEST <= 5** evaluates to true, then statements 1, 2 and 3 are compiled, otherwise statements 4, 5 and 6 are compiled. In place of the expression **TEST <= 5**, other expressions like **(LEVEL == HIGH || LEVEL == LOW)** or **ADAPTER == VGA** can also be used.

If we so desire, we can have nested conditional compilation directives. An example that uses such directives is shown below.

```
# if ADAPTER == VGA
    code for video graphics array
# else
    # if ADAPTER == SVGA
        code for super video graphics array
    # else
        code for extended graphics adapter
    # endif
# endif
```

The above program segment can be made more compact by using another conditional compilation directive called **#elif**. The same program using this directive can be rewritten as shown below. Observe

that by using the **#elif** directives, the number of **#endifs** used in the program get reduced.

```
# if ADAPTER == VGA
    code for video graphics array
# elif ADAPTER == SVGA
    code for super video graphics array
# else
    code for extended graphics adapter
# endif
```

Miscellaneous Directives

There are two more preprocessor directives available, though they are not very commonly used. They are:

- (a) **#undef**
- (b) **#pragma**

#undef Directive

On some occasions, it may be desirable to cause a defined name to become 'undefined'. This can be accomplished by means of the **#undef** directive. In order to undefine a macro that has been earlier **#defined**, the directive,

```
# undef macro template
```

can be used. Thus the statement,

```
# undef PENTIUM
```

would cause the definition of **PENTIUM** to be removed from the system. All subsequent **#ifdef PENTIUM** statements would evaluate to false. In practice, seldom are you required to undefine a macro, but for some reason if you are required to, then you know that there is something to fall back upon.

#pragma Directive

This directive is another special-purpose directive that you can use to turn on or off certain features. Pragmas vary from one compiler to another. There are certain pragmas available with Microsoft C compiler that deal with formatting source listings and placing comments in the object file generated by the compiler. Turbo C/C++ compiler has got a

pragma that allows you to suppress warnings generated by the compiler. Some of these pragmas are discussed below.

- (a) **#pragma startup** and **#pragma exit**: These directives allow us to specify functions that are called upon program startup (before **main()**) or program exit (just before the program terminates). Their usage is as follows:

```
# include <stdio.h>
void fun1( ) ;
void fun2( ) ;

# pragma startup fun1
# pragma exit fun2

int main( )
{
    printf ( "Inside main\n" ) ;
    return 0 ;
}

void fun1( )
{
    printf ( "Inside fun1\n" ) ;
}

void fun2( )
{
    printf ( "Inside fun2\n" ) ;
}
```

And here is the output of the program.

```
Inside fun1
Inside main
Inside fun2
```

Note that the functions **fun1()** and **fun2()** should neither receive nor return any value. If we want two functions to get executed at startup then their pragmas should be defined in the reverse order in which you want to get them called.

- (b) **#pragma warn**: On compilation the compiler reports Errors and Warnings in the program, if any. Errors provide the programmer with no options, apart from correcting them. Warnings, on the other hand, offer the programmer a hint or suggestion that something may be *wrong* with a particular piece of code. Two most common situations when warnings are displayed are as under:
- If you have written code that the compiler's designers (or the ANSI-C specification) consider bad C programming practice. For example, if a function does not return a value then it should be declared as **void**.
 - If you have written code that might cause run-time errors, such as assigning a value to an uninitialized pointer.

The **#pragma warn** directive tells the compiler whether or not we want to suppress a specific warning. Usage of this pragma is shown below.

```
#include <stdio.h>
#pragma warn -rvl /* return value */
#pragma warn -par /* parameter not used */
#pragma warn -rch /* unreachable code */

int f1( )
{
    int a = 5 ;
}

void f2 ( int x )
{
    printf ( "Inside f2" ) ;
}

int f3( )
{
    int x = 6 ;
    return x ;
    x++ ;
}

int main( )
{
```

```
f1( );  
f2 ( 7 );  
f3( );  
return 0 ;  
}
```

If you go through the program, you can notice three problems immediately. These are:

- (a) Though promised, **f1()** doesn't return a value.
- (b) The parameter **x** that is passed to **f2()** is not being used anywhere in **f2()**.
- (c) The control can never reach **x++** in **f3()**.

If we compile the program, we should expect warnings indicating the above problems. However, this does not happen since we have suppressed the warnings using the **#pragma** directives. If we replace the '-' sign with a '+', then these warnings would be flashed on compilation. Though it is a bad practice to suppress warnings, at times, it becomes useful to suppress them. For example, if you have written a huge program and are trying to compile it, then to begin with, you are more interested in locating the errors, rather than the warnings. At such times, you may suppress the warnings. Once you have located all errors, then you may turn on the warnings and sort them out.

The Build Process

There are many steps involved in converting a C program into an executable form. Figure 12.1 shows these different steps along with the files created during each stage.

Many software development tools, like TC++ and Visual Studio, hide many of the steps shown in Figure 12.1 from us. However, it is important to understand these steps for two reasons:

- (a) It would help you understand the process much better, than just believing that, some kind of 'magic' generates the executable code.
- (b) If you are to alter any of these steps, then you would know how to do it.

Let us now understand the steps mentioned in Figure 12.1 in detail.

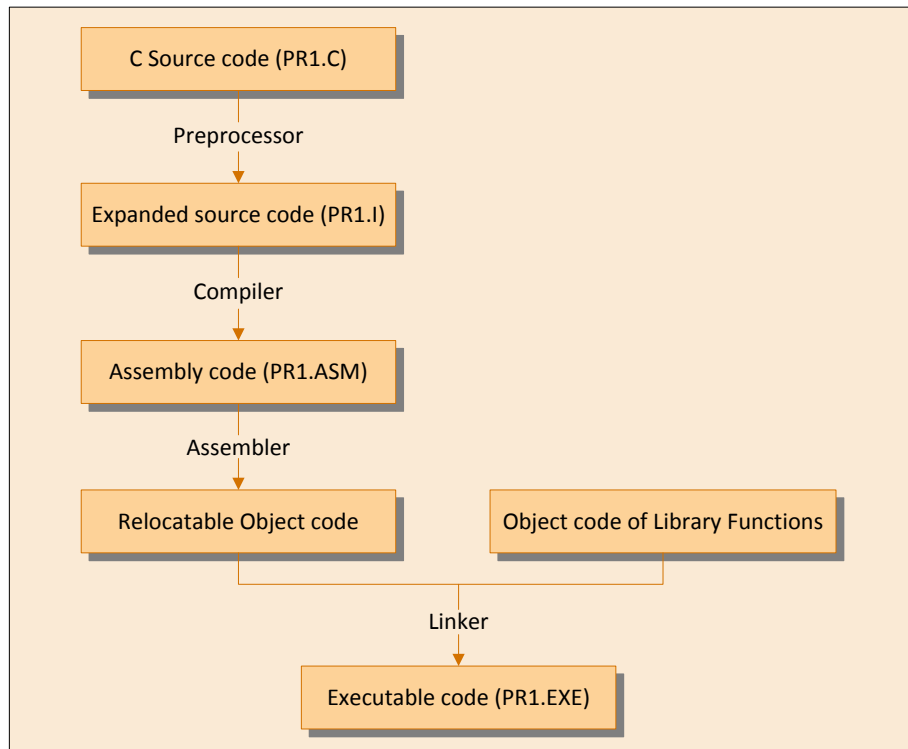


Figure 12.1

Preprocessing

During this step, the C source code is expanded based on the preprocessor directives like **#define**, **#include**, **#ifdef**, etc. The expanded source code is stored in an intermediate file with **.i** extension. Thus, if our source code is stored in PR1.C then the expanded source code is stored in PR1.I. The expanded source code is also in C language. Note that the **'i'** extension may vary from one compiler to another.

Compilation

The expanded source code is then passed to the compiler, which identifies the syntax errors in the expanded source code. These errors are displayed along with warnings, if any. As we saw in the last section, using the **#pragma** directive we can control which warnings are to be displayed / hidden.

If the expanded source code is error-free, then the compiler translates the expanded source code in C, into an equivalent assembly language program. Different processors (CPUs) support different set of assembly instructions, using which they can be programmed. Hence, the compiler

targeted for Intel Itanium platform would generate the assembly code using the instructions understood by Intel Itanium. Likewise, the same C program, when compiled using a compiler targeted for Intel Pentium V, is likely to generate a different assembly language code. The assembly code is typically stored in .ASM file. Thus, for PR1.1 the assembly code would be stored in PR1.ASM.

Assembling

The job of the Assembler is to translate .ASM program into Relocatable Object code. Thus, if the assembly language instructions are present in PR1.ASM then the relocatable object code gets stored in PR1.OBJ.

Here the word 'Relocatable' means that the program is complete except for one thing—no specific memory addresses have yet been assigned to the code and data sections in the relocatable code. All the addresses are relative offsets.

The .OBJ file that gets created is a specially formatted binary file. The object file contains a header and several sections. The header describes the sections that follow it. These sections are:

- (a) Text section – This section contains machine language code equivalent to the expanded source code.
- (b) Data Section – This section contains global variables and their initial values.
- (c) Bss (Block Started by Symbol) section – This section contains uninitialized global variables.
- (d) Symbol Table – This section contains information about symbols found during assembling of the program. Typical information present in the symbol table includes:
 - Names, types and sizes of global variables.
 - Names and addresses of functions defined in source code.
 - Names of external functions like **printf()** and **scanf()**.

Although there are machine language instructions in the .OBJ file it cannot be executed directly. This is because of following reasons:

- (a) The external functions like **printf()** are not present in the .OBJ file.

- (b) The .OBJ file may use some global variables defined in another .OBJ file. For example, PR1.OBJ may use a global variable **count2** which is defined in the file PR2.OBJ.
- (c) The .OBJ file may use a function defined in another .OBJ file. For example, PR2.OBJ may use a function **display()** defined in the file PR1.OBJ.

Note that parts of the symbol table may be incomplete because all the variables and functions may not be defined in the same file. The references to such variables and functions (symbols) that are defined in other source files are later on resolved by the linker.

Linking

Linking is the final stage in creating an executable program. It has to do the following important things:

- (a) Find definition of all external functions—those which are defined in other .OBJ files, and those which are defined in Libraries (like **printf()**).
- (b) Find definition of all global variables—those which are defined in other .OBJ files, and those which are defined in Libraries (like **errno** which is a commonly used global variable that is defined in Standard C Library).
- (c) Combine Data Sections of different .OBJ files into a single Data Section.
- (d) Combine Code Sections of different .OBJ files into a single Code Section.

While combining different .OBJ files the linker has to address one problem. Addresses of all variables and functions in the Symbol Table of the .OBJ file are Relative addresses. This means that address of a Symbol (variable or function), is in fact only an offset from start of the Section (Data or Code Section) to which it belongs. For example, if there are two 4-byte wide global integer variables **count1** and **index1** in PR1.OBJ, then the Symbol Table will have addresses 0 and 4 for them. Similarly, if PR2.OBJ has two similar variables **count2** and **index2** then Symbol Table in PR2.OBJ will have addresses 0 and 4 for these variables. Same addressing scheme is used with functions present in each .OBJ file.

When linker combines the two .OBJ files, it has to re-adjust the addresses of global variables and functions. Thus, the variables **count1**, **index1**, **count2** and **index2** will now enjoy addresses 0, 4, 8 and 12, respectively. Similar re-adjustment of addresses will be done for functions. Even after re-adjustment the addresses of variables and functions are still 'relative' in the combined Data and Code sections of the .EXE file.

In the .EXE file machine language code from all of the input object files will be in the Text section. Similarly, all initialized and uninitialized variables will reside in the new Data and Bss sections, respectively.

During linking if the linker detects errors such as mis-spelling the name of a library function in the source code, or using the incorrect number or type of parameters for a function, it stops the linking process and doesn't create the binary executable file.

Loading

Once the .EXE file is created and stored on the disk, it is ready for execution. When we execute it, it is first brought from the disk into the memory (RAM) by an Operating System component called Program Loader. Program Loader can place the .EXE anywhere in memory depending on its availability. Since all the addresses in the .EXE file are 'relative' addresses, exact 'position' where .EXE is loaded in memory doesn't matter. No further adjustment of addresses is necessary. Thus, the Code and Data in .EXE file are 'Position Independent'. Once the loading process is completed, the execution begins from the first instruction in Code section of the file loaded in memory.

Modern Operating Systems like Windows and Linux permit loading and execution of multiple programs in memory. One final word before we end this topic. Like a .OBJ file, a .EXE file is also a formatted binary file. The format of these binary file differs from one Operating System to another. For example, Windows Operating System uses Portable Executable (PE) file format, whereas, Linux uses Executable and Linking Format (ELF). Hence an .OBJ or .EXE file created for Windows cannot be used on Linux and vice-versa.

Figure 12.2 summarizes the role played by each processor program during the build process.

Processor	Input	Output
Editor	Program typed from keyboard	C source code containing program and preprocessor commands
Preprocessor	C source code file	Expanded Source code file created after processing preprocessor commands
Compiler	Expanded Source code file	Assembly language code
Assembler	Assembly language code	Relocatable Object code in machine language
Linker	Object code of our program and object code of library functions	Executable code in machine language
Loader	Executable file	-

Figure 12.2

Summary

- (a) The preprocessor directives enable the programmer to write programs that are easy to develop, read, modify and transport to a different computer system.
- (b) We can make use of various preprocessor directives, such as **#define**, **#include**, **#ifdef** - **#else** - **#endif**, **#if** and **#elif** in our program.
- (c) The directives like **#undef** and **#pragma** are also useful although they are seldom used.
- (d) A good understanding of entire build process is important as it gives a good insight into the conversion of source code to executable code.

Exercise

[A] Answer the following:

- (a) A preprocessor directive is:
 1. A message from compiler to the programmer
 2. A message from compiler to the linker

3. A message from programmer to the preprocessor
4. A message from programmer to the microprocessor

(b) Which of the following are correctly formed **#define** statements:

```
#define INCH PER FEET 12
#define SQR (X) ( X * X )
#define SQR(X) X * X
#define SQR(X) ( X * X )
```

(c) State True or False:

1. A macro must always be written in capital letters.
2. A macro should always be accommodated in a single line.
3. After preprocessing when the program is sent for compilation the macros are removed from the expanded source code.
4. Macros with arguments are not allowed.
5. Nested macros are allowed.
6. In a macro call the control is passed to the macro.

(d) How many **#include** directives can be there in a given program file?

(e) What is the difference between the following two **#include** directives:

```
#include "conio.h"
#include <conio.h>
```

(f) A header file is:

1. A file that contains standard library functions
2. A file that contains definitions and macros
3. A file that contains user-defined functions
4. A file that is present in current working directory

(g) Which of the following is not a preprocessor directive

1. **#if**
2. **#elseif**
3. **#undef**
4. **#pragma**

(h) All macro substitutions in a program are done:

1. Before compilation of the program

2. After compilation
3. During execution
4. None of the above

(i) In a program the statement:

```
#include "filename"
```

(j) is replaced by the contents of the file "filename":

1. Before compilation
2. After Compilation
3. During execution
4. None of the above

[B] What will be the output of the following programs:

(a) # include <stdio.h>

```
int main( )
{
    int i = 2 ;
    # ifdef DEF
        i *= i ;
    # else
        printf ( "%d\n", i ) ;
    # endif
    return 0 ;
}
```

(b) # include <stdio.h>

```
# define PRODUCT(x) ( x * x )
int main( )
{
    int i = 3, j, k, l ;
    j = PRODUCT( i + 1 ) ;
    k = PRODUCT( i++ ) ;
    l = PRODUCT ( ++i ) ;
    printf ( "%d %d %d %d\n", i, j, k, l ) ;
    return 0 ;
}
```

(c) # include <stdio.h>

```
# define PI 3.14
# define AREA( x, y, z ) ( PI * x * x + y * z ) ;
```

```
int main( )
{
    float a = AREA ( 1, 5, 8 ) ;
    float b = AREA ( AREA ( 1, 5, 8 ), 4, 5 ) ;
    printf ( " a = %f\n", a ) ;
    printf ( " b = %f\n", b ) ;
    return 0 ;
}
```

[C] Attempt the following:

- (a) If a macro is not getting expanded as per your expectation, how will you find out how is it being expanded by the preprocessor.
- (b) Write down macro definitions for the following:
 - 1. To test whether a character is a small case letter or not.
 - 2. To test whether a character is a upper case letter or not.
 - 3. To test whether a character is an alphabet or not. Make use of the macros you defined in 1 and 2 above.
 - 4. To obtain the bigger of two numbers.
- (c) Write macro definitions with arguments for calculation of area and perimeter of a triangle, a square and a circle. Store these macro definitions in a file called "areaperi.h". Include this file in your program, and call the macro definitions for calculating area and perimeter for different squares, triangles and circles.
- (d) Write down macro definitions for the following:
 - 1. To find arithmetic mean of two numbers.
 - 2. To find absolute value of a number.
 - 3. To convert a upper case alphabet to lower case.
 - 4. To obtain the bigger of two numbers.
- (e) Write macro definitions with arguments for calculation of Simple Interest and Amount. Store these macro definitions in a file called "interest.h". Include this file in your program, and use the macro definitions for calculating simple interest and amount.

- **What are Arrays?**
A Simple Program using Array
- **More on Arrays**
Array Initialization
Array Elements in Memory
Bounds Checking
Passing Array Elements to a Function
- **Pointers and Arrays**
Passing an Entire Array to a Function
The Real Thing
- **Summary**
- **Exercise**



13

Arrays

- What are Arrays
 - A Simple Program Using Array
- More on Arrays
 - Array Initialisation
 - Array Elements in Memory
 - Bounds Checking
 - Passing Array Elements to a Function
- Pointers and Arrays
 - Passing an Entire Array to a Function
 - The Real Thing
- Summary
- Exercise

The C language provides a capability that enables the user to design a set of similar data types, called array. This chapter describes how arrays can be created and manipulated in C.

We should note that, in many C books and courses, arrays and pointers are taught separately. I feel it is worthwhile to deal with these topics together. This is because pointers and arrays are so closely related that discussing arrays without discussing pointers would make the discussion incomplete and wanting. In fact, all arrays make use of pointers internally. Hence, it is all too relevant to study them together rather than as isolated topics.

What are Arrays?

For understanding the arrays properly, let us consider the following program:

```
#include <stdio.h>
int main( )
{
    int x;
    x = 5;
    x = 10;
    printf ( "x = %d\n", x );
    return 0;
}
```

No doubt, this program will print the value of **x** as 10. Why so? Because, when a value 10 is assigned to **x**, the earlier value of **x**, i.e., 5 is lost. Thus, ordinary variables (the ones which we have used so far) are capable of holding only one value at a time (as in this example). However, there are situations in which we would want to store more than one value at a time in a single variable.

For example, suppose we wish to arrange the percentage marks obtained by 100 students in ascending order. In such a case, we have two options to store these marks in memory:

- (a) Construct 100 variables to store percentage marks obtained by 100 different students, i.e., each variable containing one student's marks.
- (b) Construct one variable (called array or subscripted variable) capable of storing or holding all the hundred values.

Obviously, the second alternative is better. A simple reason for this is, it would be much easier to handle one variable than handling 100 different variables. Moreover, there are certain logics that cannot be dealt with, without the use of an array. Now a formal definition of an array—An array is a collective name given to a group of ‘similar quantities’. These similar quantities could be percentage marks of 100 students, or salaries of 300 employees, or ages of 50 employees. What is important is that the quantities must be ‘similar’. Each member in the group is referred to by its position in the group. For example, assume the following group of numbers, which represent percentage marks obtained by five students.

```
per = { 48, 88, 34, 23, 96 }
```

If we want to refer to the second number of the group, the usual notation used is per_2 . Similarly, the fourth number of the group is referred as per_4 . However, in C, the fourth number is referred as **per[3]**. This is because, in C, the counting of elements begins with 0 and not with 1. Thus, in this example **per[3]** refers to 23 and **per[4]** refers to 96. In general, the notation would be **per[i]**, where, **i** can take a value 0, 1, 2, 3, or 4, depending on the position of the element being referred. Here **per** is the subscripted variable (array), whereas **i** is its subscript.

Thus, an array is a collection of similar elements. These similar elements could be all **ints**, or all **floats**, or all **chars**, etc. Usually, the array of characters is called a ‘string’, whereas an array of **ints** or **floats** is called simply an array. Remember that all elements of any given array must be of the same type, i.e., we cannot have an array of 10 numbers, of which 5 are **ints** and 5 are **floats**.

A Simple Program using Array

Let us try to write a program to find average marks obtained by a class of 30 students in a test.

```
#include <stdio.h>
int main( )
{
    int avg, sum = 0 ;
    int i ;
    int marks[ 30 ] ; /* array declaration */
}
```

```
for ( i = 0 ; i <= 29 ; i++ )
{
    printf ( "Enter marks " ) ;
    scanf ( "%d", &marks[ i ] ) ; /* store data in array */
}

for ( i = 0 ; i <= 29 ; i++ )
    sum = sum + marks[ i ] ; /* read data from an array*/

avg = sum / 30 ;
printf ( "Average marks = %d\n", avg ) ;
return 0 ;
}
```

There is a lot of new material in this program, so let us take it apart slowly.

Array Declaration

To begin with, like other variables, an array needs to be declared so that the compiler will know what kind of an array and how large an array we want. In our program, we have done this with the statement:

```
int marks[ 30 ] ;
```

Here, **int** specifies the type of the variable, just as it does with ordinary variables and the word **marks** specifies the name of the variable. The **[30]** however is new. The number 30 tells how many elements of the type **int** will be in our array. This number is often called the 'dimension' of the array. The bracket (**[]**) tells the compiler that we are dealing with an array.

Accessing Elements of an Array

Once an array is declared, let us see how individual elements in the array can be referred. This is done with subscript, the number in the brackets following the array name. This number specifies the element's position in the array. All the array elements are numbered, starting with 0. Thus, **marks[2]** is not the second element of the array, but the third. In our program, we are using the variable **i** as a subscript to refer to various elements of the array. This variable can take different values and hence can refer to the different elements in the array in turn. This ability to use variables to represent subscripts is what makes arrays so useful.

Entering Data into an Array

Here is the section of code that places data into an array:

```
for ( i = 0 ; i <= 29 ; i++ )
{
    printf ( "Enter marks " ) ;
    scanf ( "%d", &marks[ i ] ) ;
}
```

The **for** loop causes the process of asking for and receiving a student's marks from the user to be repeated 30 times. The first time through the loop, **i** has a value 0, so the **scanf()** function will cause the value typed to be stored in the array element **marks[0]**, the first element of the array. This process will be repeated until **i** becomes 29. This is last time through the loop, which is a good thing, because there is no array element like **marks[30]**.

In **scanf()** function, we have used the "address of" operator (&) on the element **marks[i]** of the array, just as we have used it earlier on other variables (**&rate**, for example). In so doing, we are passing the address of this particular array element to the **scanf()** function, rather than its value; which is what **scanf()** requires.

Reading Data from an Array

The balance of the program reads the data back out of the array and uses it to calculate the average. The **for** loop is much the same, but now the body of the loop causes each student's marks to be added to a running total stored in a variable called **sum**. When all the marks have been added up, the result is divided by 30, the number of students, to get the average.

```
for ( i = 0 ; i <= 29 ; i++ )
    sum = sum + marks[ i ] ;

avg = sum / 30 ;
printf ( "Average marks = %d\n", avg ) ;
```

To fix our ideas, let us revise whatever we have learnt about arrays:

- (a) An array is a collection of similar elements.
- (b) The first element in the array is numbered 0, so the last element is 1 less than the size of the array.

- (c) An array is also known as a subscripted variable.
- (d) Before using an array, its type and dimension must be declared.
- (e) However big an array, its elements are always stored in contiguous memory locations. This is a very important point which we would discuss in more detail later on.

More on Arrays

Array is a very popular data type with C programmers. This is because of the convenience with which arrays lend themselves to programming. The features which make arrays so convenient to program would be discussed below, along with the possible pitfalls in using them.

Array Initialization

So far we have used arrays that did not have any values in them to begin with. We managed to store values in them during program execution. Let us now see how to initialize an array while declaring it. Following are a few examples that demonstrate this:

```
int num[ 6 ] = { 2, 4, 12, 5, 45, 5 };  
int n[ ] = { 2, 4, 12, 5, 45, 5 };  
float press[ ] = { 12.3, 34.2, -23.4, -11.3 };
```

Note the following points carefully:

- (a) Till the array elements are not given any specific values, they are supposed to contain garbage values.
- (b) If the array is initialised where it is declared, mentioning the dimension of the array is optional as in the 2nd and 3rd examples above.

Array Elements in Memory

Consider the following array declaration:

```
int arr[ 8 ];
```

What happens in memory when we make this declaration? 32 bytes get immediately reserved in memory, 4 bytes each for the 8 integers (under TC/TC++ the array would occupy 16 bytes as each integer would occupy 2 bytes). And since the array is not being initialized, all eight values present in it would be garbage values. This so happens because the storage class of this array is assumed to be **auto**. If the storage class is declared to be **static**, then all the array elements would have a default

initial value as zero. Whatever be the initial values, all the array elements would always be present in contiguous memory locations. This arrangement of array elements in memory is shown in Figure 13.1.

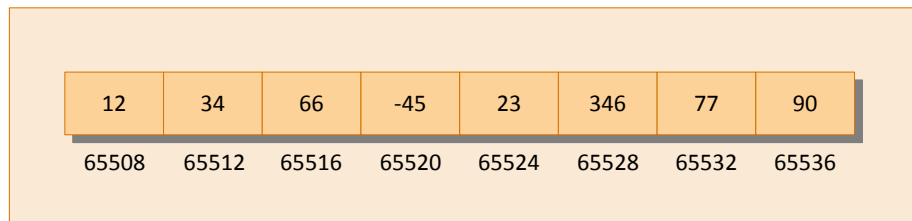


Figure 13.1

Bounds Checking

In C, there is no check to see if the subscript used for an array exceeds the size of the array. Data entered with a subscript exceeding the array size will simply be placed in memory outside the array; probably on top of other data, or on the program itself. This will lead to unpredictable results, to say the least, and there will be no error message to warn you that you are going beyond the array size. In some cases, the computer may just hang. Thus, the following program may turn out to be suicidal:

```
#include <stdio.h>
int main( )
{
    int num[ 40 ], i ;

    for ( i = 0 ; i <= 100 ; i++ )
        num[ i ] = i ;
    return 0 ;
}
```

Thus, to see to it that we do not reach beyond the array size, is entirely the programmer's botheration and not the compiler's.

Passing Array Elements to a Function

Array elements can be passed to a function by calling the function by value, or by reference. In the call by value, we pass values of array elements to the function, whereas in the call by reference, we pass addresses of array elements to the function. These two calls are illustrated below.

```
/* Demonstration of call by value */
# include <stdio.h>
void display ( int );
int main( )
{
    int i;
    int marks[ ] = { 55, 65, 75, 56, 78, 78, 90 };
    for ( i = 0 ; i <= 6 ; i++ )
        display ( marks[ i ] );
    return 0 ;
}
void display ( int m )
{
    printf ( "%d ", m );
}
```

And here's the output...

```
55 65 75 56 78 78 90
```

Here, we are passing an individual array element at a time to the function **display()** and getting it printed in the function **display()**. Note that, since at a time only one element is being passed, this element is collected in an ordinary integer variable **m**, in the function **display()**.

And now the call by reference.

```
/* Demonstration of call by reference */
# include <stdio.h>
void disp ( int * );
int main( )
{
    int i;
    int marks[ ] = { 55, 65, 75, 56, 78, 78, 90 };
    for ( i = 0 ; i <= 6 ; i++ )
        disp ( &marks[ i ] );
    return 0 ;
}
void disp ( int *n )
{
    printf ( "%d ", *n );
}
```

And here's the output...

```
55 65 75 56 78 78 90
```

Here, we are passing addresses of individual array elements to the function **disp()**. Hence, the variable in which this address is collected (**n**), is declared as a pointer variable. And since **n** contains the address of array element, to print out the array element, we are using the 'value at address' operator (*****).

Read the following program carefully. The purpose of the function **disp()** is just to display the array elements on the screen. The program is only partly complete. You are required to write the function **show()** on your own. Try your hand at it.

```
# include <stdio.h>
void disp ( int * );
int main( )
{
    int i ;
    int marks[ ] = { 55, 65, 75, 56, 78, 78, 90 };
    for ( i = 0 ; i <= 6 ; i++ )
        disp ( &marks[ i ] );
    return 0 ;
}
void disp ( int *n )
{
    show ( &n );
}
```

Pointers and Arrays

To be able to see what pointers have got to do with arrays, let us first learn some pointer arithmetic. Consider the following example:

```
# include <stdio.h>
int main( )
{
    int i = 3, *x ;
    float j = 1.5, *y ;
    char k = 'c', *z ;
    printf ( "Value of i = %d\n", i );
    printf ( "Value of j = %f\n", j );
```



```
printf ( "Value of k = %c\n", k );  
x = &i ;  
y = &j ;  
z = &k ;  
printf ( "Original address in x = %u\n", x ) ;  
printf ( "Original address in y = %u\n", y ) ;  
printf ( "Original address in z = %u\n", z ) ;  
x++ ;  
y++ ;  
z++ ;  
printf ( "New address in x = %u\n", x ) ;  
printf ( "New address in y = %u\n", y ) ;  
printf ( "New address in z = %u\n", z ) ;  
return 0 ;  
}
```

Here is the output of the program.

```
Value of i = 3  
Value of j = 1.500000  
Value of k = c  
Original address in x = 65524  
Original address in y = 65520  
Original address in z = 65519  
New address in x = 65528  
New address in y = 65524  
New address in z = 65520
```

Observe the last three lines of the output. 65528 is original value in **x** plus 4, 65524 is original value in **y** plus 4, and 65520 is original value in **z** plus 1. This so happens because every time a pointer is incremented, it points to the immediately next location of its type. That is why, when the integer pointer **x** is incremented, it points to an address four locations after the current location, since an **int** is always 4 bytes long (under TC/TC++, since **int** is 2 bytes long, new value of **x** would be 65526). Similarly, **y** points to an address 4 locations after the current location and **z** points 1 location after the current location. This is a very important result and can be effectively used while passing the entire array to a function.

The way a pointer can be incremented, it can be decremented as well, to point to earlier locations. Thus, the following operations can be performed on a pointer:

- (a) Addition of a number to a pointer. For example,

```
int i = 4, *j, *k ;
j = &i ;
j = j + 1 ;
j = j + 9 ;
k = j + 3 ;
```

- (b) Subtraction of a number from a pointer. For example,

```
int i = 4, *j, *k ;
j = &i ;
j = j - 2 ;
j = j - 5 ;
k = j - 6 ;
```

- (c) Subtraction of one pointer from another.

One pointer variable can be subtracted from another provided both variables point to elements of the same array. The resulting value indicates the number of elements separating the corresponding array elements. This is illustrated in the following program:

```
# include <stdio.h>
int main( )
{
    int arr[ ] = { 10, 20, 30, 45, 67, 56, 74 } ;
    int *i, *j ;

    i = &arr[ 1 ] ;
    j = &arr[ 5 ] ;
    printf ( "%d %d\n", j - i, *j - *i ) ;
    return 0 ;
}
```

Here **i** and **j** have been declared as integer pointers holding addresses of first and fifth element of the array, respectively.

Suppose the array begins at location 65502, then the elements **arr[1]** and **arr[5]** would be present at locations 65506 and 65522 respectively, since each integer in the array occupies 2 bytes in memory. The expression **j - i** would print a value 4 and not 8. This is because **j** and **i** are pointing to locations that are 4 integers apart. What will be the result of the expression ***j - *i**? 36, since ***j** and ***i** return the values present at addresses contained in the pointers **j** and **i**.

(d) Comparison of two pointer variables

Pointer variables can be compared provided both variables point to objects of the same data type. Such comparisons can be useful when both pointer variables point to elements of the same array. The comparison can test for either equality or inequality. Moreover, a pointer variable can be compared with zero (usually expressed as NULL). The following program illustrates how the comparison is carried out:

```
#include <stdio.h>
int main( )
{
    int arr[ ] = { 10, 20, 36, 72, 45, 36 };
    int *j, *k;

    j = &arr[ 4 ];
    k = ( arr + 4 );
    if ( j == k )
        printf ( "The two pointers point to the same location\n" );
    else
        printf ( "The two pointers do not point to the same
location\n" );

    return 0 ;
}
```

A word of caution! Do not attempt the following operations on pointers... they would never work out.

- (a) Addition of two pointers
- (b) Multiplication of a pointer with a constant
- (c) Division of a pointer with a constant

Now we will try to correlate the following two facts, which we have learnt above:

- (a) Array elements are always stored in contiguous memory locations.
- (b) A pointer when incremented always points to an immediately next location of its type.

Suppose we have an array **num[]** = { 24, 34, 12, 44, 56, 17 }. Figure 13.2 shows how this array is located in memory.

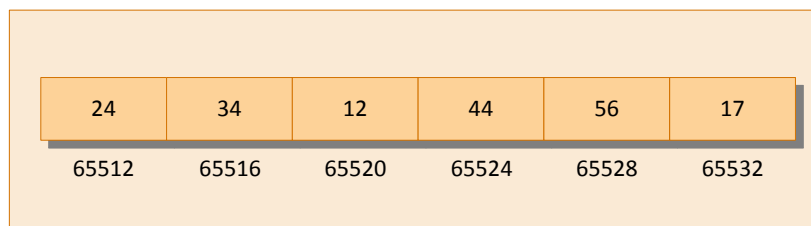


Figure 13.2

Here is a program that prints out the memory locations in which the elements of this array are stored.

```
# include <stdio.h>
int main( )
{
    int num[ ] = { 24, 34, 12, 44, 56, 17 };
    int i;
    for ( i = 0 ; i <= 5 ; i++ )
    {
        printf ( "element no. %d ", i );
        printf ( "address = %u\n", &num[ i ] );
    }
    return 0 ;
}
```

The output of this program would look like this:

```
element no. 0 address = 65512
element no. 1 address = 65516
element no. 2 address = 65520
element no. 3 address = 65524
element no. 4 address = 65528
element no. 5 address = 65532
```

Note that the array elements are stored in contiguous memory locations, each element occupying 2 bytes, since it is an integer array. When you run this program, you may get different addresses, but what is certain is that each subsequent address would be 4 bytes (2 bytes in Turbo C/C++) greater than its immediate predecessor.

Our next two programs show ways in which we can access the elements of this array.

```
#include <stdio.h>
int main( )
{
    int num[ ] = { 24, 34, 12, 44, 56, 17 };
    int i;
    for ( i = 0 ; i <= 5 ; i++ )
    {
        printf ( "address = %u ", &num[ i ] );
        printf ( "element = %d\n", num[ i ] );
    }
    return 0 ;
}
```

The output of this program would be:

```
address = 65512 element = 24
address = 65516 element = 34
address = 65520 element = 12
address = 65524 element = 44
address = 65528 element = 56
address = 65532 element = 17
```

This method of accessing array elements by using subscripted variables is already known to us. This method has, in fact, been given here for easy comparison with the next method, which accesses the array elements using pointers.

```
#include <stdio.h>
int main( )
{
    int num[ ] = { 24, 34, 12, 44, 56, 17 };
    int i, *j;
```

```
j = &num[ 0 ]; /* assign address of zeroth element */
for ( i = 0 ; i <= 5 ; i++ )
{
    printf ( "address = %u ", j );
    printf ( "element = %d\n", *j );
    j++ ; /* increment pointer to point to next location */
}
return 0 ;
}
```

The output of this program would be:

```
address = 65512 element = 24
address = 65516 element = 34
address = 65520 element = 12
address = 65524 element = 44
address = 65528 element = 56
address = 65532 element = 17
```

In this program, to begin with, we have collected the base address of the array (address of the 0th element) in the variable `j` using the statement,

```
j = &num[ 0 ]; /* assigns address 65512 to j */
```

When we are inside the loop for the first time, `j` contains the address 65512, and the value at this address is 24. These are printed using the statements,

```
printf ( "address = %u ", j );
printf ( "element = %d\n", *j );
```

On incrementing `j`, it points to the next memory location of its type (that is location no. 65516). But location no. 65516 contains the second element of the array, therefore when the `printf( )` statements are executed for the second time, they print out the second element of the array and its address (i.e., 34 and 65516)... and so on till the last element of the array has been printed.

Obviously, a question arises as to which of the above two methods should be used when? Accessing array elements by pointers is **always** faster than accessing them by subscripts. However, from the point of view of convenience in programming, we should observe the following:

Array elements should be accessed using pointers, if the elements are to be accessed in a fixed order, say from beginning to end, or from end to beginning, or every alternate element or any such definite logic.

Instead, it would be easier to access the elements using a subscript if there is no fixed logic in accessing the elements. However, in this case also, accessing the elements by pointers would work faster than subscripts.

Passing an Entire Array to a Function

In the previous section, we saw two programs—one in which we passed individual elements of an array to a function, and another in which we passed addresses of individual elements to a function. Let us now see how to pass an entire array to a function rather than its individual elements. Consider the following example:

```
/* Demonstration of passing an entire array to a function */
#include <stdio.h>
void display ( int *, int );
int main( )
{
    int num[ ] = { 24, 34, 12, 44, 56, 17 };
    display ( &num[ 0 ], 6 );
    return 0 ;
}
void display ( int *j, int n )
{
    int i ;
    for ( i = 0 ; i <= n - 1 ; i++ )
    {
        printf ( "element = %d\n", *j );
        j++ ; /* increment pointer to point to next element */
    }
}
```

Here, the **display()** function is used to print out the array elements. Note that the address of the zeroth element is being passed to the **display()** function. The **for** loop is same as the one used in the earlier program to access the array elements using pointers. Thus, just passing the address of the zeroth element of the array to a function is as good as passing the entire array to the function. It is also necessary to pass the total number of elements in the array, otherwise the **display()** function

would not know when to terminate the **for** loop. Note that the address of the zeroth element (many a time called the base address) can also be passed by just passing the name of the array. Thus, the following two function calls are same:

```
display ( &num[ 0 ], 6 );  
display ( num, 6 );
```

The Real Thing

If you have grasped the concept of storage of array elements in memory and the arithmetic of pointers, here is some real food for thought. Once again consider the following array:

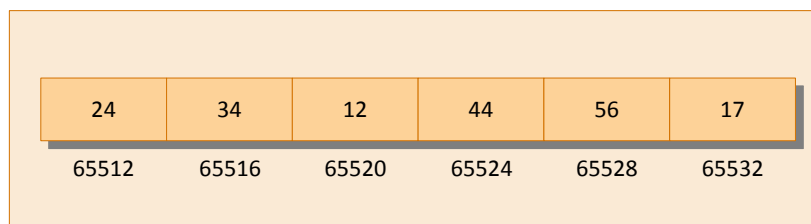


Figure 13.3

This is how we would declare the above array in C,

```
int num[ ] = { 24, 34, 12, 44, 56, 17 };
```

We also know, that on mentioning the name of the array, we get its base address. Thus, by saying ***num**, we would be able to refer to the zeroth element of the array, that is, 24. One can easily see that ***num** and ***(num + 0)** both refer to 24.

Similarly, by saying ***(num + 1)**, we can refer the first element of the array, that is, 34. In fact, this is what the C compiler does internally. When we say, **num[i]**, the C compiler internally converts it to ***(num + i)**. This means that all the following notations are same:

```
num[ i ]  
*( num + i )  
*( i + num )  
i[ num ]
```

And here is a program to prove my point.


```
/* Accessing array elements in different ways */
#include <stdio.h>
int main( )
{
    int num[ ] = { 24, 34, 12, 44, 56, 17 };
    int i;
    for ( i = 0 ; i <= 5 ; i++ )
    {
        printf ( "address = %u ", &num[ i ] );
        printf ( "element = %d %d ", num[ i ], *( num + i ) );
        printf ( "%d %d\n", *( i + num ), i[ num ] );
    }
    return 0 ;
}
```

The output of this program would be:

```
address = 65512 element = 24 24 24 24
address = 65516 element = 34 34 34 34
address = 65520 element = 12 12 12 12
address = 65524 element = 44 44 44 44
address = 65528 element = 56 56 56 56
address = 65532 element = 17 17 17 17
```

Summary

- (a) An array is similar to an ordinary variable except that it can store multiple elements of similar type.
- (b) Compiler doesn't perform bounds checking on an array.
- (c) The array variable acts as a pointer to the zeroth element of the array. In a 1-D array, zeroth element is a single value, whereas, in a 2-D array this element is a 1-D array.
- (d) On incrementing a pointer it points to the next location of its type.
- (e) Array elements are stored in contiguous memory locations and so they can be accessed using pointers.
- (f) Only limited arithmetic can be done on pointers.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    int num[ 26 ], temp ;
    num[ 0 ] = 100 ;
    num[ 25 ] = 200 ;
    temp = num[ 25 ] ;
    num[ 25 ] = num[ 0 ] ;
    num[ 0 ] = temp ;
    printf ( "%d %d\n", num[ 0 ], num[ 25 ] ) ;
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
{
    int array[ 26 ], i ;
    for ( i = 0 ; i <= 25 ; i++ )
    {
        array[ i ] = 'A' + i ;
        printf ( "%d %c\n", array[ i ], array[ i ] ) ;
    }
    return 0 ;
}
```
- (c)

```
#include <stdio.h>
int main( )
{
    int sub[ 50 ], i ;
    for ( i = 0 ; i <= 48 ; i++ ) ;
    {
        sub[ i ] = i ;
        printf ( "%d\n", sub[ i ] ) ;
    }
    return 0 ;
}
```

[B] Point out the errors, if any, in the following program segments:

- (a)

```
/* mixed has some char and some int values */
```

```
# include <stdio.h>
int char mixed[ 100 ];
int main( )
{
    int a[ 10 ], i;
    for ( i = 1 ; i <= 10 ; i++ )
    {
        scanf ( "%d", a[ i ] );
        printf ( "%d\n", a[ i ] );
    }
    return 0 ;
}
```

(b) # include <stdio.h>

```
int main( )
{
    int size ;
    scanf ( "%d", &size ) ;
    int arr[ size ] ;
    for ( i = 1 ; i <= size ; i++ )
    {
        scanf ( "%d", &arr[ i ] ) ;
        printf ( "%d\n", arr[ i ] ) ;
    }
    return 0 ;
}
```

(c) # include <stdio.h>

```
int main( )
{
    int i, a = 2, b = 3 ;
    int arr[ 2 + 3 ] ;
    for ( i = 0 ; i < a+b ; i++ )
    {
        scanf ( "%d", &arr[ i ] ) ;
        printf ( "%d\n", arr[ i ] ) ;
    }
    return 0 ;
}
```

[C] Answer the following:

(a) An array is a collection of:

1. Different data types scattered throughout memory
2. The same data type scattered throughout memory
3. The same data type placed next to each other in memory
4. Different data types placed next to each other in memory

(b) Are the following array declarations correct?

```
int a (25) ;  
int size = 10, b[ size ] ;  
int c = { 0,1,2 } ;
```

(c) Which element of the array does this expression reference?

```
num[ 4 ]
```

(d) What is the difference between the 5's in these two expressions?
(Select the correct answer)

```
int num[ 5 ] ;  
num[ 5 ] = 11 ;
```

1. First is particular element, second is type
2. First is array size, second is particular element
3. First is particular element, second is array size
4. Both specify array size

(e) State whether the following statements are True or False:

1. The array **int num[26]** has twenty-six elements.
2. The expression **num[1]** designates the first element in the array.
3. It is necessary to initialize the array at the time of declaration.
4. The expression **num[27]** designates the twenty-eighth element in the array.

[D] Attempt the following:

- (a) Twenty-five numbers are entered from the keyboard into an array. The number to be searched is entered through the keyboard by the user. Write a program to find if the number to be searched is present in the array and if it is present, display the number of times it appears in the array.
- (b) Implement the Selection Sort, Bubble Sort and Insertion sort algorithms on a set of 25 numbers. (Refer Figures 13.4 (a), 13.4 (b), 13.4 (c) for the logic of the algorithms)
 - Selection sort

- Bubble Sort
- Insertion Sort

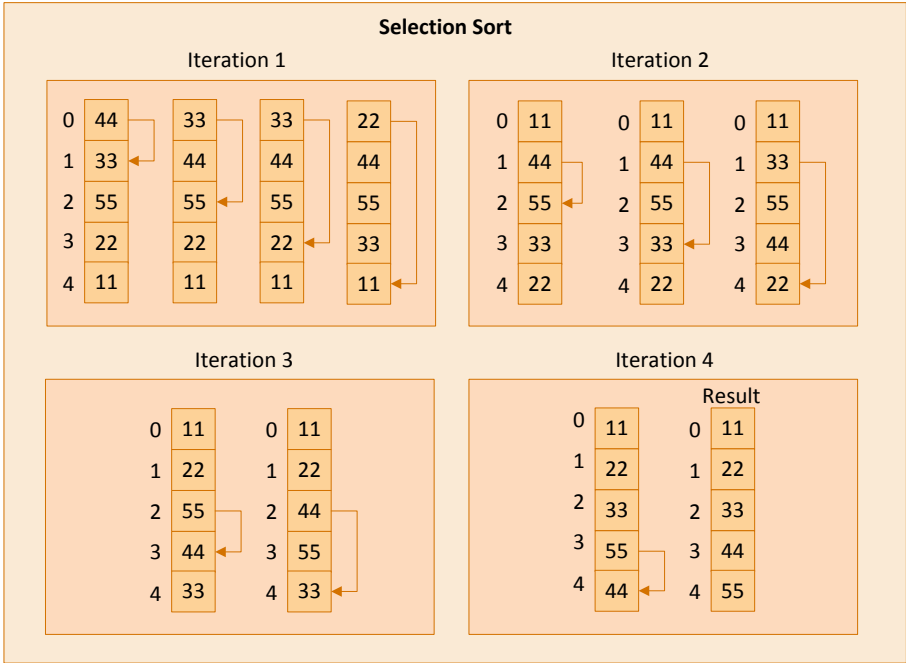


Figure 13.4 (a)

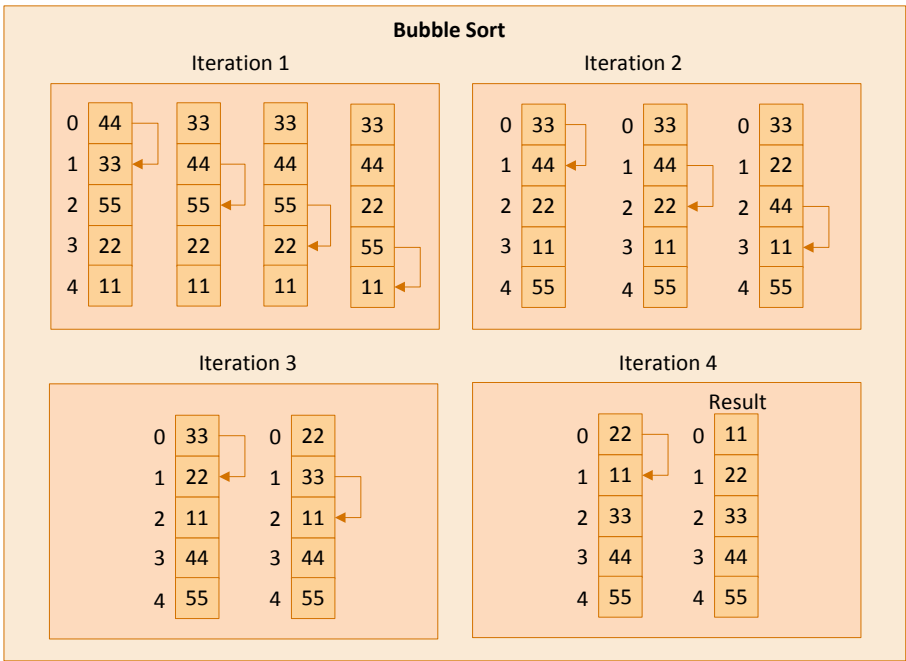


Figure 13.4 (b)

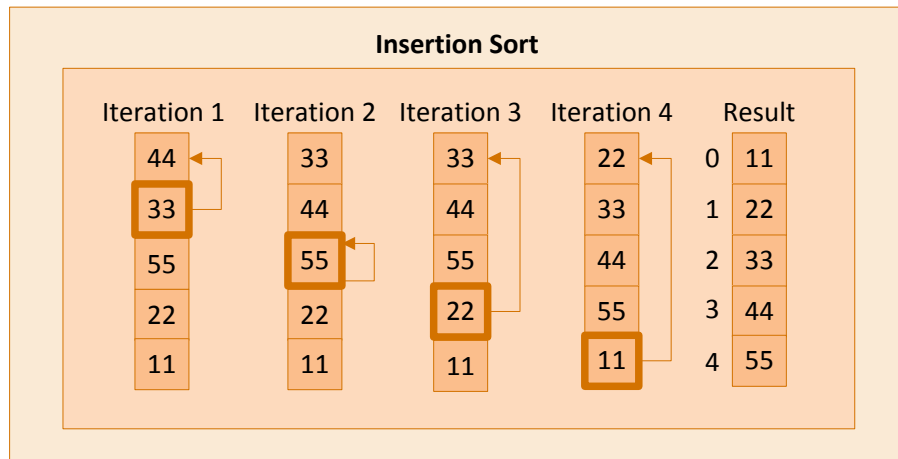


Figure 13.4 (c)

- (c) Implement in a program the following procedure to generate prime numbers from 1 to 100. This procedure is called sieve of Eratosthenes.

- Step 1 Fill an array **num[100]** with numbers from 1 to 100.
- Step 2 Starting with the second entry in the array, set all its multiples to zero.
- Step 3 Proceed to the next non-zero element and set all its multiples to zero.
- Step 4 Repeat Step 3 till you have set up the multiples of all the non-zero elements to zero.
- Step 5 At the conclusion of Step 4, all the non-zero entries left in the array would be prime numbers, so print out these numbers.

- (d) Twenty-five numbers are entered from the keyboard into an array. Write a program to find out how many of them are positive, how many are negative, how many are even and how many odd.
- (e) Write a program that interchanges the odd and even elements of an array.

[E] What will be the output of the following programs:

- (a) `# include <stdio.h>`
`int main( )`

```
{
    int b[] = { 10, 20, 30, 40, 50 };
    int i;
    for ( i = 0 ; i <= 4 ; i++ )
        printf ( "%d\n" *( b + i ) );
    return 0 ;
}
```

(b) # include <stdio.h>

```
int main( )
{
    int b[] = { 0, 20, 0, 40, 5 };
    int i, *k;
    k = b;
    for ( i = 0 ; i <= 4 ; i++ )
    {
        printf ( "%d\n" *k );
        k++;
    }
    return 0 ;
}
```

(c) # include <stdio.h>

```
void change ( int *, int );
int main( )
{
    int a[] = { 2, 4, 6, 8, 10 };
    int i;
    change ( a, 5 );
    for ( i = 0 ; i <= 4 ; i++ )
        printf ( "%d\n", a[ i ] );
    return 0 ;
}

void change ( int *b, int n )
{
    int i;
    for ( i = 0 ; i < n ; i++ )
        *( b + i ) = *( b + i ) + 5 ;
}
```

(d) # include <stdio.h>

```
int main( )
```

```
{
    static int a[ 5 ];
    int i;
    for ( i = 0 ; i <= 4 ; i++ )
        printf ( "%d\n", a[ i ] );
    return 0 ;
}
```

```
(e) # include <stdio.h>
int main( )
{
    int a[ 5 ] = { 5, 1, 15, 20, 25 };
    int i, j, k = 1, m ;
    i = ++a[ 1 ];
    j = a[ 1 ]++;
    m = a[ i++ ];
    printf ( "%d %d %d\n", i, j, m );
}
```

[F] Point out the errors, if any, in the following programs:

```
(a) # include <stdio.h>
int main( )
{
    int array[ 6 ] = { 1, 2, 3, 4, 5, 6 };
    int i;
    for ( i = 0 ; i <= 25 ; i++ )
        printf ( "%d\n", array[ i ] );
    return 0 ;
}
```

```
(b) # include <stdio.h>
int main( )
{
    int sub[ 50 ], i ;
    for ( i = 1 ; i <= 50 ; i++ )
    {
        sub[ i ] = i ;
        printf ( "%d\n", sub[ i ] );
    }
    return 0 ;
}
```

```
(c) # include <stdio.h>
```



```

int main( )
{
    int a[ ] = { 10, 20, 30, 40, 50 };
    int j;
    j = a; /* store the address of zeroth element */
    j = j + 3;
    printf ( "%d\n" *j );
    return 0;
}

```

(d) # include <stdio.h>

```

int main( )
{
    float a[ ] = { 13.24, 1.5, 1.5, 5.4, 3.5 };
    float *j;
    j = a;
    j = j + 4;
    printf ( "%d %d %d\n", j, *j, a[ 4 ] );
    return 0;
}

```

(e) # include <stdio.h>

```

int main( )
{
    int max = 5;
    float arr[ max ];
    for ( i = 0 ; i < max ; i++ )
        scanf ( "%f", &arr[ i ] );
    return 0;
}

```

[G] Answer the following:

- (a) What will happen if you try to put so many values into an array when you initialize it that the size of the array is exceeded?
1. Nothing
 2. Possible system malfunction
 3. Error message from the compiler
 4. Other data may be overwritten
- (b) In an array **int arr[12]** the word **arr** represents the a _____ of the array.

- (c) What will happen if you put too few elements in an array when you initialize it?
 - 1. Nothing
 - 2. Possible system malfunction
 - 3. Error message from the compiler
 - 4. Unused elements will be filled with 0's or garbage
- (d) What will happen if you assign a value to an element of an array whose subscript exceeds the size of the array?
 - 1. The element will be set to 0
 - 2. Nothing, it's done all the time
 - 3. Other data may be overwritten
 - 4. Error message from the compiler
- (e) When you pass an array as an argument to a function, what actually gets passed?
 - 1. Address of the array
 - 2. Values of the elements of the array
 - 3. Address of the first element of the array
 - 4. Number of elements of the array
- (f) Which of these are reasons for using pointers?
 - 1. To manipulate parts of an array
 - 2. To refer to keywords, such as **for** and **if**
 - 3. To return more than one value from a function
 - 4. To refer to particular programs more conveniently
- (g) If you don't initialize a static array, what will be the elements set to?
 - 1. 0
 - 2. an undetermined value
 - 3. a floating point number
 - 4. the character constant '\0'

[H] State True or False:

- (a) Address of a floating-point variable is always a whole number.
- (b) Which of the following is the correct way of declaring a float pointer:
 - 1. `float ptr ;`
 - 2. `float *ptr ;`

3. *float ptr ;
4. None of the above

(c) Add the missing statement for the following program to print 35:

```
#include <stdio.h>
int main( )
{
    int j, *ptr ;
    *ptr = 35 ;
    printf ( "%d\n", j ) ;
    return 0 ;
}
```

(d) if **int s[5]** is a one-dimensional array of integers, which of the following refers to the third element in the array?

1. *(s + 2)
2. *(s + 3)
3. s + 3
4. s + 2

[I] Attempt the following:

- (a) Write a program to copy the contents of one array into another in the reverse order.
- (b) If an array **arr** contains **n** elements, then write a program to check if **arr[0] = arr[n-1]**, **arr[1] = arr[n - 2]** and so on.
- (c) Write a program using pointers to find the smallest number in an array of 25 integers.
- (d) Write a program which performs the following tasks:
 - Initialize an integer array of 10 elements in **main()**
 - Pass the entire array to a function **modify()**
 - In **modify()** multiply each element of array by 3
 - Return the control to **main()** and print the new array elements in **main()**

- **Two-Dimensional Arrays**
 - Initializing a Two-Dimensional Array
 - Memory Map of a Two-Dimensional Array
 - Pointers and Two-Dimensional Arrays
 - Pointer to an Array
 - Passing 2-D Array to a Function
- **Array of Pointers**
- **Three-Dimensional Array**
- **Summary**
- **Exercise**



14 *Multidimensional Arrays*

- Two Dimensional Arrays
 - Initialising a 2-Dimensional Array
 - Memory Map of a 2-Dimensional Array
 - Pointers and 2-Dimensional Arrays
 - Pointer to an Array
 - Passing 2-D Array to a Function
- Array of Pointers
- Three-Dimensional Array
- Summary
- Exercise

In the last chapter we have explored arrays with only one dimension. It is also possible for arrays to have two or more dimensions. This chapter describes how multidimensional arrays can be created and manipulated in C.

Two-Dimensional Arrays

The two-dimensional array is also called a matrix. Let us see how to create this array and work with it. Here is a sample program that stores roll number and marks obtained by a student side-by-side in a matrix.

```
#include <stdio.h>
int main( )
{
    int stud[ 4 ][ 2 ];
    int i, j ;

    for ( i = 0 ; i <= 3 ; i++ )
    {
        printf ( "Enter roll no. and marks" ) ;
        scanf ( "%d %d", &stud[ i ][ 0 ], &stud[ i ][ 1 ] ) ;
    }
    for ( i = 0 ; i <= 3 ; i++ )
        printf ( "%d %d\n", stud[ i ][ 0 ], stud[ i ][ 1 ] ) ;

    return 0 ;
}
```

There are two parts to the program—in the first part, through a **for** loop, we read in the values of roll no. and marks, whereas, in the second part through another **for** loop, we print out these values.

Look at the **scanf()** statement used in the first **for** loop:

```
scanf ( "%d %d", &stud[ i ][ 0 ], &stud[ i ][ 1 ] ) ;
```

In **stud[i][0]** and **stud[i][1]**, the first subscript of the variable **stud**, is row number which changes for every student. The second subscript tells which of the two columns are we talking about—the zeroth column which contains the roll no. or the first column which contains the marks. Remember the counting of rows and columns begin with zero. The complete array arrangement is shown in Figure 14.1.

	column no. 0	column no. 1
row no. 0	1234	56
row no. 1	1212	33
row no. 2	1434	80
row no. 3	1312	78

Figure 14.1

Thus, 1234 is stored in **stud[0][0]**, 56 is stored in **stud[0][1]** and so on. The above arrangement highlights the fact that a two- dimensional array is nothing but a collection of a number of one- dimensional arrays placed one below the other.

In our sample program, the array elements have been stored row-wise and accessed row-wise. However, you can access the array elements column-wise as well. Traditionally, the array elements are being stored and accessed row-wise; therefore we would also stick to the same strategy.

Initializing a Two-Dimensional Array

How do we initialize a two-dimensional array? As simple as this...

```
int stud[ 4 ][ 2 ] = {  
    { 1234, 56 },  
    { 1212, 33 },  
    { 1434, 80 },  
    { 1312, 78 }  
};
```

or even this would work...

```
int stud[ 4 ][ 2 ] = { 1234, 56, 1212, 33, 1434, 80, 1312, 78 };
```

of course, with a corresponding loss in readability.

It is important to remember that, while initializing a 2-D array, it is necessary to mention the second (column) dimension, whereas the first dimension (row) is optional.

Thus the declarations,

```
int arr[ 2 ][ 3 ] = { 12, 34, 23, 45, 56, 45 };
int arr[ ][ 3 ] = { 12, 34, 23, 45, 56, 45 };
```

are perfectly acceptable,

whereas,

```
int arr[ 2 ][ ] = { 12, 34, 23, 45, 56, 45 };
int arr[ ][ ] = { 12, 34, 23, 45, 56, 45 };
```

would never work.

Memory Map of a Two-Dimensional Array

Let us reiterate the arrangement of array elements in a two-dimensional array of students, which contains roll nos. in one column and the marks in the other.

The array arrangement shown in Figure 14.1 is only conceptually true. This is because memory doesn't contain rows and columns. In memory, whether it is a one-dimensional or a two-dimensional array, the array elements are stored in one continuous chain. The arrangement of array elements of a two-dimensional array in memory is shown in Figure 14.2:

s[0][0]	s[0][1]	s[1][0]	s[1][1]	s[2][0]	s[2][1]	s[3][0]	s[3][1]
1234	56	1212	33	1434	80	1312	78
65508	65512	65516	65520	65524	65528	65532	65536

Figure 14.2

We can easily refer to the marks obtained by the third student using the subscript notation as shown below.

```
printf ( "Marks of third student = %d", stud[ 2 ][ 1 ] );
```

Can we not refer to the same element using pointer notation, the way we did in one-dimensional arrays? Answer is yes. Only the procedure is slightly difficult to understand. So, read on...

Pointers and Two-Dimensional Arrays

The C language embodies an unusual but powerful capability—it can treat parts of arrays as arrays. More specifically, each row of a two-dimensional array can be thought of as a one-dimensional array. This is a very important fact if we wish to access array elements of a two-dimensional array using pointers.

Thus, the declaration,

```
int s[ 5 ][ 2 ] ;
```

can be thought of as setting up an array of 5 elements, each of which is a one-dimensional array containing 2 integers. We refer to an element of a one-dimensional array using a single subscript. Similarly, if we can imagine **s** to be a one-dimensional array, then we can refer to its zeroth element as **s[0]**, the next element as **s[1]** and so on. More specifically, **s[0]** gives the address of the zeroth one-dimensional array, **s[1]** gives the address of the first one-dimensional array and so on. This fact can be demonstrated by the following program:

```
/* Demo: 2-D array is an array of arrays */
#include <stdio.h>
int main( )
{
    int s[ 4 ][ 2 ] = {
        { 1234, 56 },
        { 1212, 33 },
        { 1434, 80 },
        { 1312, 78 }
    };

    int i ;
    for ( i = 0 ; i <= 3 ; i++ )
        printf ( "Address of %d th 1-D array = %u\n", i, s[ i ] ) ;
    return 0 ;
}
```

And here is the output...

```
Address of 0 th 1-D array = 65508
Address of 1 th 1-D array = 65516
Address of 2 th 1-D array = 65524
Address of 3 th 1-D array = 65532
```

Let's figure out how the program works. The compiler knows that **s** is an array containing 4 one-dimensional arrays, each containing 2 integers. Each one-dimensional array occupies 4 bytes (two bytes for each integer). These one-dimensional arrays are placed linearly (zeroth 1-D array followed by first 1-D array, etc.). Hence, each one-dimensional array starts 4 bytes further along than the last one, as can be seen in the memory map of the array shown in Figure 14.3.

s[0][0]	s[0][1]	s[1][0]	s[1][1]	s[2][0]	s[2][1]	s[3][0]	s[3][1]
1234	56	1212	33	1434	80	1312	78
65508	65512	65516	65520	65524	65528	65532	65536

Figure 14.3

We know that the expressions **s[0]** and **s[1]** would yield the addresses of the zeroth and first one-dimensional array respectively. From Figure 14.3 these addresses turn out to be 65508 and 65516.

Now, we have been able to reach each one-dimensional array. What remains is to be able to refer to individual elements of a one-dimensional array. Suppose we want to refer to the element **s[2][1]** using pointers. We know (from the above program) that **s[2]** would give the address 65524, the address of the second one-dimensional array. Obviously (65524 + 1) would give the address 65528. Or (**s[2] + 1**) would give the address 65528. And the value at this address can be obtained by using the value at address operator, saying ***(s[2] + 1)**. But, we have already studied while learning one-dimensional arrays that **num[i]** is same as ***(num + i)**. Similarly, ***(s[2] + 1)** is same as, ***(*(s + 2) + 1)**. Thus, all the following expressions refer to the same element:

```
s[ 2 ][ 1 ]
*( s[ 2 ] + 1 )
*( *( s + 2 ) + 1 )
```

Using these concepts, the following program prints out each element of a two-dimensional array using pointer notation:

```
/* Pointer notation to access 2-D array elements */
#include <stdio.h>
int main( )
{
```

```
int s[ 4 ][ 2 ] = {
    { 1234, 56 },
    { 1212, 33 },
    { 1434, 80 },
    { 1312, 78 }
};

int i, j;

for ( i = 0 ; i <= 3 ; i++ )
{
    for ( j = 0 ; j <= 1 ; j++ )
        printf ( "%d ", *( s + i ) + j ) ;
    printf ( "\n" ) ;
}
return 0 ;
}
```

And here is the output...

```
1234 56
1212 33
1434 80
1312 78
```

Pointer to an Array

If we can have a pointer to an integer, a pointer to a float, a pointer to a char, then can we not have a pointer to an array? We certainly can. The following program shows how to build and use it:

```
/* Usage of pointer to an array */
# include <stdio.h>
int main( )
{
    int s[ 4 ][ 2 ] = {
        { 1234, 56 },
        { 1212, 33 },
        { 1434, 80 },
        { 1312, 78 }
    };

    int ( *p )[ 2 ] ;
```

```

int i, j, *pint ;
for ( i = 0 ; i <= 3 ; i++ )
{
    p = &s[ i ] ;
    pint = ( int * ) p ;
    printf ( "\n" ) ;
    for ( j = 0 ; j <= 1 ; j++ )
        printf ( "%d ", *( pint + j ) ) ;
}
return 0 ;
}

```

And here is the output...

```

1234 56
1212 33
1434 80
1312 78

```

Here **p** is a pointer to an array of two integers. Note that the parentheses in the declaration of **p** are necessary. Absence of them would make **p** an array of 2 integer pointers. Array of pointers is covered in a later section in this chapter. In the outer **for** loop, each time we store the address of a new one-dimensional array. Thus first time through this loop, **p** would contain the address of the zeroth 1-D array. This address is then assigned to an integer pointer **pint**. Lastly, in the inner **for** loop using the pointer **pint**, we have printed the individual elements of the 1-D array to which **p** is pointing.

But why should we use a pointer to an array to print elements of a 2-D array. Is there any situation where we can appreciate its usage better? The entity pointer to an array is immensely useful when we need to pass a 2-D array to a function. This is discussed in the next section.

Passing 2-D Array to a Function

There are three ways in which we can pass a 2-D array to a function. These are illustrated in the following program:

```

/* Three ways of accessing a 2-D array */
# include <stdio.h>
void display ( int *q, int , int ) ;
void show ( int ( *q )[ 4 ], int, int ) ;

```

```
void print ( int q[ ][ 4 ], int , int );
int main( )
{
    int a[ 3 ][ 4 ] = {
        1, 2, 3, 4,
        5, 6, 7, 8,
        9, 0, 1, 6
    };
    display ( a, 3, 4 );
    show ( a, 3, 4 );
    print ( a, 3, 4 );
    return 0 ;
}

void display ( int *q, int row, int col )
{
    int i, j ;
    for ( i = 0 ; i < row ; i++ )
    {
        for ( j = 0 ; j < col ; j++ )
            printf ( "%d ", * ( q + i * col + j ) );
        printf ( "\n" );
    }
    printf ( "\n" );
}

void show ( int ( *q )[ 4 ], int row, int col )
{
    int i, j ;
    int *p ;

    for ( i = 0 ; i < row ; i++ )
    {
        p = q + i ;
        for ( j = 0 ; j < col ; j++ )
            printf ( "%d ", * ( p + j ) );

        printf ( "\n" );
    }
    printf ( "\n" );
}
```

```

void print ( int q[ ][ 4 ], int row, int col )
{
    int i, j;

    for ( i = 0 ; i < row ; i++ )
    {
        for ( j = 0 ; j < col ; j++ )
            printf ( "%d ", q[ i ][ j ] );
        printf ( "\n" );
    }
    printf ( "\n" );
}

```

And here is the output...

```

1 2 3 4
5 6 7 8
9 0 1 6

1 2 3 4
5 6 7 8
9 0 1 6

1 2 3 4
5 6 7 8
9 0 1 6

```

In the **display()** function, we have collected the base address of the 2-D array being passed to it in an ordinary **int** pointer. Then, through the two **for** loops using the expression *** (q + i * col + j)**, we have reached the appropriate element in the array. Suppose **i** is equal to 2 and **j** is equal to 3, then we wish to reach the element **a[2][3]**. Let us see whether the expression *** (q + i * col + j)** does give this element or not. Refer Figure 14.4 to understand this.

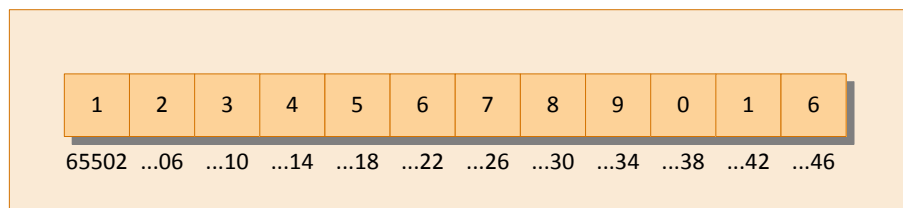


Figure 14.4

The expression $*(q + i * \text{col} + j)$ becomes $*(65502 + 2 * 4 + 3)$. This turns out to be $*(65502 + 11)$. Since 65502 is the address of an integer, $*(65502 + 11)$ turns out to be $*(65546)$. Value at this address is 6. This is indeed same as $a[2][3]$. A more general formula for accessing each array element would be:

```
*( base address + row no. * no. of columns + column no. )
```

In the **show()** function, we have defined **q** to be a pointer to an array of 4 integers through the declaration:

```
int (*q)[4];
```

To begin with, **q** holds the base address of the zeroth 1-D array, i.e. 65502 (refer Figure 14.4). This address is then assigned to **p**, an **int** pointer, and then using this pointer, all elements of the zeroth 1-D array are accessed. Next time through the loop, when **i** takes a value 1, the expression **q + i** fetches the address of the first 1-D array. This is because, **q** is a pointer to zeroth 1-D array and adding 1 to it would give us the address of the next 1-D array. This address is once again assigned to **p**, and using it all elements of the next 1-D array are accessed.

In the third function **print()**, the declaration of **q** looks like this:

```
int q[][4];
```

This is same as **int (*q)[4]**, where **q** is pointer to an array of 4 integers. The only advantage is that, we can now use the more familiar expression **q[i][j]** to access array elements. We could have used the same expression in **show()** as well.

Array of Pointers

The way there can be an array of **ints** or an array of **floats**, similarly, there can be an array of pointers. Since a pointer variable always contains an address, an array of pointers would be nothing but a collection of addresses. The addresses present in the array of pointers can be addresses of isolated variables or addresses of array elements or any other addresses. All rules that apply to an ordinary array apply to the array of pointers as well. I think a program would clarify the concept.

```
#include <stdio.h>
int main( )
{
```

```

int *arr[ 4 ]; /* array of integer pointers */
int i = 31, j = 5, k = 19, l = 71, m ;

arr[ 0 ] = &i ;
arr[ 1 ] = &j ;
arr[ 2 ] = &k ;
arr[ 3 ] = &l ;
for ( m = 0 ; m <= 3 ; m++ )
    printf ( "%d\n", * ( arr[ m ] ) ) ;
return 0 ;
}

```

Figure 14.5 shows the contents and the arrangement of the array of pointers in memory. As you can observe, **arr** contains addresses of isolated **int** variables **i**, **j**, **k** and **l**. The **for** loop in the program picks up the addresses present in **arr** and prints the values present at these addresses.

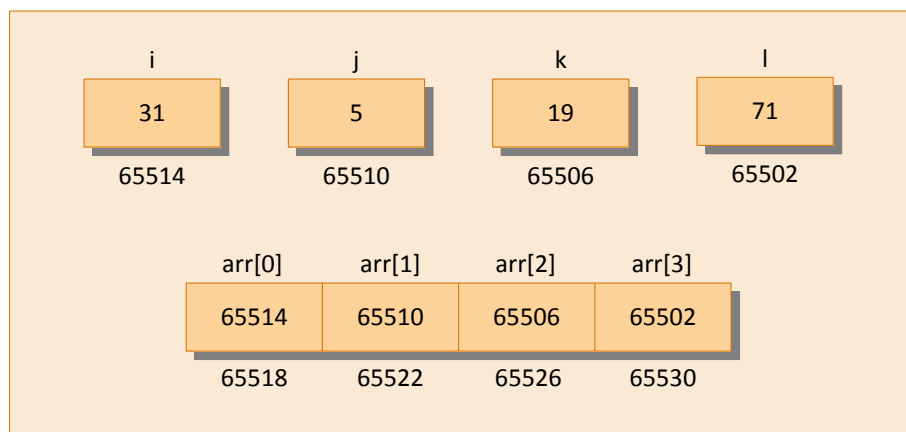


Figure 14.5

An array of pointers can even contain the addresses of other arrays. The following program would justify this:

```

# include <stdio.h>
int main( )
{
    static int a[ ] = { 0, 1, 2, 3, 4 };
    int *p[ ] = { a, a + 1, a + 2, a + 3, a + 4 };
}

```



```
printf ( "%u %u %d\n", p, *p, * ( *p ) );  
return 0 ;  
}
```

I would leave it for you to figure out the output of this program.

Three-Dimensional Array

We aren't going to show a programming example that uses a three-dimensional array. This is because, in practice, one rarely uses this array. However, an example of initializing a three-dimensional array will consolidate your understanding of subscripts.

```
int arr[ 3 ][ 4 ][ 2 ] = {  
    {  
        { 2, 4 },  
        { 7, 8 },  
        { 3, 4 },  
        { 5, 6 }  
    },  
    {  
        { 7, 6 },  
        { 3, 4 },  
        { 5, 3 },  
        { 2, 3 }  
    },  
    {  
        { 8, 9 },  
        { 7, 2 },  
        { 3, 4 },  
        { 5, 1 },  
    }  
};
```

A 3-D array can be thought of as an array of arrays of arrays. The outer array has three elements, each of which is a 2-D array of four 1-D arrays, each of which contains two integers. In other words, a 1-D array of two elements is constructed first. Then four such 1-D arrays are placed one below the other to give a 2-D array containing four rows. Then, three such 2-D arrays are placed one behind the other to yield a 3-D array containing three 2-D arrays. In the array declaration, note how the

commas have been given. Figure 14.6 would possibly help you in visualizing the situation better.

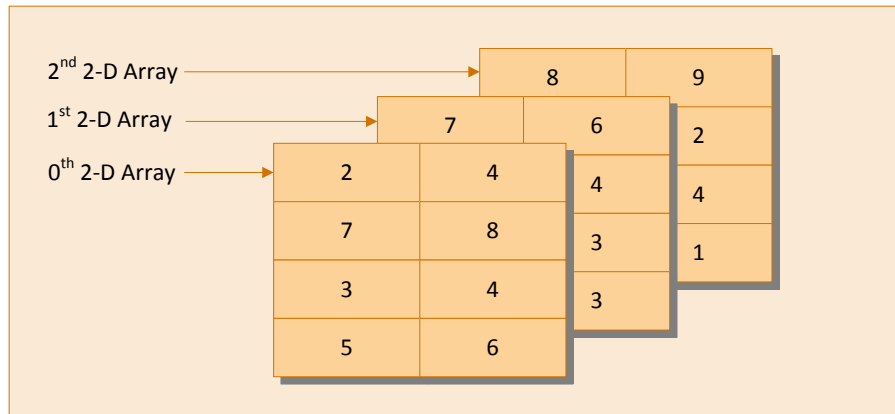


Figure 14.6

Again remember that the arrangement shown above is only conceptually true. In memory, the same array elements are stored linearly as shown in Figure 14.7.

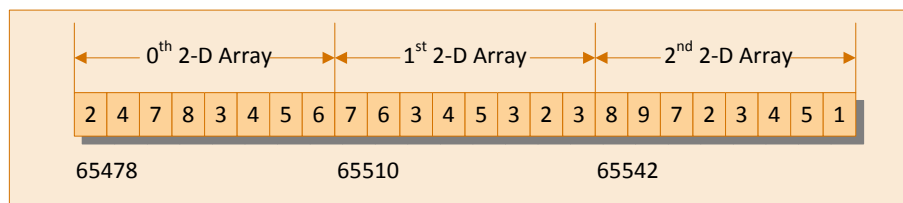


Figure 14.7

How would you refer to the array element 1 in the above array? The first subscript should be [2], since the element is in third two-dimensional array; the second subscript should be [3] since the element is in fourth row of the two-dimensional array; and the third subscript should be [1] since the element is in second position in the one-dimensional array. We can, therefore, say that the element 1 can be referred as **arr[2][3][1]**. It may be noted here that the counting of array elements even for a 3-D array begins with zero. Can we not refer to this element using pointer notation? Of course, yes. For example, the following two expressions refer to the same element in the 3-D array:

```
arr[ 2 ][ 3 ][ 1 ]
*( *( *( arr + 2 ) + 3 ) + 1 )
```

Summary

- (a) It is possible to construct multidimensional arrays.
- (b) A 2-D array is a collection of several 1-D arrays.
- (c) A 3-D array is a collection of several 2-D arrays.
- (d) All elements of a 2-D or a 3-D array are internally accessed using pointers.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    int n[ 3 ][ 3 ] = {
                                2, 4, 3,
                                6, 8, 5,
                                3, 5, 1
                            };
    printf ( "%d %d %d\n", *n, n[ 3 ][ 3 ], n[ 2 ][ 2 ] );
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
{
    int n[ 3 ][ 3 ] = {
                                2, 4, 3,
                                6, 8, 5,
                                3, 5, 1
                            };

    int i, *ptr ;
    ptr = n ;
    for ( i = 0 ; i <= 8 ; i++ )
        printf ( "%d\n", *( ptr + i ) );
    return 0 ;
}
```
- (c)

```
#include <stdio.h>
int main( )
```

```

{
    int n[ 3 ][ 3 ] = {
                        2, 4, 3,
                        6, 8, 5,
                        3, 5, 1
                    };

    int i, j;
    for ( i = 0 ; i <= 2 ; i++ )
        for ( j = 0 ; j <= 2 ; j++ )
            printf ( "%d %d\n", n[ i ][ j ], *( *( n + i ) + j ) );
    return 0 ;
}

```

[B] Point out the errors, if any, in the following programs:

(a) # include <stdio.h>

```

int main( )
{
    int twod[ ][ ] = {
                        2, 4,
                        6, 8
                    };

    printf ( "%d\n", twod );
    return 0 ;
}

```

(b) # include <stdio.h>

```

int main( )
{
    int three[ 3 ][ ] = {
                        2, 4, 3,
                        6, 8, 2,
                        2, 3, 1
                    };

    printf ( "%d\n", three[ 1 ][ 1 ] );
    return 0 ;
}

```

[C] Attempt the following:

- (a) How will you initialize a three-dimensional array **threed[3][2][3]**?
How will you refer the first and last element in this array?

- (b) Write a program to pick up the largest number from any 5 row by 5 column matrix.
- (c) Write a program to obtain transpose of a 4 x 4 matrix. The transpose of a matrix is obtained by exchanging the elements of each row with the elements of the corresponding column.
- (d) Very often in fairs we come across a puzzle that contains 15 numbered square pieces mounted on a frame. These pieces can be moved horizontally or vertically. A possible arrangement of these pieces is shown in Figure 14.8:

1	4	15	7
8	10	2	11
14	3	6	13
12	9	5	

Figure 14.8

As you can see there is a blank at bottom right corner. Implement the following procedure through a program:

Draw the boxes as shown above. Display the numbers in the above order. Allow the user to hit any of the arrow keys (up, down, left, or right). If you are using Turbo C/C++, use the library function **gotoxy()** to position the cursor on the screen while drawing the boxes. If you are using Visual Studio then use the following function to position the cursor:

```
#include <windows.h>
void gotoxy ( short col, short row )
{
    HANDLE h = GetStdHandle ( STD_OUTPUT_HANDLE );
    COORD position = { col, row };
    SetConsoleCursorPosition ( h, position );
}
```

If user hits say, right arrow key then the piece with a number 5 should move to the right and blank should replace the original

position of 5. Similarly, if down arrow key is hit, then 13 should move down and blank should replace the original position of 13. If left arrow key or up arrow key is hit then no action should be taken.

The user would continue hitting the arrow keys till the numbers aren't arranged in ascending order.

Keep track of the number of moves in which the user manages to arrange the numbers in ascending order. The user who manages it in minimum number of moves is the one who wins.

How do we tackle the arrow keys? We cannot receive them using **scanf()** function. Arrow keys are special keys which are identified by their 'scan codes'. Use the following function in your program. It would return the scan code of the arrow key being hit. The scan codes for the arrow keys are:

up arrow key – 72 down arrow key – 80
left arrow key – 75 right arrow key – 77

```
#include <conio.h>
int getkey( )
{
    int ch ;
    ch = getch( ) ;
    if ( ch == 0 )
    {
        ch = getch( ) ;
        return ch ;
    }
    return ch ;
}
```

- (e) Match the following with reference to the program segment given below:

```
int i, j, = 25;
int *pi, *pj = & j;
.....
..... /* more lines of program */
.....
*pj = j + 5;
j = *pj + 5 ;
pj = pj ;
```

*pi = i + j

Each integer quantity occupies 2 bytes of memory. The value assigned to *i* begins at (hexadecimal) address F9C and the value assigned to *j* begins at address F9E. Match the value represented by left hand side quantities with the right.

- | | |
|------------------|----------------|
| 1. &i | a. 30 |
| 2. &j | b. F9E |
| 3. pj | c. 35 |
| 4. *pj | d. FA2 |
| 5. i | e. F9C |
| 6. pi | f. 67 |
| 7. *pi | g. unspecified |
| 8. (pi + 2) | h. 65 |
| 9. (*pi + 2) | i. F9E |
| 10. * (pi + 2) | j. F9E |
| | k. FA0 |
| | l. F9D |

- (f) Match the following with reference to the following program segment:

```
int x[ 3 ][ 5 ] = {
    { 1, 2, 3, 4, 5 },
    { 6, 7, 8, 9, 10 },
    { 11, 12, 13, 14, 15 }
}, *n = &x;
```

- | | |
|-------------------------|-------|
| 1. *(*(x + 2) + 1) | a. 9 |
| 2. *(*x + 2) + 5 | b. 13 |
| 3. *(*(x + 1)) | c. 4 |
| 4. *(*(x) + 2) + 1 | d. 3 |
| 5. * (*(x + 1) + 3) | e. 2 |
| 6. *n | f. 12 |
| 7. *(n + 2) | g. 14 |
| 8. *(n + 3) + 1 | h. 7 |
| 9. *(n + 5)+1 | i. 1 |
| 10. ++*n | j. 8 |
| | k. 5 |
| | l. 10 |
| | m. 6 |

- (g) Match the following with reference to the following program segment:

```
unsigned int arr[ 3 ][ 3 ] = {
                                2, 4, 6,
                                9, 1, 10,
                                16, 64, 5
                                };
```

1. **arr	a. 64
2. **arr < *(*arr + 2)	b. 18
3. *(arr + 2) / (*(*arr + 1) > **arr)	c. 6
4. *(arr[1] + 1) arr[1][2]	d. 3
5. *(arr[0]) *(arr[2])	e. 0
6. arr[1][1] < arr[0][1]	f. 16
7. arr[2][1] & arr[2][0]	g. 1
8. arr[2][2] arr[0][1]	h. 11
9. arr[0][1] ^ arr[0][2]	i. 20
10. ++**arr + --arr[1][1]	j. 2
	k. 5
	l. 4

- (h) Write a program to find if a square matrix is symmetric.
- (i) Write a program to add two 6 x 6 matrices.
- (j) Write a program to multiply any two 3 x 3 matrices.
- (k) Given an array **p[5]**, write a function to shift it circularly left by two positions. Thus, if p[0] = 15, p[1] = 30, p[2] = 28, p[3] = 19 and p[4] = 61 then after the shift p[0] = 28, p[1] = 19, p[2] = 61, p[3] = 15 and p[4] = 30. Call this function for a (4 x 5) matrix and get its rows left shifted.
- (l) A 6 x 6 matrix is entered through the keyboard. Write a program to obtain the Determinant value of this matrix.
- (m) For the following set of sample data, compute the standard deviation and the mean.

-6, -12, 8, 13, 11, 6, 7, 2, -6, -9, -10, 11, 10, 9, 2

The formula for standard deviation is

$$\frac{\sqrt{(x_i - \bar{x})^2}}{n}$$

where x_i is the data item and $\bar{x}$ is the mean.

- (n) The area of a triangle can be computed by the sine law when 2 sides of the triangle and the angle between them are known.

$$\text{Area} = (1 / 2) ab \sin (\text{angle})$$

Given the following 6 triangular pieces of land, write a program to find their area and determine which is largest.

Plot No.	a	b	angle
1	137.4	80.9	0.78
2	155.2	92.62	0.89
3	149.3	97.93	1.35
4	160.0	100.25	9.00
5	155.6	68.95	1.25
6	149.7	120.0	1.75

- (o) For the following set of n data points (x, y) , compute the correlation coefficient r , given by

$$r = \frac{\sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

x	y
34.22	102.43
39.87	100.93
41.85	97.43
43.23	97.81
40.06	98.32
53.29	98.32
53.29	100.07
54.14	97.08
49.12	91.59
40.71	94.85
55.15	94.65

- (p) For the following set of point given by (x, y) fit a straight line given by $y = a + bx$

where,

$$a = \bar{y} - b\bar{x} \text{ and}$$

$$b = \frac{n \sum yx - \sum x \sum y}{[n \sum x^2 - (\sum x)^2]}$$

x	Y
3.0	1.5
4.5	2.0
5.5	3.5
6.5	5.0
7.5	6.0
8.5	7.5
8.0	9.0
9.0	10.5
9.5	12.0
10.0	14.0

- (q) The **X** and **Y** coordinates of 10 different points are entered through the keyboard. Write a program to find the distance of last point from the first point (sum of distances between consecutive points).
- (r) A dequeue is an ordered set of elements in which elements may be inserted or retrieved from either end. Using an array simulate a dequeue of characters and the operations retrieve left, retrieve right, insert left, insert right. Exceptional conditions such as dequeue full or empty should be indicated. Two pointers (namely, left and right) are needed in this simulation.
- (s) Sudoku is a popular number-placement puzzle (refer Figure 14.9). The objective is to fill a 9×9 grid with digits so that each column, each row, and each of the nine 3×3 sub-grids that compose the grid contains all of the digits from 1 to 9. The puzzle setter provides a partially completed grid, which typically has a unique solution. One such solution is given below. Write a program to check whether the solution is correct or not.

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Figure 14.9

- What are Strings?
- More about Strings
- Pointers and Strings
- Standard Library String Functions
 - strlen()*
 - strcpy()*
 - strcat()*
 - strcmp()*
- Summary
- Exercise



15 ***Strings***

- What are Strings
- More about Strings
- Pointers and Strings
- Standard Library String Functions
 - strlen()*
 - strcpy()*
 - strcat()*
 - strcmp()*
- Summary
- Exercise

In the last chapter, you learnt how to define arrays of various sizes and dimensions, how to initialize arrays, how to pass arrays to a function, etc. With this knowledge under your belt, you should be ready to handle strings, which are, simply put, a special kind of array. And strings, the ways to manipulate them, and how pointers are related to strings are going to be the topics of discussion in this chapter.

What are Strings?

The way a group of integers can be stored in an integer array, similarly a group of characters can be stored in a character array. Character arrays are many a time also called strings. Many languages internally treat strings as character arrays, but somehow conceal this fact from the programmer. Character arrays or strings are used by programming languages to manipulate text, such as words and sentences.

A string constant is a one-dimensional array of characters terminated by a null (`'\0'`). For example,

```
char name[] = { 'H', 'A', 'E', 'S', 'L', 'E', 'R', '\0' };
```

Each character in the array occupies 1 byte of memory and the last character is always `'\0'`. What character is this? It looks like two characters, but it is actually only one character, with the `\` indicating that what follows it is something special. `'\0'` is called null character. Note that `'\0'` and `'0'` are not same. ASCII value of `'\0'` is 0, whereas ASCII value of `'0'` is 48. Figure 15.1 shows the way a character array is stored in memory. Note that the elements of the character array are stored in contiguous memory locations.

The terminating null (`'\0'`) is important, because it is the only way the functions that work with a string can know where the string ends. In fact, a string not terminated by a `'\0'` is not really a string, but merely a collection of characters.

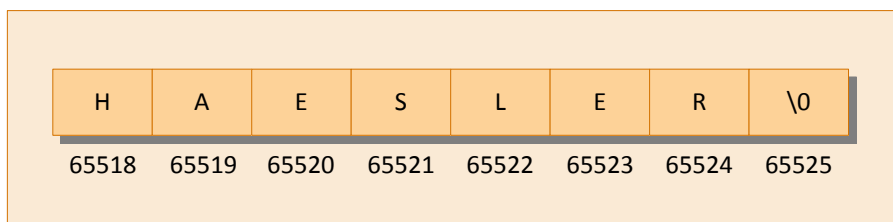


Figure 15.1

C concedes the fact that you would use strings very often and hence provides a shortcut for initializing strings. For example, the string used above can also be initialized as,

```
char name[ ] = "HAESLER" ;
```

Note that, in this declaration `'\0'` is not necessary. C inserts the null character automatically.

More about Strings

In what way are character arrays different from numeric arrays? Can elements in a character array be accessed in the same way as the elements of a numeric array? Do I need to take any special care of `'\0'`? Why numeric arrays don't end with a `'\0'`? Declaring strings is okay, but how do I manipulate them? Questions galore!! Well, let us settle some of these issues right away with the help of sample programs.

```
/* Program to demonstrate printing of a string */
#include <stdio.h>
int main( )
{
    char name[ ] = "Klinsman" ;
    int i = 0 ;

    while ( i <= 7 )
    {
        printf ( "%c", name[ i ] ) ;
        i++ ;
    }
    printf ( "\n" ) ;
    return 0 ;
}
```

And here is the output...

```
Klinsman
```

No big deal. We have initialized a character array, and then printed out the elements of this array within a **while** loop. Can we write the **while** loop without using the final value 7? We can; because we know that each character array always ends with a `'\0'`. Following program illustrates this:


```
# include <stdio.h>
int main( )
{
    char name[ ] = "Klinsman" ;
    int i = 0 ;
    while ( name[ i ] != '\0' )
    {
        printf ( "%c", name[ i ] ) ;
        i++ ;
    }
    printf ( "\n" ) ;
    return 0 ;
}
```

And here is the output...

Klinsman

This program doesn't rely on the length of the string (number of characters in it) to print out its contents and hence is definitely more general than the earlier one. Here is another version of the same program; this one uses a pointer to access the array elements.

```
# include <stdio.h>
int main( )
{
    char name[ ] = "Klinsman" ;
    char *ptr ;
    ptr = name ; /* store base address of string */
    while ( *ptr != '\0' )
    {
        printf ( "%c", *ptr ) ;
        ptr++ ;
    }
    printf ( "\n" ) ;
    return 0 ;
}
```

As with the integer array, by mentioning the name of the array, we get the base address (address of the zeroth element) of the array. This base address is stored in the variable **ptr** using,

```
ptr = name ;
```

Once the base address is obtained in **ptr**, ***ptr** would yield the value at this address, which gets printed promptly through,

```
printf ( "%c", *ptr ) ;
```

Then, **ptr** is incremented to point to the next character in the string. This derives from two facts: array elements are stored in contiguous memory locations and on incrementing a pointer, it points to the immediately next location of its type. This process is carried out until **ptr** points to the last character in the string, that is, `'\0'`.

In fact, the character array elements can be accessed exactly in the same way as the elements of an integer array. Thus, all the following notations refer to the same element:

```
name[ i ]  
*( name + i )  
*( i + name )  
i[ name ]
```

Even though there are so many ways (as shown above) to refer to the elements of a character array, rarely is any one of them used. This is because **printf()** function has got a sweet and simple way of doing it, as shown below. Note that **printf()** doesn't print the `'\0'`.

```
# include <stdio.h>  
int main( )  
{  
    char name[ ] = "Klinsman" ;  
    printf ( "%s", name ) ;  
}
```

The **%s** used in **printf()** is a format specification for printing out a string. The same specification can be used to receive a string from the keyboard, as shown below.

```
# include <stdio.h>  
int main( )  
{  
    char name[ 25 ] ;
```

```
printf ( "Enter your name " ) ;  
scanf ( "%s", name ) ;  
printf ( "Hello %s!\n", name ) ;  
return 0 ;  
}
```

And here is a sample run of the program...

```
Enter your name Debashish  
Hello Debashish!
```

Note that the declaration **char name[25]** sets aside 25 bytes under the array **name[]**, whereas the **scanf()** function fills in the characters typed at keyboard into this array until the Enter key is hit. Once enter is hit, **scanf()** places a '\0' in the array. Naturally, we should pass the base address of the array to the **scanf()** function.

While entering the string using **scanf()**, we must be cautious about two things:

- (a) The length of the string should not exceed the dimension of the character array. This is because the C compiler doesn't perform bounds checking on character arrays. Hence, if you carelessly exceed the bounds, there is always a danger of overwriting something important, and in that event, you would have nobody to blame but yourselves.
- (b) **scanf()** is not capable of receiving multi-word strings. Therefore, names such as 'Debashish Roy' would be unacceptable. The way to get around this limitation is by using the function **gets()**. The usage of functions **gets()** and its counterpart **puts()** is shown below.

```
# include <stdio.h>  
int main( )  
{  
    char name[ 25 ] ;  
    printf ( "Enter your full name: " ) ;  
    gets ( name ) ;  
    puts ( "Hello!" ) ;  
    puts ( name ) ;  
    return 0 ;  
}
```

And here is the output...

```
Enter your full name: Debashish Roy
Hello!
Debashish Roy
```

The program and the output are self-explanatory except for the fact that, **puts()** can display only one string at a time (hence the use of two **puts()** in the program above). Also, on displaying a string, unlike **printf()**, **puts()** places the cursor on the next line. Though **gets()** is capable of receiving only one string at a time, the plus point with **gets()** is that it can receive a multi-word string.

If we are prepared to take the trouble, we can make **scanf()** accept multi-word strings by writing it in this manner:

```
char name[ 25 ] ;
printf ( "Enter your full name " ) ;
scanf ( "%[ ^\n ]s", name ) ;
```

Here, **[^\n]** indicates that **scanf()** will keep receiving characters into **name[]** until **\n** is encountered. Though workable, this is not the best of the ways to call a function, you would agree.

Pointers and Strings

Suppose we wish to store “Hello”. We may either store it in a string or we may ask the C compiler to store it at some location in memory and assign the address of the string in a **char** pointer. This is shown below.

```
char str[ ] = "Hello" ;
char *p = "Hello" ;
```

There is a subtle difference in usage of these two forms. For example, we cannot assign a string to another, whereas, we can assign a **char** pointer to another **char** pointer. This is shown in the following program:

```
int main( )
{
    char str1[ ] = "Hello" ;
    char str2[ 10 ] ;
    char *s = "Good Morning" ;
    char *q ;
    str2 = str1 ; /* error */
}
```

```
    q = s ; /* works */  
    return 0 ;  
}
```

Also, once a string has been defined, it cannot be initialized to another set of characters. Unlike strings, such an operation is perfectly valid with **char** pointers.

```
int main( )  
{  
    char str1[ ] = "Hello" ;  
    char *p = "Hello" ;  
    str1 = "Bye" ; /* error */  
    p = "Bye" ; /* works */  
}
```

Standard Library String Functions

With every C compiler, a large set of useful string handling library functions are provided. Figure 15.2 lists the more commonly used functions along with their purpose.

Function	Use
strlen	Finds length of a string
strlwr	Converts a string to lowercase
strupr	Converts a string to uppercase
strcat	Appends one string at the end of another
strncat	Appends first n characters of a string at the end of another
strcpy	Copies a string into another
strncpy	Copies first n characters of one string into another
strcmp	Compares two strings
strncmp	Compares first n characters of two strings
strcmpi	Compares two strings by ignoring the case
stricmp	Compares two strings without regard to case (identical to strcmpi)
strnicmp	Compares first n characters of two strings without regard to case
strdup	Duplicates a string
strchr	Finds first occurrence of a given character in a string
strrchr	Finds last occurrence of a given character in a string
strstr	Finds first occurrence of a given string in another string
strset	Sets all characters of string to a given character
strnset	Sets first n characters of a string to a given character
strrev	Reverses string

Figure 15.2

From the list given in Figure 15.2, we shall discuss functions **strlen()**, **strcpy()**, **strcat()** and **strcmp()**, since these are the most commonly used. This will also illustrate how the library functions in general handle strings. Let us study these functions one-by-one.

strlen()

This function counts the number of characters present in a string. Its usage is illustrated in the following program:

```
#include <stdio.h>
#include <string.h>
int main( )
{
```

```
char arr[ ] = "Bamboozled" ;
int len1, len2 ;

len1 = strlen ( arr ) ;
len2 = strlen ( "Humpty Dumpty" ) ;
printf ( "string = %s length = %d\n", arr, len1 ) ;
printf ( "string = %s length = %d\n", "Humpty Dumpty", len2 ) ;
return 0 ;
}
```

The output would be...

```
string = Bamboozled length = 10
string = Humpty Dumpty length = 13
```

Note that, in the first call to the function **strlen()**, we are passing the base address of the string, and the function, in turn, returns the length of the string. While calculating the length, it doesn't count '\0'. Even in the second call,

```
len2 = strlen ( "Humpty Dumpty" ) ;
```

what gets passed to **strlen()** is the address of the string and not the string itself. Can we not write a function **xstrlen()**, which imitates the standard library function **strlen()**? Let us give it a try...

```
/* A look-alike of the function strlen( ) */
# include <stdio.h>
int xstrlen ( char * ) ;
int main( )
{
    char arr[ ] = "Bamboozled" ;
    int len1, len2 ;

    len1 = xstrlen ( arr ) ;
    len2 = xstrlen ( "Humpty Dumpty" ) ;

    printf ( "string = %s length = %d\n", arr, len1 ) ;
    printf ( "string = %s length = %d\n", "Humpty Dumpty", len2 ) ;
    return 0 ;
}

int xstrlen ( char *s )
```

```
{
    int length = 0 ;

    while ( *s != '\0' )
    {
        length++ ;
        s++ ;
    }
    return ( length ) ;
}
```

The output would be...

```
string = Bamboozled length = 10
string = Humpty Dumpty length = 13
```

The function **xstrlen()** is fairly simple. All that it does is to keep counting the characters till the end of string is met. Or in other words, keep counting characters till the pointer **s** points to **'\0'**.

strcpy()

This function copies the contents of one string into another. The base addresses of the source and target strings should be supplied to this function. Here is an example of **strcpy()** in action...

```
# include <stdio.h>
# include <string.h>
int main( )
{
    char source[ ] = "Sayonara" ;
    char target[ 20 ] ;
    strcpy ( target, source ) ;
    printf ( "source string = %s\n", source ) ;
    printf ( "target string = %s\n", target ) ;
    return 0 ;
}
```

And here is the output...

```
source string = Sayonara
target string = Sayonara
```


On supplying the base addresses, **strcpy()** goes on copying the characters in source string into the target string till it encounters the end of source string ('\\0'). It is our responsibility to see to it that the target string's dimension is big enough to hold the string being copied into it. Thus, a string gets copied into another, piece-meal, character-by-character. There is no short-cut for this. Let us now attempt to mimic **strcpy()**, via our own string copy function, which we will call **xstrcpy()**.

```
#include <stdio.h>
void xstrcpy ( char *, char * );
int main( )
{
    char source[ ] = "Sayonara";
    char target[ 20 ];
    xstrcpy ( target, source );
    printf ( "source string = %s\\n", source );
    printf ( "target string = %s\\n", target );
    return 0 ;
}
void xstrcpy ( char *t, char *s )
{
    while ( *s != '\\0' )
    {
        *t = *s ;
        s++ ;
        t++ ;
    }
    *t = '\\0' ;
}
```

The output of the program would be...

```
source string = Sayonara
target string = Sayonara
```

Note that having copied the entire source string into the target string, it is necessary to place a '\\0' into the target string, to mark its end.

If you look at the prototype of **strcpy()** standard library function, it looks like this...

```
strcpy ( char *t, const char *s ) ;
```

We didn't use the keyword `const` in our version of `xstrcpy( )` and still our function worked correctly. So what is the need of the `const` qualifier?

What would happen if we add the following lines beyond the last statement of `xstrcpy( )`?

```
s = s - 8 ;  
*s = 'K' ;
```

This would change the source string to "Kayonara". Can we not ensure that the source string doesn't change even accidentally in `xstrcpy( )`? We can, by changing the definition as follows:

```
void xstrcpy ( char *t, const char *s )  
{  
    while ( *s != '\0' )  
    {  
        *t = *s ;  
        s++ ;  
        t++ ;  
    }  
    *t = '\0' ;  
}
```

By declaring `char *s` as `const`, we are declaring that the source string should remain constant (should not change). Thus the `const` qualifier ensures that your program does not inadvertently alter a variable that you intended to be a constant. It also reminds anybody reading the program listing that the variable is not intended to change.

Let us understand the difference between the following two statements:

```
char str[ ] = "Quest" ;  
char *p = "Quest" ;
```

Here `str` acts as a constant pointer to a string, whereas, `p` acts as a pointer to a constant string. As a result, observe which operations are permitted, and which are not:

```
str++ ; /* error, constant pointer cannot change */  
*str = 'Z' ; /* works, because string is not constant */  
p++ ; /* works, because pointer is not constant */  
*p = 'M' ; /* error, because string is constant */
```

The keyword **const** can also be used in context of ordinary variables like **int**, **float**, etc. The following program shows how this can be done:

```
#include <stdio.h>
int main( )
{
    float r, a ;
    const float pi = 3.14 ;
    printf ( "Enter radius of circle " ) ;
    scanf ( "%f", &r ) ;
    a = pi * r * r ;
    printf ( "Area of circle = %f\n", a ) ;
    return 0 ;
}
```

strcat()

This function concatenates the source string at the end of the target string. For example, “Bombay” and “Nagpur” on concatenation would result into a string “BombayNagpur”. Here is an example of **strcat()** at work.

```
#include <stdio.h>
#include <string.h>
int main( )
{
    char source[ ] = "Folks!" ;
    char target[ 30 ] = "Hello" ;
    strcat ( target, source ) ;
    printf ( "source string = %s\n", source ) ;
    printf ( "target string = %s\n", target ) ;
    return 0 ;
}
```

And here is the output...

```
source string = Folks!
target string = HelloFolks!
```

Note that the target string has been made big enough to hold the final string. I leave it to you to develop your own **xstrcat()** on lines of **xstrlen()** and **xstrcpy()**.

strcmp()

This is a function which compares two strings to find out whether they are same or different. The two strings are compared character-by-character until there is a mismatch or end of one of the strings is reached, whichever occurs first. If the two strings are identical, **strcmp()** returns a value zero. If they're not, it returns the numeric difference between the ASCII values of the first non-matching pair of characters. Here is a program which puts **strcmp()** in action.

```
#include <stdio.h>
#include <string.h>
int main( )
{
    char string1[ ] = "Jerry";
    char string2[ ] = "Ferry";
    int i, j, k;
    i = strcmp ( string1, "Jerry" );
    j = strcmp ( string1, string2 );
    k = strcmp ( string1, "Jerry boy" );
    printf ( "%d %d %d\n", i, j, k );
    return 0;
}
```

And here is the output...

```
0 4 -32
```

In the first call to **strcmp()**, the two strings are identical—"Jerry" and "Jerry"—and the value returned by **strcmp()** is zero. In the second call, the first character of "Jerry" doesn't match with the first character of "Ferry" and the result is 4, which is the numeric difference between ASCII value of 'J' and ASCII value of 'F'. In the third call to **strcmp()**, "Jerry" doesn't match with "Jerry boy", because the null character at the end of "Jerry" doesn't match the blank in "Jerry boy". The value returned is -32, which is the value of null character minus the ASCII value of space, i.e., '\0' minus ' ', which is equal to -32.

The exact value of mismatch rarely concerns us. All that we usually want to know is whether or not the first string is alphabetically before the second string. If it is, a negative value is returned; if it isn't, a positive value is returned. Try to implement this logic in a user-defined function **xstrcmp()**.

Summary

- (a) A string is nothing but an array of characters terminated by '\0'.
- (b) Being an array, all the characters of a string are stored in contiguous memory locations.
- (c) Though **scanf()** can be used to receive multi-word strings, **gets()** can do the same job in a cleaner way.
- (d) Both **printf()** and **puts()** can handle multi-word strings.
- (e) Strings can be operated upon using several standard library functions like **strlen()**, **strcpy()**, **strcat()** and **strcmp()** which can manipulate strings.

Exercise

[A] What will be the output of the following programs:

- (a)

```
#include <stdio.h>
int main( )
{
    char c[ 2 ] = "A" ;
    printf ( "%c\n", c[ 0 ] ) ;
    printf ( "%s\n", c ) ;
    return 0 ;
}
```
- (b)

```
#include <stdio.h>
int main( )
{
    char s[ ] = "Get organised! learn C!!" ;
    printf ( "%s\n", &s[ 2 ] ) ;
    printf ( "%s\n", s ) ;
    printf ( "%s\n", &s ) ;
    printf ( "%c\n", s[ 2 ] ) ;
    return 0 ;
}
```
- (c)

```
#include <stdio.h>
int main( )
{
    char s[ ] = "No two viruses work similarly" ;
```

```
int i = 0 ;
while ( s[ i ] != 0 )
{
    printf ( "%c %c\n", s[ i ], *( s + i ) );
    printf ( "%c %c\n", i[ s ], *( i + s ) );
    i++ ;
}
return 0 ;
}
```

```
(d) # include <stdio.h>
int main( )
{
    char s[ ] = "Churchgate: no church no gate" ;
    char t[ 25 ] ;
    char *ss, *tt ;
    ss = s ;
    while ( *ss != '\0' )
        *tt++ = *ss++ ;
    printf ( "%s\n", t ) ;
    return 0 ;
}
```

```
(e) # include <stdio.h>
int main( )
{
    char str1[ ] = { 'H', 'e', 'l', 'l', 'o', 0 } ;
    char str2[ ] = "Hello" ;
    printf ( "%s\n", str1 ) ;
    printf ( "%s\n", str2 ) ;
    return 0 ;
}
```

```
(f) # include <stdio.h>
void main( )
{
    printf ( 5 + "Good Morning " ) ;
    return 0 ;
}
```

```
(g) # include <stdio.h>
int main( )
{
```

```
    printf ( "%c\n", "abcdefgh"[ 4 ] );  
    return 0 ;  
}
```

```
(h) # include <stdio.h>  
int main( )  
{  
    printf ( "%d %d %d\n", sizeof ( '3' ), sizeof ( "3" ), sizeof ( 3 ) );  
    return 0 ;  
}
```

[B] Point out the errors, if any, in the following programs:

- ```
(a) # include <stdio.h>
 # include <string.h>
 int main()
 {
 char *str1 = "United" ;
 char *str2 = "Front" ;
 char *str3 ;
 str3 = strcat (str1, str2) ;
 printf ("%s\n", str3) ;
 return 0 ;
 }

(b) # include <stdio.h>
 int main()
 {
 int arr[] = { 'A', 'B', 'C', 'D' } ;
 int i ;
 for (i = 0 ; i <= 3 ; i++)
 printf ("%d", arr[i]) ;
 printf ("\n") ;
 return 0 ;
 }

(c) # include <stdio.h>
 int main()
 {
 char arr[8] = "Rhombus" ;
 int i ;
 for (i = 0 ; i <= 7 ; i++)
 printf ("%d", *arr) ;
 arr++ ;
```

```
 return 0 ;
}
```

**[C] Fill in the blanks:**

- (a) "A" is a \_\_\_\_\_ whereas 'A' is a \_\_\_\_\_.
- (b) A string is terminated by a \_\_\_\_\_ character, which is written as \_\_\_\_\_.
- (c) The array **char name[ 10 ]** can consist of a maximum of \_\_\_\_\_ characters.
- (d) The array elements are always stored in \_\_\_\_\_ memory locations.

**[D] Attempt the following:**

- (a) Which is more appropriate for reading in a multi-word string?  
`gets( )`    `printf( )`    `scanf( )`    `puts( )`
- (b) If the string "Alice in wonder land" is fed to the following **scanf( )** statement, what will be the contents of the arrays **str1**, **str2**, **str3** and **str4**?  
`scanf ( "%s%s%s%s", str1, str2, str3, str4 ) ;`
- (c) Write a program that extracts part of the given string from the specified position. For example, if the sting is "Working with strings is fun", then if from position 4, 4 characters are to be extracted then the program should return string as "king". If the number of characters to be extracted is 0 then the program should extract entire string from the specified position.
- (d) Write a program that converts a string like "124" to an integer 124.
- (e) Write a program that generates and prints the Fibonacci words of order 0 through 5. If  $f(0) = "a"$ ,  $f(1) = "b"$ ,  $f(2) = "ba"$ ,  $f(3) = "bab"$ ,  $f(4) = "babba"$ , etc.
- (f) To uniquely identify a book a 10-digit ISBN (International Standard Book Number) is used. The rightmost digit is a checksum digit. This digit is determined from the other 9 digits using the condition that  $d_1 + 2d_2 + 3d_3 + \dots + 10d_{10}$  must be a multiple of 11 (where  $d_i$  denotes the  $i^{\text{th}}$  digit from the right). The checksum digit  $d_1$  can be any value from 0 to 10: the ISBN convention is to use the value X to denote 10. Write a program that receives a 10-digit integer,



computes the checksum, and reports whether the ISBN number is correct or not.

- (g) A Credit Card number is usually a 16-digit number. A valid Credit Card number would satisfy a rule explained below with the help of a dummy Credit Card number—4567 1234 5678 9129. Start with the rightmost - 1 digit and multiply every other digit by 2.

4 5 6 7   1 2 3 4   5 6 7 8   9 1 2 9

8 12   2   6   10   14   18   4

Then subtract 9 from any number larger than 10. Thus we get:

8 3 2 6 1 5 9 4

Add them all up to get 38.

Add all the other digits to get 42.

Sum of 38 and 42 is 80. Since 80 is divisible by 10, the Credit Card number is valid.

Write a program that receives a Credit Card number and checks using the above rule whether the Credit Card number is valid.

# 16

## Handling Multiple Strings

- **Two-Dimensional Array of Characters**
- **Array of Pointers to Strings**
- **Limitation of Array of Pointers to Strings**
- **Solution**
- **Summary**
- **Exercise**



---

# 16

## *Handling Multiple Strings*

---

- Two-Dimensional Array of Characters
- Array of Pointers to Strings
- Limitation of Array of Pointers to String  
Solution
- Summary
- Exercise

In the last chapter, you learnt how to deal with individual strings. But often we are required to deal with a set of strings rather than an isolated string. This chapter discusses how such situations can be handled effectively.

## Two-Dimensional Array of Characters

In Chapter 14 we saw several examples of two-dimensional integer arrays. Let's now look at a similar entity, but one dealing with characters. Our example program asks you to type your name. When you do so, it checks your name against a master list to see if you are worthy of entry to the palace. Here's the program...

```
#include <stdio.h>
#include <string.h>
#define FOUND 1
#define NOTFOUND 0

int main()
{
 char masterlist[6][10] = {
 "akshay",
 "parag",
 "raman",
 "srinivas",
 "gopal",
 "rajesh"
 };

 int i, flag, a ;
 char yourname[10] ;

 printf ("Enter your name ") ;
 scanf ("%s", yourname) ;

 flag = NOTFOUND ;
 for (i = 0 ; i <= 5 ; i++)
 {
 a = strcmp (&masterlist[i][0], yourname) ;
 if (a == 0)
 {
 printf ("Welcome, you can enter the palace\n") ;
 flag = FOUND ;
 }
 }
}
```

```
 break ;
 }
}
if (flag == NOTFOUND)
 printf ("Sorry, you are a trespasser\n") ;
return 0 ;
}
```

And here is the output for two sample runs of this program...

```
Enter your name dinesh
Sorry, you are a trespasser
Enter your name raman
Welcome, you can enter the palace
```

Notice how the two-dimensional character array has been initialized. The order of the subscripts in the array declaration is important. The first subscript gives the number of names in the array, while the second subscript gives the length of each item in the array.

Instead of initializing names, had these names been supplied from the keyboard, the program segment would have looked like this...

```
for (i = 0 ; i <= 5 ; i++)
 scanf ("%s", &masterlist[i][0]) ;
```

While comparing the strings through **strcmp( )**, note that the addresses of the strings are being passed to **strcmp( )**. As seen in the last section, if the two strings match, **strcmp( )** would return a value 0, otherwise it would return a non-zero value.

The variable **flag** is used to keep a record of whether the control did reach inside the if or not. To begin with, we set **flag** to NOTFOUND. Later through the loop, if the names match, **flag** is set to FOUND. When the control reaches beyond the **for** loop, if **flag** is still set to NOTFOUND, it means none of the names in the **masterlist[ ][ ]** matched with the one supplied from the keyboard.

The names would be stored in the memory as shown in Figure 16.1. Note that each string ends with a '\0'. The arrangement, as you can appreciate, is similar to that of a two-dimensional numeric array.

|       |   |   |   |   |   |    |    |   |    |                          |
|-------|---|---|---|---|---|----|----|---|----|--------------------------|
| 65454 | a | k | s | h | a | y  | \0 |   |    |                          |
| 65464 | p | a | r | a | g | \0 |    |   |    |                          |
| 65474 | r | a | m | a | n | \0 |    |   |    |                          |
| 65484 | s | r | i | n | i | v  | a  | s | \0 |                          |
| 65494 | g | o | p | a | l | \0 |    |   |    |                          |
| 65504 | r | a | j | e | s | h  | \0 |   |    |                          |
|       |   |   |   |   |   |    |    |   |    | 65513<br>(last location) |

Figure 16.1

Here, 65454, 65464, 65474, etc., are the base addresses of successive names. As seen from the above pattern, some of the names do not occupy all the bytes reserved for them. For example, even though 10 bytes are reserved for storing the name “akshay”, it occupies only 7 bytes. Thus, 3 bytes go waste. Similarly, for each name, there is some amount of wastage. In fact, more the number of names, more would be the wastage. Can this not be avoided? Yes, it can be... by using what is called an ‘array of pointers’, which is our next topic of discussion.

### Array of Pointers to Strings

As we know, a pointer variable always contains an address. Therefore, if we construct an array of pointers, it would contain a number of addresses. Let us see how the names in the earlier example can be stored in the array of pointers.

```
char *names[] = {
 "akshay",
 "parag",
 "raman",
 "srinivas",
 "gopal",
 "rajesh"
};
```

In this declaration, **names[ ]** is an array of pointers. It contains base addresses of respective names. That is, base address of “akshay” is stored in **names[ 0 ]**, base address of “parag” is stored in **names[ 1 ]** and so on. This is depicted in Figure 16.2.

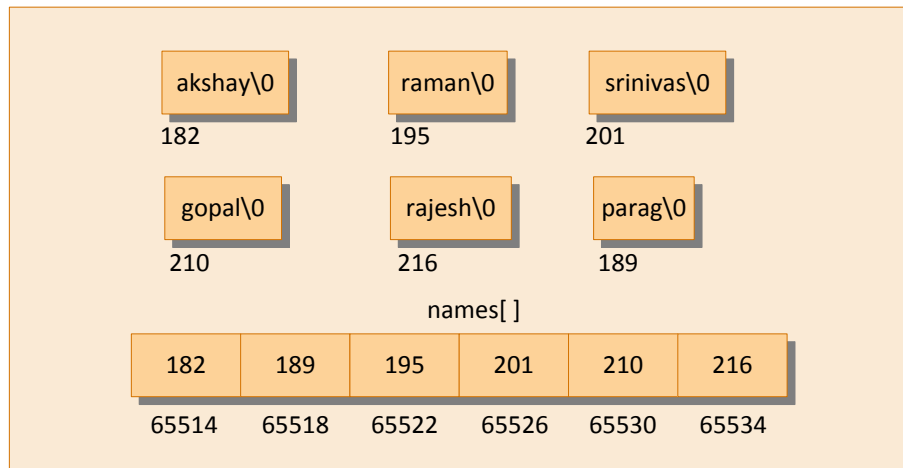


Figure 16.2

In the two-dimensional array of characters, the strings occupied 60 bytes. As against this, in array of pointers, the strings occupy only 41 bytes—a net saving of 19 bytes. A substantial saving, you would agree. Thus, one reason for storing strings in an array of pointers is to make a more efficient use of available memory.

Another reason to use an array of pointers to store strings is to obtain greater ease in manipulation of the strings. This is shown by the following programs. The first one uses a two-dimensional array of characters to store the names, whereas the second uses an array of pointers to strings. The purpose of both the programs is very simple. We want to exchange the position of the names “raman” and “srinivas”.

```
/* Exchange names using 2-D array of characters */
#include <stdio.h>
int main()
{
 char names[][10] = {
 "akshay",
 "parag",
 "raman",
 "srinivas",
 "gopal",
 "rajesh"
 };

 int i;
 char t;
```

```
printf ("Original: %s %s\n", &names[2][0], &names[3][0]);

for (i = 0 ; i <= 9 ; i++)
{
 t = names[2][i];
 names[2][i] = names[3][i];
 names[3][i] = t ;
}

printf ("New: %s %s\n", &names[2][0], &names[3][0]);
return 0 ;
}
```

And here is the output...

```
Original: raman srinivas
New: srinivas raman
```

Note that in this program to exchange the names, we are required to exchange corresponding characters of the two names. In effect, 10 exchanges are needed to interchange two names.

Let us see, if the number of exchanges can be reduced by using an array of pointers to strings. Here is the program...

```
include <stdio.h>
int main()
{
 char *names[] = {
 "akshay",
 "parag",
 "raman",
 "srinivas",
 "gopal",
 "rajesh"
 };

 char *temp ;
 printf ("Original: %s %s\n", names[2], names[3]);
 temp = names[2];
 names[2] = names[3];
 names[3] = temp ;
 printf ("New: %s %s\n", names[2], names[3]);
 return 0 ;
}
```



```
}
```

And here is the output...

Original: raman srinivas

New: srinivas raman

The output is same as the earlier program. In this program, all that we are required to do is to exchange the addresses (of the names) stored in the array of pointers, rather than the names themselves. Thus, by effecting just one exchange, we are able to interchange names. This makes handling strings very convenient.

Thus, from the point of view of efficient memory usage and ease of programming, an array of pointers to strings definitely scores over a two-dimensional character array. That is why, even though, in principle, strings can be stored and handled through a two-dimensional array of characters, in actual practice, it is the array of pointers to strings, which is more commonly used.

### Limitation of Array of Pointers to Strings

When we are using a two-dimensional array of characters, we are at liberty to either initialize the strings where we are declaring the array, or receive the strings using **scanf( )** function. However, when we are using an array of pointers to strings, we can initialize the strings at the place where we are declaring the array, but we cannot receive the strings from keyboard using **scanf( )**. Thus, the following program would never work out:

```
#include <stdio.h>
int main()
{
 char *names[6];
 int i ;
 for (i = 0 ; i <= 5 ; i++)
 {
 printf ("Enter name ");
 scanf ("%s", names[i]);
 }
 return 0 ;
}
```

The program doesn't work because; when we are declaring the array, it is containing garbage values. And it would be definitely wrong to send these garbage values to **scanf( )** as the addresses where it should keep the strings received from the keyboard.

### Solution

If we are bent upon receiving the strings from keyboard using **scanf( )** function and then storing their addresses in an array of pointers to strings, we can do it in a slightly roundabout manner as shown below.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main()
{
 char *names[6];
 char n[50];
 int len, i ;
 char *p ;
 for (i = 0 ; i <= 5 ; i++)
 {
 printf ("Enter name ");
 scanf ("%s", n);
 len = strlen (n);
 p = (char *) malloc (len + 1); /* +1 for accommodating \0 */
 strcpy (p, n);
 names[i] = p ;
 }
 for (i = 0 ; i <= 5 ; i++)
 printf ("%s\n", names[i]);
 return 0 ;
}
```

Here we have first received a name using **scanf( )** in a string **n[ ]**. Then we have found out its length using **strlen( )** and allocated space for making a copy of this name. This memory allocation has been done using a standard library function called **malloc( )**. This function requires the number of bytes to be allocated and returns the base address of the chunk of memory that it allocates. The address returned by this function is always of the type **void \***. This is because **malloc( )** doesn't know what for did we allocate the memory. A **void \*** means a pointer which is a

legal address but it is not address of a **char**, or address of an **int**, or address of any other datatype. Hence it has been converted into **char \*** using a C language feature called typecasting. Typecasting will be discussed in detail in Chapter 22. The prototype of this function has been declared in the header file 'stdlib.h'. Hence we have **#included** this file.

But why did we not use array to allocate memory? This is because, with arrays, we have to commit to the size of the array at the time of writing the program. Moreover, there is no way to increase or decrease the array size during execution of the program. In other words, when we use arrays, static memory allocation takes place. Unlike this, using **malloc( )**, we can allocate memory dynamically, during execution. The argument that we pass to **malloc( )** can be a variable whose value can change during execution.

Once we have allocated the memory using **malloc( )**, we have copied the name received through the keyboard into this allocated space and finally stored the address of the allocated chunk in the appropriate element of **names[ ]**, the array of pointers to strings.

This solution suffers in performance because we need to allocate memory and then do the copying of string for each name received through the keyboard.

## Summary

- (a) Though in principle a 2-D array can be used to handle several strings, in practice an array of pointers to strings is preferred since it takes less space and is efficient in processing strings.
- (b) **malloc( )** function can be used to allocate space in memory on the fly during execution of the program.

## Exercise

**[A]** Answer the following:

- (a) How many bytes in memory would be occupied by the following array of pointers to strings? How many bytes would be required to store the same strings, if they are stored in a two-dimensional character array?

```
char *mess[] = {
 "Hammer and tongs",
 "Tooth and nail",
 "Spit and polish",
 "You and C"
};
```

- (b) Can an array of pointers to strings be used to collect strings from the keyboard? If yes, how? If not, why not?

**[B]** Attempt the following:

- (a) Write a program that uses an array of pointers to strings **str[ ]**. Receive two strings **str1** and **str2** and check if **str1** is embedded in any of the strings in **str[ ]**. If **str1** is found, then replace it with **str2**.

```
char *str[] = {
 "We will teach you how to...",
 "Move a mountain",
 "Level a building",
 "Erase the past",
 "Make a million",
 "...all through C!"
};
```

For example if **str1** contains "mountain" and **str2** contains "car", then the second string in **str** should get changed to "Move a car".

- (b) Write a program to sort a set of names stored in an array in alphabetical order.
- (c) Write a program to reverse the strings stored in the following array of pointers to strings:

```
char *s[] = {
 "To err is human...",
 "But to really mess things up...",
 "One needs to know C!"
};
```

- (d) Develop a program that receives the month and year from the keyboard as integers and prints the calendar in the following format.

| April 2015 |     |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|-----|
| Mon        | Tue | Wed | Thu | Fri | Sat | Sun |
|            |     | 1   | 2   | 3   | 4   | 5   |
| 6          | 7   | 8   | 9   | 10  | 11  | 12  |
| 13         | 14  | 15  | 16  | 17  | 18  | 19  |
| 20         | 21  | 22  | 23  | 24  | 25  | 26  |
| 27         | 28  | 29  | 30  |     |     |     |

Note that according to the Gregorian calendar 01/01/01 was Monday. With this as the base, the calendar should be generated.

- (e) A factory has 3 divisions and stocks 4 categories of products. An inventory table is updated for each division and for each product as they are received. There are three independent suppliers of products to the factory.
1. Design a data format to represent each transaction.
  2. Write a program to take a transaction and update the inventory.
  3. If the cost per item is also given write a program to calculate the total inventory values.
- (f) Modify the above program suitably so that once the calendar for a particular month and year has been displayed on the screen, then using arrow keys the user must be able to change the calendar in the following manner:
- Up arrow key : Next year, same month  
 Down arrow key : Previous year, same month  
 Right arrow key : Same year, next month  
 Left arrow key : Same year, previous month
- If the Escape (Esc) key is hit then the procedure should stop.
- Hint: Use the **getkey( )** function discussed in Chapter 14, problem number [C](d).
- (g) Write a program to delete all vowels from a sentence. Assume that the sentence is not more than 80 characters long.
- (h) Write a program that will read a line and delete from it all occurrences of the word 'the'.

- (i) Write a program that takes a set of names of individuals and abbreviates the first, middle and other names except the last name by their first letter.
- (j) Write a program to count the number of occurrences of any two vowels in succession in a line of text. For example, in the following sentence:  
 “Please read this application and give me gratuity”  
 such occurrences are ea, ea, ui.
- (k) Write a program that receives an integer (less than or equal to nine digits in length) and prints out the number in words. For example, if the number input is 12342, then the output should be Twelve Thousand Three Hundred Forty Two.
- (l) Write a program that receives a 5-digit number and prints it out in large size as shown below.

```
#####
 # # ## # #
 # # # # #
#####
 # ##### # # ### #
 # # # # #
 # # # # #
#####
```

- **Why use Structures?**
  - Declaring a Structure
  - Accessing Structure Elements
  - How Structure Elements are Stored?
- **Array of Structures**
- **Additional Features of Structures**
- **Uses of Structures**
- **Summary**
- **Exercise**



---

# ***17 Structures***

---

- Why Use Structures
  - Declaring a Structure
  - Accessing Structure Elements
  - How Structure Elements are Stored
- Array of Structures
- Additional Features of Structures
- Uses of Structures
- Summary
- Exercise



Which mechanic is good enough who knows how to repair only one type of vehicle? None. Same thing is true about C language. It wouldn't have been so popular had it been able to handle only all **ints**, or all **floats** or all **chars** at a time. In fact, when we handle real world data, we don't usually deal with little atoms of information by themselves—things like integers, characters and such. Instead, we deal with entities that are collections of things, each thing having its own attributes, just as the entity we call a 'book' is a collection of things, such as title, author, call number, publisher, number of pages, date of publication, etc. As you can see, all this data is dissimilar, like author is a string, whereas number of pages is an integer. For dealing with such collections, C provides a data type called 'structure'. A structure gathers together, different atoms of information that comprise a given entity. And structure is the topic of this chapter.

### Why use Structures?

We have seen earlier how ordinary variables can hold one piece of information and how arrays can hold a number of pieces of information of the same data type. These two data types can handle a great variety of situations. But quite often we deal with entities that are collection of dissimilar data types.

For example, suppose you want to store data about a book. You might want to store its name (a string), its price (a float) and number of pages in it (an int). If data about say 3 such books is to be stored, then we can follow two approaches:

- (a) Construct individual arrays, one for storing names, another for storing prices and still another for storing number of pages.
- (b) Use a structure variable.

Let us examine these two approaches one-by-one. For the sake of programming convenience, assume that the names of books would be single character long. Let us begin with a program that uses arrays.

```
#include <stdio.h>
int main()
{
 char name[3];
 float price[3];
 int pages[3], i ;
```

```
printf ("Enter names, prices and no. of pages of 3 books\n");

for (i = 0 ; i <= 2 ; i++)
 scanf ("%c %f %d", &name[i], &price[i], &pages[i]);

printf ("\nAnd this is what you entered\n");
for (i = 0 ; i <= 2 ; i++)
 printf ("%c %f %d\n", name[i], price[i], pages[i]);
return 0 ;
}
```

And here is the sample run...

```
Enter names, prices and no. of pages of 3 books
A 100.00 354
C 256.50 682
F 233.70 512
```

```
And this is what you entered
A 100.000000 354
C 256.500000 682
F 233.700000 512
```

This approach, no doubt, allows you to store names, prices and number of pages. But as you must have realized, it is an unwieldy approach that obscures the fact that you are dealing with a group of characteristics related to a single entity—the book.

The program becomes more difficult to handle as the number of items relating to the book goes on increasing. For example, we would be required to use a number of arrays, if we also decide to store name of the publisher, date of purchase of book, etc. To solve this problem, C provides a special data type—the structure.

A structure contains a number of data types grouped together. These data types may or may not be of the same type. The following example illustrates the use of this data type:

```
#include <stdio.h>
int main()
{
 struct book
 {
```

```
 char name ;
 float price ;
 int pages ;
};
struct book b1, b2, b3 ;

printf ("Enter names, prices & no. of pages of 3 books\n") ;
scanf ("%c %f %d", &b1.name, &b1.price, &b1.pages) ;
scanf ("%c %f %d", &b2.name, &b2.price, &b2.pages) ;
scanf ("%c %f %d", &b3.name, &b3.price, &b3.pages) ;
printf ("And this is what you entered\n") ;
printf ("%c %f %d\n", b1.name, b1.price, b1.pages) ;
printf ("%c %f %d\n", b2.name, b2.price, b2.pages) ;
printf ("%c %f %d\n", b3.name, b3.price, b3.pages) ;
return 0 ;
}
```

And here is the output...

```
Enter names, prices and no. of pages of 3 books
A 100.00 354
C 256.50 682
F 233.70 512
And this is what you entered
A 100.000000 354
C 256.500000 682
F 233.700000 512
```

This program demonstrates two fundamental aspects of structures:

- (a) Declaration of a structure
- (b) Accessing of structure elements

Let us now look at these concepts one-by-one.

### Declaring a Structure

In our example program, the following statement declares the structure type:

```
struct book
{
```

```
char name ;
float price ;
int pages ;
};
```

This statement defines a new data type called **struct book**. Each variable of this data type will consist of a character variable called **name**, a float variable called **price** and an integer variable called **pages**. The general form of a structure declaration statement is given below.

```
struct <structure name>
{
 structure element 1 ;
 structure element 2 ;
 structure element 3 ;

};
```

Once the new structure data type has been defined, one or more variables can be declared to be of that type. For example, the variables **b1, b2, b3** can be declared to be of the type **struct book**, as,

```
struct book b1, b2, b3 ;
```

This statement sets aside space in memory. It makes available space to hold all the elements in the structure—in this case, 7 bytes—one for **name**, four for **price** and two for **pages**. These bytes are always in adjacent memory locations.

If we so desire, we can combine the declaration of the structure type and the structure variables in one statement.

For example,

```
struct book
{
 char name ;
 float price ;
 int pages ;
};
struct book b1, b2, b3 ;
```

is same as...

```
struct book
{
 char name ;
 float price ;
 int pages ;
} b1, b2, b3 ;
```

or even...

```
struct
{
 char name ;
 float price ;
 int pages ;
} b1, b2, b3 ;
```

Like primary variables and arrays, structure variables can also be initialized where they are declared. The format used is quite similar to that used to initialize arrays.

```
struct book
{
 char name[10] ;
 float price ;
 int pages ;
};
struct book b1 = { "Basic", 130.00, 550 };
struct book b2 = { "Physics", 150.80, 800 };
struct book b3 = { 0 } ;
```

Note the following points while declaring a structure type:

- (a) The closing brace ( } ) in the structure type declaration must be followed by a semicolon ( ; ).
- (b) It is important to understand that a structure type declaration does not tell the compiler to reserve any space in memory. All a structure declaration does is, it defines the 'form' of the structure.
- (c) Usually structure type declaration appears at the top of the source code file, before any variables or functions are defined. In very large programs they are usually put in a separate header file, and the file

is included (using the preprocessor directive **#include**) in whichever program we want to use this structure type.

- (d) If a structure variable is initiated to a value `{ 0 }`, then all its elements are set to value 0, as in **b3** above. This is a handy way of initializing structure variables. In absence of this, we would have been required to initialize each individual element to a value 0.

### Accessing Structure Elements

Having declared the structure type and the structure variables, let us see how the elements of the structure can be accessed.

In arrays, we can access individual elements of an array using a subscript. Structures use a different scheme. They use a dot (.) operator. So to refer to **pages** of the structure defined in our sample program, we have to use,

```
b1.pages
```

Similarly, to refer to **price**, we would use,

```
b1.price
```

Note that before the dot, there must always be a structure variable and after the dot, there must always be a structure element.

### How Structure Elements are Stored?

Whatever be the elements of a structure, they are always stored in contiguous memory locations. The following program would illustrate this:

```
/* Memory map of structure elements */
include <stdio.h>
int main()
{
 struct book
 {
 char name ;
 float price ;
 int pages ;
 };
 struct book b1 = { 'B', 130.00, 550 } ;
```

```
printf ("Address of name = %u\n", &b1.name) ;
printf ("Address of price = %u\n", &b1.price) ;
printf ("Address of pages = %u\n", &b1.pages) ;
return 0 ;
}
```

Here is the output of the program...

```
Address of name = 65518
Address of price = 65519
Address of pages = 65523
```

Actually, the structure elements are stored in memory as shown in the Figure 17.1.

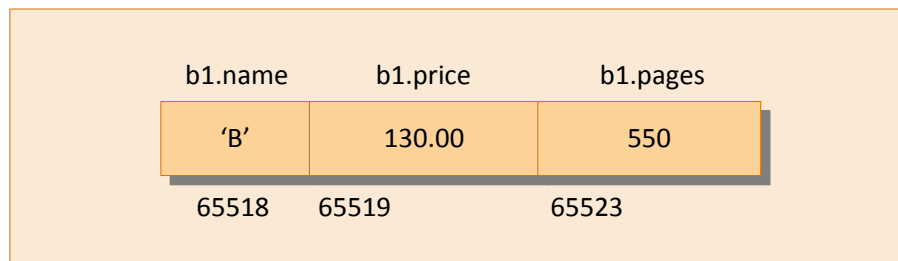


Figure 17.1

## Array of Structures

Our sample program showing usage of structure is rather simple minded. All it does is, it receives values into various structure elements and output these values. But that's all we intended to do anyway... show how structure types are created, how structure variables are declared and how individual elements of a structure variable are referenced.

In our sample program, to store data of 100 books, we would be required to use 100 different structure variables from **b1** to **b100**, which is definitely not very convenient. A better approach would be to use an array of structures. Following program shows how to use an array of structures:

```
/* Usage of an array of structures */
#include <stdio.h>
void linkfloat() ;
int main()
{
```

```
struct book
{
 char name ;
 float price ;
 int pages ;
};

struct book b[100] ;
int i ;

for (i = 0 ; i <= 99 ; i++)
{
 printf ("Enter name, price and pages ") ;
 fflush (stdin) ;
 scanf ("%c %f %d", &b[i].name, &b[i].price, &b[i].pages) ;
}

for (i = 0 ; i <= 99 ; i++)
 printf ("%c %f %d\n", b[i].name, b[i].price, b[i].pages) ;

return 0 ;
}

void linkfloat()
{
 float a = 0, *b ;
 b = &a ; /* cause emulator to be linked */
 a = *b ; /* suppress the warning - variable not used */
}
```

Now a few comments about the program:

- (a) Notice how the array of structures is declared...

```
struct book b[100] ;
```

This provides space in memory for 100 structures of the type **struct book**.

- (b) The syntax we use to reference each element of the array **b** is similar to the syntax used for arrays of **ints** and **chars**. For example, we refer to zeroth book's price as **b[ 0 ].price**. Similarly, we refer first book's pages as **b[ 1 ].pages**.



- (c) It should be appreciated what careful thought Dennis Ritchie has put into C language. He first defined array as a collection of similar elements; then realized that dissimilar data types that are often found in real life cannot be handled using arrays, therefore created a new data type called structure. But even using structures, programming convenience could not be achieved, because a lot of variables (**b1** to **b100** for storing data about hundred books) needed to be handled. Therefore, he allowed us to create an array of structures; an array of similar data types which themselves are a collection of dissimilar data types. Hats off to the genius!
- (d) In an array of structures, all elements of the array are stored in adjacent memory locations. Since each element of this array is a structure, and since all structure elements are always stored in adjacent locations, you can very well visualize the arrangement of array of structures in memory. In our example, **b[ 0 ]**'s **name**, **price** and **pages** in memory would be immediately followed by **b[ 1 ]**'s **name**, **price** and **pages**, and so on.
- (e) What is the function **linkfloat( )** doing here? If you don't define it, you are likely to get an error "Floating-Point Formats Not Linked" with many C Compilers. What causes this error to occur? When parsing our source file, if the compiler encounters a reference to the address of a float, it sets a flag to have the linker link in the floating-point emulator. A floating-point emulator is used to manipulate floating-point numbers in functions like **scanf( )** and **atof( )**. There are some cases in which the reference to the **float** is a bit obscure and the compiler does not detect the need for the emulator. The most common is using **scanf( )** to read a **float** in an array of structures as shown in our program.

How can we force the formats to be linked? That's where the **linkfloat( )** function comes in. It forces linking of the floating-point emulator into an application. There is no need to call this function, just define it anywhere in your program.

### Additional Features of Structures

Let us now explore the intricacies of structures with a view of programming convenience. We would highlight these intricacies with suitable examples.

- (a) The values of a structure variable can be assigned to another structure variable of the same type using the assignment operator.

It is not necessary to copy the structure elements piece-meal. Obviously, programmers prefer assignment to piece-meal copying. This is shown in the following example:

```
include <stdio.h>
include <string.h>
int main()
{
 struct employee
 {
 char name[10] ;
 int age ;
 float salary ;
 };
 struct employee e1 = { "Sanjay", 30, 5500.50 };
 struct employee e2, e3 ;

 /* piece-meal copying */
 strcpy (e2.name, e1.name) ;
 /* e2.name = e1. name is wrong */
 e2.age = e1.age ;
 e2.salary = e1.salary ;

 /* copying all elements at one go */
 e3 = e2 ;

 printf ("%s %d %f\n", e1.name, e1.age, e1.salary) ;
 printf ("%s %d %f\n", e2.name, e2.age, e2.salary) ;
 printf ("%s %d %f\n", e3.name, e3.age, e3.salary) ;
 return 0 ;
}
```

The output of the program would be...

```
Sanjay 30 5500.500000
Sanjay 30 5500.500000
Sanjay 30 5500.500000
```

Ability to copy the contents of all structure elements of one variable into the corresponding elements of another structure variable is rather surprising, since C does not allow assigning the contents of one array to another just by equating the two. As we saw earlier,

for copying arrays, we have to copy the contents of the array element-by-element.

This copying of all structure elements at one go has been possible only because the structure elements are stored in contiguous memory locations. Had this not been so, we would have been required to copy structure variables element by element. And who knows, had this been so, structures would not have become popular at all.

- (b) One structure can be nested within another structure. Using this facility, complex data types can be created. The following program shows nested structures at work:

```
#include <stdio.h>
int main()
{
 struct address
 {
 char phone[15];
 char city[25];
 int pin ;
 };

 struct emp
 {
 char name[25];
 struct address a ;
 };
 struct emp e = { "jeru", "531046", "nagpur", 10 };

 printf ("name = %s phone = %s\n", e.name, e.a.phone) ;
 printf ("city = %s pin = %d\n", e.a.city, e.a.pin) ;
 return 0 ;
}
```

And here is the output...

```
name = jeru phone = 531046
city = nagpur pin = 10
```

Notice the method used to access the element of a structure that is part of another structure. For this, the dot operator is used twice, as in the expression,

`e.a.pin` or `e.a.city`

Of course, the nesting process need not stop at this level. We can nest a structure within a structure, within another structure, which is in still another structure and so on... till the time we can comprehend the structure ourselves. Such construction, however, gives rise to variable names that can be surprisingly self-descriptive, for example:

`maruti.engine.bolt.large.qty`

This clearly signifies that we are referring to the quantity of large sized bolts that fit on an engine of a Maruti car.

- (c) Like an ordinary variable, a structure variable can also be passed to a function. We may either pass individual structure elements or the entire structure variable at one shot. Let us examine both the approaches one-by-one using suitable programs.

```
/* Passing individual structure elements */
#include <stdio.h>
void display (char *, char *, int);
int main()
{
 struct book
 {
 char name[25];
 char author[25];
 int callno ;
 };
 struct book b1 = { "Let us C", "YPK", 101 };

 display (b1.name, b1.author, b1.callno);
 return 0 ;
}

void display (char *s, char *t, int n)
{
```

```
printf ("%s %s %d\n", s, t, n) ;
}
```

And here is the output...

Let us C YPK 101

Observe that in the declaration of the structure, **name** and **author** have been declared as arrays. Therefore, when we call the function **display( )** using,

```
display (b1.name, b1.author, b1.callno) ;
```

we are passing the base addresses of the arrays **name** and **author**, but the value stored in **callno**. Thus, this is a mixed call—a call by reference as well as a call by value.

It can be immediately realized that to pass individual elements would become more tedious as the number of structure elements goes on increasing. A better way would be to pass the entire structure variable at a time. This method is shown in the following program:

```
include <stdio.h>
struct book
{
 char name[25] ;
 char author[25] ;
 int callno ;
};
void display (struct book) ;

int main()
{
 struct book b1 = { "Let us C", "YPK", 101 } ;
 display (b1) ;
 return 0 ;
}

void display (struct book b)
{
 printf ("%s %s %d\n", b.name, b.author, b.callno) ;
}
```

And here is the output...

```
Let us C YPK 101
```

Note that here the calling of function **display( )** becomes quite compact,

```
display (b1) ;
```

Having collected what is being passed to the **display( )** function, the question comes, how do we define the formal arguments in the function. We cannot say,

```
struct book b1 ;
```

because the data type **struct book** is not known to the function **display( )**. Therefore, it becomes necessary to declare the structure type **struct book** outside **main( )**, so that it becomes known to all functions in the program.

- (d) The way we can have a pointer pointing to an **int**, or a pointer pointing to a **char**, similarly we can have a pointer pointing to a **struct**. Such pointers are known as 'structure pointers'.

Let us look at a program that demonstrates the usage of a structure pointer.

```
include <stdio.h>
int main()
{
 struct book
 {
 char name[25];
 char author[25];
 int callno ;
 };
 struct book b1 = { "Let us C", "YPK", 101 };
 struct book *ptr ;
 ptr = &b1 ;
 printf ("%s %s %d\n", b1.name, b1.author, b1.callno) ;
 printf ("%s %s %d\n", ptr->name, ptr->author, ptr->callno) ;
 return 0 ;
}
```

The first `printf( )` is as usual. The second `printf( )` however is peculiar. We can't use `ptr.name` or `ptr.callno` because `ptr` is not a structure variable but a pointer to a structure, and the dot operator requires a structure variable on its left. In such cases C provides an operator `->`, called an arrow operator to refer to the structure elements. Remember that on the left hand side of the `'.'` structure operator, there must always be a structure variable, whereas on the left hand side of the `'->'` operator, there must always be a pointer to a structure. The arrangement of the structure variable and pointer to structure in memory is shown in the Figure 17.2.

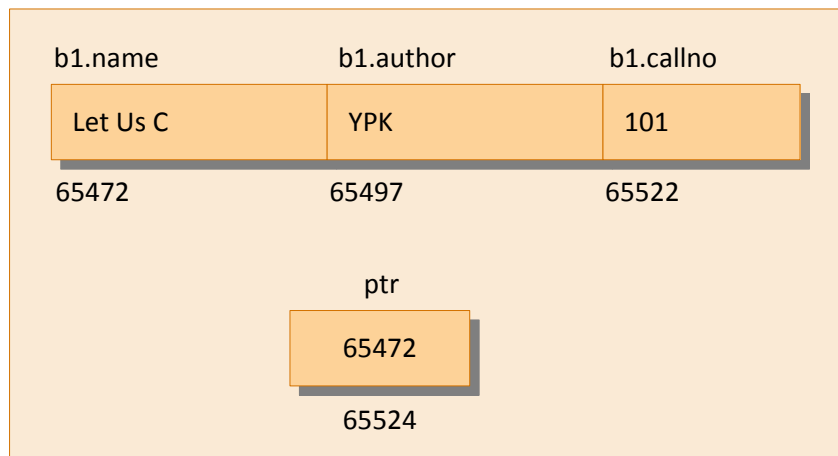


Figure 17.2

Can we not pass the address of a structure variable to a function? We can. The following program demonstrates this:

```
/* Passing address of a structure variable */
#include <stdio.h>
struct book
{
 char name[25];
 char author[25];
 int callno ;
};
void display (struct book *);

int main()
{
 struct book b1 = { "Let us C", "YPK", 101 };
 display (&b1);
}
```

```
 return 0 ;
}

void display (struct book *b)
{
 printf ("%s %s %d\n", b->name, b->author, b->callno) ;
}
```

And here is the output...

```
Let us C YPK 101
```

Again note that, to access the structure elements using pointer to a structure, we have to use the ‘->’ operator.

Also, the structure **struct book** should be declared outside **main( )** such that this data type is available to **display( )** while declaring pointer to the structure.

(e) Consider the following code snippet:

```
include <stdio.h>
struct emp
{
 int a ;
 char ch ;
 float s ;
};
int main()
{
 struct emp e ;
 printf ("%u %u %u\n", &e.a, &e.ch, &e.s) ;
 return 0 ;
}
```

If we execute this program using TC/TC++ Compiler we get the addresses as:

```
65518 65520 65521
```

As expected, in memory the **char** begins immediately after the **int** and **float** begins immediately after the **char**.



However, if we run the same program using Visual Studio compiler then the output turns out to be:

```
1245044 1245048 1245052
```

It can be observed from this output that the **float** doesn't get stored immediately after the **char**. In fact there is a hole of three bytes after the **char**. Let us understand the reason for this. Visual Studio is a 32-bit compiler targeted to generate code for a 32-bit microprocessor. The architecture of this microprocessor is such that it is able to fetch the data that is present at an address, which is a multiple of four much faster than the data present at any other address. Hence the Visual Studio compiler aligns every element of a structure at an address that is multiple of four. That's the reason why there were three holes created between the **char** and the **float**.

However, some programs need to exercise precise control over the memory areas where data is placed. For example, suppose we wish to read the contents of the boot sector (first sector on the hard disk) into a structure. For this the byte arrangement of the structure elements must match the arrangement of various fields in the boot sector of the disk. The **#pragma pack** directive offers a way to fulfil this requirement. This directive specifies packing alignment for structure members. The pragma takes effect at the first structure declaration after the pragma is seen.

Visual Studio compiler supports this feature, whereas Turbo C/C++ doesn't. The following code shows how to use this directive:

```
#include <stdio.h>
#pragma pack(1)
struct emp
{
 int a ;
 char ch ;
 float s ;
};
#pragma pack()

int main()
{
 struct emp e ;
```

```
printf ("%u %u %u\n", &e.a, &e.ch, &e.s) ;
return 0 ;
}
```

Here, **#pragma pack ( 1 )** lets each structure element to begin on a 1-byte boundary as justified by the output of the program given below.

```
1245044 1245048 1245049
```

## Uses of Structures

Where are structures useful? The immediate application that comes to the mind is Database Management. That is, to maintain data about employees in an organization, books in a library, items in a store, financial accounting transactions in a company, etc. But mind you, use of structures stretches much beyond database management. They can be used for a variety of purposes like:

- (a) Changing the size of the cursor
- (b) Clearing the contents of the screen
- (c) Placing the cursor at an appropriate position on screen
- (d) Drawing any graphics shape on the screen
- (e) Receiving a key from the keyboard
- (f) Checking the memory size of the computer
- (g) Finding out the list of equipment attached to the computer
- (h) Formatting a floppy
- (i) Hiding a file from the directory
- (j) Displaying the directory of a disk
- (k) Sending the output to printer
- (l) Interacting with the mouse

And that is certainly a very impressive list! At least impressive enough to make you realize how important a data type a structure is and to be thorough with it if you intend to program any of the above applications.

## Summary

- (a) A structure is usually used when we wish to store dissimilar data together.
- (b) Structure elements can be accessed through a structure variable using a dot (.) operator.

- (c) Structure elements can be accessed through a pointer to a structure using the arrow (->) operator.
- (d) All elements of one structure variable can be assigned to another structure variable using the assignment (=) operator.
- (e) It is possible to pass a structure variable to a function either by value or by address.
- (f) It is possible to create an array of structures.

## Exercise

**[A]** What will be the output of the following programs:

- (a) 

```
#include <stdio.h>
#include <string.h>
int main()
{
 struct gospel
 {
 int num ;
 char mess1[50] ;
 char mess2[50] ;
 } m ;

 m.num = 1 ;
 strcpy (m.mess1, "If all that you have is hammer") ;
 strcpy (m.mess2, "Everything looks like a nail") ;

 /* assume that the strucure is located at address 1004 */
 printf ("%u %u %u\n", &m.num, m.mess1, m.mess2) ;
 return 0 ;
}
```
- (b) 

```
#include <stdio.h>
#include <string.h>
int main()
{
 struct part
 {
 char partname[50] ;
 int partnumber ;
 } ;
```

```
 struct part p, *ptrp ;

 ptrp = &p ;
 strcpy (p.partname, "CrankShaft") ;
 p.partnumber = 102133 ;

 printf ("%s %d\n", p.partname, p.partnumber) ;
 printf ("%s %d\n", (*ptrp).partname, (*ptrp).partnumber) ;
 printf ("%s %d\n", ptrp->partname, ptrp->partnumber) ;
 return 0 ;
}

(c) # include <stdio.h>
 struct gospel
 {
 int num ;
 char mess1[50] ;
 char mess2[50] ;
 } m1 = {
 2, "If you are driven by success",
 "make sure that it is a quality drive"
 };

 int main()
 {
 struct gospel m2, m3 ;
 m2 = m1 ;
 m3 = m2 ;
 printf ("%d %s %s\n", m1.num, m2.mess1, m3.mess2) ;
 return 0 ;
 }
```

**[B]** Point out the errors, if any, in the following programs:

```
(a) # include <stdio.h>
 # include <string.h>
 int main()
 {
 struct employee
 {
 char name[25] ;
 int age ;
 float salary ;
 };
 }
```

```
 struct employee e ;
 strcpy (e.name, "Shailesh") ;
 age = 25 ;
 salary = 15500.00 ;
 printf ("%s %d %f\n", e.name, age, salary) ;
 return 0 ;
}
```

```
(b) #include <stdio.h>
int main()
{
 struct
 {
 char bookname[25] ;
 float price ;
 };
 struct book b = { "Go Embedded!", 240.00 } ;
 printf ("%s %f\n", b.bookname, b.price) ;
 return 0 ;
}
```

```
(c) #include <stdio.h>
struct virus
{

 char signature[25] ;
 char status[20] ;
 int size ;
} v[2] = {
 "Yankee Doodle", "Deadly", 1813,
 "Dark Avenger", "Killer", 1795
};
int main()
{
 int i ;
 for (i = 0 ; i <= 1 ; i++)
 printf ("%s %s\n", v.signature, v.status) ;
 return 0 ;
}
```

```
(d) #include <stdio.h>
struct s
{
```

```
 char ch ;
 int i ;
 float a ;
};
void f (struct s) ;
void g (struct s *) ;

int main()
{
 struct s var = { 'C', 100, 12.55 } ;
 f (var) ;
 g (&var) ;
 return 0 ;
}
void f (struct s v)
{
 printf ("%c %d %f\n", v -> ch, v -> i, v -> a) ;
}
void g (struct s *v)
{
 printf ("%c %d %f\n", v->ch, v->i, v->a) ;
}
```

```
(e) # include <stdio.h>
struct s
{
 int i ;
 struct s *p ;
};
int main()
{
 struct s var1, var2 ;

 var1.i = 100 ;
 var2.i = 200 ;
 var1.p = &var2 ;
 var2.p = &var1 ;
 printf ("%d %d\n", var1.p -> i, var2.p -> i) ;
 return 0 ;
}
```

[C] Answer the following:

(a) Ten floats are to be stored in memory. What would you prefer, an array or a structure?

(b) Given the statement,

```
maruti.engine.bolts = 25 ;
```

which of the following is True?

1. Structure bolts is nested within structure engine
2. Structure engine is nested within structure maruti
3. Structure maruti is nested within structure engine
4. Structure maruti is nested within structure bolts

(c) State True or False:

1. All structure elements are stored in contiguous memory locations.
2. An array should be used to store dissimilar elements, and a structure to store similar elements.
3. In an array of structures, not only are all structures stored in contiguous memory locations, but the elements of individual structures are also stored in contiguous locations.

(d) struct time

```
{
 int hours ;
 int minutes ;
 int seconds ;
} t ;
struct time *pt ;
pt = &t ;
```

With reference to the above declarations which of the following refers to **seconds** correctly:

1. pt.seconds
2. ( \*pt ).seconds
3. time.seconds
4. pt -> seconds

(e) Match the following with reference to the program segment given below.

```
struct
```

```

{
 int x, y;
} s[] = { 10, 20, 15, 25, 8, 75, 6, 2 };
int *i;
i = s;

```

- |                                        |        |
|----------------------------------------|--------|
| 1. $*(i + 3)$                          | a. 85  |
| 2. $s[i[7]].x$                         | b. 2   |
| 3. $s[(s + 2) \rightarrow y / 3[l]].y$ | c. 6   |
| 4. $i[i[1] - i[2]]$                    | d. 7   |
| 5. $i[s[3].y]$                         | e. 16  |
| 6. $(s + 1) \rightarrow x + 5$         | f. 15  |
| 7. $*(1 + i) ** (i + 4) / *i$          | g. 25  |
| 8. $s[i[0] - i[4]].y + 10$             | h. 8   |
| 9. $(*(s + *(i + 1) / *i)).x + 2$      | i. 1   |
| 10. $++i[i[6]]$                        | j. 100 |
|                                        | k. 10  |
|                                        | l. 20  |

**[D] Attempt the following:**

- (a) Create a structure to specify data on students given below:

Roll number, Name, Department, Course, Year of joining

Assume that there are not more than 450 students in the college.

- (1) Write a function to print names of all students who joined in a particular year.
- (2) Write a function to print the data of a student whose roll number is received by the function.

- (b) Create a structure to specify data of customers in a bank. The data to be stored is: Account number, Name, Balance in account. Assume maximum of 200 customers in the bank.

- (1) Write a function to print the Account number and name of each customer with balance below Rs. 100.
- (2) If a customer requests for withdrawal or deposit, the form contains the fields:

Acct. no, amount, code (1 for deposit, 0 for withdrawal)

Write a program to give a message, "The balance is insufficient for the specified withdrawal", if on withdrawal the balance falls below Rs. 100.



- (c) An automobile company has serial number for engine parts starting from AA0 to FF9. The other characteristics of parts are year of manufacture, material and quantity manufactured.
  - (1) Specify a structure to store information corresponding to a part.
  - (2) Write a program to retrieve information on parts with serial numbers between BB1 and CC6.
- (d) A record contains name of cricketer, his age, number of test matches that he has played and the average runs that he has scored in each test match. Create an array of structures to hold records of 20 such cricketers and then write a program to read these records and arrange them in ascending order by average runs. Use the **qsort( )** standard library function.
- (e) There is a structure called **employee** that holds information like employee code, name and date of joining. Write a program to create an array of structures and enter some data into it. Then ask the user to enter current date. Display the names of those employees whose tenure is greater than equal to 3 years.
- (f) Create a structure called **library** to hold accession number, title of the book, author name, price of the book, and flag indicating whether book is issued or not. Write a menu-driven program that implements the working of a library. The menu options should be:
  - 1. Add book information
  - 2. Display book information
  - 3. List all books of given author
  - 4. List the title of specified book
  - 5. List the count of books in the library
  - 6. List the books in the order of accession number
  - 7. Exit
- (g) Write a function that compares two given dates. To store a date use a structure that contains three members namely day, month and year. If the dates are equal the function should return 0, otherwise it should return 1.
- (h) Linked list is a very common data structure that is often used to store similar data in memory. The individual elements of a linked list are stored “somewhere” in memory. The order of the elements is maintained by explicit links between them. Thus, a linked list is a

collection of elements called nodes, each of which stores two item of information—an element of the list, and a link, i.e., a pointer or an address that indicates explicitly the location of the node containing the successor of this list element.

Write a program to build a linked list by adding new nodes at the beginning, at the end or in the middle of the linked list. Also write a function **display( )** which displays all the nodes present in the linked list.

- (i) A stack is a data structure in which addition of new element or deletion of existing element always takes place at the same end known as 'top' of stack. Write a program to implement a stack using a linked list.
- (j) In a data structure called queue the addition of new element takes place at the end (called 'rear' of queue) whereas deletion takes place at the other end (called 'front' of queue). Write a program to implement a queue using a linked list.
- (k) Write a program to implement an ascending order linked list. This means that any new element that is added to the linked list gets inserted at a place in the linked list such that its ascending order nature remains intact.
- (l) Write a program that receives wind speed as input and categorizes the hurricane as per the following table:

| Wind Speed in miles / hour | Hurricane Category |
|----------------------------|--------------------|
| 74 – 95                    | I                  |
| 96 – 110                   | II                 |
| 111 – 130                  | III                |
| 131 – 155                  | IV                 |
| 155                        | V                  |

- (m) There are five players from which the Most Valuable Player (MVP) is to be chosen. Each player is to be judged by 3 judges, who would assign a rank to each player. The player whose sum of ranks is highest is chosen as MVP. Write a program to implement this scheme.



# 18

## Console Input/Output

- **Types of I/O**
- **Console I/O Functions**
  - Formatted Console I/O Functions  
*sprintf( )* and *sscanf( )* Functions
  - Unformatted Console I/O Functions
- **Summary**
- **Exercise**



---

# ***18 Console Input/Output***

---

- Types of I/O
- Console I/O Functions
  - Formatted Console I/O Functions
  - sprintf( )* and *sscanf( )* Functions
  - Unformatted Console I/O Functions
- Summary
- Exercise

As mentioned in the first chapter, Dennis Ritchie wanted C to remain compact. In keeping with this intention, he deliberately omitted everything related with Input/Output (I/O) from his definition of the language. Thus, C simply has no provision for receiving data from any of the input devices (like say keyboard, disk, etc.), or for sending data to the output devices (like say monitor, disk, etc.). Then how do we manage I/O, and how is it that we were able to use **printf( )** and **scanf( )** if C has nothing to offer for I/O? This is what we intend to explore in this chapter.

### Types of I/O

Though C has no provision for I/O, it, of course, has to be dealt with at some point or the other. There is not much use of writing a program that spends all its time telling itself a secret. Each Operating System (OS) has its own facility for inputting and outputting data from and to the files and devices. It's a simple matter for a system programmer to write a few small programs that would link the C Compiler for a particular operating system's I/O facilities.

The developers of C Compilers do just that. They write several standard I/O functions and put them in libraries. These libraries are available with all C Compilers. Whichever C Compiler you are using, it's almost certain that you have access to a library of I/O functions.

Do understand that the I/O facilities with different operating systems would be different. Thus, the way one OS displays output on screen may be different than the way another OS does it. For example, the standard library function **printf( )** for DOS-based C compiler has been written keeping in mind the way DOS outputs characters to screen. Similarly, the **printf( )** function for a UNIX-based compiler has been written keeping in mind the way UNIX outputs characters to screen. We, as programmers, do not have to bother about which **printf( )** has been written in what manner. We should just use **printf( )** and it would take care of the rest of the details that are OS dependent. Same is true about all other standard library functions available for I/O.

There are numerous library functions available for I/O. These can be classified into two broad categories:

- (a) Console I/O functions - Functions to receive input from keyboard and write output to VDU.
- (b) File I/O functions - Functions to perform I/O operations on a floppy disk or a hard disk.

In this chapter we would be discussing only Console I/O functions. File I/O functions would be discussed in Chapter 19.

Console I/O Functions

The screen and keyboard together are called a console. Console I/O functions can be further classified into two categories—formatted and unformatted console I/O functions. The basic difference between them is that the formatted functions allow the input read from the keyboard or the output displayed on the VDU to be formatted as per our requirements. For example, if values of average marks and percentage marks are to be displayed on the screen, then the details like where this output would appear on the screen, how many spaces would be present between the two values, the number of places after the decimal points, etc., can be controlled using formatted functions. The functions available under each of these two categories are shown in Figure 18.1. Now let us discuss these console I/O functions in detail.

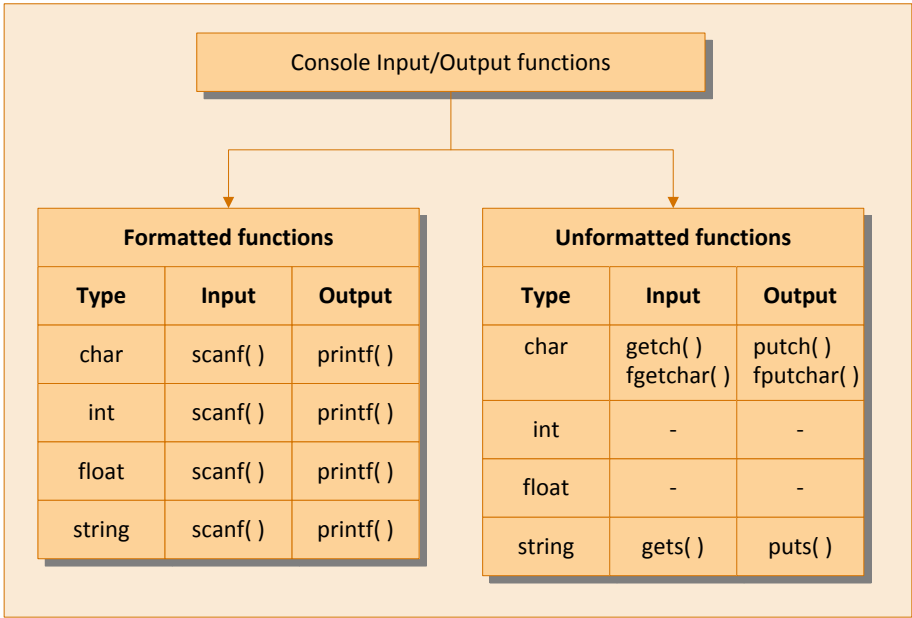


Figure 18.1

Formatted Console I/O Functions

As can be seen from Figure 18.1, the functions **printf( )**, and **scanf( )** fall under the category of formatted console I/O functions. These functions allow us to supply the input in a fixed format and let us obtain the output in the specified form. Let us discuss these functions one-by-one.

We have talked a lot about **printf( )**, used it regularly, but without having introduced it formally. Well, better late than never. Its general form looks like this...

```
printf ("format string", list of variables) ;
```

The format string can contain:

- (a) Characters that are simply printed as they are
- (b) Format specifications that begin with a % sign
- (c) Escape sequences that begin with a \ sign

For example, look at the following program:

```
include <stdio.h>
int main()
{
 int avg = 346 ;
 float per = 69.2 ;
 printf ("Average = %d\nPercentage = %f\n", avg, per) ;
 return 0 ;
}
```

The output of the program would be...

```
Average = 346
Percentage = 69.200000
```

How does **printf( )** function interpret the contents of the format string? For this, it examines the format string from left to right. So long as it doesn't come across either a % or a \, it continues to dump the characters that it encounters, on to the screen. In this example, **Average** = is dumped on the screen. The moment it comes across a format specification in the format string, it picks up the first variable in the list of variables and prints its value in the specified format. In this example, the moment **%d** is met, the variable **avg** is picked up and its value is printed. Similarly, when an escape sequence is met, it takes the appropriate action. In this example, the moment **\n** is met, it places the cursor at the beginning of the next line. This process continues till the end of format string is reached.

### Format Specifications

The **%d** and **%f** used in the **printf( )** are called format specifiers. They tell **printf( )** to print the value of **avg** as a decimal integer and the value of



**per** as a float. Following is the list of format specifiers that can be used with the **printf( )** function.

| Data type |                      | Format specifier |
|-----------|----------------------|------------------|
| Integer   | short signed         | %d or %l         |
|           | short unsigned       | %u               |
|           | long signed          | %ld              |
|           | long unsigned        | %lu              |
|           | unsigned hexadecimal | %x               |
|           | unsigned octal       | %o               |
| Real      | float                | %f               |
|           | double               | %lf              |
|           | long double          | %Lf              |
| Character | signed character     | %c               |
|           | unsigned character   | %c               |
| String    |                      | %s               |

Figure 18.2

We can provide optional specifiers shown in Figure 18.3 in the format specifications.

| Specifier | Description                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------|
| w         | Digits specifying field width                                                                                  |
| .         | Decimal point separating field width from precision (precision means number of places after the decimal point) |
| p         | Digits specifying precision                                                                                    |
| -         | Minus sign for left justifying the output in the specified field width                                         |

Figure 18.3

Now a short explanation about these optional format specifiers. The field-width specifier tells **printf( )** how many columns on screen should

be used while printing a value. For example, **%10d** says, “print the variable as a decimal integer in a field of 10 columns”. If the value to be printed happens not to fill up the entire field, the value is right justified and is padded with blanks on the left. If we include the minus sign in format specifier (as in **%-10d**), this means left justification is desired and the value will be padded with blanks on the right. If the field-width used turns out to be less than what is required to print the number, the field-width is ignored and the complete number is printed. Here is an example that illustrates all these features.

```
#include <stdio.h>
int main()
{
 int weight = 63 ;
 printf ("weight is %d kg\n", weight) ;
 printf ("weight is %2d kg\n", weight) ;
 printf ("weight is %4d kg\n", weight) ;
 printf ("weight is %6d kg\n", weight) ;
 printf ("weight is %-6d kg\n", weight) ;
 printf ("weight is %1d kg\n", weight) ;
 return 0 ;
}
```

The output of the program would look like this ...

| Columns   | 0123456789012345678901234567890 |
|-----------|---------------------------------|
| weight is | 63 kg                           |
| weight is | 63 kg                           |
| weight is | 63 kg                           |
| weight is | 63 kg                           |
| weight is | 63 kg                           |
| weight is | 63 kg                           |

Specifying the field width can be useful in creating tables of numeric values, as the following program demonstrates:

```
#include <stdio.h>
int main()
{
 printf ("%f %f %f\n", 5.0, 13.5, 133.9) ;
 printf ("%f %f %f\n", 305.0, 1200.9, 3005.3) ;
 return 0 ;
}
```

```
}
```

And here is the output...

```
5.000000 13.500000 133.900000
305.000000 1200.900000 3005.300000
```

Even though the numbers have been printed, the numbers have not been lined up properly and hence are hard to read. A better way would be something like this...

```
include <stdio.h>
int main()
{
 printf ("%10.1f %10.1f %10.1f\n", 5.0, 13.5, 133.9) ;
 printf ("%10.1f %10.1f %10.1f\n", 305.0, 1200.9, 3005.3) ;
 return 0 ;
}
```

This results into a much better output...

```
01234567890123456789012345678901
 5.0 13.5 133.9
 305.0 1200.9 3005.3
```

Note that the specifier **%10.1f** specifies that a float be printed left-aligned within 10 columns with only one place beyond the decimal point.

The format specifiers can be used even while displaying a string of characters. The following program would clarify this point:

```
include <stdio.h>
int main()
{
 char firstname1[] = "Sandy" ;
 char surname1[] = "Malya" ;
 char firstname2[] = "AjayKumar" ;
 char surname2[] = "Gurubaxani" ;
 printf ("%20s%20s\n", firstname1, surname1) ;
 printf ("%20s%20s\n", firstname2, surname2) ;
 return 0 ;
}
```

And here's the output...

```
012345678901234567890123456789012345678901234567890
 Sandy Malya
 AjayKumar Gurubaxani
```

The format specifier **%20s** reserves 20 columns for printing a string and then prints the string in these 20 columns with right justification. This helps lining up names of different lengths properly. Obviously, the format **%-20s** would have left-justified the string. Had we used **%20.10s** it would have meant left justify the string in 20 columns and print only first 10 characters of the string.

### Escape Sequences

We saw earlier how the newline character, **\n**, when inserted in a **printf()**'s format string, takes the cursor to the beginning of the next line. The newline character is an 'escape sequence', so called because the backslash symbol (**\**) is considered as an 'escape' character—it causes an escape from the normal interpretation of a string, so that the next character is recognized as one having a special meaning.

The following example shows usage of **\n** and a new escape sequence **\t**, called 'Tab'. A **\t** moves the cursor to the next Tab stop. A 80-column screen usually has 10 Tab stops. In other words, the screen is divided into 10 zones of 8 columns each. Printing a Tab takes the cursor to the beginning of next printing zone. For example, if cursor is positioned in column 5, then printing a Tab takes it to column 8.

```
include <stdio.h>
int main()
{
 printf ("You\tmust\tbe\tcrazy\nto\tthate\tthis\tbook\n");
 return 0 ;
}
```

And here's the output...

```
01234567890123456789012345678901234567890
You must be crazy
to hate this book
```

The **\n** character causes a new line to begin following 'crazy'. The Tab and newline are probably the most commonly used escape sequences,

but there are others as well. Figure 18.4 shows a complete list of these escape sequences.

| Escape Seq. | Purpose      | Escape Seq. | Purpose         |
|-------------|--------------|-------------|-----------------|
| \n          | New line     | \t          | Tab             |
| \b          | Backspace    | \r          | Carriage return |
| \f          | Form feed    | \a          | Alert           |
| \'          | Single quote | \"          | Double quote    |
| \\          | Backslash    |             |                 |

Figure 18.4

The first few of these escape sequences are more or less self-explanatory. **\b** moves the cursor one position to the left of its current position. **\r** takes the cursor to the beginning of the line in which it is currently placed. **\a** alerts the user by sounding the speaker inside the computer. Form feed advances the computer stationery attached to the printer to the top of the next page. Characters that are ordinarily used as delimiters... the single quote, double quote, and the backslash can be printed by preceding them with the backslash. Thus, the statement,

```
printf ("He said, \"Let's do it!\"");
```

will print...

```
He said, "Let's do it!"
```

So far we have been describing **printf( )**'s specification as if we are forced to use only **%d** for an integer, only **%c** for a char, only **%s** for a string and so on. This is not true at all. In fact, **printf( )** uses the specification that we mention and attempts to perform the specified conversion, and does its best to produce a proper result. Sometimes the result is nonsensical, as in case when we ask it to print a string using **%d**. Sometimes the result is useful, as in the case we ask **printf( )** to print ASCII value of a character using **%d**. Sometimes the result is disastrous and the entire program blows up.

The following program shows a few of these conversions, some sensible, some weird:

And here's the output ...

I would leave it to you to analyze the results by yourselves. Some of the conversions you would find are quite sensible.

The general form of **scanf( )** statement is as follows:

For example:

Note that we are sending addresses of variables (addresses are obtained by using '&' the 'address of' operator) to **scanf( )** function. This is necessary because the values received from keyboard must be dropped into variables corresponding to these addresses. The values that are supplied through the keyboard must be separated by either blank(s),

Tab(s), or newline(s). Do not include these escape sequences in the format string.

All the format specifications that we learnt in **printf( )** function are applicable to **scanf( )** function as well.

### **sprintf( ) and sscanf( ) Functions**

The **sprintf( )** function works similar to the **printf( )** function except for one small difference. Instead of sending the output to the screen as **printf( )** does, this function writes the output to an array of characters. The following program illustrates this:

```
#include <stdio.h>
int main()
{
 int i = 10 ;
 char ch = 'A' ;
 float a = 3.14 ;
 char str[20] ;

 printf ("%d %c %f\n", i, ch, a) ;
 sprintf (str, "%d %c %f", i, ch, a) ;
 printf ("%s\n", str) ;
 return 0 ;
}
```

In this program, the **printf( )** prints out the values of **i**, **ch** and **a** on the screen, whereas **sprintf( )** stores these values in the character array **str**. Since the string **str** is present in memory, what is written into **str** using **sprintf( )** doesn't get displayed on the screen. Once **str** has been built, its contents can be displayed on the screen. In our program this was achieved by the second **printf( )** statement.

The counterpart of **sprintf( )** is the **sscanf( )** function. It allows us to read characters from a string and to convert and store them in C variables according to specified formats. The **sscanf( )** function comes in handy for in-memory conversion of characters to values. You may find it convenient to read in strings from a file and then extract values from a string by using **sscanf( )**. The usage of **sscanf( )** is same as **scanf( )**, except that the first argument is the string from which reading is to take place.

### Unformatted Console I/O Functions

There are several standard library functions available under this category—those that can deal with a single character and those that can deal with a string of characters. For openers, let us look at those which handle one character at a time.

So far, for input we have consistently used the **scanf( )** function. However, for some situations, the **scanf( )** function has one glaring weakness... you need to hit the Enter key before the function can digest what you have typed. However, we often want a function that will read a single character the instant it is typed without waiting for the Enter key to be hit. **getch( )** and **getche( )** are two functions which serve this purpose. These functions return the character that has been most recently typed. The 'e' in **getche( )** function means it echoes (displays) the character that you typed to the screen. As against this, **getch( )** just returns the character that you typed without echoing it on the screen. **getchar( )** works similarly and echoes the character that you typed on the screen, but unfortunately requires Enter key to be typed following the character that you typed. The difference between **getchar( )** and **fgetchar( )** is that the former is a macro whereas the latter is a function.

The prototypes of **getch( )** and **getche( )** are present in the header file **conio.h**. The macro **getchar( )** and the prototype of **fgetchar( )** are present in **stdio.h**. Here is a sample program that illustrates the use of these functions and macro.

```
#include <stdio.h>
#include <conio.h>
int main()
{
 char ch;

 printf ("Press any key to continue");
 getch(); /* will not echo the character */

 printf ("\nType any character");
 ch = getche(); /* will echo the character typed */

 printf ("\nType any character");
 getchar(); /* will echo character, must be followed by enter key */
 printf ("\nContinue Y/N");
 fgetchar(); /* will echo character, must be followed by enter key */
}
```



```
 return 0 ;
}
```

And here is a sample run of this program...

```
Press any key to continue
Type any character B
Type any character W
Continue Y/N Y
```

**putch( )** and **putchar( )** form the other side of the coin. They print a character on the screen. As far as the working of **putch( )** **putchar( )** and **fputchar( )** is concerned, it's exactly same. The following program illustrates this:

```
include <stdio.h>
include <conio.h>
int main()
{
 char ch = 'A' ;

 putch (ch) ;
 putchar (ch) ;
 fputchar (ch) ;
 putch ('Z') ;
 putchar ('Z') ;
 fputchar ('Z') ;
 return 0 ;
}
```

And here is the output...

```
AAAZZZ
```

The limitation of **putch( )**, **putchar( )** and **fputchar( )** is that they can output only one character at a time.

### **gets( ) and puts( )**

**gets( )** receives a string from the keyboard. Why is it needed? Because **scanf( )** function has some limitations while receiving string of characters, as the following example illustrates:

```
include <stdio.h>
```

```
int main()
{
 char name[50];

 printf ("Enter name ");
 scanf ("%s", name);
 printf ("%s\n", name);
 return 0 ;
}
```

And here is the output...

```
Enter name Jonty Rhodes
Jonty
```

Surprised? Where did “Rhodes” go? It never got stored in the array **name[ ]**, because the moment the blank was typed after “Jonty”, **scanf( )** assumed that the name being entered has ended. The result is that there is no way (at least not without a lot of trouble on the programmer’s part) to enter a multi-word string into a single variable (**name** in this case) using **scanf( )**. The solution to this problem is to use **gets( )** function. As said earlier, it gets a string from the keyboard. It is terminated when an Enter key is hit. Thus, spaces and tabs are perfectly acceptable as part of the input string. More exactly, **gets( )** function gets a newline (**\n**) terminated string of characters from the keyboard and replaces the **\n** with a **\0**.

The **puts( )** function works exactly opposite to **gets( )** function. It outputs a string to the screen.

Here is a program which illustrates the usage of these functions.

```
include <stdio.h>
int main()
{
 char footballer[40];

 puts ("Enter name");
 gets (footballer); /* sends base address of array */
 puts ("Happy footballing!");
 puts (footballer);
 return 0 ;
}
```

Following is the sample output:

```
Enter name
Jonty Rhodes
Happy footballing!
Jonty Rhodes
```

Why did we use two **puts( )** functions to print “Happy footballing!” and “Jonty Rhodes”? Because, unlike **printf( )**, **puts( )** can output only one string at a time. If we attempt to print two strings using **puts( )**, only the first one gets printed. Similarly, unlike **scanf( )**, **gets( )** can be used to read only one string at a time.

## Summary

- (a) There is no keyword available in C for doing input/output.
- (b) All I/O in C is done using standard library functions.
- (c) Apart from **printf( )** and **scanf( )**, there are several functions available for performing console input/output.
- (d) The formatted console I/O functions can force the user to receive the input in a fixed format and display the output in a fixed format.
- (e) There are several format specifiers and escape sequences available to format input and output.
- (f) Unformatted console I/O functions work faster since they do not have the overheads of formatting the input or output.

## Exercise

**[A]** What will be the output of the following programs:

- (a) 

```
#include <stdio.h>
#include <ctype.h>
int main()
{
 char ch ;
 ch = getchar() ;
 if (islower (ch))
 putchar (toupper (ch)) ;
 else
 putchar (tolower (ch)) ;
```

```
 return 0 ;
}
```

(b) #include <stdio.h>

```
int main()
{
 int i = 2 ;
 float f = 2.5367 ;
 char str[] = "Life is like that" ;

 printf ("%4d\t%3.3f\t%4s\n", i, f, str) ;
 return 0 ;
}
```

(c) #include <stdio.h>

```
int main()
{
 printf ("More often than \b\b not \rthe person who \\
 wins is the one who thinks he can!\n") ;
 return 0 ;
}
```

(d) #include <conio.h>

```
char p[] = "The sixth sick sheikh's sixth ship is sick" ;
int main()
{
 int i = 0 ;
 while (p[i] != '\0')
 {
 putchar (p[i]) ;
 i++ ;
 }
 return 0 ;
}
```

**[B]** Point out the errors, if any, in the following programs:

(a) #include <stdio.h>

```
int main()
{
 int i ;
 char a[] = "Hello" ;
 while (a != '\0')
 {
```

```
 printf ("%c", *a);
 a++;
 }
 return 0 ;
}
```

- (b) # include <stdio.h>  
int main( )  
{  
 double dval ;  
 scanf ( "%f", &dval ) ;  
 printf ( "Double Value = %lf\n", dval ) ;  
 return 0 ;  
}
- (c) # include <stdio.h>  
int main( )  
{  
 int ival ;  
 scanf ( "%d\n", &n ) ;  
 printf ( "Integer Value = %d\n", ival ) ;  
 return 0 ;  
}
- (d) # include <stdio.h>  
int main( )  
{  
 char \*mess[ 5 ] ;  
 int i ;  
 for ( i = 0 ; i < 5 ; i++ )  
 scanf ( "%s", mess[ i ] ) ;  
 return 0 ;  
}
- (e) # include <stdio.h>  
int main( )  
{  
 int dd, mm, yy ;  
 printf ( "Enter day, month and year\n" ) ;  
 scanf ( "%d%c%d%c%d", &dd, &mm, &yy ) ;  
 printf ( "The date is: %d - %d - %d\n", dd, mm, yy ) ;  
 return 0 ;  
}

- ```
(f) #include <stdio.h>
int main( )
{
    char text ;
    sprintf ( text, "%4d\t%2.2f\n%s", 12, 3.452, "Merry Go Round" ) ;
    printf ( "%s\n", text ) ;
    return 0 ;
}

(g) #include <stdio.h>
int main( )
{
    char buffer[ 50 ] ;
    int no = 97;
    double val = 2.34174 ;
    char name[ 10 ] = "Shweta" ;

    sprintf ( buffer, "%d %lf %s", no, val, name ) ;
    printf ( "%s\n", buffer ) ;
    sscanf ( buffer, "%4d %2.2lf %s", &no, &val, name ) ;
    printf ( "%s\n", buffer ) ;
    printf ( "%d %lf %s\n", no, val, name ) ;
    return 0 ;
}
```

[C] Answer the following:

- (a) To receive the string "We have got the guts, you get the glory!!" in an array **char str[100]** which of the following functions would you use?
1. scanf ("%s", str) ;
 2. gets (str) ;
 3. getche (str) ;
 4. fgetchar (str) ;
- (b) Which function would you use if a single key were to be received through the keyboard?
1. scanf()
 2. gets()
 3. getche()
 4. getchar()
- (c) If an integer is to be entered through the keyboard, which function would you use?

1. `scanf( )`
 2. `gets( )`
 3. `getche( )`
 4. `getchar( )`
- (d) What is the difference between **`getchar( )`**, **`fgetchar( )`**, **`getch( )`** and **`getche( )`**?
- (e) Which of the following can a format string of a **`printf( )`** function contain:
1. Characters, format specifications and escape sequences
 2. Character, integers and floats
 3. Strings, integers and escape sequences
 4. Inverted commas, percentage sign and backslash character
- (f) The purpose of the field-width specifier in a **`printf( )`** function is to:
1. Control the margins of the program listing
 2. Specify the maximum value of a number
 3. Control the size of font used to print numbers
 4. Specify how many columns would be used to print the number

[D] Answer the following:

- (a) Define two functions **`xgets( )`** and **`xputs( )`** which work similar to the standard library functions **`gets( )`** and **`puts( )`**.
- (b) Define a function **`getint( )`**, which would receive a numeric string from the keyboard, convert it to an integer number and return the integer to the calling function. A sample usage of **`getint( )`** is shown below.

```
#include <stdio.h>
int main( )
{
    int a ;
    a = getint( ) ;
    printf ( "you entered %d\n", a ) ;
    return 0 ;
}
```

- (c) Define a function **`getfloat( )`**, which would receive a numeric string from the keyboard, convert it to a float value and return the float to the calling function.

- (d) If we are to display the following output properly aligned which format specifiers would you use?

Discovery of India	Jawaharlal Nehru	425.50
My Experiments with Truth	Mahatma Gandhi	375.50
Sunny Days	Sunil Gavaskar	95.50
One More Over	Erapalli Prasanna	85.00

- **Data Organization**
- **File Operations**
 - Opening a File
 - Reading from a File
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 - Closing the File
- **Counting Characters, Tabs, Spaces, ...**
- **A File-Copy Program**
 - Writing to a File
- **File Opening Modes**
- **String (Line) I/O in Files**
 - The Awkward Newline
- **Record I/O in Files**
- **Text Files and Binary Files**
- **Record I/O Revisited**
- **Database Management**
- **Low-Level File I/O**
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- **I/O under Windows**
- **Summary**
- **Exercise**



19 File Input/Output

- Data Organization
- File Operations
 - Opening a File
 - Reading from a File
 - Trouble in Opening a File
 - Closing the File
- Counting Characters, Tabs, Spaces, ...
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- I/O Under Windows
- Summary
- Exercise

Often it is not enough to just display the data on the screen. This is because if the data is large, only a limited amount of it can be stored in memory and only a limited amount of it can be displayed on the screen. It would be inappropriate to store this data in memory for one more reason. Memory is volatile and its contents would be lost once the program is terminated. So if we need the same data again it would have to be either entered through the keyboard again or would have to be regenerated programmatically. Obviously both these operations would be tedious. At such times it becomes necessary to store the data in a manner that can be later retrieved and displayed either in part or in whole. This medium is usually a 'file' on the disk. This chapter discusses how file I/O (Input/Output) operations can be performed.

Data Organization

Before we start doing file input/output let us first find out how data is organized on the disk. All data stored on the disk is in binary form. How this binary data is stored on the disk varies from one OS to another. However, this does not affect the C programmer since he has to use only the library functions written for the particular OS to be able to perform input/output. It is the compiler vendor's responsibility to correctly implement these library functions by taking the help of OS. This is illustrated in Figure 19.1.

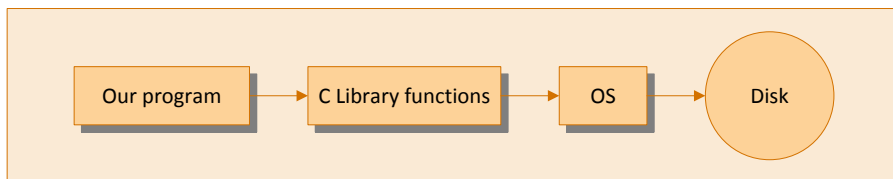


Figure 19.1

File Operations

There are different operations that can be carried out on a file. These are:

- (a) Creation of a new file
- (b) Opening an existing file
- (c) Reading from a file
- (d) Writing to a file
- (e) Moving to a specific location in a file (seeking)
- (f) Closing a file

Let us now write a program to read a file and display its contents on the screen. We will first list the program and show what it does, and then dissect it line-by-line. Here is the listing...

```
/* Display contents of a file on screen. */
#include <stdio.h>
int main( )
{
    FILE *fp ;
    char ch ;

    fp = fopen ( "PR1.C", "r" ) ;
    while ( 1 )
    {
        ch = fgetc ( fp ) ;
        if ( ch == EOF )
            break ;

        printf ( "%c", ch ) ;
    }
    printf ( "\n" ) ;
    fclose ( fp ) ;
    return 0 ;
}
```

On execution of this program it displays the contents of the file 'PR1.C' on the screen. Let us now understand how it does the same.

Opening a File

Before we can read (or write) information from (to) a file on a disk we must open the file. To open the file we have called the function **fopen()**. It would open a file "PR1.C" in 'read' mode, which tells the C compiler that we would be reading the contents of the file. Note that "r" is a string and not a character; hence the double quotes and not single quotes. In fact **fopen()** performs three important tasks when you open the file in "r" mode:

- (a) Firstly it searches on the disk the file to be opened.
- (b) Then it loads the file from the disk into a place in memory called buffer.
- (c) It sets up a character pointer that points to the first character of the buffer.

Why do we need a buffer at all? Imagine how inefficient it would be to actually access the disk every time we want to read a character from it. Every time we read something from a disk, it takes some time for the disk drive to position the read/write head correctly. On a floppy disk system, the drive motor has to actually start rotating the disk from a standstill position every time the disk is accessed. If this were to be done for every character we read from the disk, it would take a long time to complete the reading operation. This is where a buffer comes in. It would be more sensible to read the contents of the file into the buffer while opening the file and then read the file character by character from the buffer rather than from the disk. This is shown in Figure 19.2.

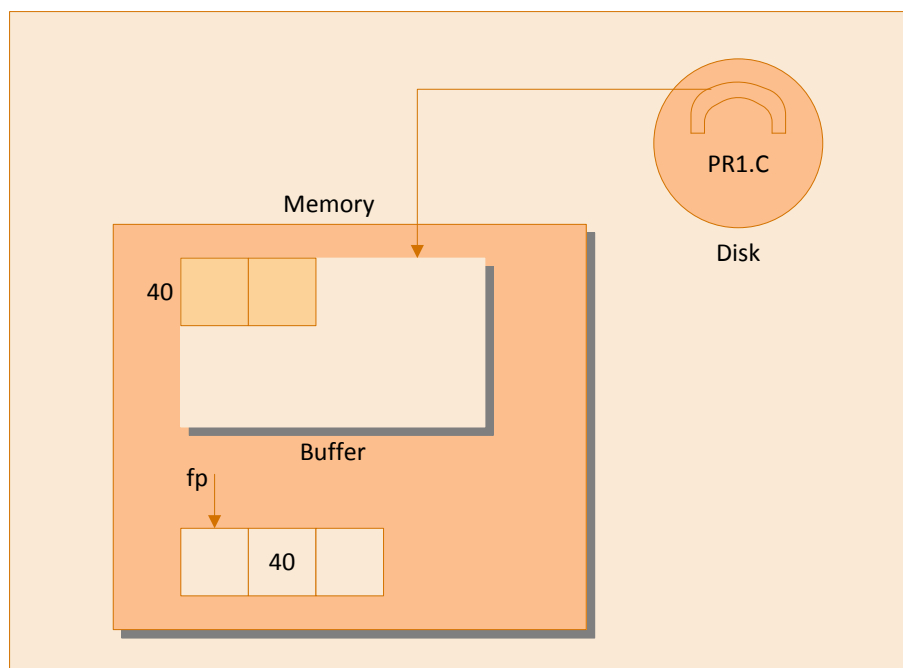


Figure 19.2

Same argument also applies to writing information in a file. Instead of writing characters in the file on the disk one character at a time it would be more efficient to write characters in a buffer and then finally transfer the contents from the buffer to the disk.

To be able to successfully read from a file information like mode of opening, size of file, place in the file from where the next read operation would be performed, etc., has to be maintained. Since all this information is inter-related, all of it is gathered together by **fopen()** in a structure called **FILE**. **fopen()** returns the address of this structure,

which we have collected in the structure pointer called **fp**. We have declared **fp** as follows:

```
FILE *fp ;
```

The **FILE** structure has been defined in the header file “stdio.h” (standing for standard input/output header file). Therefore, it is necessary to **#include** this file.

Reading from a File

Once the file has been opened for reading using **fopen()**, as we have seen, the file’s contents are brought into buffer (partly or wholly) and a pointer is set up that points to the first character in the buffer. This pointer is one of the elements of the structure to which **fp** is pointing (refer Figure 19.2).

To read the file’s contents from memory, there exists a function called **fgetc()**. This has been used in our program as,

```
ch = fgetc ( fp ) ;
```

fgetc() reads the character from the current pointer position, advances the pointer position so that it now points to the next character, and returns the character that is read, which we collected in the variable **ch**. Note that once the file has been opened, we no longer refer to the file by its name, but through the file pointer **fp**.

We have used the function **fgetc()** within an indefinite **while** loop. There has to be a way to break out of this **while**. When shall we break out... when all the characters from the file have been read. But how would we know this? Well, **fgetc()** returns a macro EOF (End of File) once all the characters have been read and we attempt to read one more character. The EOF macro has been defined in the file “stdio.h”. In place of the function **fgetc()**, we could have as well used the macro **getc()** with the same effect.

In our program we go on reading each character from the file till end of file is not met. As each character is read, we display it on the screen. Once out of the loop, we close the file.

Trouble in Opening a File

There is a possibility that when we try to open a file using the function **fopen()**, the file may not be opened. While opening the file in “r” mode,

this may happen because the file being opened may not be present on the disk at all. And you obviously cannot read a file that doesn't exist. Similarly, while opening the file for writing, **fopen()** may fail due to a number of reasons, like, disk space may be insufficient to create a new file, or the disk may be write protected or the disk is damaged and so on.

Crux of the matter is that it is important for any program that accesses disk files to check whether a file has been opened successfully before trying to read or write to the file. If the file opening fails due to any of the several reasons mentioned above, the **fopen()** function returns a value NULL (defined in "stdio.h" as **#define NULL 0**). Here is how this can be handled in a program...

```
#include <stdio.h>
#include <stdlib.h>
int main( )
{
    FILE *fp ;
    fp = fopen ( "PR1.C", "r" ) ;
    if ( fp == NULL )
    {
        puts ( "cannot open file" ) ;
        exit ( 1 ) ;
    }
    return 0 ;
}
```

The call to the function **exit()** terminates the execution of the program. Usually, a value 0 is passed to **exit()** if the program termination is normal. A non-zero value indicates an abnormal termination of the program. If there are multiple exit points in the program, then the value passed to **exit()** can be used to find out from where the execution of the program got terminated. There are different ways of examining this value in different programming environments. For example, in Visual Studio this value is shown in the Output Window at the bottom. In Turbo C++ this value can be seen through the Compile | Information menu item. The prototype of **exit()** function is declared in the header file **stdlib.h**.

Closing the File

When we have finished reading from the file, we need to close it. This is done using the function **fclose()** through the statement,

```
fclose ( fp ) ;
```

Once we close the file, we can no longer read from it using **getc()** unless we reopen the file. Note that to close the file, we don't use the filename but the file pointer **fp**. On closing the file, the buffer associated with the file is removed from memory.

In this program we have opened the file for reading. Suppose we open a file with an intention to write characters into it. This time too, a buffer would get associated with it. When we attempt to write characters into this file using **fputc()** the characters would get written to the buffer. When we close this file using **fclose()** two operations would be performed:

- (d) The characters in the buffer would be written to the file on the disk.
- (e) The buffer would be eliminated from memory.

You can imagine a possibility when the buffer may become full before we close the file. In such a case the buffer's contents would be written to the disk the moment it becomes full. All this buffer management is done for us by the library functions.

Counting Characters, Tabs, Spaces, ...

Having understood the first file I/O program in detail let us now try our hand at one more. Let us write a program that will read a file and count how many characters, spaces, tabs and newlines are present in it. Here is the program...

```
/* Count chars, spaces, tabs and newlines in a file */
#include <stdio.h>
int main( )
{
    FILE *fp ;
    char ch ;
    int nol = 0, not = 0, nob = 0, noc = 0 ;

    fp = fopen ( "PR1.C", "r" ) ;
    while ( 1 )
    {
```



```
    ch = fgetc ( fp ) ;
    if ( ch == EOF )
        break ;
    noc++ ;
    if ( ch == ' ' )
        nob++ ;
    if ( ch == '\n' )
        nol++ ;
    if ( ch == '\t' )
        not++ ;
}
fclose ( fp ) ;
printf ( "Number of characters = %d\n", noc ) ;
printf ( "Number of blanks = %d\n", nob ) ;
printf ( "Number of tabs = %d\n", not ) ;
printf ( "Number of lines = %d\n", nol ) ;
return 0 ;
}
```

Here is a sample run...

```
Number of characters = 125
Number of blanks = 25
Number of tabs = 13
Number of lines = 22
```

The above statistics are true for a file “PR1.C”, which I had on my disk. You may give any other filename and obtain different results. I believe the program is self-explanatory.

In this program too, we have opened the file for reading and then read it character-by-character. Let us now try a program that needs to open a file for writing.

A File-Copy Program

We have already used the function **fgetc()** which reads characters from a file. Its counterpart is a function called **fputc()** which writes characters to a file. As a practical use of these character I/O functions, we can copy the contents of one file into another, as demonstrated in the following program. This program takes the contents of a file and copies them into another file, character-by-character.

```
# include <stdio.h>
# include <stdlib.h>
int main( )
{
    FILE *fs, *ft ;
    char ch ;

    fs = fopen ( "pr1.c", "r" ) ;
    if ( fs == NULL )
    {
        puts ( "Cannot open source file" ) ;
        exit ( 1 ) ;
    }
    ft = fopen ( "pr2.c", "w" ) ;
    if ( ft == NULL )
    {
        puts ( "Cannot open target file" ) ;
        fclose ( fs ) ;
        exit ( 2 ) ;
    }
    while ( 1 )
    {
        ch = fgetc ( fs ) ;

        if ( ch == EOF )
            break ;
        else
            fputc ( ch, ft ) ;
    }
    fclose ( fs ) ;
    fclose ( ft ) ;
    return 0 ;
}
```

I hope most of the stuff in the program can be easily understood, since it has already been dealt with in the earlier section. What is new is only the function **fputc()**. Let us see how it works.

Writing to a File

The **fputc()** function is similar to the **putch()** function, in the sense that both output characters. However, **putch()** function always writes to the VDU, whereas, **fputc()** writes to the file. Which file? The file signified by **ft**. The writing process continues till all characters from the source file have been written to the target file, following which the **while** loop terminates.

Note that our sample file-copy program is capable of copying only text files. To copy files with extension .EXE or .JPG, we need to open the files in binary mode, a topic that would be dealt with in sufficient detail in a later section.

File Opening Modes

In our first program on file I/O, we have opened the file in read ("r") mode. However, "r" is but one of the several modes in which we can open a file. Following is a list of all possible modes in which a file can be opened. The tasks performed by **fopen()**, when a file is opened in each of these modes, are also mentioned.

"r" Searches file. If the file is opened successfully **fopen()** loads it into memory and sets up a pointer which points to the first character in it. If the file cannot be opened, **fopen()** returns NULL.

Operations possible – reading from the file.

"w" Searches file. If the file exists, its contents are overwritten. If the file doesn't exist, a new file is created. Returns NULL, if unable to open file.

Operations possible – writing to the file.

"a" Searches file. If the file is opened successfully **fopen()** loads it into memory and sets up a pointer that points to the last character in it. If the file doesn't exist, a new file is created. Returns NULL, if unable to open file.

Operations possible - adding new contents at the end of file.

"r+" Searches file. If is opened successfully **fopen()** loads it into memory and sets up a pointer which points to the first character in it. Returns NULL, if unable to open the file.

Operations possible - reading existing contents, writing new contents, modifying existing contents of the file.

"w+" Searches file. If the file exists, its contents are overwritten. If the file doesn't exist, a new file is created. Returns NULL, if unable to open file.

Operations possible - writing new contents, reading them back and modifying existing contents of the file.

"a+" Searches file. If the file is opened successfully **fopen()** loads it into memory and sets up a pointer which points to the first character in it. If the file doesn't exist, a new file is created. Returns NULL, if unable to open file.

Operations possible - reading existing contents, appending new contents to end of file. Cannot modify existing contents.

String (Line) I/O in Files

For many purposes, character I/O is just what is needed. However, in some situations, the usage of functions that read or write entire strings might turn out to be more efficient.

Reading or writing strings of characters from and to files is as easy as reading and writing individual characters. Here is a program that writes strings to a file using the function **fputs()**.

```
/* Receives strings from keyboard and writes them to file */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
int main( )
{
    FILE *fp ;
    char s[ 80 ] ;

    fp = fopen ( "POEM.TXT", "w" ) ;
    if ( fp == NULL )
    {
        puts ( "Cannot open file" ) ;
        exit ( 1 ) ;
    }
}
```

```
printf ( "\nEnter a few lines of text:\n" );  
while ( strlen ( gets ( s ) ) > 0 )  
{  
    fputs ( s, fp );  
    fputs ( "\n", fp );  
}  
fclose ( fp );  
return 0 ;  
}
```

And here is a sample run of the program...

```
Enter a few lines of text:  
Shining and bright, they are forever,  
so true about diamonds,  
more so of memories,  
especially yours!
```

Note that each string is terminated by hitting Enter. To terminate the execution of the program, hit Enter at the beginning of a line. This creates a string of zero length, which the program recognizes as the signal to close the file and exit.

We have set up a character array to receive the string; the **fputs()** function then writes the contents of the array to the disk. Since **fputs()** does not automatically add a newline character to the end of the string, we must do this explicitly to make it easier to read the string back from the file.

Here is a program that reads strings from a disk file.

```
/* Reads strings from the file and displays them on screen */  
# include <stdio.h>  
# include <stdlib.h>  
int main( )  
{  
    FILE *fp ;  
    char s[ 80 ] ;  
  
    fp = fopen ( "POEM.TXT", "r" ) ;  
    if ( fp == NULL )  
    {  
        puts ( "Cannot open file" ) ;  
    }
```

```
        exit ( 1 ) ;  
    }  
  
    while ( fgets ( s, 79, fp ) != NULL )  
        printf ( "%s", s ) ;  
  
    printf ( "\n" ) ;  
    fclose ( fp ) ;  
    return 0 ;  
}
```

And here is the output...

```
Shining and bright, they are forever,  
so true about diamonds,  
more so of memories,  
especially yours !
```

The function **fgets()** takes three arguments. The first is the address where the string is stored, and the second is the maximum length of the string. This argument prevents **fgets()** from reading in too long a string and overflowing the array. The third argument, as usual, is the pointer to the structure **FILE**. On reading a line from the file, the string **s** would contain the line contents a `\n` followed by a `\0`. Thus the string is terminated by **fgets()** and we do not have to terminate it specifically. When all the lines from the file have been read, we attempt to read one more line, in which case **fgets()** returns a **NULL**.

The Awkward Newline

We had earlier written a program that counts the total number of characters present in a file. If we use that program to count the number of characters present in the above poem (stored in the file **"POEM.TXT"**), it would give us the character count as 101. The same file if seen in the directory, would be reported to contain 105 characters. You can verify this by running the **DIR** command in the command window (invoked using the **"cmd"** command).

This discrepancy occurs because when we attempt to write a `"\n"` to the file using **fputs()**, **fputs()** converts the `\n` to `\r\n` combination. Here `\r` stands for carriage return and `\n` for linefeed. If we read the same line back using **fgets()** the reverse conversion happens. Thus when we write

the first line of the poem and a “\n” using two calls to **fputs()**, what gets written to the file is

```
Shining and bright, they are forever,\r\n
```

When the same line is read back into the array **s[]** using **fgets()**, the array contains

```
Shining and bright, they are forever,\n\0
```

Thus conversion of **\n** to **\r\n** during writing and **\r\n** conversion to **\n** during reading is a feature of the standard library functions and not that of the OS. Hence the OS counts **\r** and **\n** as separate characters. In our poem there are four lines, therefore there is a discrepancy of four characters (105 - 101).

Record I/O in Files

So far, we have dealt with reading and writing only characters and strings. What if we want to read or write numbers from/to file? Furthermore, what if we desire to read/write a combination of characters, strings and numbers? For this first we would organize this dissimilar data together in a structure and then use **fprintf()** and **fscanf()** library functions to read/write data from/to file. Following program illustrates the use of structures for writing records of employees:

```
/* Writes records to a file using structure */
#include <stdio.h>
#include <conio.h>
int main( )
{
    FILE *fp ;
    char another = 'Y' ;
    struct emp
    {
        char name[ 40 ] ;
        int age ;
        float bs ;
    };
    struct emp e ;

    fp = fopen ( "EMPLOYEE.DAT", "w" ) ;
```

```
if ( fp == NULL )
{
    puts ( "Cannot open file" );
    exit ( 1 );
}

while ( another == 'Y' )
{
    printf ( "\nEnter name, age and basic salary: " );
    scanf ( "%s %d %f", e.name, &e.age, &e.bs );
    fprintf ( fp, "%s %d %f\n", e.name, e.age, e.bs );

    printf ( "Add another record (Y/N) " );
    fflush ( stdin );
    another = getche( );
}

fclose ( fp );
return 0 ;
}
```

And here is the output of the program...

```
Enter name, age and basic salary: Sunil 34 1250.50
Add another record (Y/N) Y
Enter name, age and basic salary: Sameer 21 1300.50
Add another record (Y/N) Y
Enter name, age and basic salary: Rahul 34 1400.55
Add another record (Y/N) N
```

In this program we are just reading the data into a structure variable using **scanf()**, and then dumping it into a disk file using **fprintf()**. The user can input as many records as he/she desires. The procedure ends when the user supplies 'N' for the question 'Add another record (Y/N)'.

The key to this program is the function **fprintf()**, which writes the values in the structure variable to the file. This function is similar to **printf()**, except that a **FILE** pointer is included as the first argument. As in **printf()**, we can format the data in a variety of ways, by using **fprintf()**. In fact, all the format conventions of **printf()** function work with **fprintf()** as well.

Perhaps you are wondering what for have we used the function **fflush()**. The reason is to get rid of a peculiarity of **scanf()**. After supplying data for one employee, we would hit the Enter key. What **scanf()** does is it assigns name, age and salary to appropriate variables and keeps the Enter key unread in the keyboard buffer. So when it's time to supply Y or N for the question 'Another employee (Y/N)', **getch()** will read the Enter key from the buffer thinking that user has entered the Enter key. To avoid this problem, we use the function **fflush()**. It is designed to remove or 'flush out' any data remaining in the buffer. The argument to **fflush()** must be the buffer which we want to flush out. Here we have used 'stdin', which means buffer related with standard input device—keyboard.

Let us now write a program that reads the employee records created by the above program. Here is how it can be done...

```
/* Read records from a file using structure */
#include <stdio.h>
#include <stdlib.h>
int main( )
{
    FILE *fp ;
    struct emp
    {
        char name[ 40 ] ;
        int age ;
        float bs ;
    };
    struct emp e ;

    fp = fopen ( "EMPLOYEE.DAT", "r" ) ;

    if ( fp == NULL )
    {
        puts ( "Cannot open file" ) ;
        exit ( 1 ) ;
    }

    while ( fscanf ( fp, "%s %d %f", e.name, &e.age, &e.bs ) != EOF )
        printf ( "%s %d %f\n", e.name, e.age, e.bs ) ;

    fclose ( fp ) ;
}
```

```
    return 0 ;  
}
```

And here is the output of the program...

```
Sunil 34 1250.500000  
Sameer 21 1300.500000  
Rahul 34 1400.500000
```

Text Files and Binary Files

All the programs that we wrote in this chapter so far worked on text files. Some of them would not work correctly on binary files. A text file contains only textual information like alphabets, digits and special symbols. In actuality the ASCII codes of these characters are stored in text files. A good example of a text file is any C program, say PR1.C.

As against this, a binary file is merely a collection of bytes. This collection might be a compiled version of a C program (say PR1.EXE), or music data stored in a MP3 file or a picture stored in a JPG file. A very easy way to find out whether a file is a text file or a binary file is to open that file in Notepad. If on opening the file you can make out what is displayed then it is a text file, otherwise it is a binary file.

As mentioned while explaining the file-copy program, the program cannot copy binary files successfully. We can improve the same program to make it capable of copying text as well as binary files as shown below.

```
#include <stdio.h>  
#include <stdlib.h>  
int main( )  
{  
    FILE *fs, *ft ;  
    int ch ;  
  
    fs = fopen ( "pr1.exe", "rb" ) ;  
    if ( fs == NULL )  
    {  
        puts ( "Cannot open source file" ) ;  
        exit ( 1 ) ;  
    }  
  
    ft = fopen ( "newpr1.exe", "wb" ) ;
```

```
if ( ft == NULL )
{
    puts ( "Cannot open target file" );
    fclose ( fs );
    exit ( 2 );
}

while ( 1 )
{
    ch = fgetc ( fs );

    if ( ch == EOF )
        break ;
    else
        fputc ( ch, ft );
}

fclose ( fs );
fclose ( ft );
return 0 ;
}
```

Using this program we can comfortably copy text as well as binary files. Note that here we have opened the source and target files in “rb” and “wb” modes, respectively. While opening the file in text mode we can use either “r” or “rt”, but since text mode is the default mode we usually drop the ‘t’.

From the programming angle there are two main areas where text and binary mode files are different. These are:

- (a) Handling of newlines
- (b) Storage of numbers

Let us explore these three differences.

Text versus Binary Mode: Newlines

We have already seen that, in text mode, a newline character is converted into the carriage return-linefeed combination before being written to the disk. Likewise, the carriage return-linefeed combination on the disk is converted back into a newline when the file is read by a C

program. However, if a file is opened in binary mode, as opposed to text mode, these conversions will not take place.

Text versus Binary Mode: Storage of Numbers

The only function that is available for storing numbers in a disk file is the **fprintf()** function. It is important to understand how numerical data is stored on the disk by **fprintf()**. Text and characters are stored one character per byte, as we would expect. Are numbers stored as they are in memory, 4 bytes for an integer, 4 bytes for a float, and so on? No.

Numbers are stored as strings of characters. Thus, 12579, even though it occupies 4 bytes in memory, when transferred to the disk using **fprintf()**, would occupy 5 bytes, 1 byte per character. Similarly, the floating-point number 1234.56 would occupy 7 bytes on disk. Thus, numbers with more digits would require more disk space.

Hence if large amount of numerical data is to be stored in a disk file, using text mode may turn out to be inefficient. The solution is to open the file in binary mode and use those functions (**fread()** and **fwrite()** which are discussed in the next section) which store the numbers in binary format. It means each number would occupy same number of bytes on disk as it occupies in memory.

Record I/O Revisited

The record I/O program that we did in an earlier section has two disadvantages:

- (a) The numbers (basic salary) would occupy more number of bytes, since the file has been opened in text mode. This is because when the file is opened in text mode, each number is stored as a character string.
- (b) If the number of fields in the structure increase (say, by adding address, house rent allowance, etc.), writing structures using **fprintf()**, or reading them using **fscanf()**, becomes quite clumsy.

Let us now see a more efficient way of reading/writing records (structures). This makes use of two functions **fread()** and **fwrite()**. We will write two programs, first one would write records to the file, and the second would read these records from the file and display them on the screen.

```
/* Receives records from keyboard and writes them to a file in binary
   mode */
#include <stdio.h>
#include <stdlib.h>
int main( )
{
    FILE *fp ;
    char another = 'Y' ;
    struct emp
    {
        char name[ 40 ] ;
        int age ;
        float bs ;
    };
    struct emp e ;

    fp = fopen ( "EMP.DAT", "wb" ) ;

    if ( fp == NULL )
    {
        puts ( "Cannot open file" ) ;
        exit ( 1 ) ;
    }

    while ( another == 'Y' )
    {
        printf ( "\nEnter name, age and basic salary: " ) ;
        scanf ( "%s %d %f", e.name, &e.age, &e.bs ) ;
        fwrite ( &e, sizeof ( e ), 1, fp ) ;

        printf ( "Add another record (Y/N) " ) ;
        fflush ( stdin ) ;
        another = getche( ) ;
    }

    fclose ( fp ) ;
    return 0 ;
}
```

And here is the output...

```
Enter name, age and basic salary: Suresh 24 1250.50
Add another record (Y/N) Y
Enter name, age and basic salary: Ranjan 21 1300.60
Add another record (Y/N) Y
Enter name, age and basic salary: Harish 28 1400.70
Add another record (Y/N) N
```

Most of this program is similar to the one that we wrote earlier, which used **fprintf()** instead of **fwrite()**. Note, however, that the file “EMP.DAT” has now been opened in binary mode.

The information obtained from the keyboard about the employee is placed in the structure variable **e**. Then, the following statement writes the structure to the file:

```
fwrite ( &e, sizeof ( e ), 1, fp ) ;
```

Here, the first argument is the address of the structure to be written to the disk.

The second argument is the size of the structure in bytes. Instead of counting the bytes occupied by the structure ourselves, we let the program do it for us by using the **sizeof()** operator. The **sizeof()** operator gives the size of the variable in bytes. This keeps the program unchanged in the event of change in the elements of the structure.

The third argument is the number of such structures that we want to write at one time. In this case, we want to write only one structure at a time. Had we had an array of structures, for example, we might have wanted to write the entire array at once.

The last argument is the pointer to the file we want to write to.

Now, let us write a program to read back the records written to the disk by the previous program.

```
/* Reads records from binary file and displays them on screen */
#include <stdio.h>
#include <stdlib.h>
int main( )
{
    FILE *fp ;
    struct emp
    {
```

```
    char name[ 40 ];
    int age ;
    float bs ;
};
struct emp e ;

fp = fopen ( "EMP.DAT", "rb" );

if ( fp == NULL )
{
    puts ( "Cannot open file" );
    exit ( 1 );
}

while ( fread ( &e, sizeof ( e ), 1, fp ) == 1 )
    printf ( "%s %d %f\n", e.name, e.age, e.bs );

fclose ( fp );
return 0 ;
}
```

Here, the **fread()** function causes the data read from the disk to be placed in the structure variable **e**. The format of **fread()** is same as that of **fwrite()**. The function **fread()** returns the number of records read. Ordinarily, this should correspond to the third argument, the number of records we asked for... 1 in this case. If we have reached the end of file, since **fread()** cannot read anything, it returns a 0. By testing for this situation, we know when to stop reading.

As you can now appreciate, any database management application in C must make use of **fread()** and **fwrite()** functions, since they store numbers more efficiently, and make writing/reading of structures quite easy. Note that even if the number of elements belonging to the structure increases, the format of **fread()** and **fwrite()** remains same.

Database Management

So far we have learnt record I/O in bits and pieces. However, in any serious database management application, we will have to combine all that we have learnt in a proper manner to make sense. I have attempted to do this in the following menu-driven program. There is a provision to Add, Modify, List and Delete records, the operations that are imperative

in any database management. Following comments would help you in understanding the program easily:

- Addition of records must always take place at the end of existing records in the file, much in the same way you would add new records in a register manually.
- Listing records means displaying the existing records on the screen. Naturally, records should be listed from first record to last record.
- While modifying records, first we must ask the user which record he/she intends to modify. Instead of asking the record number to be modified, it would be more meaningful to ask for the name of the employee whose record is to be modified. On modifying the record, the existing record gets overwritten by the new record.
- In deleting records, except for the record to be deleted, rest of the records must first be written to a temporary file, then the original file must be deleted, and the temporary file must be renamed back to original.
- Observe carefully the way the file has been opened, first for reading and writing, and if this fails (the first time you run this program it would certainly fail, because that time the file is not existing), for writing and reading. It is imperative that the file should be opened in binary mode.
- Note that the file is being opened only once and closed only once, which is quite logical.
- **system("cls")** function call clears the contents of the screen and **gotoxy()** places the cursor at appropriate position on the screen. The parameters passed to **gotoxy()** are column number followed by row number.

Given below is the complete listing of the program.

```
/* A menu-driven program for elementary database management */  
#include <stdio.h>  
#include <stdlib.h>  
#include <conio.h>  
#include <string.h>  
#include <system.h>  
void gotoxy ( short int col, short int row );
```



```
int main( )
{
    FILE *fp, *ft ;
    char another, choice ;
    struct emp
    {
        char name[ 40 ] ;
        int age ;
        float bs ;
    };
    struct emp e ;
    char empname[ 40 ] ;
    long int recsize ;

    fp = fopen ( "EMP.DAT", "rb+" ) ;

    if ( fp == NULL )
    {
        fp = fopen ( "EMP.DAT", "wb+" ) ;

        if ( fp == NULL )
        {
            puts ( "Cannot open file" ) ;
            exit ( 1 ) ;
        }
    }

    recsize = sizeof ( e ) ;

    while ( 1 )
    {
        system ( "cls" ) ;
        /* This will work in Visual Studio.
        In Turbo C / C++ use function clrscr( ) */

        gotoxy ( 30, 10 ) ;
        printf ( "1. Add Records" ) ;
        gotoxy ( 30, 12 ) ;
        printf ( "2. List Records" ) ;
        gotoxy ( 30, 14 ) ;
        printf ( "3. Modify Records" ) ;
```

```
gotoxy ( 30, 16 ) ;
printf ( "4. Delete Records" ) ;
gotoxy ( 30, 18 ) ;
printf ( "0. Exit" ) ;
gotoxy ( 30, 20 ) ;
printf ( "Your choice" ) ;

fflush ( stdin ) ;
choice = getche( ) ;
switch ( choice )
{
    case '1' :

        fseek ( fp, 0 , SEEK_END ) ;
        another = 'Y' ;

        while ( another == 'Y' )
        {
            printf ( "\nEnter name, age and basic salary " ) ;
            scanf ( "%s %d %f", e.name, &e.age, &e.bs ) ;
            fwrite ( &e, recsize, 1, fp ) ;
            printf ( "\nAdd another Record (Y/N) " ) ;
            fflush ( stdin ) ;
            another = getche( ) ;
        }

        break ;

    case '2' :

        rewind ( fp ) ;

        while ( fread ( &e, recsize, 1, fp ) == 1 )
            printf ( "\n%s %d %f", e.name, e.age, e.bs ) ;

        break ;

    case '3' :

        another = 'Y' ;
        while ( another == 'Y' )
```

```
        {
            printf ( "\nEnter name of employee to modify " );
            scanf ( "%s", empname );

            rewind ( fp );
            while ( fread ( &e, recsize, 1, fp ) == 1 )
            {
                if ( strcmp ( e.name, empname ) == 0 )
                {
                    printf ( "\nEnter new name, age & bs " );
                    scanf ( "%s %d %f", e.name, &e.age,
                        &e.bs );
                    fseek ( fp, - recsize, SEEK_CUR );
                    fwrite ( &e, recsize, 1, fp );
                    break ;
                }
            }

            printf ( "\nModify another Record (Y/N) " );
            fflush ( stdin );
            another = getche( );
        }

        break ;

    case '4' :

        another = 'Y' ;
        while ( another == 'Y' )
        {
            printf ( "\nEnter name of employee to delete " );
            scanf ( "%s", empname );

            ft = fopen ( "TEMP.DAT", "wb" );

            rewind ( fp );
            while ( fread ( &e, recsize, 1, fp ) == 1 )
            {
                if ( strcmp ( e.name, empname ) != 0 )
                    fwrite ( &e, recsize, 1, ft );
            }
        }
    }
```

```

        fclose ( fp ) ;
        fclose ( ft ) ;
        remove ( "EMP.DAT" ) ;
        rename ( "TEMP.DAT", "EMP.DAT" ) ;

        fp = fopen ( "EMP.DAT", "rb+" ) ;

        printf ( "Delete another Record (Y/N) " ) ;
        fflush ( stdin ) ;
        another = getche ( ) ;
    }
    break ;

    case '0' :
        fclose ( fp ) ;
        exit ( 0 ) ;
    }
}
return 0 ;
}

/* Use this function in Visual Studio */
/* In TC/TC++ use the library function gotoxy( ) declared in "conio.h" */
void gotoxy ( short int col, short int row )
{
    HANDLE h Stdout = GetStdHandle ( STD_OUTPUT_HANDLE ) ;
    COORD position = { col, row } ;
    SetConsoleCursorPosition ( hStdout, position ) ;
}

```

To understand how this program works, you need to be familiar with the concept of pointers. A pointer is initiated whenever we open a file. On opening a file, a pointer is set up which points to the first record in the file. To be precise this pointer is present in the structure to which the file pointer returned by **fopen()** points to. On using the function either **fread()** or **fwrite()**, the pointer moves to the beginning of the next record. On closing a file the pointer is deactivated. Note that the pointer movement is of utmost importance since **fread()** always reads that record where the pointer is currently placed. Similarly, **fwrite()** always writes the record where the pointer is currently placed.

The **rewind()** function places the pointer to the beginning of the file, irrespective of where it is present right now.

The **fseek()** function lets us move the pointer from one record to another. In the program above, to move the pointer to the previous record from its current position, we used the function,

```
fseek ( fp, -recsize, SEEK_CUR );
```

Here, **-recsize** moves the pointer back by **recsize** bytes from the current position. **SEEK_CUR** is a macro defined in “stdio.h”.

Similarly, the following **fseek()** would place the pointer beyond the last record in the file.

```
fseek ( fp, 0, SEEK_END );
```

In fact, **-recsize** or **0** are just the offsets that tell the compiler by how many bytes should the pointer be moved from a particular position. The third argument could be **SEEK_END**, **SEEK_CUR** or **SEEK_SET**. All these act as a reference from which the pointer should be offset. **SEEK_END** means move the pointer from the end of the file, **SEEK_CUR** means move the pointer with reference to its current position and **SEEK_SET** means move the pointer with reference to the beginning of the file.

If we wish to know where the pointer is positioned right now, we can use the function **ftell()**. It returns this position as a **long int** which is an offset from the beginning of the file. The value returned by **ftell()** can be used in subsequent calls to **fseek()**. A sample call to **ftell()** is shown below.

```
position = ftell ( fp );
```

where **position** is a **long int**.

Low-Level File I/O

In low-level File I/O, data cannot be written as individual characters, or as strings or as formatted data. There is only one way data can be written or read in low-level file I/O functions—as a buffer full of bytes.

Writing a buffer full of data resembles the **fwrite()** function. However, unlike **fwrite()**, the programmer must set up the buffer for the data, place the appropriate values in it before writing, and take them out after writing. Thus, the buffer in the low-level I/O functions is very much a

part of the program, rather than being invisible as in high-level file I/O functions.

Low-level file I/O functions offer following advantages:

- (a) Since these functions parallel the methods that the OS uses to write to the disk, they are more efficient than the high-level file I/O functions.
- (b) Since there are fewer layers of routines to go through, low-level I/O functions operate faster than their high-level counterparts.

Let us now write a program that uses low-level file input/output functions.

A Low-Level File-Copy Program

Earlier we had written a program to copy the contents of one file to another. In that program we had read the file character-by-character using **fgetc()**. Each character that was read was written into the target file using **fputc()**. Instead of performing the I/O on a character-by-character basis we can read a chunk of bytes from the source file and then write this chunk into the target file. While doing so the chunk would be read into the buffer and would be written to the file from the buffer. While doing so we would manage the buffer ourselves, rather than relying on the library functions to do so. This is what is low-level about this program. Here is a program which shows how this can be done.

```
/* File-copy program which copies text, .com and .exe files */
#include <fcntl.h>
#include <types.h> /* if present in sys directory use "sys\\types.h" */
#include <stat.h> /* if present in sys directory use "sys\\stat.h" */
#include <stdlib.h>
#include <stdio.h>
int main( )
{
    char buffer[ 512 ], source [ 128 ], target [ 128 ];
    int inhandle, outhandle, bytes ;

    printf ( "\nEnter source file name" ) ;
    gets ( source ) ;

    inhandle = open ( source, O_RDONLY | O_BINARY ) ;
```

```
if ( inhandle == -1 )
{
    puts ( "Cannot open file" ) ;
    exit ( 1 ) ;
}

printf ( "\nEnter target file name" ) ;
gets ( target ) ;

outhandle = open ( target, O_CREAT | O_BINARY | O_WRONLY,
                  S_IWRITE ) ;
if ( outhandle == -1 )
{
    puts ( "Cannot open file" ) ;
    close ( inhandle ) ;
    exit ( 2 ) ;
}

while ( 1 )
{
    bytes = read ( inhandle, buffer, 512 ) ;

    if ( bytes > 0 )
        write ( outhandle, buffer, bytes ) ;
    else
        break ;
}

close ( inhandle ) ;
close ( outhandle ) ;
return 0 ;
}
```

Declaring the Buffer

The first difference that you will notice in this program is that we declare a character buffer,

```
char buffer[ 512 ] ;
```

This is the buffer in which the data read from the file will be placed. The size of this buffer is important for efficient operation. Depending on the

operating system, buffers of certain sizes are handled more efficiently than others.

Opening a File

We have opened two files in our program, one is the source file from which we read the information, and the other is the target file into which we write the information read from the source file.

As in high-level file I/O, the file must be opened before we can access it. This is done using the statement,

```
inhandle = open ( source, O_RDONLY | O_BINARY ) ;
```

We open the file for the same reason as we did earlier—to establish communication with operating system about the file. As usual, we have to supply to **open()**, the filename and the mode in which we want to open the file. The possible file opening modes are given below.

- O_APPEND - Opens a file for appending
- O_CREAT - Creates a new file for writing (has no effect if file already exists)
- O_RDONLY - Opens a new file for reading only
- O_RDWR - Creates a file for both reading and writing
- O_WRONLY - Creates a file for writing only
- O_BINARY - Opens a file in binary mode
- O_TEXT - Opens a file in text mode

These 'O-flags' are defined in the file "fcntl.h". So this file must be included in the program while using low-level file I/O. Note that the file "stdio.h" is not necessary for low-level file I/O. When two or more O-flags are used together, they are combined using the bitwise OR operator (|). Chapter 21 discusses bitwise operators in detail.

The other statement used in our program to open the file is,

```
outhandle = open ( target, O_CREAT | O_BINARY | O_WRONLY,  
                  S_IWRITE ) ;
```

Note that since the target file doesn't exist when it is being opened, we have used the O_CREAT flag, and since we want to write to the file and not read from it, therefore we have used O_WRONLY. And finally, since we want to open the file in binary mode we have used O_BINARY.

Whenever `O_CREAT` flag is used, another argument must be added to **`open( )`** function to indicate the read/write status of the file to be created. This argument is called 'permission argument'. Permission arguments could be any of the following:

- `S_IWRITE` - Writing to the file permitted
- `S_IREAD` - Reading from the file permitted

To use these permissions, both the files "`types.h`" and "`stat.h`" must be **#included** in the program alongwith "`fcntl.h`".

File Handles

Instead of returning a `FILE` pointer as **`fopen( )`** did, in low-level file I/O, **`open( )`** returns an integer value called 'file handle'. This is a number assigned to a particular file, which is used thereafter to refer to the file. If **`open( )`** returns a value of -1, it means that the file couldn't be successfully opened.

Interaction between Buffer and File

The following statement reads the file or as much of it as will fit into the buffer:

```
bytes = read ( inhandle, buffer, 512 );
```

The **`read( )`** function takes three arguments. The first argument is the file handle, the second is the address of the buffer and the third is the maximum number of bytes we want to read.

The **`read( )`** function returns the number of bytes actually read. This is an important number, since it may very well be less than the buffer size (512 bytes), and we will need to know just how full the buffer is before we can do anything with its contents. In our program we have assigned this number to the variable **`bytes`**.

For copying the file, we must use both the **`read( )`** and the **`write( )`** functions in a **`while`** loop. The **`read( )`** function returns the number of bytes actually read. This is assigned to the variable **`bytes`**. This value will be equal to the buffer size (512 bytes) until the end of file, when the buffer may only be partially full. The variable **`bytes`** therefore is used to tell **`write( )`**, as to how many bytes to write from the buffer to the target file.

Note that the buffers are created in the stack, which is of limited size. Hence, when large buffers are used, they must be made global variables, otherwise stack overflow would occur.

I/O under Windows

As said earlier, I/O in C is carried out using functions present in the library that comes with the C compiler targeted for a specific OS. Windows permits several applications to use the same screen simultaneously. Hence there is a possibility that what is written by one application to the console may get overwritten by the output sent by another application to the console. To avoid such situations, Windows has completely abandoned console I/O functions. It uses a separate mechanism to send output to a window representing an application.

Though under Windows, console I/O functions are not used, still functions like **fprintf()**, **fscanf()**, **fread()**, **fwrite()**, **sprintf()**, **sscanf()** work exactly same under Windows as well.

Summary

- (a) File I/O can be performed on a character-by-character basis, a line-by-line basis, a record-by-record basis or a chunk-by-chunk basis.
- (b) Different operations that can be performed on a file are—creation of a new file, opening an existing file, reading from a file, writing to a file, moving to a specific location in a file (seeking) and closing a file.
- (c) File I/O is done using a buffer to improve the efficiency.
- (d) A file can be a text file or a binary file depending upon its contents.
- (e) Library functions convert **\n** to **\r\n** or vice versa while writing/reading to/from a file.
- (f) Many library functions convert a number to a numeric string before writing it to a file, thereby using more space on disk. This can be avoided using functions **fread()** and **fwrite()**.
- (g) In low-level file I/O we can do the buffer management ourselves.

Exercise

[A] Point out the errors, if any, in the following programs:

```
(a) # include <stdio.h>
    void openfile ( char *, FILE ** );
    int main( )
    {
        FILE *fp ;
        openfile ( "Myfile.txt", fp ) ;
        if ( fp == NULL )
            printf ( "Unable to open file...\n" ) ;
        return 0 ;
    }
    void openfile ( char *fn, FILE **f )
    {
        *f = fopen ( fn, "r" ) ;
    }
```

```
(b) # include <stdio.h>
    # include <stdlib.h>
    int main( )
    {
        FILE *fp ;
        char c ;
        fp = fopen ( "TRY.C" ,"r" ) ;
        if ( fp == null )
        {
            puts ( "Cannot open file\n" ) ;
            exit( ) ;
        }
        while ( ( c = getc ( fp ) ) != EOF )
            putchar ( c ) ;
        fclose ( fp ) ;
        return 0 ;
    }
```

```
(c) # include <stdio.h>
    int main( )
    {
        char fname[ ] = "c:\\students.dat" ;
        FILE *fp ;
```

```
        fp = fopen ( fname, "tr" ) ;
        if ( fp == NULL )
            printf ( "Unable to open file...\n" ) ;
        return 0 ;
    }

(d) # include <stdio.h>
    int main( )
    {
        FILE *fp ;
        char str[ 80 ] ;
        fp = fopen ( "TRY.C", "r" ) ;
        while ( fgets ( str, 80, fp ) != EOF )
            fputs ( str ) ;
        fclose ( fp ) ;
        return 0 ;
    }

(e) # include <stdio.h>
    int main( )
    {
        unsigned char ;
        FILE *fp ;

        fp = fopen ( "trial", 'r' ) ;
        while ( ( ch = getc ( fp ) ) != EOF )
            printf ( "%c", ch ) ;

        fclose ( *fp ) ;
        return 0 ;
    }

(f) # include <stdio.h>
    int main( )
    {
        FILE *fp ;
        char names[ 20 ] ;
        int i ;
        fp = fopen ( "students.dat", "wb" ) ;
        for ( i = 0 ; i <= 10 ; i++ )
        {
            puts ( "\nEnter name: " ) ;
            gets ( name ) ;
```

```
        fwrite ( name, size of ( name ), 1, fp );
    }
    close ( fp );
    return 0 ;
}
```

- (g) # include <stdio.h>
int main()
{
 FILE *fp ;
 char name[20] = "Ajay" ;
 int i ;
 fp = fopen ("students.dat", "r") ;
 for (i = 0 ; i <= 10 ; i++)
 fwrite (name, sizeof (name), 1, fp) ;
 close (fp) ;
 return 0 ;
}
- (h) # include <fcntl.h>
include <stdio.h>
int main()
{
 int fp ;
 fp = open ("pr22.c" , "r") ;
 if (fp == -1)
 puts ("cannot open file\n") ;
 else
 close (fp) ;
 return 0 ;
}
- (i) # include <stdio.h>
int main()
{
 int fp ;
 fp = fopen ("students.dat", READ | BINARY) ;
 if (fp == -1)
 puts ("cannot open file\n") ;
 else
 close (fp) ;
 return 0 ;
}

[B] Answer the following:

- (a) The FILE structure is defined in which of the following files:
1. `stdlib.h`
 2. `stdio.c`
 3. `io.h`
 4. `stdio.h`
- (b) If a file contains the line "I am a boy\r\n" then on reading this line into the array `str[ ]` using `fgets( )` what would `str[ ]` contain?
1. I am a boy\r\n\0
 2. I am a boy\r\0
 3. I am a boy\n\0
 4. I am a boy
- (c) State True or False:
1. The disadvantage of high-level file I/O functions is that the programmer has to manage the file buffers.
 2. If a file is opened for reading, it is necessary that the file must exist.
 3. If a file opened for writing already exists, its contents would be overwritten.
 4. For opening a file in append mode it is necessary that the file should exist.
- (d) On opening a file for reading which of the following activities are performed:
1. The disk is searched for existence of the file.
 2. The file is brought into memory.
 3. A pointer is set up which points to the first character in the file.
 4. All the above.
- (e) Is it necessary that a file created in text mode must always be opened in text mode for subsequent operations?
- (f) While using the statement,
- ```
fp = fopen ("myfile.c", "r");
```
- what happens if,
- 'myfile.c' does not exist on the disk
  - 'myfile.c' exists on the disk

- (g) What is the purpose of the library function **fflush( )**?
- (h) While using the statement,  

```
fp = fopen ("myfile.c", "wb") ;
```

what happens if,
  - 'myfile.c' does not exist on the disk
  - 'myfile.c' exists on the disk
- (i) A floating-point array contains percentage marks obtained by students in an examination. To store these marks in a file 'marks.dat', in which mode would you open the file and why?

**[C] Attempt the following:**

- (a) Write a program to read a file and display its contents along with line numbers before each line.
- (b) Write a program to append the contents of one file at the end of another.
- (c) Suppose a file contains student's records with each record containing name and age of a student. Write a program to read these records and display them in sorted order by name.
- (d) Write a program to copy contents of one file to another. While doing so replace all lowercase characters to their equivalent uppercase characters.
- (e) Write a program that merges lines alternately from two files and writes the results to new file. If one file has less number of lines than the other, the remaining lines from the larger file should be simply copied into the target file.
- (f) Write a program to display the contents of a text file on the screen. Make following provisions:  
  
Display the contents inside a box drawn with opposite corner co-ordinates being (0, 1) and (79, 23). Display the name of the file whose contents are being displayed, and the page numbers in the zeroth row. The moment one screenful of file has been displayed, flash a message 'Press any key...' in 24<sup>th</sup> row. When a key is hit, the next page's contents should be displayed, and so on till the end of file.
- (g) Write a program to encrypt/decrypt a file using:

- (1) An offset cipher: In an offset cipher each character from the source file is offset with a fixed value and then written to the target file.

For example, if character read from the source file is 'A', then convert this into a new character by offsetting 'A' by a fixed value, say 128, and then writing the new character to the target file.

- (2) A substitution cipher: In this each character read from the source file is substituted by a corresponding predetermined character and this character is written to the target file.

For example, if character 'A' is read from the source file, and if we have decided that every 'A' is to be substituted by '!', then a '!' would be written to the target file in place of every 'A'. Similarly, every 'B' would be substituted by '5' and so on.

- (h) In the file 'CUSTOMER.DAT' there are 100 records with the following structure:

```
struct customer
{
 int accno ;
 char name[30] ;
 float balance ;
};
```

In another file 'TRANSACTIONS.DAT' there are several records with the following structure:

```
struct trans
{
 int accno ;
 char trans_type ;
 float amount ;
};
```

The element **trans\_type** contains D/W indicating deposit or withdrawal of amount. Write a program to update 'CUSTOMER.DAT' file, i.e., if the **trans\_type** is 'D' then update the **balance** of 'CUSTOMER.DAT' by adding **amount** to balance for the corresponding **accno**. Similarly, if **trans\_type** is 'W' then subtract the **amount** from **balance**. However, while subtracting the amount



ensure that the amount should not get overdrawn, i.e., at least 100 Rs. Should remain in the account.

- (i) There are 100 records present in a file with the following structure:

```
struct date
{
 int d, m, y ;
};

struct employee
{
 int empcode[6] ;
 char empname[20] ;
 struct date join_date ;
 float salary ;
};
```

Write a program to read these records, arrange them in ascending order by **join\_date** and write them to a target file.

- (j) A hospital keeps a file of blood donors in which each record has the format:

Name: 20 Columns  
Address: 40 Columns  
Age: 2 Columns  
Blood Type: 1 Column (Type 1, 2, 3 or 4)

Write a program to read the file and print a list of all blood donors whose age is below 25 and whose blood type is 2.

- (k) Given a list of names of students in a class, write a program to store the names in a file on disk. Make a provision to display the  $n^{\text{th}}$  name in the list ( $n$  is data to be read) and to display all names starting with S.

- (l) Assume that a Master file contains two fields, roll number and name of the student. At the end of the year, a set of students join the class and another set leaves. A Transaction file contains the roll numbers and an appropriate code to add or delete a student.

Write a program to create another file that contains the updated list of names and roll numbers. Assume that the Master file and the Transaction file are arranged in ascending order by roll numbers. The updated file should also be in ascending order by roll numbers.

- (m) In a small firm employee numbers are given in serial numerical order, that is 1, 2, 3, etc.
- Create a file of employee data with following information: employee number, name, sex, gross salary.
  - If more employees join, append their data to the file.
  - If an employee with serial number 25 (say) leaves, delete the record by making gross salary 0.
  - If some employee's gross salary increases, retrieve the record and update the salary.

Write a program to implement the above operations.

- (n) Given a text file, write a program to create another text file deleting the words "a", "the", "an" and replacing each one of them with a blank space.
- (o) You are given a data file EMPLOYEE.DAT with the following record structure:

```
struct employee
{
 int empno ;
 char name[30] ;
 int basic, grade ;
};
```

Every employee has a unique **empno** and there are supposed to be no gaps between employee numbers. Records are entered into the data file in ascending order of employee number. It is intended to check whether there are missing employee numbers. Write a program to read the data file records sequentially and display the list of missing employee numbers.

- (p) Write a program to carry out the following operations:
- Read a text file "TRIAL.TXT" consisting of a maximum of 50 lines of text, each line with a maximum of 80 characters.
  - Count and display the number of words contained in the file.
  - Display the total number of four letter words in the text file.

Assume that the end of a word may be a space, comma or a full stop followed by one or more spaces or a newline character.

- (q) Write a program to read a list of words from a file, sort the words in alphabetical order and display them one word per line. Also give the total number of words in the list. Output format should be:

Total Number of words in the list is \_\_\_\_\_

Alphabetical listing of words is:

.....

.....

.....

Assume the end of the list is indicated by **ZZZZZZ** and there are maximum of 25 words in the Text file.

(r) Write a program to carry out the following:

(a) Read a text file 'INPUT.TXT'

(b) Print each word in reverse order

Example:

Input: INDIA IS MY COUNTRY

Output: AIDNI SI YM YRTNUOC

Assume that maximum word length is 10 characters and words are separated by newline/blank characters.

(s) Write a C program to read a large text file 'NOTES.TXT' and print it on the printer in cut-sheets, introducing page breaks at the end of every 50 lines and a pause message on the screen at the end of every page for the user to change the paper.

- Using *argc* and *argv*
- Detecting Errors in Reading/Writing
- Standard I/O Devices
- I/O Redirection
  - Redirecting the Output
  - Redirecting the Input
  - Both Ways at Once
- Summary
- Exercise



---

## ***20 More Issues In Input/Output***

---

- Using *argc* and *argv*
- Detecting Errors in Reading/Writing
- Standard I/O Devices
- I/O Redirection
  - Redirecting the Output
  - Redirecting the Input
  - Both Ways at Once
- Summary
- Exercise

In Chapters 18 and 19 we saw how Console I/O and File I/O operations are done in C. There are still some more issues related with input/output that remain to be understood. These issues help in making the I/O operations more elegant.

### Using *argc* and *argv*

To execute the file-copy programs that we saw in Chapter 18, we are required to first type the program, compile it, and then execute it. This program can be improved in two ways:

- (a) There should be no need to compile the program every time to use the file-copy utility. It means the program must be executable at command prompt (C:\> if you are using command window, Start | Run dialog if you are using Windows and \$ prompt if you are using UNIX).
- (b) Instead of the program prompting us to enter the source and target filenames, we must be able to supply them at command prompt, in the form:

```
filecopy PR1.C PR2.C
```

where, PR1.C is the source filename and PR2.C is the target filename.

The first improvement is simple. In Visual Studio the executable file (the one which can be executed at command prompt and has an extension .EXE) can be created by using F7 to compile the program. In Turbo C/C++ the same can be done using F9. In UNIX this is not required since in UNIX, every time we compile a program, we always get an executable file.

The second improvement is possible by passing the source filename and target filename to the function **main( )**. This is illustrated in the program given below.

```
#include <stdio.h>
#include <stdlib.h>
int main (int argc, char *argv[])
{
 FILE *fs, *ft ;
 char ch ;
```

```
if (argc != 3)
{
 puts ("Improper number of arguments\n");
 exit (1);
}

fs = fopen (argv[1], "r");
if (fs == NULL)
{
 puts ("Cannot open source file\n");
 exit (2);
}

ft = fopen (argv[2], "w");
if (ft == NULL)
{
 puts ("Cannot open target file\n");
 fclose (fs);
 exit (3);
}

while (1)
{
 ch = fgetc (fs);

 if (ch == EOF)
 break ;
 else
 fputc (ch, ft);
}

fclose (fs);
fclose (ft);
return 0 ;
}
```

The arguments that we pass on to **main( )** at the command prompt are called command-line arguments. The function **main( )** can have two arguments, traditionally named as **argc** and **argv**. Out of these, **argv** is an array of pointers to strings and **argc** is an **int** whose value is equal to the number of strings to which **argv** points. When the program is executed, the strings on the command line are passed to **main( )**. More precisely,

the strings at the command line are stored in memory and address of the first string is stored in **argv[ 0 ]**, address of the second string is stored in **argv[ 1 ]** and so on. The argument **argc** is set to the number of strings given on the command line. For example, in our sample program, if at the command prompt we give,

```
filecopy PR1.C PR2.C
```

then,

```
argc would contain 3
argv[0] would contain base address of the string "filecopy"
argv[1] would contain base address of the string "PR1.C"
argv[2] would contain base address of the string "PR2.C"
```

Whenever we pass arguments to **main( )**, it is a good habit to check whether the correct number of arguments have been passed on to **main( )** or not. In our program this has been done through,

```
if (argc != 3)
{
 puts ("Improper number of arguments\n");
 exit (1);
}
```

Rest of the program is same as the earlier file-copy program. This program is better than the earlier file-copy program on two counts:

- (a) There is no need to recompile the program every time we want to use this utility. It can be executed at command prompt.
- (b) We are able to pass source file name and target file name to **main( )**, and utilize them in **main( )**.

One final comment... the **while** loop that we have used in our program can be written in a more compact form, as shown below.

```
while ((ch = fgetc (fs)) != EOF)
 fputc (ch, ft);
```

This avoids the usage of an indefinite loop and a **break** statement to come out of this loop. Here, first **fgetc ( fs )** gets the character from the file, assigns it to the variable **ch**, and then **ch** is compared against **EOF**. Remember that it is necessary to put the expression



```
ch = fgetc (fs)
```

within a pair of parentheses, so that first the character read is assigned to variable **ch** and then it is compared with **EOF**.

There is one more way of writing the **while** loop. It is shown below.

```
while (!feof (fs))
{
 ch = fgetc (fs) ;
 fputc (ch, ft) ;
}
```

Here, **feof( )** is a macro which returns a 0 if end of file is not reached. Hence we use the **!** operator to negate this 0 to the truth value. When the end of file is reached, **feof( )** returns a non-zero value, **!** makes it 0 and since now the condition evaluates to false, the **while** loop gets terminated.

Note that in each one of them, the following three methods for opening a file are same, since in each one of them, essentially a base address of the string (pointer to a string) is being passed to **fopen( )**.

```
fs = fopen ("PR1.C" , "r") ;
fs = fopen (filename, "r") ;
fs = fopen (argv[1] , "r") ;
```

## Detecting Errors in Reading/Writing

Not at all times when we perform a read or write operation on a file, are we successful in doing so. Naturally there must be a provision to test whether our attempt to read/write was successful or not.

The standard library function **ferror( )** reports any error that might have occurred during a read/write operation on a file. It returns a zero if the read/write is successful and a non-zero value in case of a failure. The following program illustrates the usage of **ferror( )**:

```
include <stdio.h>
int main()
{
 FILE *fp ;
 char ch ;
```

```
fp = fopen ("TRIAL", "w");

while (!feof (fp))
{
 ch = fgetc (fp);
 if (ferror())
 {
 printf ("Error in reading file\n");
 break ;
 }
 else
 printf ("%c", ch);
}

fclose (fp);
return 0 ;
}
```

In this program, the **fgetc( )** function would obviously fail first time around since the file has been opened for writing, whereas **fgetc( )** is attempting to read from the file. The moment the error occurs, **ferror( )** returns a non-zero value and the **if** block gets executed. Instead of printing the error message using **printf( )**, we can use the standard library function **perror( )** which prints the error message specified by the compiler. Thus in the above program the **perror( )** function can be used as shown below.

```
if (ferror())
{
 perror ("TRIAL");
 break ;
}
```

Note that when the error occurs, the error message that is displayed is:

TRIAL: Permission denied

This means we can precede the system error message with any message of our choice. In our program, we have just displayed the filename in place of the error message.

## Standard I/O Devices

To perform reading or writing operations on a file, we need to use the function **fopen( )**, which sets up a file pointer to refer to this file. Most OSs also predefine pointers for three standard files. To access these pointers, we need not use **fopen( )**. These standard file pointers are shown in Figure 20.1

| Standard File pointer | Description                      |
|-----------------------|----------------------------------|
| stdin                 | Standard input device (Keyboard) |
| stdout                | Standard output device (Monitor) |
| stderr                | Standard error device (Monitor)  |

Figure 20.1

Thus the statement **ch = fgetc ( stdin )** would read a character from the keyboard rather than from a file. We can use this statement without any need to use **fopen( )** or **fclose( )** function calls.

## I/O Redirection

Most operating systems incorporate a powerful feature that allows a program to read and write files, even when such a capability has not been incorporated in the program. This is done through a process called 'redirection'.

Normally a C program receives its input from the standard input device, which is assumed to be the keyboard, and sends its output to the standard output device, which is assumed to be the VDU. In other words, the OS makes certain assumptions about where input should come from and where output should go. Redirection permits us to change these assumptions.

For example, using redirection the output of the program that normally goes to the VDU can be sent to the disk or the printer without really making a provision for it in the program. This is often a more convenient and flexible approach than providing a separate function in the program to write to the disk or printer. Similarly, redirection can be used to read information from disk file directly into a program, instead of receiving the input from keyboard.

To use redirection facility is to execute the program from the command prompt, inserting the redirection symbols at appropriate places. Let us understand this process with the help of a program.

### Redirecting the Output

Let's see how we can redirect the output of a program, from the screen to a file. We'll start by considering the simple program shown below.

```
/* File name: util.c */
#include <stdio.h>
int main()
{
 char ch ;
 while ((ch = getc (stdin)) != EOF)
 putc (ch, stdout) ;
 return 0 ;
}
```

On compiling this program, we would get an executable file UTIL.EXE. Normally, when we execute this file, the **putc( )** function will cause whatever we type to be printed on screen, until we type Ctrl-Z, at which point the program will terminate, as shown in the following sample run. The Ctrl-Z character is often called end of file character.

```
C>UTIL.EXE
perhaps I had a wicked childhood,
perhaps I had a miserable youth,
but somewhere in my wicked miserable past,
there must have been a moment of truth ^Z
C>
```

Now let's see what happens when we invoke this program from in a different way, using redirection:

```
C>UTIL.EXE > POEM.TXT
C>
```

Here we are causing the output to be redirected to the file POEM.TXT. Can we prove that this output has indeed gone to the file POEM.TXT? Yes, by using the TYPE command as follows:

```
C>TYPE POEM.TXT
perhaps I had a wicked childhood,
```

```
perhaps I had a miserable youth,
but somewhere in my wicked miserable past,
there must have been a moment of truth
C>
```

There's the result of our typing sitting in the file. The redirection operator, '>', causes any output intended for the screen to be written to the file whose name follows the operator.

Note that the data to be redirected to a file doesn't need to be typed by a user at the keyboard; the program itself can generate it. Any output normally sent to the screen can be redirected to a disk file. As an example, consider the following program for generating the ASCII table on screen:

```
/* File name: ascii.c */
#include <stdio.h>
int main()
{
 int ch ;
 for (ch = 0 ; ch <= 255 ; ch++)
 printf ("%d %c\n", ch, ch) ;
 return 0 ;
}
```

When this program is compiled and then executed at command prompt using the redirection operator,

```
C>ASCII.EXE > TABLE.TXT
```

the output is written to the file. This can be a useful capability any time you want to capture the output in a file, rather than displaying it on the screen.

### Redirecting the Input

We can also redirect input to a program so that, instead of reading a character from the keyboard, a program reads it from a file. Let us now see how this can be done.

To redirect the input, we need to have a file containing something to be displayed. Suppose we use a file called NEWPOEM.TXT containing the following lines:

Let's start at the very beginning,  
A very good place to start!

We'll assume that using some text editor these lines have been placed in the file NEWPOEM.TXT. Now, we use the input redirection operator '<' before the file, as shown below.

```
C>UTIL.EXE < NEWPOEM.TXT
Let's start at the very beginning,
A very good place to start!
C>
```

The lines are printed on the screen with no further effort on our part. Using redirection, we've made our program UTIL.C perform the work of the TYPE command.

### Both Ways at Once

Redirection of input and output can be used together; the input for a program can come from a file via redirection, at the same time its output can be redirected to a file. Such a program is called a filter. The following command demonstrates this process:

```
C>UTIL.EXE < NEWPOEM.TXT > POETRY.TXT
```

In this case, our program receives the redirected input from the file NEWPOEM.TXT and instead of sending the output to the screen; it would redirect it to the file POETRY.TXT.

While using such multiple redirections, don't try to send output to the same file from which you are receiving input. This is because the output file is erased before it is written to. So by the time we manage to receive the input from a file, it is already erased.

Redirection can be a powerful tool for developing utility programs to examine or alter data in files. Thus, redirection is used to establish a relationship between a program and a file. Another OS operator can be used to relate two programs directly, so that the output of one is fed directly into another, with no files involved. This is called 'piping', and is done using the operator '|', called pipe. We won't pursue this topic, but you can read about it in the OS Help.

## Summary

- (a) We can pass parameters to a program at command line using the concept of 'command-line arguments'.
- (b) The command line argument **argv** contains values passed to the program, whereas, **argc** contains number of arguments.
- (c) We can use the standard file pointer **stdin** to take input from standard input device, such as keyboard.
- (d) We can use the standard file pointer **stdout** to send output to the standard output device, such as a monitor.
- (e) Redirection allows a program to read from or write to files at command prompt.
- (f) The operators **<** and **>** are called redirection operators.

## Exercise

**[A]** Answer the following:

- (a) How will you use the following program to:
  - Copy the contents of one file into another.
  - Create a new file and add some text to it.
  - Display the contents of an existing file.

```
#include <stdio.h>
int main()
{
 char ch, str[10] ;
 while ((ch = getc (stdin)) != -1)
 putc (ch, stdout) ;
 return 0 ;
}
```

- (b) State True or False:
  - 1. We can send arguments at command line even if we define **main( )** function without parameters.
  - 2. To use standard file pointers we don't need to open the file using **fopen( )**.
  - 3. The zeroth element of the **argv** array is always the name of the executable file.

- (c) Point out the errors, if any, in the following program:

```
#include <stdio.h>
int main (int ac, char (*) av[])
{
 printf ("%d\n", ac);
 printf ("%s\n", av[0]);
 return 0 ;
}
```

**[B]** Attempt the following:

- (a) Write a program using command line arguments to search for a word in a file and replace it with the specified word. The usage of the program is shown below.

C> change <old word> <new word> <filename>

- (b) Write a program that can be used at command prompt as a calculating utility. The usage of the program is shown below.

C> calc <switch> <n> <m>

Where, **n** and **m** are two integer operands. **switch** can be any one of the arithmetic or comparison operators. If arithmetic operator is supplied, the output should be the result of the operation. If comparison operator is supplied then the output should be **True** or **False**.



# 21

## Operations On Bits

- Bit Numbering and Conversion
- Bit Operations
- One's Complement Operator
- Right Shift Operator
- Left Shift Operator
  - Utility of Left Shift Operator
- Bitwise AND Operator
  - Utility of AND Operator
- Bitwise OR Operator
- Bitwise XOR Operator
- The *showbits( )* Function
- Bitwise Compound Assignment Operators
- Summary
- Exercise



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# 21 Operations On Bits

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- Bit Numbering and Conversion
- Bit Operations
- One's Complement Operator
- Right Shift Operator
- Left Shift Operator
  - Utility of Left Shift Operator
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  - Utility of AND Operator
- Bitwise OR Operator
- Bitwise XOR Operator
- The *showbits( )* Function
- Bitwise Compound Assignment Operators
- Summary
- Exercise

So far we have dealt with characters, integers, floats and their variations. The smallest element in memory on which we are able to operate as yet is a byte; and we operated on it by making use of the data type `char`. However, we haven't attempted to look within these data types to see how they are constructed out of individual bits, and how these bits can be manipulated. Being able to operate on a bit-level, can be very important in programming, especially when a program must interact directly with the hardware. This is because, the programming languages are byte-oriented, whereas hardware tends to be bit-oriented. Let us now delve inside the byte and see how it is constructed and how it can be manipulated effectively. So let us take apart the byte... bit-by-bit.

### Bit Numbering and Conversion

A bit (short for Binary Digit) is the most basic unit of information. It can take a value 0 or 1. 4 bits together form a nibble, 8 bits form a byte, 16 bits form a word and 32 bits form a double-word. Bits are numbered from zero onwards, increasing from right to left as shown in Figure 21.1.

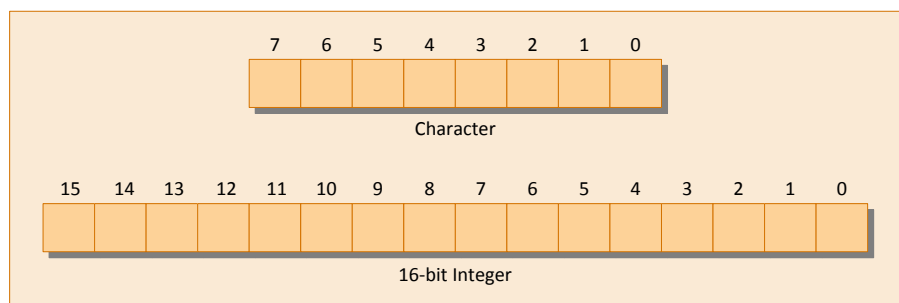


Figure 21.1

Suppose we wish to store binary values 10110110 and 00111100 in 2 bytes. If we are required to do this through a C program we won't be able to use these bit patterns directly. This is because C language doesn't understand binary numbering system. So we need to convert these binary numbers into decimal numbers. This conversion is shown in Figure 21.2.

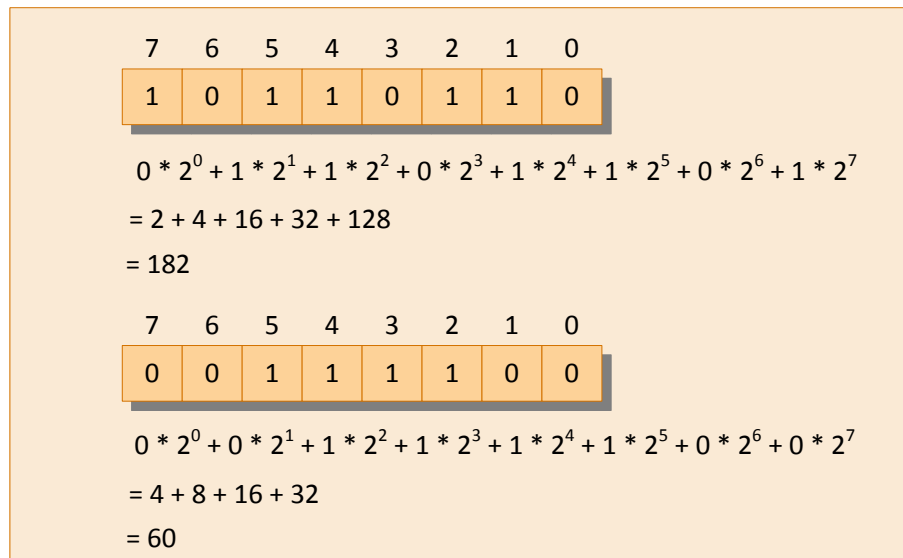


Figure 21.2

As you can see from Figure 21.2 the binary to decimal conversion process involves remembering powers of 2. This is alright if the binary number is a 8-bit number, but if it is a 16-bit number then remembering powers like  $2^{15}$ ,  $2^{14}$ ,  $2^{13}$ , etc., is difficult. A much easier method is to convert binary numbers to hexadecimal numbers. As you might be aware, in hexadecimal numbering system each number is built using a combination of digits 0 to 9 and A to F. Digits A to F are symbols used to represent value 10 to 15. Each hexadecimal digit can be represented using a 4-bit nibble as shown in Figure 21.3.

| Hex | Binary | Hex | Binary |
|-----|--------|-----|--------|
| 0   | 0000   | 8   | 1000   |
| 1   | 0001   | 9   | 1001   |
| 2   | 0010   | A   | 1010   |
| 3   | 0011   | B   | 1011   |
| 4   | 0100   | C   | 1100   |
| 5   | 0101   | D   | 1101   |
| 6   | 0110   | E   | 1110   |
| 7   | 0111   | F   | 1111   |

Figure 21.3

Using Figure 21.3, it is very easy to convert binary values into their equivalent hexadecimal values. This is shown in Figure 21.4.

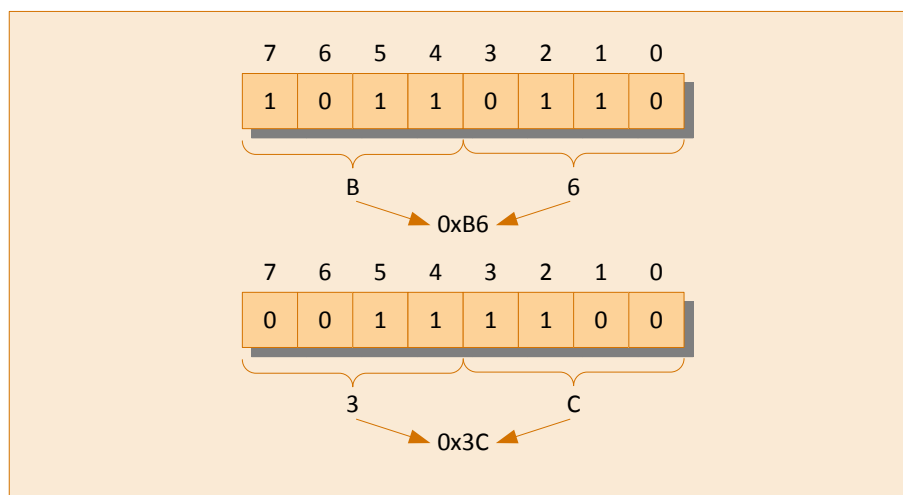


Figure 21.4

You would agree this is an easier way to represent the binary number than to find its decimal equivalent. In this method, neither multiplication nor addition is needed. In fact, since there are only 16 hex digits, it's fairly easy to memorize the binary equivalent of each one. Quick now, what's binary 1100 in hex? That's right—C. You are already getting the

feel of it. With a little practice, it is easy to translate even long numbers into hex. Thus, 1100 0101 0011 1010 binary is C53A hex.

As it happens with many unfamiliar subjects, learning hexadecimal numbers requires a little practice. Try your hand at converting some binary numbers and vice versa. Soon you will be talking hexadecimal as if you had known it all your life.

## Bit Operations

Now that we have understood the bit numbering and the binary to hex conversion process it is time to access or manipulate bits. Here are some examples of operations that we may wish to perform on bits:

- (a) Set bit 3 to 0
- (b) Set bit 5 to 1
- (c) Check whether bit 6 is 1 (on) or 0 (off)

As you can see, in the first two examples we are manipulating (writing) a bit, whereas, in the third example we are accessing (reading) a bit. To be able to access or manipulate individual bits C language provides a powerful set of bit manipulation operators. These are shown in Figure 21.5.

| Operator | Meaning                   |
|----------|---------------------------|
| ~        | One's complement          |
| >>       | Right shift               |
| <<       | Left shift                |
| &        | Bitwise AND               |
|          | Bitwise OR                |
| ^        | Bitwise XOR(Exclusive OR) |

Figure 21.5

These operators can operate on **ints** and **chars** but not on **floats** and **doubles**. Before we examine each of these operators, let me introduce you to a function called **showbits( )**. Throughout this discussion about bitwise operators, we are going to use this function, but we are not going to discuss the details of this function immediately. The task of **showbits( )** is to display the binary representation of any integer or

character value that it receives. Let us begin with a plain-jane example with **showbits( )** in action.

```
/* Print binary equivalent of characters using showbits() function */
include <stdio.h>
void showbits (unsigned char) ;

int main()
{
 unsigned char num ;

 for (num = 0 ; num <= 5 ; num++)
 {
 printf ("\nDecimal %d is same as binary ", num) ;
 showbits (num) ;
 }
 return 0 ;
}

void showbits (unsigned char n)
{
 int i ;
 unsigned char j, k, andmask ;

 for (i = 7 ; i >= 0 ; i--)
 {
 j = i ;
 andmask = 1 << j ;
 k = n & andmask ;
 k == 0 ? printf ("0") : printf ("1") ;
 }
}
```

On execution the program produces the following output...

```
Decimal 0 is same as binary 00000000
Decimal 1 is same as binary 00000001
Decimal 2 is same as binary 00000010
Decimal 3 is same as binary 00000011
Decimal 4 is same as binary 00000100
Decimal 5 is same as binary 00000101
```

Let us now explore the various bitwise operators one-by-one.

### One's Complement Operator

On taking one's complement of a number, all 1's present in the number are changed to 0's and all 0's are changed to 1's. For example, one's complement of 1010 is 0101. Similarly, one's complement of 1111 is 0000. Note that, here when we talk of a number, we are talking of binary equivalent of the number. Thus, one's complement of 65 means one's complement of 0100 0001, which is binary equivalent of 65. One's complement of 65 therefore would be 1011 1110. One's complement operator is represented by the symbol `~` (tilde). Following program shows one's complement operator in action:

```
#include <stdio.h>
int main()
{
 unsigned char ch = 32 ;
 unsigned char dh ;

 dh = ~ch ;
 printf ("~ch = %d\n", dh) ;
 printf ("~ch = %x\n", dh) ;
 printf ("~ch = %X\n", dh) ;

 return 0 ;
}
```

On execution the program produces the following output:

```
~ch = 223
~ch = df
~ch = DF
```

Here **ch** contains a value 32, whose binary equivalent is 00100000. On taking one's complement of it, we get 11011111, which in decimal is 223. As we learnt earlier, hexadecimal equivalent of 11011111 is DF. The hexadecimal equivalent gets printed in smallcase if we use `%x` and in capital if we use `%X`.

Let us try one more program that prints one's complement of different numbers in a loop. Once again to print the binary equivalent of a number we have used the **showbits( )** function.



```
include <stdio.h>
void showbits (unsigned char) ;

int main()
{
 unsigned char num, k ;

 for (num = 0 ; num <= 3 ; num++)
 {
 printf ("\nDecimal %d is same as binary ", num) ;
 showbits (num) ;

 k = ~num ;
 printf ("\nOne's complement of %d is ", num) ;
 showbits (k) ;
 }
 return 0 ;
}

void showbits (unsigned char n)
{
 int i ;
 unsigned char j, k, andmask ;

 for (i = 7 ; i >= 0 ; i--)
 {
 j = i ;
 andmask = 1 << j ;
 k = n & andmask ;
 k == 0 ? printf ("0") : printf ("1") ;
 }
}
```

And here is the output of the above program...

```
Decimal 0 is same as binary 00000000
One's complement of 0 is 11111111
Decimal 1 is same as binary 00000001
One's complement of 1 is 11111110
Decimal 2 is same as binary 00000010
One's complement of 2 is 11111101
```

Decimal 3 is same as binary 00000011  
One's complement of 3 is 11111100

## Right Shift Operator

The right shift operator is represented by `>>`. It needs two operands. It shifts each bit in its left operand to the right. The number of places the bits are shifted depends on the number following the operator (i.e., its right operand). Thus, `ch >> 3` would shift all bits in `ch` three places to the right. Similarly, `ch >> 5` would shift all bits 5 places to the right.

If the variable `ch` contains the bit pattern 11010111, then, `ch >> 1` would give 01101011 and `ch >> 2` would give 00110101.

Note that as the bits are shifted to the right, blanks are created on the left. These blanks must be filled somehow. They are always filled with zeros. The following program demonstrates the effect of right shift operator:

```
include <stdio.h>
void showbits (unsigned char);

int main()
{
 unsigned char num = 225, i, k ;

 printf ("\nDecimal %d is same as binary ", i) ;
 showbits (i) ;

 for (i = 0 ; i <= 5 ; i++)
 {
 k = num >> i ;
 printf ("\n%d right shift %d gives ", num, i) ;
 showbits (k) ;
 }
 return 0 ;
}

void showbits (unsigned char n)
{
 int i ;
 unsigned char j, k, andmask ;
```

```
for (i = 7 ; i >= 0 ; i--)
{
 j = i ;
 andmask = 1 << j ;
 k = n & andmask ;
 k == 0 ? printf ("0") : printf ("1") ;
}
```

The output of the above program would be...

Decimal 225 is same as binary 11100001  
5225 right shift 0 gives 11100001  
5225 right shift 1 gives 01110000  
5225 right shift 2 gives 00111000  
5225 right shift 3 gives 00011100  
5225 right shift 4 gives 00001110  
5225 right shift 5 gives 00000111

Note that if the operand is a multiple of 2, then shifting the operand 1 bit to right is same as dividing it by 2 and ignoring the remainder. Thus,

64 >> 1 gives 32  
64 >> 2 gives 16  
128 >> 2 gives 32

but,

27 >> 1 is 13  
49 >> 1 is 24 .

### A Word of Caution

In the expression **a >> b** if **b** is negative the result is unpredictable. If **a** is negative then its left most bit (sign bit) would be 1. On right shifting **a** it would result in extending the sign bit. For example, if **a** contains -1, its binary representation would be 11111111. If we right shift it by 4 then the result would still be 11111111. Similarly, if **a** contains -5, then its binary equivalent would be 11111011. On right shifting it by 1 we would get 11111101, which is equal to -3. Thus on right shifting 11111011 the right-most bit, i.e. 1, would be lost; other bits would be shifted one position to the right and the sign which was negative (1) will be

extended, i.e., it would be preserved as 1. The following program, would help you get a clear picture of this:

```
include <stdio.h>
void showbits (unsigned char) ;

int main()
{
 char num = -5, j, k ;

 printf ("\nDecimal %d is same as binary ", num) ;
 showbits (num) ;

 for (j = 1 ; j <= 3 ; j++)
 {
 k = num >> j ;
 printf ("\n%d right shift %d gives ", num, j) ;
 showbits (k) ;
 }
 return 0 ;
}

void showbits (unsigned char n)
{
 int i ;
 unsigned char j, k, andmask ;

 for (i = 7 ; i >= 0 ; i--)
 {
 j = i ;
 andmask = 1 << j ;
 k = n & andmask ;
 k == 0 ? printf ("0") : printf ("1") ;
 }
}
```

The output of the above program would be...

```
Decimal -5 is same as binary 11111011
-5 right shift 1 gives 11111101
-5 right shift 2 gives 11111110
-5 right shift 3 gives 11111111
```

## Left Shift Operator

The left shift operator (<<) is similar to the right shift operator (>>), the only difference being that the bits are shifted to the left, and for each bit shifted, a 0 is added to the right of the number. The following program should clarify this point:

```
#include <stdio.h>
void showbits (unsigned char) ;

int main()
{
 unsigned char num = 225, j, k ;

 printf ("\nDecimal %d is same as binary ", num) ;
 showbits (num) ;

 for (j = 0 ; j <= 4 ; j++)
 {
 k = num << j ;
 printf ("\n%d left shift %d gives ", num, j) ;
 showbits (k) ;
 }
 return 0 ;
}

void showbits (unsigned char n)
{
 int i ;
 unsigned char j, k, andmask ;

 for (i = 7 ; i >= 0 ; i--)
 {
 j = i ;
 andmask = 1 << j ;
 k = n & andmask ;
 k == 0 ? printf ("0") : printf ("1") ;
 }
}
```

The output of the above program would be...

```
Decimal 225 is same as binary 11100001
225 left shift 0 gives 11100001
225 left shift 1 gives 11000010
225 left shift 2 gives 10000100
225 left shift 3 gives 00001000
```

### Utility of Left Shift Operator

The left shift operator is often used to create a number with a particular bit in it set to 1. For example, we can create a number with its 3<sup>rd</sup> bit set to 1. The following program shows how this can be achieved:

```
include <stdio.h>
int main()
{
 unsigned char a ;

 a = 1 << 3 ;
 printf ("a = %02x", a) ;
 return 0 ;
}
```

Binary value of 1 is 00000001. On left-shifting this by 3 we get 00001000. Thus we are able to create a value with its 3<sup>rd</sup> bit set to 1. Such operations are required quite often while writing programs that interact with hardware or while building embedded systems. Hence it is often done using a macro as shown below.

```
define _BV(x) (1 << x)
include <stdio.h>

int main()
{
 unsigned char a ;

 a = _BV(3) ;
 printf ("a = %02x", a) ;
 return 0 ;
}
```

The **\_BV** macro stands for **Bit Value**. Its argument indicates which bit in the number would be set when this macro is used. As you must have guessed, during processing the macro **\_BV( 3 )** would get expanded to **1 << 3**.

Note the use of the format specifier **%02x** in the **printf( )** function. This ensures that the output is printed in 2 columns, with a leading 0, if required. Thus the output of both the programs would be 08.

### Bitwise AND Operator

This operator is represented as **&**. Remember it is different than **&&**, the logical AND operator. The **&** operator operates on two operands. While operating upon these two operands they are compared on a bit-by-bit basis. Hence both the operands must be of the same type (either **char** or **int**). The second operand is often called an AND mask. The **&** operator operates on a pair of bits to yield a resultant bit. The rules that decide the value of the resultant bit are shown in Figure 21.6.

| First bit | Second bit | First bit & Second bit |
|-----------|------------|------------------------|
| 0         | 0          | 0                      |
| 0         | 1          | 0                      |
| 1         | 0          | 0                      |
| 1         | 1          | 1                      |

Figure 21.6

This can be represented in a more understandable form as a 'Truth Table' shown in Figure 21.7.

| & | 0 | 1 |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |

Figure 21.7

The example given in Figure 21.8 shows more clearly what happens while ANDing one operand with another. The rules given in the Figure 21.7 are applied to each pair of bits one-by-one.

|   |   |   |   |   |   |   |   |                                 |
|---|---|---|---|---|---|---|---|---------------------------------|
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |                                 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | This operand when ANDed bitwise |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |                                 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | with this operand yields        |
| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |                                 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | this result                     |

Figure 21.8

Work through the Truth Table and confirm that the result obtained is really correct.

Thus, it must be clear that the operation is being performed on individual bits, and the operation performed on one pair of bits is completely independent of the operation performed on the other pairs.

### Utility of AND Operator

AND operator is used in two situations:

- To check whether a particular bit of an operand is ON or OFF.
- To turn OFF a particular bit.

Both these uses are discussed in the following example.

Suppose, from the bit pattern 10101101 (0xAD) of an operand, we want to check whether bit number 5 is ON (1) or OFF (0). Since we want to check the bit number 5, the second operand for the AND operation should be 00100000. This second operand is often known as AND mask. The ANDing operation is shown below.

|          |                       |
|----------|-----------------------|
| 10101101 | Original bit pattern  |
| 00100000 | AND mask              |
| -----    |                       |
| 00100000 | Resulting bit pattern |



The resulting value we get in this case is 32 (or 0x20), i.e., the value of the second operand. The result turned out to be 32 (or 0x20) since the fifth bit of the first operand was ON. Had it been OFF, the bit number 5 in the resulting bit pattern would have evaluated to 0 and the complete bit pattern would have been 00000000.

Thus, depending upon the bit number to be checked in the first operand, we decide the second operand, and on ANDing these two operands the result decides whether the bit was ON or OFF. If the bit is ON (1), the resulting value turns out to be a non-zero value, which is equal to the value of second operand. If the bit is OFF (0), the result is zero, as seen above.

Let us now turn our attention to the second use of the AND operator. As you can see, in the bit pattern 10101101 (0xAD), 3<sup>rd</sup> bit is ON. To put it off, we need to AND the 3<sup>rd</sup> bit with 0. While doing so the values of other bits in the pattern should not get disturbed. For this we need to AND the other bits with 1. This operation is shown below.

|          |                       |
|----------|-----------------------|
| 10101101 | Original bit pattern  |
| 11110111 | AND mask              |
| -----    |                       |
| 10100101 | Resulting bit pattern |

The following program puts this logic into action:

```
include <stdio.h>
void showbits (unsigned char) ;

int main()
{
 unsigned char num = 0xAD, j ;

 printf ("\nValue of num = ") ;
 showbits (num) ;

 j = num & 0x20 ;

 if (j == 0)
 printf ("\nIts fifth bit is off") ;
 else
 printf ("\nIts fifth bit is on") ;
}
```

```
j = num & 0x08 ;

if (j == 0)
 printf ("\nIts third bit is off") ;
else
{
 printf ("\nIts third bit is on") ;

 num = num & 0xF7 ;
 printf ("\nNew value of num = ") ;
 showbits (num) ;
 j = num & 0x08 ;

 if (j == 0)
 printf ("\nNow its third bit is turned off") ;
}
return 0 ;
}

void showbits (unsigned char n)
{
 int i ;
 unsigned char j, k, andmask ;

 for (i = 7 ; i >= 0 ; i--)
 {
 j = i ;
 andmask = 1 << j ;
 k = n & andmask ;
 k == 0 ? printf ("0") : printf ("1") ;
 }
}
```

And here is the output...

```
Value of num = 10101101
Its fifth bit is on
Its third bit is on
New value of num = 10100101
Now its third bit is turned off
```

Note the use of **&** operator in the statements:

```
j = num & 0x20 ;
j = num & 0x08 ;
num = num & 0xF7 ;
```

A quick glance at these statements does not indicate what operation is being carried out through them. Hence a better idea is to use the macro **\_BV** as shown below.

```
define _BV(x) (1 << x)

j = num & _BV(5) ;
j = num & _BV(3) ;
num = num & ~ _BV(3) ;
```

In the last statement **\_BV( 3 )** would yield 00001000 and one's complement of this number would fetch 11110111.

## Bitwise OR Operator

Another important bitwise operator is the OR operator which is represented as **|**. The rules that govern the value of the resulting bit obtained after ORing of two bits is shown in the Truth Table shown in Figure 21.9.

|   | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 1 |

Figure 21.9

Using the Truth Table confirm the result obtained on ORing the two operands as shown below.

```
11010000 Original bit pattern
00000111 OR mask

```

**11010111 Resulting bit pattern**

Bitwise OR operator is usually used to put ON a particular bit in a number.

Let us consider the bit pattern 11000011. If we want to put ON bit number 3, then the OR mask to be used would be 00001000. Note that all the other bits in the mask are set to 0 and only the bit, which we want to set ON in the resulting value is set to 1. The code snippet which will achieve this is given below.

```
define _BV(x) (1 << x)

unsigned char num = 0xC3 ;
num = num | _BV(3) ;
```

**Bitwise XOR Operator**

The XOR operator is represented as ^ and is also called an Exclusive OR Operator. The OR operator returns 1, when any one of the two bits or both the bits are 1, whereas XOR returns 1 only if one of the two bits is 1. The Truth Table for the XOR operator is shown in Figure 21.10.

| ^ |   |   |
|---|---|---|
|   | 0 | 1 |
| 0 | 0 | 1 |
| 1 | 1 | 0 |

Figure 21.10

XOR operator is used to toggle (change) a bit ON or OFF. A number XORed with another number twice gives the original number. This is shown in the following program:

```
include <stdio.h>
int main()
{
 unsigned char b = 0x32 ; /* Binary 00110010 */
```

```
b = b ^ 0x0C ;
printf ("\n%02x", b); /* this will print 0x3E */

b = b ^ 0x0C ;
printf ("\n%02x", b); /* this will print 0x32 */

return 0 ;
}
```

### The *showbits( )* Function

We have used this function quite often in this chapter. Now we have sufficient knowledge of bitwise operators and hence are in a position to understand it. The function is given below followed by brief explanation.

```
void showbits (unsigned char n)
{
 unsigned char i, k, andmask ;

 for (i = 7 ; i >= 0 ; i--)
 {
 andmask = 1 << i ;
 k = n & andmask ;

 k == 0 ? printf ("0") : printf ("1") ;
 }
}
```

All that is being done in this function is, using an AND operator and a variable **andmask**, we are checking the status of individual bits of **n**. If the bit is OFF we print a 0, otherwise we print a 1.

First time through the loop, the variable **andmask** will contain the value 10000000, which is obtained by left-shifting 1, seven places. If the variable **n**'s most significant bit (leftmost bit) is 0, then **k** would contain a value 0, otherwise it would contain a non-zero value. If **k** contains 0, then **printf( )** will print out 0, otherwise it will print out 1.

In the second go-around of the loop, the value of **i** is decremented and hence the value of **andmask** changes, which will now be 01000000. This checks whether the next most significant bit is 1 or 0, and prints it out accordingly. The same operation is repeated for all bits in the number.

## Bitwise Compound Assignment Operators

Consider the following bitwise operations:

```
unsigned char a = 0xFA, b = 0xA7, c = 0xFF, d = 0xA3, e = 0x43 ;
a = a << 1 ;
b = b >> 2 ;
c = c | 0x2A ;
d = d & 0x4A ;
e = e ^ 0x21 ;
```

These operations can be written more elegantly and in a compact fashion as shown below.

```
unsigned char a = 0xFA, b = 0xA7, c = 0xFF, d = 0xA3, e = 0x43 ;
a <<= 1 ;
b >>= 2 ;
c |= 0x2A ;
d &= 0x4A ;
e ^= 0x21 ;
```

The operators `<<=`, `>>=`, `|=`, `&=` and `^=` are called bitwise compound assignment operators. Note that there does not exist an operator `~=`. This is because `~` is a unary operator and needs only one operand.

## Summary

- (a) To help manipulate hardware oriented data—individual bits rather than bytes a set of bitwise operators are used.
- (b) It is convenient to convert binary numbers into their hexadecimal equivalents than converting them to their decimal equivalents.
- (c) The bitwise operators include operators like one's complement, right-shift, left-shift, bitwise AND, OR, and XOR.
- (d) The one's complement converts all 0s in its operand to 1s and all 1s to 0s.
- (e) The right-shift and left-shift operators are useful in eliminating bits from a number—either from the left or from the right.
- (f) The bitwise AND operators is useful in testing whether a bit is on/off and in putting off a particular bit.
- (g) The bitwise OR operator is used to turn on a particular bit.

- (h) The XOR operator works almost same as the OR operator except one minor variation.

## Exercise

[A] Answer the following:

- (a) The information about colors is to be stored in bits of a **char** variable called **color**. The bit number 0 to 6, each represent 7 colors of a rainbow, i.e., bit 0 represents violet, 1 represents indigo, and so on. Write a program that asks the user to enter a number and based on this number it reports which colors in the rainbow do the number represents.
- (b) A company planning to launch a new newspaper in market conducts a survey. The various parameters considered in the survey were, the economic status (upper, middle, and lower class) the languages readers prefer (English, Hindi, Regional language) and category of paper (daily, supplement, tabloid). Write a program, which reads data of 10 respondents through keyboard, and stores the information in an array of integers. The bit-wise information to be stored in an integer is given below:

| Bit Number | Information       |
|------------|-------------------|
| 0          | Upper class       |
| 1          | Middle class      |
| 2          | Lower class       |
| 3          | English           |
| 4          | Hindi             |
| 5          | Regional Language |
| 6          | Daily             |
| 7          | Supplement        |
| 8          | Tabloid           |

At the end give the statistical data for number of persons who read English daily, number of Upper class people who read Tabloid and number of Regional Language readers.

- (c) In an inter-college competition, various sports like cricket, basketball, football, hockey, lawn tennis, table tennis, carom and chess are played between different colleges. The information regarding the games won by a particular college is stored in bit numbers 0, 1, 2, 3, 4, 5, 6, 7 and 8, respectively of an integer

variable called **game**. The college that wins in 5 or more than 5 games is awarded the Champion of Champions trophy. If a number representing the bit pattern mentioned above is entered through the keyboard then write a program to find out whether the college won the Champion of the Champions trophy or not, along with the names of the games won by the college.

- (d) An animal could be a canine (dog, wolf, fox, etc.), a feline (cat, lynx, jaguar, etc.), a cetacean (whale, narwhal, etc.) or a marsupial (koala, wombat, etc.). The information whether a particular animal is canine, feline, cetacean, or marsupial is stored in bit number 0, 1, 2 and 3, respectively of a integer variable called **type**. Bit number 4 of the variable **type** stores the information about whether the animal is Carnivore or Herbivore.

For the following animal, complete the program to determine whether the animal is a herbivore or a carnivore. Also determine whether the animal is a canine, feline, cetacean or a marsupial.

```
struct animal
{
 char name[30];
 int type ;
}
struct animal a = { "OCELOT", 18 };
```

- (e) The time field in a structure is 2 bytes long. Distribution of different bits which account for hours, minutes and seconds is given in Figure 21.11. Write a function which would receive the 2-byte time and return to the calling function, the hours, minutes and seconds.

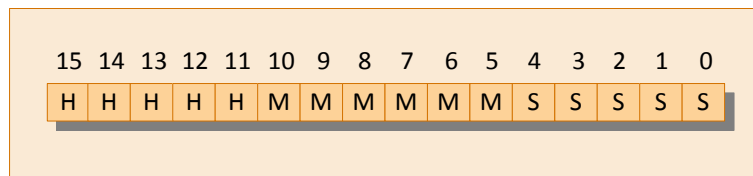


Figure 21.11

- (f) In order to save disk space, information about student is stored in an integer variable. If bit number 0 is on then it indicates I<sup>st</sup> year student, bit number 1 to 3 stores II<sup>nd</sup> year, III<sup>rd</sup> year and IV<sup>th</sup> year student, respectively. Bits 4 to 7 store the stream Mechanical, Chemical, Electronics and IT. Rest of the bits store room number. Based on the given data, write a program that asks for the room



number and displays the information about the student, if its data exists in the array. The contents of array are,

```
int data[] = { 273, 548, 786, 1096 } ;
```

- (g) What will be the output of the following program:

```
include <stdio.h>
int main()
{
 int i = 32, j = 65, k, l, m, n, o, p ;
 k = i | 35 ;
 l = ~k ;
 m = i & j ;
 n = j ^ 32 ;
 o = j << 2 ;
 p = i >> 5 ;
 printf ("k = %d l = %d m = %d\n", k, l, m) ;
 printf ("n = %d o = %d p = %d\n", n, o, p) ;
 return 0 ;
}
```

**[B] Answer the following:**

- (a) What is hexadecimal equivalent of the following binary numbers:

```
0101 1010
11000011
1010101001110101
1111000001011010
```

- (b) Rewrite the following expressions using bitwise compound assignment operators:

```
a = a | 3
a = a & 0x48
b = b ^ 0x22
c = c << 2
d = d >> 4
```

- (c) Consider an unsigned integer in which rightmost bit is numbered as 0. Write a function **checkbits ( x, p, n )** which returns true if all "n" bits starting from position "p" are turned on. For example, **checkbits ( x, 4, 3 )** will return true if bits 4, 3 and 2 are 1 in number x.

- (d) Write a program to scan a 8-bit number into a variable and check whether its 3<sup>rd</sup>, 6<sup>th</sup> and 7<sup>th</sup> bit is on.
- (e) Write a program to receive an unsigned 16-bit integer and then exchange the contents of its 2 bytes using bitwise operators.
- (f) Write a program to receive a 8-bit number into a variable and then exchange its higher 4 bits with lower 4 bits.
- (g) Write a program to receive a 8-bit number into a variable and then set its odd bits to 1.
- (h) Write a program to receive a 8-bit number into a variable and then if its 3<sup>rd</sup> and 5<sup>th</sup> bit are on. If these bits are found to be on then put them off.
- (i) Write a program to receive a 8-bit number into a variable and then if its 3<sup>rd</sup> and 5<sup>th</sup> bit are off. If these bits are found to be off then put them on.
- (j) Rewrite the **showbits( )** function used in this chapter using the **\_BV** macro.



- **Enumerated Data Type**
  - Uses of Enumerated Data Type
  - Are Enums Necessary?
- **Renaming Data Types with *typedef***
- **Typecasting**
- **Bit Fields**
- **Pointers to Functions**
- **Functions Returning Pointers**
- **Functions with Variable Number of Arguments**
- **Unions**
  - Union of Structures
- **Utility of Unions**
- **The *volatile* Qualifier**
- **Summary**
- **Exercise**



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# 22 *Miscellaneous Features*

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- Enumerated Data Type
  - Uses of Enumerated Data Type
  - Are Enums Necessary?
- Renaming Data Types with *typedef*
- Typecasting
- Bit fields
- Pointers to Functions
- Functions Returning Pointers
- Functions with Variable Number of Arguments
- Unions
  - Union of Structures
- Utility of Unions
- The *volatile* Qualifier
- Summary
- Exercise

The topics discussed in this chapter were either too large or far too removed from the mainstream C programming for inclusion in the earlier chapters. These topics provide certain useful programming features, and could prove to be of immense help in certain programming strategies. In this chapter, we would examine enumerated data types, the **typedef** keyword, typecasting, bit fields, function pointers, functions with variable number of arguments and unions.

## Enumerated Data Type

The enumerated data type gives you an opportunity to invent your own data type and define what values the variable of this data type can take. This can help in making the program listings more readable, which can be an advantage when a program gets complicated or when more than one programmer would be working on it. Using enumerated data type can also help you reduce programming errors.

As an example, one could invent a data type called **mar\_status** which can have four possible values—single, married, divorced or widowed. Don't confuse these values with variable names; married, for instance, has the same relationship to the variable **mar\_status** as the number 15 has with a variable of type **int**.

The format of the **enum** definition is similar to that of a structure. Here's how the example stated above can be implemented.

```
enum mar_status
{
 single, married, divorced, widowed
};
enum mar_status person1, person2 ;
```

Like structures, this declaration has two parts:

- (a) The first part declares the data type and specifies its possible values. These values are called 'enumerators'.
- (b) The second part declares variables of this data type.

Now we can give values to these variables:

```
person1 = married ;
person2 = divorced ;
```

Remember, we can't use values that aren't in the original declaration.

Thus, the following expression would cause an error:

```
person1 = unknown ;
```

Internally, the compiler treats the enumerators as integers. Each value on the list of permissible values corresponds to an integer, starting with 0. Thus, in our example, single is stored as 0, married is stored as 1, divorced as 2 and widowed as 3.

This way of assigning numbers can be overridden by the programmer by initializing the enumerators to different integer values as shown below.

```
enum mar_status
{
 single = 100, married = 200, divorced = 300, widowed = 400
};
enum mar_status person1, person2 ;
```

### Uses of Enumerated Data Type

Enumerated variables are usually used to clarify the operation of a program. For example, if we need to use employee departments in a payroll program, it makes the listing easier to read if we use values like Assembly, Manufacturing, Accounts rather than the integer values 0, 1, 2, etc. The following program illustrates the point I am trying to make:

```
#include <stdio.h>
#include <string.h>
int main()
{
 enum emp_dept
 {
 assembly, manufacturing, accounts, stores
 };
 struct employee
 {
 char name[30];
 int age ;
 float bs ;
 enum emp_dept department ;
 };
 struct employee e ;
```

```
strcpy (e.name, "Lothar Mattheus") ;
e.age = 28 ;
e.bs = 5575.50 ;
e.department = manufacturing ;

printf ("Name = %s\n", e.name) ;
printf ("Age = %d\n", e.age) ;
printf ("Basic salary = %f\n", e.bs) ;
printf ("Dept = %d\n", e.department) ;

if (e.department == accounts)
 printf ("%s is an accountant\n", e.name) ;
else
 printf ("%s is not an accountant\n", e.name) ;
return 0 ;
}
```

And here is the output of the program...

```
Name = Lothar Mattheus
Age = 28
Basic salary = 5575.50
Dept = 1
Lothar Mattheus is not an accountant
```

Let us now dissect the program. We first defined the data type **enum emp\_dept** and specified the four possible values, namely, assembly, manufacturing, accounts and stores. Then we defined a variable **department** of the type **enum emp\_dept** in a structure. The structure **employee** has three other elements containing employee information.

The program first assigns values to the variables in the structure. The statement,

```
e.department = manufacturing ;
```

assigns the value manufacturing to **e.department** variable. This is much more informative to anyone reading the program than a statement like,

```
e.department = 1 ;
```



The next part of the program shows an important weakness of using **enum** variables... there is no way to use the enumerated values directly in input/output functions like **scanf( )** and **printf( )**.

The **printf( )** function is not smart enough to perform the translation; the department is printed out as 1 and not manufacturing. Of course, we can write a function to print the correct enumerated values, using a **switch** statement, but that would reduce the clarity of the program. Even with this limitation, however, there are many situations in which enumerated variables are god sent!

### Are Enums Necessary?

Is there a way to achieve what was achieved using Enums in the previous program? Yes, using macros as shown below.

```
include <string.h>
define ASSEMBLY 0
define MANUFACTURING 1
define ACCCOUNTS 2
define STORES 3

int main()
{
 struct employee
 {
 char name[30];
 int age ;
 float bs ;
 int department ;
 };
 struct employee e ;
 strcpy (e.name, "Lothar Mattheus") ;
 e.age = 28 ;
 e.bs = 5575.50 ;
 e.department = MANUFACTURING ;
 return 0 ;
}
```

If the same effect—convenience and readability can be achieved using macros then why should we prefer enums? Because, macros have a global scope, whereas, scope of enum can either be global (if declared outside all functions) or local (if declared inside a function).

## Renaming Data types with *typedef*

There is one more technique, which, in some situations, can help to clarify the source code of a C program. This technique is to make use of the **typedef** declaration. Its purpose is to redefine the name of an existing variable type.

For example, consider the following statement in which the type **unsigned long int** is redefined to be of the type **TWOWORDS**:

```
typedef unsigned long int TWOWORDS ;
```

Now we can declare variables of the type **unsigned long int** by writing,

```
TWOWORDS var1, var2 ;
```

instead of

```
unsigned long int var1, var2 ;
```

Thus, **typedef** provides a short and meaningful way to call a data type. Usually, uppercase letters are used to make it clear that we are dealing with a renamed data type.

While the increase in readability is probably not great in this example, it can be significant when the name of a particular data type is long and unwieldy, as it often is with structure declarations. For example, consider the following structure declaration:

```
struct employee
{
 char name[30] ;
 int age ;
 float bs ;
};
struct employee e ;
```

This structure declaration can be made more handy to use when renamed using **typedef** as shown below.

```
struct employee
{
 char name[30] ;
 int age ;
```

```
 float bs ;
};
typedef struct employee EMP ;
EMP e1, e2 ;
```

Thus, by reducing the length and apparent complexity of data types, **typedef** can help to clarify source listing and save time and energy spent in understanding a program.

The above **typedef** can also be written as

```
typedef struct employee
{
 char name[30] ;
 int age ;
 float bs ;
} EMP ;
EMP e1, e2 ;
```

**typedef** can also be used to rename pointer data types as shown below.

```
struct employee
{
 char name[30] ;
 int age ;
 float bs ;
}
typedef struct employee * PEMP ;
PEMP p ;
p -> age = 32 ;
```

## Typecasting

Sometimes we are required to force the compiler to explicitly convert the value of an expression to a particular data type. This would be clear from the following example:

```
include <stdio.h>
int main()
{
 float a ;
 int x = 6, y = 4 ;
 a = x / y ;
```

```
printf ("Value of a = %f\n", a);
return 0 ;
}
```

And here is the output...

```
Value of a = 1.000000
```

The answer turns out to be 1.000000 and not 1.5. This is because, 6 and 4 are both integers and hence **6 / 4** yields an integer, 1. This 1 when stored in **a** is converted to 1.000000 . But what if we don't want the quotient to be truncated? One solution is to make either **x** or **y** as **float**. Let us say that other requirements of the program do not permit us to do this. In such a case what do we do? Use typecasting. The following program illustrates this:

```
include <stdio.h>
int main()
{
 float a ;
 int x = 6, y = 4 ;
 a = (float) x / y ;
 printf ("Value of a = %f\n", a) ;
 return 0 ;
}
```

And here is the output...

```
Value of a = 1.500000
```

This program uses typecasting. This consists of putting a pair of parentheses around the name of the data type. In this program we said,

```
a = (float) x / y ;
```

The expression ( **float** ) causes the variable **x** to be converted from type **int** to type **float** before being used in the division operation.

Here is another example of typecasting:

```
include <stdio.h>
int main()
{
 float a = 6.35 ;
```

```
printf ("Value of a on type casting = %d\n", (int) a) ;
printf ("Value of a = %f\n", a) ;
return 0 ;
}
```

And here is the output...

```
Value of a on type casting = 6
Value of a = 6.350000
```

Note that the value of **a** doesn't get permanently changed as a result of typecasting. Rather it is the value of the expression that undergoes type conversion whenever the cast appears.

## Bit Fields

If in a program a variable is to take only two values 1 and 0, we really need only a single bit to store it. Similarly, if a variable is to take values from 0 to 3, then two bits are sufficient to store these values. And if a variable is to take values from 0 through 7, then 3 bits will be enough, and so on.

Why waste an entire integer when one or two or three bits will do? Well, for one thing, there aren't any 1 bit or 2 bit or 3 bit data types available in C. However, when there are several variables whose maximum values are small enough to pack into a single memory location, we can use 'bit fields' to store several values in a single integer. To demonstrate how bit fields work, let us consider an example. Suppose we want to store the following data about an employee. Each employee can:

- (a) Be male or female
- (b) Be single, married, divorced or widowed
- (c) Have one of the eight different hobbies
- (d) Can choose from any of the fifteen different schemes proposed by the company to pursue his/her hobby.

This means we need 1 bit to store gender, 2 to store marital status, 3 for hobby, and 4 for scheme (with one value used for those who are not desirous of availing any of the schemes). We need 10 bits altogether, which means we can pack all this information into a single integer, since an integer is 16 bits long.

To do this using bit fields, we declare the following structure:

```
struct employee
{
 unsigned gender : 1 ;
 unsigned mar_stat : 2 ;
 unsigned hobby : 3 ;
 unsigned scheme : 4 ;
};
```

The colon ( : ) in the above declaration tells the compiler that we are talking about bit fields and the number after it tells how many bits to allot for the field.

Once we have established a bit field, we can reference it just like any other structure element, as shown in the program given below.

```
include <stdio.h>
define MALE 0 ;
define FEMALE 1 ;
define SINGLE 0 ;
define MARRIED 1 ;
define DIVORCED 2 ;
define WIDOWED 3 ;

int main()
{
 struct employee
 {
 unsigned gender : 1 ;
 unsigned mar_stat : 2 ;
 unsigned hobby : 3 ;
 unsigned scheme : 4 ;
 };
 struct employee e ;

 e.gender = MALE ;
 e.mar_status = DIVORCED ;
 e.hobby = 5 ;
 e.scheme = 9 ;

 printf ("Gender = %d\n", e.gender) ;
```

```
printf ("Marital status = %d\n", e.mar_status) ;
printf ("Bytes occupied by e = %d\n", sizeof (e)) ;
return 0 ;
}
```

And here is the output...

```
Gender = 0
Marital status = 2
Bytes occupied by e = 2
```

## Pointers to Functions

Every type of variable that we have discussed so far, with the exception of register, has an address. We have seen how we can reference variables of the type **char**, **int**, **float**, etc., through their addresses—that is by using pointers. Pointers can also point to C functions. And why not? C functions have addresses. If we know the function's address, we can point to it, which provides another way to invoke it. Let us see how this can be done.

```
include <stdio.h>
void display() ;
int main()
{
 printf ("Address of function display is %u\n", display) ;
 display() ; /* usual way of invoking a function */
 return 0 ;
}

void display()
{
 puts ("Long live viruses!!\n") ;
}
```

The output of the program would be:

```
Address of function display is 1125
Long live viruses!!
```

Note that, to obtain the address of a function, all that we have to do is mention the name of the function, as has been done in the **printf( )**

statement above. This is similar to mentioning the name of the array to get its base address.

Now let us see how using the address of a function, we can manage to invoke it. This is shown in the program given below.

```
/* Invoking a function using a pointer to a function */
#include <stdio.h>
void display();
int main()
{
 void (*func_ptr)();

 func_ptr = display ; /* assign address of function */
 printf ("Address of function display is %u", func_ptr) ;
 (*func_ptr)() ; /* invokes the function display() */
 return 0 ;
}
void display()
{
 puts ("\nLong live viruses!!") ;
}
```

The output of the program would be:

```
Address of function display is 1125
Long live viruses!!
```

In **main( )**, we declare the function **display( )** as a function returning nothing. But what are we to make of the declaration,

```
void (*func_ptr)() ;
```

that comes in the next line? We are obviously declaring something that, like **display( )**, will return nothing, but what is it? And why is **\*func\_ptr** enclosed in parentheses?

If we glance down a few lines in our program, we see the statement,

```
func_ptr = display ;
```

so we know that **func\_ptr** is being assigned the address of **display( )**. Therefore, **func\_ptr** must be a pointer to the function **display( )**. Thus, all that the declaration



```
void (*func_ptr)() ;
```

means is, that **func\_ptr** is a pointer to a function, which returns nothing. And to invoke the function, we are just required to write the statement,

```
(*func_ptr)() ; /* or simply, func_ptr() ; */
```

Pointers to functions are certainly awkward and off-putting. And why use them at all when we can invoke a function in a much simpler manner? What is the possible gain of using this esoteric feature of C? There are two possible uses:

- (a) In implementing callback mechanisms used popularly in Windows programming.
- (b) In binding functions dynamically, at run-time in C++ programming.

These topics are beyond the scope of this book. If you want to explore them further you can refer the book “Let Us C++” or “Test Your C++ Skills” by Yashavant Kanetkar.

## Functions Returning Pointers

The way functions return an **int**, a **float**, a **double** or any other data type, it can even return a pointer. However, to make a function return a pointer, it has to be explicitly mentioned in the calling function as well as in the function definition. The following program illustrates this:

```
int *fun() ;
int main()
{
 int *p ;
 p = fun() ;
 return 0 ;
}

int *fun()
{
 static int i = 20 ;
 return (&i) ;
}
```

This program just indicates how an integer pointer can be returned from a function. Beyond that, it doesn't serve any useful purpose. This

concept can be put to use while handling strings. For example, look at the following program which copies one string into another and returns the pointer to the target string:

```
char *copy (char *, char *);
int main()
{
 char *str;
 char source[] = "Jaded" ;
 char target[10] ;

 str = copy (target, source) ;
 printf ("%s\n", str) ;
 return 0 ;
}

char *copy (char *t, char *s)
{
 char *r ;

 r = t ;

 while (*s != '\0')
 {
 *t = *s ;
 t++ ;
 s++ ;
 }

 *t = '\0' ;
 return (r) ;
}
```

Here we have sent the base addresses of **source** and **target** strings to **copy( )**. In the **copy( )** function, the **while** loop copies the characters in the source string into the target string. Since during copying **t** is continuously incremented, before entering into the loop, the initial value of **t** is safely stored in the character pointer **r**. Once copying is over, this character pointer **r** is returned to **main( )**.

## Functions with Variable Number of Arguments

We have used **printf( )** so often without realizing how it works properly irrespective of how many arguments we pass to it. How do we go about writing such routines that can take variable number of arguments? And what have pointers got to do with it? There are three macros available in the file “**stdarg.h**” called **va\_start**, **va\_arg** and **va\_list** which allow us to handle this situation. These macros provide a method for accessing the arguments of the function when a function takes a fixed number of arguments followed by a variable number of arguments. The fixed number of arguments are accessed in the normal way, whereas the optional arguments are accessed using the macros **va\_start** and **va\_arg**. Out of these macros, **va\_start** is used to initialize a pointer to the beginning of the list of optional arguments. On the other hand, the macro **va\_arg** is used to advance the pointer to the next argument.

Let us put these concepts into action using a program. Suppose we wish to write a function **findmax( )** which would find out the maximum value from a set of values, irrespective of the number of values passed to it. Here is how we can do it...

```
#include <stdio.h>
#include <stdarg.h>
int findmax (int, ...);
int main()
{
 int max;

 max = findmax (5, 23, 15, 1, 92, 50);
 printf ("maximum = %d\n", max);

 max = findmax (3, 100, 300, 29);
 printf ("maximum = %d\n", max);
 return 0;
}

int findmax (int tot_num, ...)
{
 int max, count, num;

 va_list ptr;
```

```

 va_start (ptr, tot_num) ;
 max = va_arg (ptr, int) ;

 for (count = 1 ; count < tot_num ; count++)
 {
 num = va_arg (ptr, int) ;
 if (num > max)
 max = num ;
 }

 return (max) ;
}

```

Note how the **findmax( )** function has been declared. The ellipses ( ... ) indicate that the number of arguments after the first argument would be variable.

Here we are making two calls to **findmax( )**, first time to find maximum out of 5 values and second time to find maximum out of 3 values. Note that for each call the first argument is the count of arguments that follow the first argument. The value of the first argument passed to **findmax( )** is collected in the variable **tot\_num**. **findmax( )** begins with a declaration of a pointer **ptr** of the type **va\_list**. Observe the next statement carefully.

```

va_start (ptr, tot_num) ;

```

This statement sets up **ptr** such that it points to the first variable argument in the list. If we are considering the first call to **findmax( )**, **ptr** would now point to 23. The statement **max = va\_arg ( ptr, int )** would assign the integer being pointed to by **ptr** to **max**. Thus 23 would be assigned to **max**, and **ptr** would now start pointing to the next argument, i.e., 15. The rest of the program is fairly straightforward. We just keep picking up successive numbers in the list and keep comparing them with the latest value in **max**, till all the arguments in the list have been scanned. The final value in **max** is then returned to **main( )**.

How about another program to fix your ideas? This one calls a function **display( )** which is capable of printing any number of arguments of any type.

```

#include <stdio.h>
#include <stdarg.h>

```

```
void display (int, int, ...) ;
int main()
{
 display (1, 2, 5, 6) ;
 display (2, 4, 'A', 'a', 'b', 'c') ;
 display (3, 3, 2.5, 299.3, -1.0) ;
 return 0 ;
}

void display (int type, int num, ...)
{
 int i, j ;
 char c ;
 float f ;
 va_list ptr ;

 va_start (ptr, num) ;
 switch (type)
 {
 case 1 :
 for (j = 1 ; j <= num ; j++)
 {
 i = va_arg (ptr, int) ;
 printf ("%d ", i) ;
 }
 break ;
 case 2 :
 for (j = 1 ; j <= num ; j++)
 {
 c = va_arg (ptr, char) ;
 printf ("%c ", c) ;
 }
 break ;
 case 3 :
 for (j = 1 ; j <= num ; j++)
 {
 f = (float) va_arg (ptr, double) ;
 printf ("%f ", f) ;
 }
 }
 printf ("\n") ;
}
```

```
}
```

Here we pass two fixed arguments to the function **display( )**. The first one indicates the data type of the arguments to be printed and the second indicates the number of such arguments to be printed. Once again, through the statement **va\_start ( ptr, num )** we set up **ptr** such that it points to the first argument in the variable list of arguments. Then depending upon whether the value of type is 1, 2 or 3, we print out the arguments as **ints**, **chars** or **floats**.

In all calls to **display( )** the second argument indicates how many values are we trying to print. Contrast this with **printf( )**. To it we never pass an argument indicating how many value are we trying to print. Then how does **printf( )** figure this out? Simple. It scans the format string and counts the number of format specifiers that we have used in it to decide how many values are being printed.

## Unions

Unions are derived data types, the way structures are. Unions and structures look alike, but are engaged in totally different activities.

Both structures and unions are used to group a number of different variables together. But while a structure enables us treat a number of different variables stored at different places in memory, a union enables us to treat the same space in memory as a number of different variables. That is, a union offers a way for a section of memory to be treated as a variable of one type on one occasion, and as a different variable of a different type on another occasion.

You might wonder why it would be necessary to do such a thing, but we will be seeing several very practical applications of unions soon. First, let us take a look at a simple example.

```
/* Demo of union at work */
#include <stdio.h>
int main()
{
 union a
 {
 short int i ;
 char ch[2] ;
 };
 union a key ;
```

```
key.i = 512 ;
printf ("key.i = %d\n", key.i) ;
printf ("key.ch[0] = %d\n", key.ch[0]) ;
printf ("key.ch[1] = %d\n", key.ch[1]) ;
return 0 ;
}
```

And here is the output...

```
key.i = 512
key.ch[0] = 0
key.ch[1] = 2
```

As you can see, first we declared a data type of the type **union a**, and then a variable **key** to be of the type **union a**. This is similar to the way we first declare the structure type and then the structure variables. Also, the union elements are accessed exactly the same way in which the structure elements are accessed, using a **'.'** operator. However, the similarity ends here. To illustrate this let us compare the following data types:

```
struct a
{
 short int i ;
 char ch[2] ;
};
struct a key ;
```

This data type would occupy 4 bytes in memory, 2 for **key.i** and 1 each for **key.ch[ 0 ]** and **key.ch[ 1 ]**, as shown in Figure 22.1.

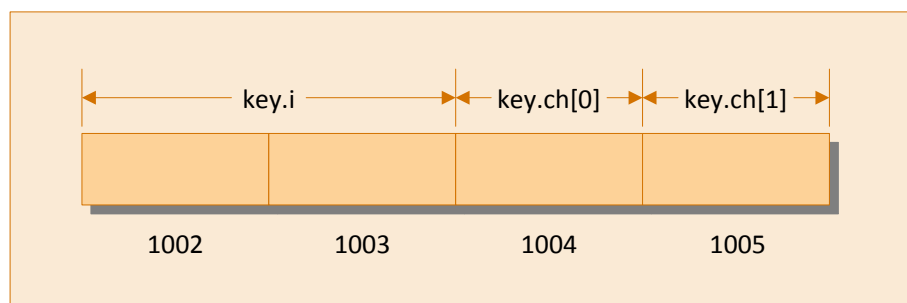


Figure 22.1

Now we declare a similar data type, but instead of using a structure we use a union.

```
union a
{
 short int i;
 char ch[2];
};
union a key;
```

Representation of this data type in memory is shown in Figure 22.2.

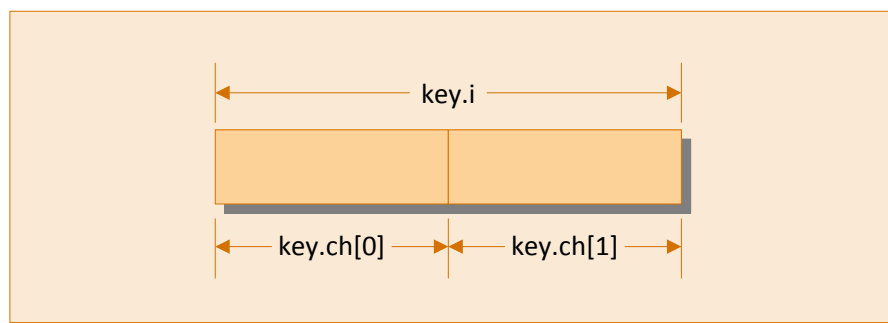


Figure 22.2

As shown in Figure 22.2, the union occupies only 2 bytes in memory. Note that the same memory locations that are used for **key.i** are also being used by **key.ch[ 0 ]** and **key.ch[ 1 ]**. It means that the memory locations used by **key.i** can also be accessed using **key.ch[ 0 ]** and **key.ch[ 1 ]**. What purpose does this serve? Well, now we can access the 2 bytes simultaneously (by using **key.i**) or the same 2 bytes individually (using **key.ch[ 0 ]** and **key.ch[ 1 ]**).

This is a frequent requirement while interacting with the hardware, i.e., sometimes we are required to access 2 bytes simultaneously and sometimes each byte individually. Faced with such a situation, using union is the answer, usually.

Perhaps you would be able to understand the union data type more thoroughly if we take a fresh look at the output of the above program. Here it is...

```
key.i = 512
key.ch[0] = 0
key.ch[1] = 2
```



Let us understand this output in detail. 512 is an integer, a 2 byte number. Its binary equivalent will be 0000 0010 0000 0000. We would expect that this binary number when stored in memory would look as shown in Figure 22.3.

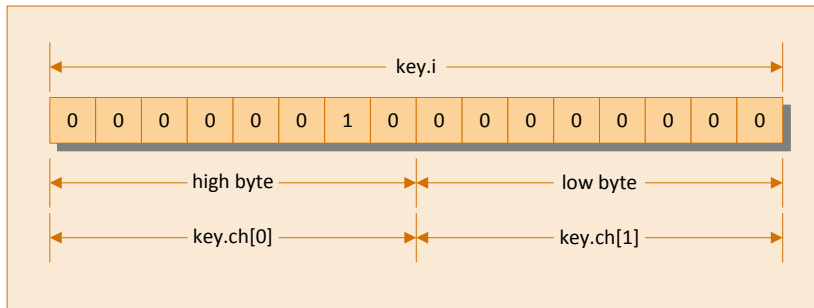


Figure 22.3

If the number is stored in this manner, then the output of **key.ch[ 0 ]** and **key.ch[ 1 ]** should have been 2 and 0, respectively. But, if you look at the output of the program written above, it is exactly the opposite. Why is it so? Because, in CPUs that follow little-endian architecture (Intel CPUs, for example), when a 2-byte number is stored in memory, the low byte is stored before the high byte. It means, actually 512 would be stored in memory as shown in Figure 22.4. In CPUs with big-endian architecture this reversal of bytes does not happen.

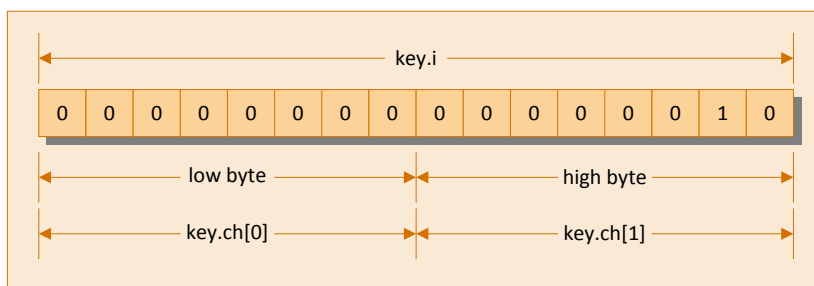


Figure 22.4

Now, we can see why value of **key.ch[ 0 ]** is printed as 0 and value of **key.ch[ 1 ]** is printed as 2.

One last thing. We can't assign different values to the different union elements at the same time. That is, if we assign a value to **key.i**, it gets automatically assigned to **key.ch[ 0 ]** and **key.ch[ 1 ]**. Vice versa, if we assign a value to **key.ch[ 0 ]** or **key.ch[ 1 ]**, it is bound to get assigned to **key.i**. Here is a program that illustrates this fact.

```
include <stdio.h>
int main()
{
 union a
 {
 short int i;
 char ch[2];
 };
 union a key;

 key.i = 512 ;
 printf ("key.i = %d\n", key.i);
 printf ("key.ch[0] = %d\n", key.ch[0]);
 printf ("key.ch[1] = %d\n", key.ch[1]);

 key.ch[0] = 50 ; /* assign a new value to key.ch[0] */
 printf ("key.i = %d\n", key.i);
 printf ("key.ch[0] = %d\n", key.ch[0]);
 printf ("key.ch[1] = %d\n", key.ch[1]);
 return 0 ;
}
```

And here is the output...

```
key.i = 512
key.ch[0] = 0
key.ch[1] = 2
key.i= 562
key.ch[0] = 50
key.ch[1] = 2
```

Before we move on to the next section, let us reiterate that a union provides a way to look at the same data in several different ways. For example, there can exist a union as shown below.

```
union b
{
 double d ;
 float f[2];
 short int i[4];
 char ch[8];
```

```
};
union b data ;
```

In what different ways can the data be accessed from it? Sometimes, as a complete set of 8 bytes (**data.d**), sometimes as two sets of 4 bytes each (**data.f[ 0 ]** and **data.f[ 1 ]**), sometimes as four sets of 2 bytes each (**data.i[ 0 ]**, **data.i[ 1 ]**, **data.i[ 2 ]** and **data.i[ 3 ]**) and sometimes as 8 individual bytes (**data.ch[ 0 ]**, **data.ch[ 1 ]**... **data.ch[ 7 ]**).

Also note that there can exist a union, each of whose elements is of different size. In such a case, the size of the union variable will be equal to the size of the longest element in the union.

## Union of Structures

Just as one structure can be nested within another, a union too can be nested in another union. Not only that, there can be a union in a structure, or a structure in a union. Here is an example of structures nested in a union.

```
#include <stdio.h>
int main()
{
 struct a
 {
 int i ;
 char c[2] ;
 };
 struct b
 {
 int j ;
 char d[2] ;
 };
 union z
 {
 struct a key ;
 struct b data ;
 };
 union z strange ;

 strange.key.i = 512 ;
 strange.data.d[0] = 0 ;
 strange.data.d[1] = 32 ;
}
```

```
printf ("%d\n", strange.key.i);
printf ("%d\n", strange.data.j);
printf ("%d\n", strange.key.c[0]);
printf ("%d\n", strange.data.d[0]);
printf ("%d\n", strange.key.c[1]);
printf ("%d\n", strange.data.d[1]);
return 0 ;
}
```

And here is the output...

```
512
512
0
0
32
32
```

Just as we do with nested structures, we access the elements of the union in this program using the ‘.’ operator twice. Thus,

```
strange.key.i
```

refers to the variable **i** in the structure **key** in the union **strange**. Analysis of the output of the above program is left to the reader.

## Utility of Unions

Suppose we wish to store information about employees in an organization. The items of information are as shown below.

```
Name
Grade
Age
If Grade = HSK (Highly Skilled)
 hobby name
 credit card no.
If Grade = SSK (Semi Skilled)
 Vehicle no.
 Distance from Co.
```

Since this is dissimilar information we can gather it together using a structure as shown below.

```
struct employee
{
 char n[20];
 char grade[4];
 int age ;
 char hobby[10];
 int crcardno ;
 char vehno[10];
 int dist ;
};
struct employee e ;
```

Though grammatically there is nothing wrong with this structure, it suffers from a disadvantage. For any employee, depending upon his/her grade, either the fields hobby and credit card no. or the fields vehicle number and distance would get used. Both sets of fields would never get used. This would lead to wastage of memory with every structure variable that we create, since every structure variable would have all the four fields apart from name, grade and age. This can be avoided by creating a **union** between these sets of fields. This is shown below.

```
struct info1
{
 char hobby[10];
 int crcardno ;
};
struct info2
{
 char vehno[10];
 int dist ;
};
union info
{
 struct info1 a ;
 struct info2 b ;
};
struct employee
{
 char n[20] ;
```

```
char grade[4];
int age ;
union info f ;
};
struct employee e ;
```

### The *volatile* Qualifier

When we define variables in a function the compiler may optimize the code that uses the variable. That is, the compiler may compile the code in a manner that will run in the most efficient way possible. The compiler achieves this by using a CPU register to store the variable's value rather than storing it in stack.

However, if we declare the variable as *volatile*, then it serves as a warning to the compiler that it should not *optimize* the code containing this variable. In such a case whenever we use the variable its value would be loaded from memory into register, operations would be performed on it and the result would be written back to the memory location allocated for the variable.

We can declare a volatile variable as:

```
volatile int j ;
```

Another place where we may want to declare a variable as *volatile* is when the variable is not within the control of the program and is likely to get altered from outside the program. For example, the variable

```
volatile float temperature ;
```

might get modified through the digital thermometer attached to the computer.

### Summary

- (a) The enumerated data type and the **typedef** declaration add to the clarity of the program.
- (b) Typecasting makes the data type conversions for specific operations.
- (c) When the information to be stored can be represented using a few bits of a byte we can use bit fields to pack more information in a byte.

- (d) Every C function has an address that can be stored in a pointer to a function. Pointers to functions provide one more way to call functions.
- (e) We can write a function that receives a variable number of arguments.
- (f) Unions permit access to same memory locations in multiple ways.

## Exercise

**[A]** What will be the output of the following programs:

- (a) 

```
#include <stdio.h>
int main()
{
 enum status { pass, fail, atkt };
 enum status stud1, stud2, stud3;
 stud1 = pass;
 stud2 = fail;
 stud3 = atkt;
 printf ("%d %d %d\n", stud1, stud2, stud3);
 return 0;
}
```
- (b) 

```
#include <stdio.h>
int main()
{
 printf ("%f\n", (float) ((int) 3.5 / 2));
 return 0;
}
```
- (c) 

```
#include <stdio.h>
int main()
{
 float i, j;
 i = (float) 3 / 2;
 j = i * 3;
 printf ("%d\n", (int) j);
 return 0;
}
```

**[B]** Point out the error, if any, in the following programs:

- (a) 

```
#include <stdio.h>
```

```
int main()
{
 typedef struct patient
 {
 char name[20] ;
 int age ;
 int systolic_bp ;
 int diastolic_bp ;
 } ptt ;
 ptt p1 = { "anil", 23, 110, 220 } ;
 printf ("%s %d\n", p1.name, p1.age) ;
 printf ("%d %d\n", p1.systolic_bp, p1.diastolic_bp) ;
 return 0 ;
}
```

```
(b) #include <stdio.h>
void show() ;
int main()
{
 void (*s)() ;
 s = show ;
 (*s)() ;
 s() ;
 return 0 ;
}
void show()
{
 printf ("don't show off. It won't pay in the long run\n") ;
}
```

```
(c) #include <stdio.h>
void show (int, float) ;
int main()
{
 void (*s)(int, float) ;
 s = show ;
 (*s)(10, 3.14) ;
 return 0 ;
}
void show (int i, float f)
{
 printf ("%d %f\n", i, f) ;
}
```



```
}
```

**[C]** Attempt the following:

- (a) Create an array of four function pointers. Each pointer should point to a different function. Each of these functions should receive two integers and return a float. Using a loop call each of these functions using the addresses present in the array.
- (b) Write a function that receives variable number of arguments, where the arguments are the coordinates of a point. Based on the number of arguments received, the function should display type of shape like a point, line, triangle, etc., that can be drawn.
- (c) Write a program, which stores information about a date in a structure containing three members—day, month and year. Using bit fields the day number should get stored in first 5 bits of day, the month number in 4 bits of month and year in 12 bits of year. Write a program to read date of joining of 10 employees and display them in ascending order of year.
- (d) Write a program to read and store information about insurance policy holder. The information contains details like gender, whether the holder is minor/major, policy name and duration of the policy. Make use of bit-fields to store this information.



- What is Linux?
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- Zombies and Orphans
- One Interesting Fact
- Communication using Signals
- Handling Multiple Signals
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- Event Driven Programming
- Where do you go from here?
- Summary
- Exercise



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# 23 *C Under Linux*

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- What is Linux
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- Exercise

**T**oday the programming world is divided into two major camps—the Windows world and the Linux world. Since its humble beginning about two decades ago, Linux has steadily drawn the attention of programmers across the globe and has successfully created a community of its own. So as a C programmer you must know how to do C programming under Linux environment. Without any further discussions let us now set out on the Linux voyage. I hope you find the journey interesting and exciting.

### What is Linux?

Linux is a clone of the UNIX operating system. Its kernel was written from scratch by Linus Trovalds with assistance from a loosely-knit team of programmers across the world on the Internet. It has all the features you would expect in a modern OS. Moreover, unlike Windows or UNIX, Linux is available completely free of cost. The kernel of Linux is available in source code form. Anybody is free to change it to suit his/her requirement, with a precondition that the changed kernel can be distributed only in the source code form. Several programs, frameworks, utilities have been built around the Linux kernel. A common user may not want the headaches of downloading the kernel, going through the complicated compilation process, then downloading the frameworks, programs and utilities. Hence many organizations have come forward to make this job easy. They distribute the precompiled kernel, programs, utilities and frameworks on a common media. Moreover, they also provide installation scripts for easy installations of the Linux OS and applications. Some of the popular distributions are RedHat, SUSE, Caldera, Debian, Mandrake, Slackware, etc. Each of them contain the same kernel but may contain different application programs, libraries, frameworks, installation scripts, utilities, etc. Which one is better than the other is only a matter of taste.

Linux was first developed for x86-based PCs (386 or higher). These days it also runs on Compaq Alpha AXP, Sun SPARC, Motorola 68000 machines (like Atari ST and Amiga), MIPS, PowerPC, ARM, Intel Itanium, SuperH, etc. Thus Linux works on literally every conceivable microprocessor architecture.

Under Linux one is faced with simply too many choices of Linux distributions, graphical shells and managers, editors, compilers, linkers, debuggers, etc. For simplicity (in my opinion), I have chosen the following combination for the programs in this chapter:

Linux Distribution – Ubuntu Linux 11.10  
Development Environment – NetBeans 8.0.2

Ubuntu Linux can be downloaded from [www.ubuntu.com](http://www.ubuntu.com) and Netbeans can be downloaded from [www.netbeans.org](http://www.netbeans.org)

## C Programming Under Linux

Is C under Linux any different than C under DOS or C under Windows? Well, it is same as well as different. It is same to the extent of using language elements like data types, control instructions and the overall syntax. The usage of standard library functions is also same even though the implementation of each might be different under different OS. For example, a **printf( )** would work under all OSs, but the way it is defined is likely to be different for different OSs. However, the programmer doesn't suffer because of this, since, he/she can continue to call **printf( )** the same way, no matter how it is implemented.

But there the similarity ends. If we are to build programs that utilize the features offered by the OS then things are bound to be different across OSs. For example, if we are to write a C program that would create a Window and display a message "hello" at the point where the user clicks the left mouse button. The architecture of this program would be very closely tied with the OS under which it is being built. This is because the mechanisms for creating a window, reporting a mouse click, handling a mouse click, displaying the message, closing the window, etc., are very closely tied with the OS for which the program is being built. In short, the programming architecture (better known as programming model) for each OS is different. Hence, naturally, the program that achieves the same task under different OS would have to be different.

## The 'Hello Linux' Program

As with any new platform, we would begin our journey in the Linux world by creating a simple 'hello world' program. Here is the source code....

```
#include <stdio.h>
int main()
{
 printf ("Hello Linux\n");
 return 0 ;
}
```

The program is exactly same as compared to a console program under DOS/Windows. It begins with **main( )** and uses **printf( )** standard library function to produce its output. So what is the difference? The difference is in the way programs are typed, compiled and executed. For this you should ideally use NetBeans as it offers an integrated development environment. Alternately, you can type the program using editor like 'vi' or 'vim' and compile and link it using 'gcc'.

If you are using NetBeans then create a new C/C++ project in it, type the above program, and build and execute it using F6.

If you are using vi – gcc combination then carry out the following steps:

- (a) Type the program and save it under the name 'hello.c'.
- (b) At the command prompt switch to the directory containing 'hello.c' using the **cd** command.
- (c) Now compile the program using the **gcc** compiler as shown below:

```
$ gcc hello.c
```

- (d) On successful compilation, **gcc** produces a file named 'a.out'. This file contains the machine code of the program which can now be executed.
- (e) Execute the program using the following command:

```
$./a.out
```

- (f) Now you should be able to see the output 'Hello Linux' on the screen.

Having created a Hello Linux program and gone through the edit-compile-execute cycle once, let us now turn our attention to Linux specific programming. We will begin with processes.

## Processes

Gone are the days when only one processor in the PC/Laptop executed only one job (task) in memory at any time. Today, the modern OSs like Windows and Linux exploit the capabilities of Dual-core and Quad-core processors and execute several tasks simultaneously. In Linux each running task is known as a 'process'.

The scheduling of different processes on multiple processors is done by a program called 'Scheduler' which is a vital component of the Linux OS.

The Kernel (core) of the OS assigns each process running in memory a unique ID to distinguish it from other running processes. This ID is often known as Process ID or simply PID. It is very simple to print the PID of a running process programmatically. Here is the program that achieves this...

```
include <stdio.h>
include <sys/types.h>
int main()
{
 printf ("Process ID = %d", getpid());
 return 0 ;
}
```

Here **getpid( )** is a library function which returns the process ID of the calling process. When the execution of the program comes to an end, the process stands terminated. Every time we run the program a new process is created. Hence the kernel assigns a new ID to the process each time. This can be verified by executing the program several times—each time it would produce a different output.

## Parent and Child Processes

As we know, our running program is a process. From this process we can create another process. There is a parent-child relationship between the two processes. The way to achieve this is by using a library function called **fork( )**. This function splits the running process into two processes, the existing one is known as parent and the new process is known as child. Here is a program that demonstrates this...

```
include <stdio.h>
include <unistd.h> /* for prototype of fork() */
int main()
{
 printf ("Before Forking\n");
 fork();
 printf ("After Forking\n");
 return 0 ;
}
```

Here is the output of the program...

Before Forking



### After Forking After Forking

Watch the output of the program. You can notice that all the statements after the **fork( )** are executed twice—once by the parent process and second time by the child process. In other words **fork( )** has managed to split our process into two.

But why on earth would we like to do this? At times, we want our program to perform two jobs simultaneously. Since these jobs may be inter-related, we may not want to create two different programs to perform them. Let me give you an example. Suppose we want to perform two jobs—copy contents of source file to target file and display an animated GIF file indicating that the file copy is in progress. The GIF file should continue to play till file copy is taking place. Once the copying is over the playing of the GIF file should be stopped. Since both these jobs are inter-related, they cannot be performed in two different programs. Also, they cannot be performed one after another. Both jobs should be performed simultaneously.

At such times, we would want to use **fork( )** to create a child process and then write the program in such a manner that file copy is done by the parent and displaying of animated GIF file is done by the child process. The following program shows how this can be achieved. Note that, the issue here is to show how to perform two different but inter-related jobs simultaneously. Hence I have skipped the actual code for file copying and playing the animated GIF file.

```
#include <stdio.h>
#include <sys/types.h>
#include <unistd.h>
int main()
{
 int pid ;

 pid = fork() ;
 if (pid == 0)
 {
 printf ("In child process\n") ;
 /* code to play animated GIF file */
 }
 else
 {

```

```
 printf ("In parent process\n");
 /* code to copy file */
 }
 return 0 ;
}
```

As we know, **fork( )** creates a child process and duplicates the code of the parent process in the child process. There onwards, the execution of the **fork( )** function continues in both the processes. Thus, the duplication code inside **fork( )** is executed once, whereas, the remaining code inside it is executed in both the parent as well as the child process. Hence, control would come back from **fork( )** twice, even though it is actually called only once. When control returns from **fork( )** of the parent process, it returns the PID of the child process. As against this, when control returns from **fork( )** of the child process, it always returns a 0. This can be exploited by our program to segregate the code that we want to execute in the parent process, from the code that we want to execute in the child process. We have done this in our program using an **if** statement. In the parent process the 'else block' would get executed, whereas, in the child process the 'if block' would get executed.

Let us now write one more program. This program would use the **fork( )** call to create a child process. In the child process we would print the PID of child and its parent, whereas, in the parent process we would print the PID of the parent and its child. Here is the program...

```
include <stdio.h>
include <sys/types.h>
include <unistd.h>
int main()
{
 int pid ;

 pid = fork() ;
 if (pid == 0)
 {
 printf ("Child : Hello I am the child process\n") ;
 printf ("Child : Child's PID: %d\n", getpid()) ;
 printf ("Child : Parent's PID: %d\n", getppid()) ;
 }
 else
 {

```

```
 printf ("Parent : Hello I am the parent process\n");
 printf ("Parent : Parent's PID: %d\n", getpid());
 printf ("Parent : Child's PID: %d\n", pid);
}
return 0 ;
}
```

Given below is the output of the program.

```
Child : Hello I am the child process
Child : Child's PID: 4706
Child : Parent's PID: 4705
Parent : Hello I am the Parent process
Parent : Parent's PID: 4705
Parent : Child's PID: 4706
```

In addition to **getpid( )**, there is another related function that we have used in this program—**getppid( )**. As the name suggests, this function returns the PID of the parent of the calling process.

You can tally the PIDs from the output and convince yourself that you have understood the **fork( )** function well. A lot of things that follow use the **fork( )** function. So make sure that you understand it thoroughly.

Note that, even Linux internally uses **fork( )** to create new child processes. Thus, there is an inverted tree-like structure of all the processes running in memory. The father of all these processes is a process called **init**. If we want to get a list of all the running processes in memory we can do so using the **ps** command as shown below.

```
$ ps -A
```

Here the switch **-A** indicates that we want to list all the running processes. We can get the **\$** prompt by starting a Terminal Window in Ubuntu Linux from Applications | Accessories menu.

## More Processes

Suppose we want to execute a program on the disk as part of a child process. For this, first we should create a child process using **fork( )** and then from within the child process we should call an **exec** function to execute the program on the disk as part of a child process. Note that, there is a family of **exec** library functions, each basically does the same job, but with a minor variation. For example, **execl( )** function permits us

to pass a list of command-line arguments to the program to be executed. **execv( )** also does the same job as **execl( )** except that the command-line arguments can be passed to it in the form of an array of pointers to strings. There also exist other variations like **execle( )** and **execvp( )**.

Let us now see a program that uses **execl( )** to run a new program in the child process.

```
#include <stdio.h>
#include <unistd.h>
int main()
{
 int pid ;

 pid = fork() ;
 if (pid == 0)
 {
 execl ("/bin/ls", "-al", "/etc", NULL) ;
 printf ("Child: After exec()\n") ;
 }
 else
 printf ("Parent process\n") ;
 return 0 ;
}
```

After forking a child process, we have called the **execl( )** function. This function accepts variable number of arguments. The first parameter to **execl( )** is the absolute path of the program to be executed. The remaining parameters describe the command-line arguments for the program to be executed. The last parameter is an end of argument marker which must always be **NULL**. Thus, in our case we have called upon the **execl( )** function to execute the **ls** program as shown below.

```
$ ls -al /etc
```

As a result, all the contents of the **/etc** directory are listed on the screen. Note that the **printf( )** below the call to **execl( )** function is not executed. This is because, the **exec** family functions overwrite the image of the calling process with the code and data of the program that is to be executed. In our case, the child process's memory was overwritten by the code and data of the **ls** program. Hence, the call to **printf( )** did not materialize.

It would make little sense in calling **execl( )** before **fork( )**. This is because, a child would not get created and **execl( )** would simply overwrite the main process itself. As a result, no statement beyond the call to **execl( )** would ever get executed. Hence **fork( )** and **execl( )** usually go hand in hand.

## Zombies and Orphans

We know that the **ps -A** command lists all the running processes. But from where does the **ps** program get this information? Well, Linux maintains a table containing information about all the processes. This table is called 'Process Table'. Apart from other information, the process table contains an entry of 'exit code' of the process. This integer value indicates the reason why the process was terminated. Even though the process comes to an end, its entry would remain in the process table until such time that the parent of the terminated process queries the exit code. This act of querying deletes the entry of the terminated process from the process table and returns the exit code to the parent that raised the query.

When we fork a new child process and the parent and the child continue to execute there are two possibilities—either the child process ends first or the parent process ends first. Let us discuss both these possibilities.

### (a) Child terminates earlier than the parent

In this case, till the time parent does not query the exit code of the terminated child the entry of the child process would continue to exist. Such a process in Linux terminology is known as a 'Zombie' process. Zombie means ghost, or in plain simple Hindi a 'Bhoot'. Moral is, a parent process should query the process table immediately after the child process has terminated. This would prevent a Zombie.

What if the parent terminates without querying. In such a case, the Zombie child process is treated as an 'Orphan' process. Immediately, the father of all processes—**init**—adopts the Orphaned process. Next, as a responsible parent, **init** queries the process table, as a result of which, the child process entry is eliminated from the process table.

### (b) Parent terminates earlier than the child

Since every parent process is launched from the Linux shell, the parent of the parent is the **shell** process. When our parent process terminates, the **shell** queries the process table. Thus a proper cleanup happens for

the parent process. However, the child process which is still running is left Orphaned. Immediately, the **init** process would adopt it and when its execution is over **init** would query the process table to clean up the entry for the child process. Note that in this case the child process does not become a Zombie.

Thus, when a Zombie or an Orphan gets created the OS takes over and ensures that a proper cleanup of the relevant process table entry happens. However, as a good programming practice our program should get the exit code of the terminated process and thereby ensure a proper cleanup. Note that, here cleanup is important (it happens anyway). Why is it important to get the exit code of the terminated process? It is because, it is the exit code that would give indication about whether the job assigned to the process was completed successfully or not. The following program shows how this can be done.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h> /* for waitpid() and WIFEXITED */
int main()
{
 unsigned int i = 0 ;
 int pid, status ;

 pid = fork() ;
 if (pid == 0)
 {
 while (i < 4294967295u)
 i++ ;
 printf ("The child is now terminating\n") ;
 }
 else
 {
 waitpid (pid, &status, 0) ;
 if (WIFEXITED (status))
 printf ("Parent: Child terminated normally\n") ;
 else
 printf ("Parent: Child terminated abnormally\n") ;
 }
 return 0 ;
}
```

In this program we have applied a big loop in the child process. This loop ensures that the child does not terminate immediately. From within the parent process we have made a call to the **waitpid( )** function. This function makes the parent process wait till the time the execution of the child process does not come to an end. This ensures that the child process never becomes Orphaned. Once the child process terminates, the **waitpid( )** function queries its exit code and returns back to the parent. As a result of querying, the child process does not become a Zombie.

The first parameter of **waitpid( )** function is the pid of the child process for which the wait has to be performed. The second parameter is the address of an integer variable which is set up with the exit status code of the child process. The third parameter is used to specify some options to control the behavior of the wait operation. We have not used this parameter and hence we have passed a **0**. Next, we have made use of the **WIFEXITED( )** macro to test if the child process exited normally or not. This macro takes the status value as a parameter and returns a non-zero value if the process terminated normally. Using this macro the parent suitably prints a message to report the termination status (normal / abnormal) of its child process.

### One Interesting Fact

When we use **fork( )** to create a child process the child process does not contain the entire data and code of the parent process. Then does it mean that the child process contains the data and code below the **fork( )** call. Even this is not so. In actuality, the code never gets duplicated. Linux internally manages to intelligently share it. As against this, some data is shared, some is not. Till the time both the processes do not change the value of the variables they keep getting shared. However, if any of the processes (either child or parent) attempt to change the value of a variable it is no longer shared. Instead a new copy of the variable is made for the process that is attempting to change it. This not only ensures data integrity but also saves precious memory.

### Communication using Signals

Communication is the essence of all progress. This is true in real life as well as in programming. In today's world a program that runs in isolation is of little use. A worthwhile program has to communicate with the outside world in general and with the OS in particular. Let us explore how this communication happens under Linux.

We have already seen how to use **fork( )** and **exec( )** functions to create a child process and to execute a new program, respectively. These library functions got the job done by communicating with the Linux OS. Thus, the direction of communication was from the program to the OS. The reverse communication—from the OS to the program—is achieved using a mechanism called ‘Signal’. Let us now write a simple program that would help you experience the signal mechanism.

```
include <stdio.h>
int main()
{
 while (1)
 printf ("Program Running\n");
 return 0 ;
}
```

The program is fairly straightforward. All that we have done here is, we have used an infinite **while** loop to print the message "Program Running" on the screen. When the program is running, we can terminate it by pressing the Ctrl+C. When we press Ctrl+C, the keyboard device driver informs the Linux kernel about pressing of this special key combination. The kernel reacts to this by sending a signal to our program. Since we have done nothing to handle this signal, the default signal handler gets called. In this default signal handler there is code to terminate the program. Hence on pressing Ctrl+C the program gets terminated.

But how on earth would the default signal handler get called. Well, it is simple. There are several signals that can be sent to a program. A unique number is associated with each signal. To avoid remembering these numbers, they have been defined as macros like **SIGINT**, **SIGKILL**, **SIGCONT**, etc., in the file ‘signal.h’. Every process contains several ‘signal ID - function pointer’ pairs indicating for which signal which function should be called. If we do not decide to handle a signal then against that signal ID the address of the default signal handler function is present. It is precisely this default signal handler for **SIGINT** that got called when we pressed Ctrl+C when the above program was executed. INT in **SIGINT** stands for interrupt.

Let us now see how we can prevent the termination of our program even after hitting Ctrl+C. This is shown in the following program:

```
include <stdio.h>
```



```
include <signal.h>

void sighandler (int signum)
{
 printf ("SIGINT received. Inside sighandler\n") ;
}

int main()
{
 signal (SIGINT, sighandler) ;
 while (1)
 printf ("Program Running\n") ;
 return 0 ;
}
```

In this program we have registered a signal handler for the SIGINT signal by using the **signal( )** library function. The first parameter of this function specifies the ID of the signal that we wish to register. The second parameter is the address of a function that should get called whenever the signal is received by our program. The signal handler function must always receive an **int** and return a **void**.

Now when we press Ctrl+C, the registered handler, namely, **sighandler( )** would get called. This function would display the message '**SIGINT received. Inside sighandler**' and return the control back to **main( )**. Note that, unlike the default handler, our handler does not terminate the execution of our program. So only way to terminate it is to kill the running process from a different terminal. For this we need to open the terminal window. Next, do a **ps -a** to obtain the list of processes running at all the command prompts that we have launched. Note down the process id of our application. Finally kill our application process by saying,

```
$ kill 23312
```

In my case the process id turned out to be **23312**. In your case the the process id might be a different number.

If we wish, we can abort the execution of the program in the signal handler itself by using the **exit ( 0 )** beyond the **printf( )**.

Note that signals work asynchronously. That is, when a signal is received no matter what our program is doing, the signal handler would

immediately get called. Once the execution of the signal handler is over, the execution of the program is resumed from the point where it left off when the signal was received.

## Handling Multiple Signals

Now that we know how to handle one signal, let us try to handle multiple signals. Here is the program to do this...

```
include <stdio.h>
include <signal.h>

void inthandler (int signum)
{
 printf ("SIGINT Received\n");
}

void termhandler (int signum)
{
 printf ("SIGTERM Received\n");
}

int main()
{
 signal (SIGINT, inthandler);
 signal (SIGTERM, termhandler);

 while (1)
 printf ("Program Running\n");

 return 0 ;
}
```

In this program, apart from **SIGINT**, we have additionally registered a new signal, **SIGTERM**. The **signal( )** function is called twice to register a different handler for the two signals. After registering the signals, we enter an infinite **while** loop to print the '**Program Running**' message on the screen.

As in the previous program, here too, when we press Ctrl+C the handler for the **SIGINT**, i.e., **inthandler( )** is called. However, when we try to kill the program from the second terminal using the **kill** command, the program does not terminate. This is because, when the **kill** command is

used, it sends the running program a **SIGTERM** signal. The default handler for the message terminates the program. Since we have handled this signal ourselves, the handler for **SIGTERM**, i.e., **termhandler( )** gets called. As a result, the **printf( )** statement in the **termhandler( )** function gets executed and the message 'SIGTERM Received' gets displayed on the screen. Once the execution of **termhandler( )** function is over, the program resumes its execution and continues to print 'Program Running'. Then how are we supposed to terminate the program? Simple. Use the following command from the another terminal window:

```
$ kill -SIGKILL 23312
```

As the command indicates, we are trying to send a **SIGKILL** signal to our program. A **SIGKILL** signal terminates the program.

Most signals may be caught by the process, but there are a few signals that the process cannot catch, and they cause the process to terminate. Such signals are often known as uncatchable signals. The **SIGKILL** signal is an uncatchable signal that forcibly terminates the execution of a process.

Note that, even if a process attempts to handle the **SIGKILL** signal by registering a handler for it, still the control would always land in the default **SIGKILL** handler, which would terminate the program.

The **SIGKILL** signal is to be used as a last resort to terminate a program that gets out of control. One such process that makes uses of this signal is a system shutdown process. It first sends a **SIGTERM** signal to all processes, waits for a while, thus giving a 'grace period' to all the running processes. However, after the grace period is over, it forcibly terminates all the remaining processes using the **SIGKILL** signal.

Note that it is also possible to handle multiple signals using a common signal handler. For this, all that we need to do is register the same function for multiple signals which calling the **signal( )** function. This is shown in the following code snippet:

```
signal (SIGINT, sighandler) ;
signal (SIGTERM, sighandler) ;
signal (SIGCONT, sighandler) ;
```

Here, during each call to the **signal( )** function we have specified the address of a common signal handler named **sighandler( )**. Thus, the

same signal handler function would get called when one of the three signals are received. This does not lead to a problem since inside the **sighandler( )** we can figure out the signal ID using the **signum** parameter of the function.

Note that, we can easily afford to mix the two methods of registering signals in a program. That is, we can register separate signal handlers for some of the signals and a common handler for some other signals. Registering a common handler makes sense if we want to react to different signals in exactly the same way.

## Blocking Signals

Sometimes, we may want that flow of execution of a critical/time-critical portion of the program should not be hampered by the occurrence of one or more signals. In such a case, we may decide to block the signal. Once we are through with the critical/time-critical code, we can unblock the signal(s). Note that, if a signal arrives when it is blocked, it is simply queued into a signal queue. When the signals are unblocked, the process immediately receives all the pending signals one after another. Thus blocking of signals defers the delivery of signals to a process till the execution of some critical/time-critical code is over. Instead of completely ignoring the signals or letting the signals interrupt the execution, it is preferable to block the signals for the moment and deliver them some time later. Let us now write a program to understand signal blocking. Here is the program...

```
#include <stdio.h>
#include <signal.h>

void sighandler (int signum)
{
 switch (signum)
 {
 case SIGTERM :
 printf ("SIGTERM Received\n");
 break ;
 case SIGINT :
 printf ("SIGINT Received\n");
 break ;
 case SIGCONT :
 printf ("SIGCONT Received\n");
 break ;
 }
}
```

```
 }
}

int main()
{
 char buffer [80] = "\0" ;
 sigset_t block ;

 signal (SIGTERM, sighandler) ;
 signal (SIGINT, sighandler) ;
 signal (SIGCONT, sighandler) ;

 sigemptyset (&block) ;
 sigaddset (&block, SIGTERM) ;
 sigaddset (&block, SIGINT) ;

 sigprocmask (SIG_BLOCK, &block, NULL) ;

 while (strcmp (buffer, "n") != 0)
 {
 printf ("Enter a String: ") ;
 gets (buffer) ;
 puts (buffer) ;
 }

 sigprocmask (SIG_UNBLOCK, &block, NULL) ;

 while (1)
 printf ("Program Running\n") ;

 return 0 ;
}
```

In this program we have registered a common handler for the **SIGINT**, **SIGTERM** and **SIGCONT** signals. Next, we want to repeatedly accept strings in a buffer and display them on the screen till the time the user does not enter an “n” from the keyboard. Additionally, we want that this activity of receiving input should not be interrupted by the **SIGINT** or the **SIGTERM** signals. However, a **SIGCONT** should be permitted. So before we proceed with the loop, we must block the **SIGINT** and **SIGTERM** signals. Once we are through with the loop, we must unblock these

signals. This blocking and unblocking of signals can be achieved using the **sigprocmask( )** library function.

Before we turn our attention to this function, let us answer one question—when does a process receive the **SIGCONT** signal? Well, a process under Linux can be suspended using the Ctrl+Z command. The process is stopped but is not terminated, i.e., it is, suspended. This gives rise to the uncatchable **SIGSTOP** signal. To resume the execution of the suspended process, we can make use of the **fg** (foreground) command. As a result of which, the suspended program resumes its execution and receives the **SIGCONT** signal (CONT means continue execution).

Let us now understand the **sigprocmask( )** function. The first parameter of the **sigprocmask( )** function specifies whether we want to block/unblock a set of signals. The next parameter is the address of a structure (typedefed as **sigset\_t**) that describes a set of signals that we want to block/unblock. The last parameter can be either NULL or the address of **sigset\_t** type variable which would be set up with the existing set of signals before blocking/unblocking signals.

There are library functions that help us to populate the **sigset\_t** structure. The **sigemptyset( )** empties a **sigset\_t** variable so that it does not refer to any signals. The only parameter that this function accepts is the address of the **sigset\_t** variable. We have used this function to quickly initialize the **sigset\_t** variable block to a known empty state. To block the **SIGINT** and **SIGTERM** we have to add the signals to the empty set of signals. This can be achieved using the **sigaddset( )** library function. The first parameter of **sigaddset( )** is the address of the **sigset\_t** variable and the second parameter is the ID of the signal that we wish to add to the existing set of signals.

After the loop, we have also used an infinite **while** loop to print the 'Program Running' message. This is done so that we can easily check that till the time the loop that receives input is not over the program cannot be terminated using Ctrl+C or **kill** command since the signals are blocked. Once the user enters "n" from the keyboard the control comes out of the **while** loop and unblocks the signals. As a result, pending signals, if any, are immediately delivered to the program. So if we press Ctrl+C or use the **kill** command when the execution of the loop that receives input is not over, these signals would be kept pending. Once we are through with the loop, the signal handlers would be called.

## Event Driven programming

Having understood the mechanism of signal processing let us now see how signaling is used by Linux-based libraries to create event driven GUI programs. As you know, in a GUI program events occur typically when we click on the window, type a character, close the window, repaint the window, etc. We have chosen the GTK library version 2.0 to create the GUI applications. Here, GTK stands for Gimp's Tool Kit. Given below is the first program that uses this Tool Kit to create a window on the screen.

```
/* mywindow.c */
#include <gtk/gtk.h>

int main (int argc, char *argv[])
{
 GtkWidget *p ;
 gtk_init (&argc, &argv) ;
 p = gtk_window_new (GTK_WINDOW_TOPLEVEL) ;
 gtk_window_set_title (p , "Sample Window") ;
 g_signal_connect (p , "destroy", gtk_main_quit, NULL) ;
 gtk_widget_set_size_request (p , 300, 300) ;
 gtk_widget_show (p) ;
 gtk_main () ;
 return 0 ;
}
```

We need to compile this program as follows:

```
$ gcc mywindow.c `pkg-config gtk+-2.0 --cflags --libs`
```

Here we are compiling the program 'mywindow.c' and then linking it with the necessary libraries from GTK. Note the quotes that we have used in the command. Here we are using the program "pkg-config", which can be obtained from [www.freedesktop.org](http://www.freedesktop.org). This program reads the .pc which comes with GTK to determine what compiler switches are needed to compile programs that use GTK. --cflags will output a list of include directories for the compiler to look in, and --libs will output the list of libraries for the compiler to link with and the directories to find them in.

Figure 23.1 shows the output of the program.

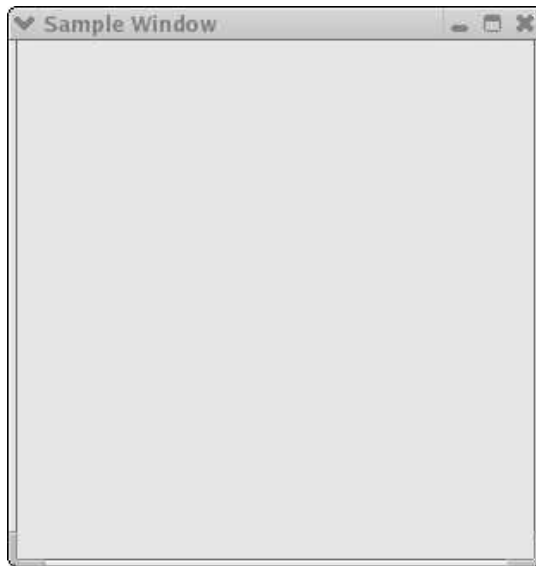


Figure 23.1

The GTK library provides a large number of functions that makes it very easy for us to create GUI programs. Every window under GTK is known as a widget. To create a simple window we have to carry out the following steps:

- (a) Initialize the GTK library with a call to **gtk\_init( )** function. This function requires the addresses of the command-line arguments received in **main( )**.
- (b) Next, call the **gtk\_window\_new( )** function to create a top-level window. The only parameter this function takes is the type of window to be created. A top-level window can be created by specifying the **GTK\_WINDOW\_TOPLEVEL** value. This call creates a window in memory and returns a pointer to the widget object. The widget object is a structure (**GtkWidget**) variable that stores lots of information including the attributes of window it represents. We have collected this pointer in a **GtkWidget** structure pointer called **p**.
- (c) Set the title for the window by making a call to **gtk\_window\_set\_title( )** function. The first parameter of this function is a pointer to the **GtkWidget** structure representing the window for which the title has to be set. The second parameter is a string describing the text to be displayed in the title of the window.



- (d) Register a signal handler for the destroy signal. The **destroy** signal is received whenever we try to close the window. The handler for the **destroy** signal should perform cleanup activities and then shut down the application. GTK provides a ready-made function called **gtk\_main\_quit( )** that does this job. We only need to associate this function with the destroy signal. This can be achieved using the **g\_signal\_connect( )** function. The first parameter of this function is the pointer to the widget for which destroy signal handler has to be registered. The second parameter is a string that specifies the name of the signal. The third parameter is the address of the signal handler routine. We have not used the fourth parameter.
- (e) Resize the window to the desired size using the **gtk\_widget\_set\_size\_request( )** function. The second and the third parameters specify the height and the width of the window, respectively.
- (f) Display the window on the screen using the function **gtk\_widget\_show( )**.
- (g) Wait in a loop to receive events for the window. This can be accomplished using the **gtk\_main( )** function.

How about another program that draws a few shapes in the window?  
Here is the program...

```
/* myshapes.c */
#include <gtk/gtk.h>

int expose_event (GtkWidget *widget, GdkEventExpose *event)
{
 GdkGC * p ;
 GdkPoint arr [5] = { 250, 150, 250, 300, 300, 350, 400, 300, 320, 190
};

 p = gdk_gc_new (widget -> window) ;
 gdk_draw_line (widget -> window, p, 10, 10, 200, 10) ;
 gdk_draw_rectangle (widget -> window, p, TRUE, 10, 20, 200, 100) ;
 gdk_draw_arc (widget -> window, p, TRUE, 200, 10, 200, 200,
 315 * 64, 90 * 64) ;
 gdk_draw_polygon (widget -> window, p, TRUE , arr, 5) ;
 /* True – fill */
 gdk_gc_unref (p) ;
```

```
 return TRUE ;
 }

int main (int argc, char *argv[])
{
 GtkWidget *p ;

 gtk_init (&argc, &argv) ;
 p = gtk_window_new (GTK_WINDOW_TOPLEVEL) ;
 gtk_window_set_title (p, "Sample Window") ;
 g_signal_connect (p, "destroy", gtk_main_quit, NULL) ;
 g_signal_connect (p, "expose_event", expose_event, NULL) ;
 gtk_widget_set_size_request (p, 500, 500) ;
 gtk_widget_show (p) ;
 gtk_main() ;
 return 0 ;
}
```

Figure 23.2 shows the output of the program.

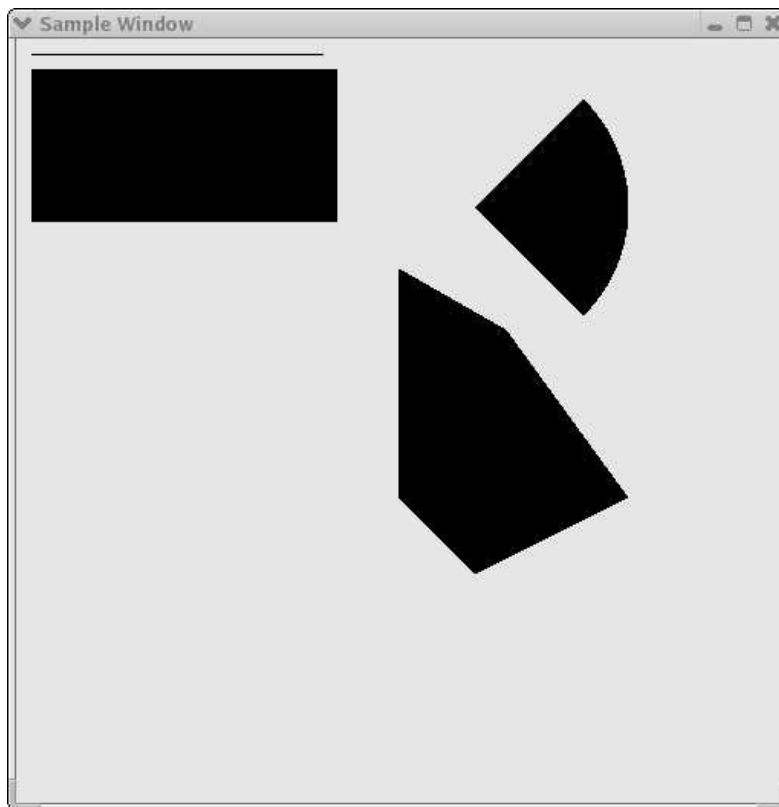


Figure 23.2

This program is similar to the previous one. The only difference is that, in addition to the destroy signal, we have registered a signal handler for the **expose\_event** using the **g\_signal\_connect( )** function. This signal is sent to our process whenever the window needs to be redrawn. By writing the code for drawing shapes in the handler for this signal we are assured that the drawing would never vanish if the window is dragged outside the screen and then brought back in, or some other window uncovers a portion of our window which was previously overlapped, and so on. This is because a **expose\_event** signal would be sent to our application, which would immediately redraw the shapes in our window.

In order to draw in the window we need to obtain a graphics context for the window using the **gdk\_gc\_new( )** function. This function returns a pointer to the graphics context structure. This pointer must be passed to the drawing functions like **gdk\_draw\_line( )**, **gdk\_draw\_rectangle( )**, **gdk\_draw\_arc( )**, **gdk\_draw\_polygon( )**, etc. The arguments passed to most of these functions are self-explanatory. Note that, the last two arguments that are passed to **gdk\_draw\_arc( )** represent the starting

angle and the ending angle of the arc. These angles are always mentioned as multiple of 64 and the ending angle is measured relative to the starting angle. Once we are through with drawing, we should release the graphics context using the **gdk\_gc\_unref( )** function.

### Where do you go from here?

You have now understood signal processing, the heart of programming under Linux. With that knowledge under your belt you are now capable of exploring the vast world of Linux on your own. Complete Linux programming deserves a book on its own. Idea here was to raise the hood and show you what lies underneath it. I am sure that if you have taken a good look at it, you can try the rest yourselves. Good luck!

### Summary

- (a) Linux is a free OS whose kernel was built by Linus Trovalds and friends.
- (b) C programs under Linux can be compiled using the popular **gcc** compiler.
- (c) **fork( )** library function can be used to create child processes.
- (d) **execl( )** library function is used to execute another program from within a running program.
- (e) **execl( )** function overwrites the image (code and data) of the calling process.
- (f) **ps** command can be used to get a list of all processes.
- (g) **kill** command can be used to terminate a process.
- (h) A 'Zombie' is a child process that has terminated, but its parent is running and has not called a function to get the exit code of the child process.
- (i) An 'Orphan' is a child process whose parent has terminated.
- (j) Orphaned processes are adopted by **init** process automatically.
- (k) A parent process can avoid creation of a Zombie and Orphan processes using **waitpid( )** function.
- (l) Programs can communicate with the Linux OS using library functions.

- (m) The Linux OS communicates with a program by means of signals.
- (n) The interrupt signal (**SIGINT**) is sent by the kernel to our program when we press Ctrl+C.
- (o) A terminate signal (**SIGTERM**) is sent to the program when we use the **kill** command.
- (p) A process cannot handle an uncatchable signal.
- (q) The **kill -SIGKILL** variation of the **kill** command generates an uncatchable **SIGKILL** signal that terminates a process.
- (r) A process can block a signal or a set of signals using the **sigprocmask( )** function.
- (s) Blocked signals are delivered to the process when the signals are unblocked.
- (t) A **SIGSTOP** signal is generated when we press Ctrl+Z.
- (u) A **SIGSTOP** signal is uncatchable signal.
- (v) A suspended process can be resumed using the **fg** command.
- (w) A process receives the **SIGCONT** signal when it resumes execution.
- (x) In GTK, the **g\_signal\_connect( )** function can be used to connect a function with an event.

## Exercise

[A] State True or False:

- (a) We can modify the kernel of Linux OS.
- (b) All distributions of Linux contain the same collection of applications, libraries and installation scripts.
- (c) Basic scheduling unit in Linux is a file.
- (d) **execl( )** library function can be used to create a new child process.
- (e) The scheduler process is the father of all processes.
- (f) A family of **fork( )** and **exec( )** functions are available, each doing basically the same job but with minor variations.
- (g) **fork( )** completely duplicates the code and data of the parent process into the child process.

- (h) **fork( )** overwrites the image (code and data) of the calling process.
- (i) **fork( )** is called twice but returns once.
- (j) Every Zombie process is essentially an Orphan process.
- (k) Every Orphan process is essentially a Zombie process.
- (l) All registered signals must have a separate signal handler.
- (m) Blocked signals are ignored by a process.
- (n) Only one signal can be blocked at a time.
- (o) Blocked signals are ignored once the signals are unblocked.
- (p) If our signal handler gets called, the default signal handler still gets called.
- (q) **gtk\_main( )** function makes use of a loop to prevent the termination of the program.
- (r) Multiple signals can be registered at a time using a single call to **signal( )** function.
- (s) The **sigprocmask( )** function can block as well as unblock signals.

**[B]** Answer the following:

- (a) If a program contains four calls to **fork( )** one after the other, how many total processes would get created?
- (b) What is the difference between a Zombie process and an Orphan process?
- (c) Write a program that prints the command-line arguments that it receives. What would be the output of the program if the command-line argument is \* ?
- (d) What purpose do the functions **getpid( )** and **getppid( )** serve?
- (e) Rewrite the program in the section 'Zombies and Orphans' in this chapter by replacing the **while** loop with a call to the **sleep( )** function. Do you observe any change in the output of the program?
- (f) How does **waitpid( )** prevent creation of Zombie or Orphan processes?
- (g) How does the Linux OS know if we have registered a signal or not?
- (h) What happens when we register a handler for a signal?

- (i) Write a program to verify whether **SIGSTOP** and **SIGKILL** signals are un-catchable signals.
- (j) Write a program to handle the **SIGINT** and **SIGTERM** signals. From inside the handler for **SIGINT** signal write an infinite loop to print the message 'Processing Signal'. Run the program and make use of Ctrl+C more than once. Run the program once again and press Ctrl+C once. Then use the **kill** command. What are your observations?
- (k) Write a program that blocks the **SIGTERM** signal during execution of the **SIGINT** signal.





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## ***24 Interview***

### ***FAQs***

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In the interview room you would be tested for three skills— Knowledge, Problem Solving Skills and Social Skills. You might be led to believe that in the interview room what matter is your personality, how smartly you answer questions, how are your mannerisms, etc. In fact the truth is much farther than that. All of these in my estimate have only 10% importance. Much more weightage is given to your knowledge and problem solving skills. If you are found good in these areas, then only the interview panel would be even interested in checking your social skills. With this in mind, I have given below questions that are very commonly asked in the interview rooms.

### Question 1

What is a Programming Paradigm?

#### Answer

Programming paradigm means the principle that is used for organizing programs. There are two major Programming Paradigms, namely, Structured Programming and Object Oriented Programming (OOP). C language uses the Structured Programming Paradigm, whereas, C++, C#, VB.NET or Java make use of OOP. OOP has lots of advantages to offer. But even while using this organizing principle you would still need a good hold over the language elements of C and the basic programming skills.

### Question 2

Is it true that Operating Systems like Windows, Linux and UNIX are written in C?

#### Answer

Major parts of popular operating systems like Windows, UNIX, Linux are still written in C. This is because even today when it comes to performance (speed of execution) nothing beats C. Also, the functions exposed by the Operating System API can be easily called through any language.

Moreover, if one is to extend the operating system to work with new devices one needs to write Device Driver programs. These programs are exclusively written in C.

### Question 3

What do you mean by scope of a variable? What are the different types of scopes that a variable can have?

**Answer**

Scope indicates the region over which the variable's declaration has an effect. The four kinds of scopes are—file, function, block and prototype.

**Question 4**

Which of the following statement is a declaration and which is a definition?

```
extern int i ;
int j ;
```

**Answer**

First is declaration, second is definition.

**Question 5**

What are the differences between a declaration and a definition?

**Answer**

There are two differences between a declaration and a definition:

In the definition of a variable space is reserved for the variable and some initial value is given to it, whereas a declaration only identifies the type of the variable. Thus definition is the place where the variable is created or assigned storage, whereas declaration refers to places where the nature of the variable is stated but no storage is allocated.

Secondly, redefinition is an error, whereas, redeclaration is not an error.

**Question 6**

Is it true that a global variable may have several declarations, but only one definition? [Yes/No]

**Answer**

Yes

**Question 7**

Is it true that a function may have several declarations, but only one definition? [Yes/No]

**Answer**

Yes

**Question 8**

When we mention the prototype of a function are we defining the function or declaring it?

**Answer**

We are declaring it. When the function, along with the statements belonging to it is mentioned, we are defining the function.

**Question 9**

Some books suggest that the following definitions should be preceded by the word *static*. Is it correct?

```
int a[] = { 2, 3, 4, 12, 32 };
struct emp e = { "sandy", 23 };
```

**Answer**

Pre-ANSI C compilers had such a requirement. Compilers which conform to ANSI C standard do not have such a requirement.

**Question 10**

If you are to share the variables or functions across several source files how would you ensure that all definitions and declarations are consistent?

**Answer**

The best arrangement is to place each definition in a relevant '.c' file. Then, put an external declaration in a header file ('.h' file) and use **#include** to bring in the declaration wherever needed. The '.c' file which contains the definition should also include the header file, so that the compiler can check that the definition matches the declaration.

**Question 11**

Global variables are available to all functions. Does there exist a mechanism by way of which I can make it available to some and not to others?

**Answer**

No. The only way this can be achieved is to define the variable locally in **main( )** instead of defining it globally and then passing it to the functions which need it.

**Question 12**

What are the different types of linkages?

**Answer**

There are three different types of linkages—external, internal and none. External linkage means global, non-static variables and functions, internal linkage means static variables and functions with file scope, and no linkage means local variables.

**Question 13**

What is **size\_t** ?

**Answer**

It is the type of the result of the **sizeof** operator. **size\_t** is used to express the size of something or the number of characters in something. For example, it is the type that you pass to **malloc( )** to indicate how many bytes you wish to allocate. Or it is the type returned by **strlen( )** to indicate the number of characters in a string.

Each implementation chooses a type like **unsigned int** or **unsigned long** (or something else) to be its **size\_t**, depending on what makes most sense. Each implementation publishes its own choice of **size\_t** in several header files like 'stdio.h', 'stdlib.h', etc. In most implementations **size\_t** is defined as:

```
typedef unsigned int size_t ;
```

This means that on this particular implementation **size\_t** is an **unsigned int**. Other implementations may make other choices.

What is important is that you should not worry about what **size\_t** looks like for a particular implementation; all you should care about is that it is the *right* type for representing object sizes and count.

**Question 14**

What is more efficient, a **switch** statement or an **if-else** chain?

**Answer**

As far as efficiency is concerned there would hardly be any difference, if at all. If the cases in a **switch** are sparsely distributed the compiler may internally use the equivalent of an **if-else** chain instead of a compact jump table. However, one should use **switch** where one can. It is definitely a cleaner way to program and certainly is not any less efficient than the **if-else** chain.

**Question 15**

Can we use a **switch** statement to switch on strings?

**Answer**

No. The cases in a **switch** must either have integer constants or constant expressions.

**Question 16**

In which order do the Relational, Arithmetic, Logical and Assignment operators get evaluated in C?

**Answer**

Arithmetic, Relational, Logical, Assignment

**Question 17**

How come that the C standard says that the expression

```
j = i++ * i++ ;
```

is undefined, whereas, the expression

```
j = i++ && i++ ;
```

is perfectly legal?

**Answer**

According to the C standard an object's stored value can be modified only once (by evaluation of expression) between two sequence points. A sequence point occurs:

- At the end of full expression (expression which is not a sub-expression in a larger expression)
- At the &&, || and ?: operators
- At a function call (after the evaluation of all arguments, just before the actual call)

Since in the first expression **i** is getting modified twice between two sequence points the expression is undefined. Also, the second expression is legal because a sequence point is occurring at **&&**, and **i** is getting modified once before and once after this sequence point.

**Question 18**

If **a[i] = i++** is undefined, then by the same reason **i = i + 1** should also be undefined. But it is not so. Why?

**Answer**

The standard says that if an object is to get modified within an expression then all accesses to it within the same expression must be for

computing the value to be stored in the object. The expression `a[ i ] = i++` is disallowed because one of the accesses of `i` (the one in `a[ i ]`) has nothing to do with the value that ends up being stored in `i`. In this case the compiler may not know whether the access should take place before or after the incremented value is stored. Since there's no good way to define it, the standard declares it as undefined. As against this, the expression `i = i + 1` is allowed because `i` is accessed to determine `i`'s final value.

### Question 19

Will the expression `*p++ = c` be disallowed by the compiler?

#### Answer

No. Because here even though the value of `p` is accessed twice it is used to modify two different objects `p` and `*p`.

### Question 20

Why should I use functions at all?

#### Answer

There are two reasons for using functions:

- (a) Writing functions avoids rewriting the same code over and over. Suppose you have a section of code in your program that calculates area of a triangle. If later in the program you want to calculate the area of a different triangle, you won't like it if you are required to write the same instructions all over again. Instead, you would prefer to jump to a 'section of code' that calculates area and then jump back to the place from where you left off. This section of code is nothing but a function.
- (b) By using functions it becomes easier to write programs and keep track of what they are doing. If the operation of a program can be divided into separate activities, and each activity placed in a different function, then each could be written and checked more or less independently. Separating the code into modular functions also makes the program easier to design and understand.

So don't try to cram the entire logic in one function. It is a very bad style of programming. Instead, break a program into small units and write functions for each of these isolated subdivisions. Don't hesitate to write functions that are called only once. What is important is that these functions perform some logically isolated task.

**Question 21**

In what form are the library functions provided?

**Answer**

Library functions are never provided in source code form. They are always made available in object code form obtained after compilation.

**Question 22**

What is the type of the variable **b** in the following declaration?

```
#define FLOATPTR float *
FLOATPTR a, b ;
```

**Answer**

**float** and not a pointer to a **float**, since on expansion the declaration becomes:

```
float *a, b ;
```

**Question 23**

Is it necessary that the header files should have a .h extension?

**Answer**

No. However, traditionally they have been given a .h extension to identify them as something different than the .c program files.

**Question 24**

What do the header files usually contain?

**Answer**

Header files contain Preprocessor directives like **#define**, **structure**, **union** and **enum** declarations, **typedef** declarations, global variable declarations and external function declarations. You should not write the actual code (i.e., function bodies) or global variable definition (that is defining or initializing instances) in header files. The **#include** directive should be used to pull in header files, not other '.c' files.

**Question 25**

Will it result into an error if a header file is included twice? [Yes/No]

**Answer**

Yes, unless the header file has taken care to ensure that if already included it doesn't get included again.



**Question 26**

How can a header file ensure that it doesn't get included more than once?

**Answer**

All declarations must be written in the manner shown below. Assume that the name of the header file is '**funcs.h**'.

```
/* funcs.h */
#ifndef _FUNCS
#define _FUNCS
/* all declarations would go here */
#endif
```

Now if we include this file twice as shown below, it will get included only once.

```
#include "funcs.h"
#include "funcs.h"
int main()
{
 /* some code */
 return 0 ;
}
```

**Question 27**

On doing **#include** where are the header files searched?

**Answer**

If **#included** using `< >` the files get searched in the predefined include path. It is possible to change the predefined include path. If **#included** with the `" "` syntax in addition to the predefined include path the files also get searched in the current directory (usually the directory from which you invoked the compiler).

**Question 28**

Can you combine the following two statements into one?

```
char *p ;
p = (char *) malloc (100) ;
```

**Answer**

```
char *p = (char *) malloc (100) ;
```

Note that the typecasting operation can be dropped completely if this program is built using gcc compiler.

**Question 29**

Are the expressions **\*ptr++** and **++\*ptr** same?

**Answer**

No. **\*ptr++** increments the pointer and not the value pointed by it, whereas **++\*ptr** increments the value being pointed to by **ptr**.

**Question 30**

Can you write another expression which does the same job as **++\*ptr** does?

**Answer**

```
(*ptr)++
```

**Question 31**

What would be the equivalent pointer expression for referring the array element **a[ i ][ j ][ k ][ l ]**?

**Answer**

```
* (* (* (* (a + i) + j) + k) + l)
```

**Question 32**

Where can one think of using pointers?

**Answer**

At lot of places, some of which are:

- Accessing array or string elements
- In passing big objects like arrays, strings and structures to functions
- Dynamic memory allocation
- Call by reference
- Implementing linked lists, trees, graphs and many other data structures

**Question 33**

How will you declare an array of three function pointers where each function receives two **ints** and returns a **float**?

**Answer**

```
float (*arr[3]) (int, int) ;
```

**Question 34**

Is the NULL pointer same as an uninitialized pointer? [Yes/No]

**Answer**

No

**Question 35**

In which header file is the NULL macro defined?

**Answer**

In files "**stdio.h**" and "**stddef.h**".

**Question 36**

Is there any difference between the following two statements?

```
char *p = 0 ;
char *t = NULL ;
```

**Answer**

No. NULL is **#defined** as 0 in the 'stdio.h' file. Thus, both **p** and **t** are null pointers.

**Question 37**

What is a null pointer?

**Answer**

For each pointer type (like say a **char** pointer) C defines a special pointer value, which is guaranteed not to point to any object or function of that type. Usually, the null pointer constant used for representing a null pointer is the integer 0.

**Question 38**

What's the difference between a null pointer, a NULL macro, the ASCII NUL character and a null string?

**Answer**

A null pointer is a pointer, which doesn't point anywhere.

A NULL macro is used to represent the null pointer in source code. It has a value 0 associated with it.

The ASCII NUL character has all its bits as 0 but doesn't have any relationship with the null pointer.

The null string is just another name for an empty string "".

**Question 39**

Is there any difference in the following two statements?

```
char *ch = "Nagpur" ;
char ch[] = "Nagpur" ;
```

**Answer**

Yes. In the first statement, the character pointer **ch** stores the address of the string "Nagpur". The pointer **ch** can be made to point to some other character string (or even nowhere). The second statement, on the other hand, specifies that space for 7 characters be allocated and that the name of the location is **ch**. Thus, it specifies the size as well as initial values of the characters in array **ch**.

**Question 40**

When are *char a[ ]* and *char \*a* treated as same by the compiler?

**Answer**

When using them as formal parameters while defining a function.

**Question 41**

What is the difference in the following declarations?

```
char *p = "Samuel" ;
char a[] = "Samuel" ;
```

**Answer**

Here **a** is an array big enough to hold the message and the '\0' following the message. Individual characters within the array can be changed but the address of the array will remain same.

On the other hand, **p** is a pointer, initialized to point to a string constant. The pointer **p** may be modified to point to another string, but if you

attempt to modify the string at which **p** is pointing the result is undefined.

**Question 42**

While handling a string do we always have to process it character- by-character or there exists a method to process the entire string as one unit.

**Answer**

A string can be processed only on a character-by-character basis.

**Question 43**

What is the similarity between a structure, union and an enumeration?

**Answer**

All of them let you define new data types.

**Question 44**

Can a structure contain a pointer to itself?

**Answer**

Certainly. Such structures are known as self-referential structures.

**Question 45**

How are structure passing and returning implemented by the compiler?

**Answer**

When structures are passed as arguments to functions, the entire structure is pushed on the stack. For big structures this is an extra overhead. This overhead can be avoided by passing pointers to structures instead of actual structures. To return structures a hidden argument generated by the compiler is passed to the function. This argument points to a location where the returned structure is copied.

**Question 46**

What is the difference between a structure and a union?

**Answer**

A union is essentially a structure in which all of the fields overlay each other. At a time only one field can be used. We can write to one field and read from another.

**Question 47**

What is the difference between an enumeration and a set of preprocessor **#defines**?

**Answer**

There is hardly any difference between the two, except that a **#define** has a global effect (throughout the file), whereas, an enumeration can have an effect local to the block, if desired. Some advantages of enumerations are that the numeric values are automatically assigned, whereas, in **#define** we have to explicitly define them. A disadvantage of enumeration is that we have no control over the sizes of enumeration variables.

**Question 48**

Is there an easy way to print enumeration values symbolically?

**Answer**

No. You can write a small function (one per enumeration) to map an enumeration constant to a string, either by using a **switch** statement or by searching an array.

**Question 49**

What is the use of bit fields in a structure declaration?

**Answer**

Bit fields are used to save space in structures having several binary flags or other small fields. Note that the colon notation for specifying the size of a field in bits is valid only in structures (and in unions); you cannot use this mechanism to specify the size of arbitrary variables.

**Question 50**

Can we have an array of bit fields? [Yes/No]

**Answer**

No.

**Question 51**

Can we specify variable field width in a **scanf( )** format string? [Yes/No]

**Answer**

No. In **scanf( )** a **\*** in format string after a **%** sign is used for suppression of assignment. That is, the current input field is scanned but not stored.

**Question 52**

Out of `fgets( )` and `gets( )` which function is safe to use?

**Answer**

`fgets( )`, because unlike `fgets( )`, `gets( )` cannot be told the size of the buffer into which the string supplied will be stored. As a result, there is always a possibility of overflow of buffer.

**Question 53**

To which numbering system can the binary number 1011011111000101 be easily converted to?

**Answer**

Hexadecimal, since each 4-digit binary represents one hexadecimal digit.

**Question 54**

Which bitwise operator is suitable for checking whether a particular bit is on or off?

**Answer**

The `&` operator.

**Question 55**

Which bitwise operator is suitable for turning off a particular bit in a number?

**Answer**

The `&` operator.

**Question 56**

Which bitwise operator is suitable for putting on a particular bit in a number?

**Answer**

The `|` operator.

**Question 57**

What is the type of *compare* in the following code segment?

```
typedef int (*ptrtofun)(char *, char *);
ptrtofun compare ;
```

**Answer**

It is a pointer to function that receives two character pointers and returns an integer.

**Question 58**

What are the advantages of using **typedef** in a program?

**Answer**

There are three main reasons for using **typedefs**:

- It makes writing of complicated declarations a lot easier. This helps in eliminating a lot of clutter in the program.
- It helps in achieving portability in programs. That is, if we use **typedefs** for data types that are machine-dependent, only the **typedefs** need change when the program is moved to a new machine platform.
- It helps in providing a better documentation for a program. For example, a node of a doubly linked list is better understood as **ptrtolist** rather than just a pointer to a complicated structure.

**Question 59**

What does the following prototype indicate?

```
void strcpy (char *target, const char *source)
```

**Answer**

We can modify the pointers **source** as well as **target**. However, the object to which **source** is pointing cannot be modified.

**Question 60**

What does the following prototype indicate?

```
const char *change (char *, int)
```

**Answer**

The function **change( )** receives a **char** pointer and an **int**, and returns a pointer to a constant **char**.

**Question 61**

What do you mean by **const** correctness?



**Answer**

A program is 'const correct' if it never changes (a more common term is mutates) a constant object.

**Question 62**

What is the difference in the following declarations?

```
const char *s ;
char const *s ;
```

**Answer**

There is no difference.

**Question 63**

To **free( )** we only pass the pointer to the block of memory that we want to deallocate. Then how does **free( )** know how many bytes it should deallocate?

**Answer**

In most implementations of **malloc( )** the number of bytes allocated is stored adjacent to the allocated block. Hence, it is simple for **free( )** to know how many bytes to deallocate.

**Question 64**

Suppose we use **realloc( )** to increase the allocated space for a 20-integer array to a 40-integer array. Will it increase the array space at the same location at which the array is present or will it try to find a different place for the bigger array?

**Answer**

Both. If the first strategy fails then it adopts the second. If the first is successful it returns the same pointer that you passed to it otherwise it returns a different pointer for the newly allocated space.

**Question 65**

When reallocating memory if any other pointers point into the same piece of memory do we have to readjust these other pointers or do they get readjusted automatically?

**Answer**

If **realloc( )** expands allocated memory at the same place then there is no need of readjustment of other pointers. However, if it allocates a

new region somewhere else the programmer has to readjust the other pointers.

### Question 66

What's the difference between **malloc( )** and **calloc( )** functions?

#### Answer

As against **malloc( )**, **calloc( )** needs two arguments, the number of elements to be allocated and the size of each element. For example,

```
p = (int *) calloc (10, sizeof (int)) ;
```

will allocate space for a 10-integer array. Additionally, **calloc( )** will also set each of this element with a value 0. Thus the above call to **calloc( )** is equivalent to:

```
p = (int *) malloc (10 * sizeof (int)) ;
memset (p, 0, 10 * sizeof (int)) ;
```

### Question 67

Which function should be used to free the memory allocated by **calloc( )**?

#### Answer

The same that we use with **malloc( )**, i.e., **free( )**.

### Question 68

How much maximum memory can we allocate in a single call to **malloc( )**?

#### Answer

The largest possible block that can be allocated using **malloc( )** depends upon the host system—particularly the size of physical memory and the OS implementation.

Theoretically the largest number of bytes that can be allocated should be the maximum value that can be held in **size\_t** which is implementation dependent. For TC/TC++ compilers the maximum number of bytes that can be allocated is equal to 64 KB.

### Question 69

What is difference between Dynamic memory allocation and Static memory allocation?

**Answer**

In Static memory allocation during compilation arrangements are made to facilitate memory allocation memory during execution. Actual allocation is done only at execution time. In Dynamic memory allocation no arrangement is done at compilation time. Memory allocation is done at execution time.

**Question 70**

Which header file should be included to dynamically allocate memory using functions like **malloc( )** and **calloc( )**?

**Answer**

stdlib.h

**Question 71**

When we dynamically allocate memory is there any way to free memory during run time?

**Answer**

Yes. Memory can be freed using **free( )** function.

**Question 72**

Is it necessary to cast the address returned by **malloc( )**?

**Answer**

It is necessary to do the typecasting if you are using TC / TC++ / Visual Studio compilers. If you are using gcc there is no need to typecast the returned address. Note that ANSI C defines an implicit type conversion between **void** pointer types (the one returned by **malloc( )**) and other pointer types.

**Question 73**

Mention any variable argument-list function that you have used and its prototype.

**Answer**

```
int printf (const char *format, ...) ;
```

**Question 74**

How can **%f** be used for both **float** and **double** arguments in **printf( )**?

**Answer**

In variable length arguments lists, types **char** and **short int** are promoted to **int**, and **float** is promoted to **double**.

**Question 75**

Can we pass a variable argument list to a function at run-time? [Yes/No]

**Answer**

No. Every actual argument list must be completely known at compile time. In that sense it is not truly a variable argument list.

**Question 76**

How can a called function determine the number of arguments that have been passed to it?

**Answer**

It cannot. Any function that takes a variable number of arguments must be able to determine the number of arguments from the arguments themselves. For example, the **printf( )** function does this by looking for format specifiers (**%**, etc.) in the format string. This is the reason why such functions fail badly if there is a mismatch in the format specifiers and the argument list.

If the arguments passed are all of same type we can pass a sentinel value like -1 or 0 or a NULL pointer at the end of the variable argument list. Alternately, we can also pass the count of number of variable arguments.

**Question 77**

Input / output function prototypes and macros are defined in which header file?

**Answer**

stdio.h

**Question 78**

What are **stdin**, **stdout** and **stderr**?

**Answer**

Standard input, standard output and standard error streams.

## Appendix A – Compilation and Execution



---

# ***A      Compilation and Execution***

---

To understand C language and gain confidence in working with it you would be required to type programs in this book and then instruct the machine to execute them. To type any programs you need another program called Editor. Once the program has been typed it needs to be converted to machine language (0s and 1s) before the machine can execute it. To carry out this conversion we need another program called Compiler. Compiler vendors provide an Integrated Development Environment (IDE) which consists of an Editor as well as the Compiler.

There are several such IDEs available in the market targeted towards different operating systems. For example, Turbo C and Turbo C++ are popular compilers that work under MS-DOS; Visual Studio and Visual Studio Express Edition are the compilers that work under Windows, whereas NetBeans compiler works under Linux. Note that Turbo C, Turbo C++ and NetBeans compilers can also be installed on machines running Windows OS.

Of these, Visual Studio Express Edition and NetBeans compilers are available free of cost. They can be downloaded from the following sites, respectively:

<http://www.microsoft.com/express/Downloads/>

<http://www.netbeans.org>

To install NetBeans on Linux you can download Ubuntu Linux from [www.ubuntu.com](http://www.ubuntu.com).

You are free to use any of the compilers mentioned above for compiling programs in this book. If you wish to know my personal choice, I would prefer Visual Studio Express Edition or NetBeans for two simple reasons—they are modern compilers with easy to use GUI and they are available free of cost.

To help you go through the installation process of Visual Studio smoothly I have also recorded a video. You can watch the same at the following link:

<http://www.kicit.com/help>

## Detailed Compilation Steps

The compilation process with each of the compilers mentioned in the previous section is a bit different. So for your benefit I am giving below

the detailed compilation and execution steps with each of these compilers.

### Compilation using Visual Studio / Visual Studio Express Edition

Carry out the following steps to compile and execute programs using any Visual Studio version:

- (a) Start Visual Studio from Start | All Programs | Microsoft Visual Studio or start Visual Studio Express Edition from Start | All Programs | Microsoft Visual C++ Express Edition.
- (b) Select File | New Project... from the File menu. Select Project Type as Visual C++ | Win32 Console Application from the dialog that pops up. Give a proper name of the project in Name TextBox (say Program1). Then click on OK and Finish.
- (c) Type the program.
- (d) Save the program using **Ctrl+S**.
- (e) Use **Ctrl+F5** to compile and execute the program.

When you use Visual Studio to create a Win32 Console Application for the above program the wizard would insert the following code by default:

```
#include "stdafx.h"
int _tmain (int argc, _TCHAR* argv[])
{
 return 0 ;
}
```

You can delete this code and type your program in its place. If you now compile the program using Ctrl+F5 you would get the following error:

```
Fatal error C1010:
unexpected end of file while looking for precompiled header.
Did you forget to add '#include "stdafx.h"' to your source?
```

If you add `#include "stdafx.h"` at the top of your program then it would compile and run successfully. However, including this file makes the program Visual Studio-centric and would not get compiled with gcc or TC / TC++ compilers. This is not good as the program no longer remains



portable. To eliminate this error, you need to make a setting in Visual Studio. To make this setting carry out the following steps:

- (a) Go to 'Solution Explorer'.
- (b) Right click on the project name and select 'Properties' from the menu that pops up. On doing so, a dialog box shown in Figure A.1 would appear.

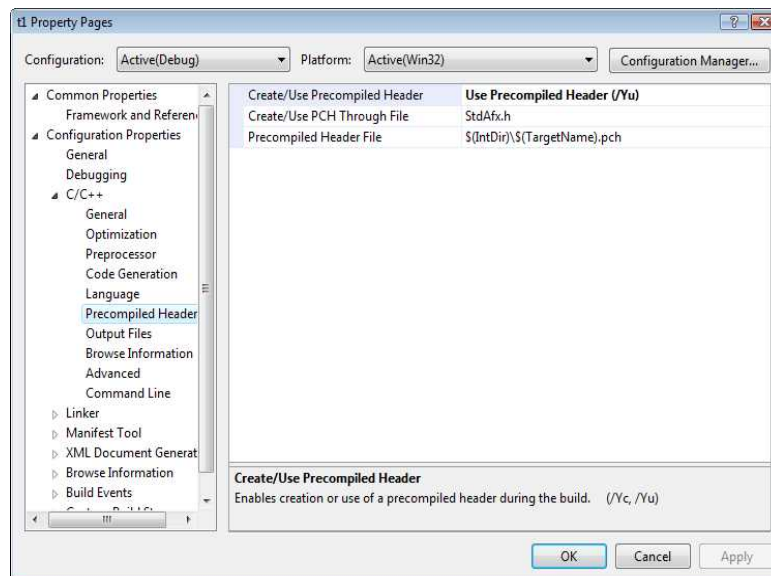


Figure A.1

- (c) From the left pane of this dialog first select 'Configuration Properties' followed by 'C/C++'.
- (d) Select 'Precompiled Headers'.
- (e) From the right pane of the dialog click on 'Create/Use Precompiled Header'. On doing so in the value for this option a triangle would appear.
- (f) Click on this triangle and a drop-down list box would appear.
- (g) From the list box select 'Not using Precompiled Header'.
- (h) Click on OK button to make the setting effective.

In addition to this, you need to make one more setting. By default Visual Studio believes that your program is a C++ program and not a C program. So by making a setting you need to tell it that your program is

a C program and not a C++ program. Carry out the following steps to make this setting:

- (a) Go to 'Solution Explorer' window.
- (b) Right click on the project name and select 'Properties' from the menu that pops up. On doing so, a dialog box shown in Figure A.2 would appear.

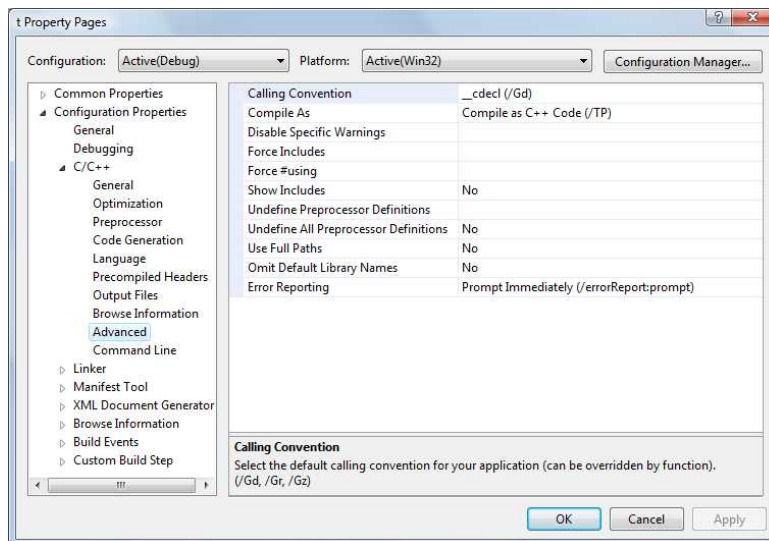


Figure A.2

- (c) From the left pane of this dialog box first select 'Configuration Properties' followed by 'C/C++'.
- (d) In C/C++ options select 'Advanced'.
- (e) Change the 'Compile As' option to 'Compile as C code (/TC)'.

Once this setting is made you can now compile the program using Ctrl+F5. This time no error would be flagged, and the program would compile and execute successfully.

### Compilation using TC / TC++

Here are the steps that you need to follow to compile and execute programs using TC/TC++...

- (a) Go to C> prompt.
- (b) The compiler (TC.EXE) is usually present in C:\TC\BIN directory. So change to that directory by typing:

C:\>CD C:\TC\BIN

- (c) Run the compiler by typing:

C:\>TC

- (d) Select **New** from the **File** menu.
- (e) Type the program.
- (f) Save the program using **F2** under a proper name (say Program1.c).
- (g) Use **Ctrl+F9** to compile and execute the program.
- (h) Use **Alt+F5** to view the output.

A word of caution! If you run this program in Turbo C++ compiler, you may get an error—"The function printf should have a prototype". To get rid of this error, perform the following steps and then recompile the program:

- (a) Select 'Options' menu and then select 'Compiler | C++ Options'. In the dialog box that pops up, select 'CPP always' in the 'Use C++ Compiler' options.
- (b) Again select 'Options' menu and then select 'Environment | Editor'. Make sure that the default extension is 'C' rather than 'CPP'.

The installation procedure of TC compiler is very simple. However, if you install TC compiler on Windows Vista or Windows 7, the window size becomes very small. So small is the screen size that you can hardly work with it. You can increase the size of TC window to occupy the entire screen. For this you will have to download an x86 emulator called DosBox from the following link:

<http://sourceforge.net/projects/dosbox/files/dosbox/0.74/DOSBox0.74-win32-installer.exe/download>

Once you have downloaded this emulator carry out the following steps:

- (a) Install the DOSBox software that you have downloaded.
- (b) Create a folder called Turbo on your C: drive.
- (c) Copy entire Turbo C / Turbo C++ software in this Turbo folder.
- (d) Start DOSBox by double clicking DOSBox icon.

- (e) You would be presented with two screens. You need to use the one which has a Z> prompt in it. Type the following command at Z> prompt:

```
mount X C:\Turbo
mount D C:\Turbo\TC
D:
cd Bin
TC
```

By typing the last command "TC" you are executing the Turbo C / Turbo C++ software. This would bring up the normal blue colored TC window. To increase the window size just press Alt+Enter. Now the TC window will accommodate the entire screen.

- (f) Go to the Options Menu (Alt+O), select 'Directories' menu item, and change the 'Include' and 'Library' directory such that they contain the following entries:

```
X:\TC\INCLUDE
X:\TC\LIB
```

- (g) Once you have typed and saved program do not compile it using the usual Ctrl+F9. Instead, compile it using the Compile Menu and execute it using Run Menu.

### Compilation using NetBeans

Carry out the following steps to compile and execute programs using any Visual Studio version:

- Start NetBeans from Start | All Programs | NetBeans.
- Select File | New Project... from the File menu. Select Project Category as C/C++ and Project Type as C/C++ Application from the dialog that pops up. Click Next button.
- Give a proper name of the project in Project Name TextBox (say Program1). Then click Finish.
- Type the program.
- Save the program using **Ctrl + S**.
- Use **F6** to compile and execute the program.

## A Word of Caution

All programs given in the book have been created assuming that you have installed Visual Studio on your machine. If you have installed TC / TC++, then you would be required to make some minor changes pertaining to clearing of screen and positioning of cursor. Let me show this to you using a simple program that first clears the screen then positions the cursor at 10<sup>th</sup> row, 20<sup>th</sup> column and prints a message at this location. If we were to use Visual Studio, the program would look like this...

```
#include <stdio.h>
#include <system.h>
void gotoxy (short int col, short int row) ;

int main()
{
 system ("cls") ; /* clear existing contents on screen */
 gotoxy (20, 10) ; /* position cursor */
 printf ("Hello") ;
 return 0 ;
}

void gotoxy (short int col, short int row)
{
 HANDLE hStdout = GetStdHandle (STD_OUTPUT_HANDLE) ;
 COORD position = { col, row } ;
 SetConsoleCursorPosition (hStdout, position) ;
}
```

A similar program in Turbo C / C++ would take the following form:

```
#include <stdio.h>
#include <conio.h>
int main()
{
 clrscr() ; /* clear existing contents on screen */
 gotoxy (20, 10) ; /* position cursor */
 printf ("Hello") ;
 return 0 ;
}
```

So if you are using Turbo C / Turbo C++ compiler then you will have to make the changes shown above to make the programs work.

## Appendix B – Precedence Table



---

# ***B***      ***Precedence Table***

---

| Description              | Operator | Associativity |
|--------------------------|----------|---------------|
| Function expression      | ( )      | Left to Right |
| Array Expression         | [ ]      | Left to Right |
| Structure operator       | ->       | Left to Right |
| Structure operator       | .        | Left to Right |
| Unary minus              | -        | Right to left |
| Increment/Decrement      | ++    -- | Right to Left |
| One's compliment         | ~        | Right to left |
| Negation                 | !        | Right to Left |
| Address of               | &        | Right to left |
| Value of address         | *        | Right to left |
| Type cast                | ( type ) | Right to left |
| Size in bytes            | sizeof   | Right to left |
| Multiplication           | *        | Left to right |
| Division                 | /        | Left to right |
| Modulus                  | %        | Left to right |
| Addition                 | +        | Left to right |
| Subtraction              | -        | Left to right |
| Left shift               | <<       | Left to right |
| Right shift              | >>       | Left to right |
| Less than                | <        | Left to right |
| Less than or equal to    | <=       | Left to right |
| Greater than             | >        | Left to right |
| Greater than or equal to | >=       | Left to right |
| Equal to                 | ==       | Left to right |
| Not equal to             | !=       | Left to right |

Continued...



Continued...

| Description          | Operator           | Associativity |
|----------------------|--------------------|---------------|
| Bitwise AND          | &                  | Left to right |
| Bitwise exclusive OR | ^                  | Left to right |
| Bitwise inclusive OR |                    | Left to right |
| Logical AND          | &&                 | Left to right |
| Logical OR           |                    | Left to right |
| Conditional          | ? :                | Right to left |
| Assignment           | =                  | Right to left |
|                      | *=      /=      %= | Right to left |
|                      | +=      -=      &= | Right to left |
|                      | ^=       =         | Right to left |
|                      | <<=      >>=       | Right to left |
| Comma                | ,                  | Right to left |

Figure B.1



## Appendix C – Chasing the Bugs



---

# **C**    *Chasing The Bugs*

---

C programmers are great innovators of our times. However, there is no shortage of horror stories about programs that took twenty times to 'debug' as they did to 'write'. Many a time programs had to be rewritten all over again because the bugs present in them could not be located. Bugs are C programmer's birthright. But how do we chase them away. No sure-shot way for that. I thought if I make a list of more common programming mistakes, it might be of help. They are not arranged in any particular order. But as you would realize, surely a great help!

- (a) Omitting the ampersand before the variables used in **scanf( )**. For example,

```
int choice ;
scanf ("%d", choice) ;
```

Here, the **&** before the variable choice is missing. Another common mistake with **scanf( )** is to give blanks either just before the format string or immediately after the format string as in,

```
int choice ;
scanf (" %d ", &choice) ;
```

Note that this is not a mistake, but till you don't understand **scanf( )** thoroughly, this is going to cause trouble. Safety is in eliminating the blanks.

- (b) Using the operator **=** instead of the operator **==**. For example, the following **while** loop becomes an infinite loop since every time, instead of checking the value of **i** against 10, it assigns the value 10 to **i**. As 10 is a non-zero value the condition will always be treated as true, forming an infinite loop.

```
int i = 10 ;
while (i = 10)
```

```
{
 printf ("got to get out");
 i++;
}
```

- (c) Ending a loop with a semicolon. Observe the following program:

```
int j = 1 ;
while (j <= 100) ;
{
 printf ("\nCompguard");
 j++;
}
```

Here, inadvertently, we have fallen in an indefinite loop. Cause is the semicolon after **while**. This semicolon is treated as a null statement by the compiler as shown below.

```
while (j <= 100)
 ;
```

This is an indefinite loop since the null statement keeps getting executed indefinitely as **j** never gets incremented.

- (d) Omitting the **break** statement at the end of a **case** in a **switch** statement. Remember that, if a **break** is not included at the end of a **case**, then execution will continue into the next **case**.

```
int ch = 1 ;
switch (ch)
{
 case 1 :
 printf ("\nGoodbye");
```

```
case 2 :
 printf ("\nLieutenant");
}
```

Here, since the **break** has not been given after the **printf( )** in **case 1**, the control runs into **case 2** and executes the second **printf( )** as well. However, this sometimes turns out to be a blessing in disguise. Especially, in cases when we want same set of statements to get executed for multiple cases.

- (e) Using **continue** in a **switch**. It is a common error to believe that the way the keyword **break** is used with loops and a **switch**; similarly the keyword **continue** can also be used with them. Remember that **continue** works only with loops, never with a **switch**.
- (f) A mismatch in the number, type and order of actual and formal arguments. Consider the following call:

```
yr = romanise (year, 1000, 'm');
```

Here, three arguments in the order **int**, **int** and **char** are being passed to **romanise( )**. When **romanise( )** receives these arguments into formal arguments, they must be received in the same order. A careless mismatch might give strange results.

- (g) Omitting provisions for returning a non-integer value from a function. If we make the following function call,

```
area = area_circle (1.5);
```

then, while defining **area\_circle( )** function later in the program, care should be taken to make it capable of returning a floating-point value. Note that unless otherwise mentioned, the compiler would assume that this function returns a value of the type **int**.

- (h) Inserting a semicolon at the end of a macro definition. This might create a problem as shown below.

```
define UPPER 25 ;
```

would lead to a syntax error if used in an expression, such as:

```
if (counter == UPPER)
```

This is because on preprocessing, the **if** statement would take the form

```
if (counter == 25 ;)
```

- (i) Omitting parentheses around a macro expansion. Consider the following macro:

```
define SQR(x) x * x
```

If we use this macro as,

```
int a ;
a = 25 / SQR (5) ;
```

we expect the value of **a** to be 1, whereas it turns out to be 25. This so happens, because, on preprocessing, the statement takes the following form:

```
a = 25 / 5 * 5 ;
```

- (j) Leaving a blank space between the macro template and the macro expansion.

```
define ABS (a) (a = 0 ? a : -a)
```



Here, the space between **ABS** and **(a)** makes the preprocessor believe that you want to expand **ABS** into **(a)**, which is certainly not what you want.

- (k) Using an expression that has side effects in a macro. Consider the following macro:

```
define SUM (a) (a + a)
```

If we use this macro as

```
int w, b = 5 ;
w = SUM(b++) ;
```

On preprocessing, the macro would be expanded to,

```
w = (b++) + (b++) ;
```

Thus, contrary to expectation, **b** will get incremented twice.

- (l) Confusing a character constant and a character string. In the statement

```
ch = 'z' ;
```

a single character is assigned to **ch**. In the statement

```
ch = "z" ;
```

a pointer to the character string "z" is assigned to **ch**.

Note that in the first case, the declaration of **ch** would be,

```
char ch ;
```

whereas in the second case it would be,

```
char *ch ;
```

- (m) Forgetting the bounds of an array.

```
int num[50], i ;
for (i = 1 ; i <= 50 ; i++)
 num[i] = i * i ;
```

Here, in the array **num**, there is no such element as **num[50]**, since array counting begins with 0 and not 1. Compiler would not give a warning if our program exceeds the bounds. If not taken care of, in extreme cases, the above code might even hang the computer.

- (n) Forgetting to reserve an extra location in a character array for the null terminator. Remember each character array ends with a '\0', therefore its dimension should be declared big enough to hold the normal characters as well as the '\0'. For example, the dimension of the array **word[ ]** should be 9 if a string "Jamboree" is to be stored in it.
- (o) Confusing the precedences of the various operators.

```
char ch ;
FILE *fp ;
fp = fopen ("text.c", "r") ;

while (ch = getc (fp) != EOF)
 putchar (ch) ;
fclose (fp) ;
```

Here, the value returned by **getc( )** will be first compared with EOF, since **!=** has a higher priority than **=**. As a result, the value that is assigned to **ch** will be the true/false result of the test—1 if the value returned by **getc( )** is not equal to **EOF**, and 0 otherwise. The correct form of the above **while** would be,

```
while ((ch = getc (fp)) != EOF)
 putchar (ch) ;
```

- (p) Confusing the operator `->` with the operator `.` while referring to a structure element. Remember, on the left of operator `.` only a structure variable can occur, whereas, on the left of operator `->` only a pointer to a structure can occur. Following example demonstrates this:

```
struct emp
{
 char name[35] ;
 int age ;
};
struct emp e = { "Dubhashi", 40 } ;
struct emp *ee ;
printf ("\n%d", e.age) ;
ee = &e ;
printf ("\n%d", ee->>age) ;
```

- (q) Exceeding the range of integers and chars. Consider the following code snippet:

```
char ch ;
for (ch = 0 ; ch <= 255 ; ch++)
 printf ("\n%c %d", ch, ch) ;
```

This is an indefinite loop. Reason is, **ch** has been declared as a **char** and its valid range is -128 to +127. Hence, the moment **ch** tries to become 128 (through **ch++**), the range is exceeded. As a result, the first number from the negative side of the range, -128, gets assigned to **ch**. Naturally, the condition is satisfied and the control remains within the loop.

## Appendix D – ASCII Chart



---

## ***D***    ***ASCII Chart***

---

There are 256 distinct characters used by PCs and Laptops. Their values range from 0 to 255. These can be grouped as shown in Figure D.1.

| Character Type     | No. of Characters |
|--------------------|-------------------|
| Capital letters    | 26                |
| Small-case Letters | 26                |
| Digits             | 10                |
| Special Symbols    | 32                |
| Control Character  | 34                |
| Graphics Character | 128               |
| Total              | 256               |

Figure D.1

Out of the 256 character set, the first 128 are often called ASCII characters and the next 128 as Extended ASCII characters. Each ASCII character has a unique appearance. The following simple program can generate the ASCII chart:

```
#include <stdio.h>
int main()
{
 int ch ;

 for (ch = 0 ; ch <= 255 ; ch++)
 printf ("%d %c\n", ch, ch) ;
 return 0 ;
}
```

This chart is shown at the end of this appendix. Out of the 128 graphic characters (Extended ASCII characters), there are characters that are used for drawing single line and double line boxes in text mode. For convenience these characters are shown in Figure D.2.

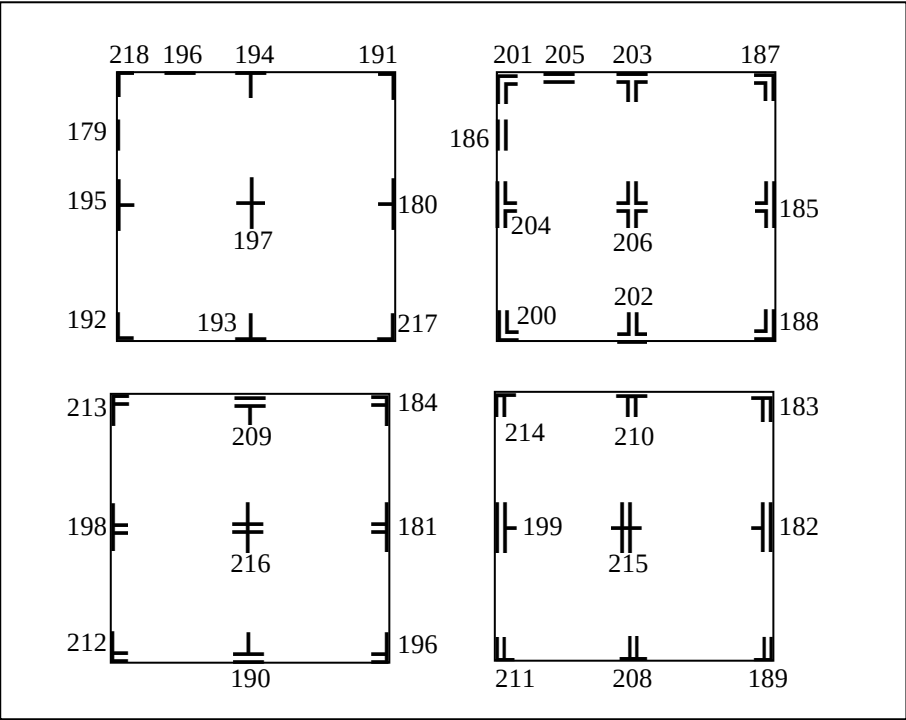


Figure D.2

| Value | Char | Value | Char | Value | Char | Value | Char | Value | Char | Value | Char           |
|-------|------|-------|------|-------|------|-------|------|-------|------|-------|----------------|
| 0     |      | 22    | —    | 44    | ,    | 66    | B    | 88    | X    | 110   | n              |
| 1     | ☺    | 23    | ↕    | 45    | -    | 67    | C    | 89    | Y    | 111   | o              |
| 2     | ☹    | 24    | ↑    | 46    | .    | 68    | D    | 90    | Z    | 112   | p              |
| 3     | ♥    | 25    | ↓    | 47    | /    | 69    | E    | 91    | [    | 113   | q              |
| 4     | ♦    | 26    | →    | 48    | 0    | 70    | F    | 92    | \    | 114   | r              |
| 5     | ♣    | 27    | ←    | 49    | 1    | 71    | G    | 93    | ]    | 115   | s              |
| 6     | ♠    | 28    | ¬    | 50    | 2    | 72    | H    | 94    | ^    | 116   | t              |
| 7     | ●    | 29    | ↔    | 51    | 3    | 73    | I    | 95    | `    | 117   | u              |
| 8     | ■    | 30    | ▲    | 52    | 4    | 74    | J    | 96    |      | 118   | v              |
| 9     | ○    | 31    | ▼    | 53    | 5    | 75    | K    | 97    | a    | 119   | w              |
| 10    | ◼    | 32    |      | 54    | 6    | 76    | L    | 98    | b    | 120   | x              |
| 11    | ♂    | 33    | !    | 55    | 7    | 77    | M    | 99    | c    | 121   | y              |
| 12    | ⊕    | 34    | "    | 56    | 8    | 78    | N    | 100   | d    | 122   | z              |
| 13    | ♪    | 35    | #    | 57    | 9    | 79    | O    | 101   | e    | 123   | {              |
| 14    | 🎵    | 36    | \$   | 58    | :    | 80    | P    | 102   | f    | 124   |                |
| 15    | ☀    | 37    | %    | 59    | ;    | 81    | Q    | 103   | g    | 125   | }              |
| 16    | ▶    | 38    | &    | 60    | <    | 82    | R    | 104   | h    | 126   | ~              |
| 17    | ◀    | 39    | '    | 61    | =    | 83    | S    | 105   | i    | 127   | <sup>M</sup> H |
| 18    | ↕    | 40    | (    | 62    | >    | 84    | T    | 106   | j    | 128   | Ç              |
| 19    | !!   | 41    | )    | 63    | ?    | 85    | U    | 107   | k    | 129   | ü              |
| 20    | ¶    | 42    | *    | 64    | @    | 86    | V    | 108   | l    | 130   | é              |
| 21    | §    | 43    | +    | 65    | A    | 87    | W    | 109   | m    | 131   | â              |



| Value | Char | Value | Char | Value | Char | Value | Char | Value | Char | Value | Char |
|-------|------|-------|------|-------|------|-------|------|-------|------|-------|------|
| 132   | ä    | 154   | Ü    | 176   | ⋮    | 198   | ƒ    | 220   | ■    | 242   | ≥    |
| 133   | à    | 155   | ç    | 177   | ⋮    | 199   | ƒ    | 221   | ■    | 243   | ≤    |
| 134   | å    | 156   | £    | 178   | ⋮    | 200   | ℓ    | 222   | ■    | 244   | ƒ    |
| 135   | ç    | 157   | ¥    | 179   |      | 201   | ƒ    | 223   | ■    | 245   |      |
| 136   | ê    | 158   | ₣    | 180   | ⌋    | 202   | ℓ    | 224   | α    | 246   | ÷    |
| 137   | ë    | 159   | ƒ    | 181   | ⌋    | 203   | ℓ    | 225   | β    | 247   | ≈    |
| 138   | è    | 160   | á    | 182   | ⌋    | 204   | ℓ    | 226   | Γ    | 248   | °    |
| 139   | ï    | 161   | í    | 183   | ⌋    | 205   | ℓ    | 227   | π    | 249   | •    |
| 140   | î    | 162   | ó    | 184   | ⌋    | 206   | ℓ    | 228   | Σ    | 250   | ·    |
| 141   | ì    | 163   | ú    | 185   | ⌋    | 207   | ℓ    | 229   | σ    | 251   | √    |
| 142   | Ä    | 164   | ñ    | 186   | ⌋    | 208   | ℓ    | 230   | u    | 252   | η    |
| 143   | Å    | 165   | Ñ    | 187   | ⌋    | 209   | ℓ    | 231   | τ    | 253   | ²    |
| 144   | É    | 166   | æ    | 188   | ⌋    | 210   | ℓ    | 232   | Φ    | 254   | ■    |
| 145   | æ    | 167   | ø    | 189   | ⌋    | 211   | ℓ    | 233   | θ    | 255   |      |
| 146   | Æ    | 168   | ç    | 190   | ⌋    | 212   | ℓ    | 234   | Ω    |       |      |
| 147   | ô    | 169   | ¬    | 191   | ⌋    | 213   | ℓ    | 235   | δ    |       |      |
| 148   | ö    | 170   | ¬    | 192   | ⌋    | 214   | ℓ    | 236   | ∞    |       |      |
| 149   | ò    | 171   | ½    | 193   | ⌋    | 215   | ℓ    | 237   | ø    |       |      |
| 150   | û    | 172   | ¼    | 194   | ⌋    | 216   | ℓ    | 238   | €    |       |      |
| 151   | ù    | 173   | í    | 195   | ⌋    | 217   | ℓ    | 239   | ∩    |       |      |
| 152   | ÿ    | 174   | «    | 196   | ⌋    | 218   | ℓ    | 240   | ≡    |       |      |
| 153   | Ö    | 175   | »    | 197   | ⌋    | 219   | ℓ    | 241   | ±    |       |      |





## Periodic Tests

- Periodic Test I
- Periodic Test II
- Periodic Test III
- Periodic Test IV

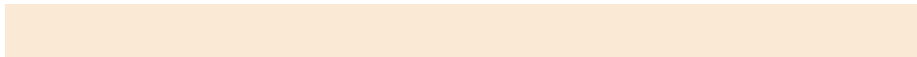


---

# **30**     *Periodic Tests*

---

- Test I
- Test II
- Test III
- Test IV



**Periodic Test I**  
**(Based on Chapters 1 to 7)**

Time: 90 Minutes

Maximum Marks: 40

**[A] Fill in the blanks:****[5 Marks, 1 Mark each]**

- (1) The expression **i++** is same as \_\_\_\_\_.
- (2) \_\_\_\_\_ type of values cannot be checked using **switch-case**.
- (3) Every instruction in a C program must end with a \_\_\_\_\_.
- (4) The size of an **int** data type is \_\_\_\_\_ bytes.
- (5) Statements written in \_\_\_\_\_ loop get executed for once even if the condition is false.

**[B] State True or False:****[5 Marks, 1 Mark each]**

- (1) The statement **for ( ; ; )** is a valid statement.
- (2) The **else** clause in an **if - else if – else** statement goes to work if all the **ifs** fail.
- (3) The **^** operator is used for performing exponentiation operations in C.
- (4) C allows only one variable on the left hand side of **=** operator.
- (5) Conditional operators cannot be nested.

**[C] What would be the output of the following programs:****[5 Marks, 1 Mark each]**

- (a) 

```
#include <stdio.h>
int main()
{
 int x = 5, y, z ;
 y = x++ ;
 z = x-- ;
 printf ("%d %d %d", x, y, z) ;
 return 0 ;
}
```
- (b) 

```
#include <stdio.h>
int main()
```

```
{
 int i = 65 ;
 char ch = i ;
 printf ("%d %c", ch, i) ;
 return 0 ;
}
```

```
(c) # include <stdio.h>
int main()
{
 int i, j ;
 for (i = 1 ; i <= 2 ; i++)
 {
 for (j = 1 ; j <= 2 ; j++)
 {
 if (i == j)
 break ;
 printf ("%d %d", i, j) ;
 }
 }
 return 0 ;
}
```

```
(d) # include <stdio.h>
int main()
{
 int x = 3, i = 1 ;
 while (i <= 2)
 {
 printf ("%d", x *= x + 4) ;
 i++ ;
 }
 return 0 ;
}
```

```
(e) # include <stdio.h>
int main()
{
 int a, b = 5 ;
 a = !b ;
 b = !a ;
 printf ("%d %d", a, b) ;
 return 0 ;
}
```

```
}
```

**[D]** Point out the error, if any, in the following programs:  
[5 Marks, 1 Mark each]

- (a) 

```
#include <stdio.h>
int main()
{
 int i = 10, j = 20 ;
 if (i = 5) && if (j = 10)
 printf ("Have a nice day") ;
 return 0 ;
}
```
- (b) 

```
#include <stdio.h>
int main()
{
 int x = 10 ;
 if (x >= 2) then
 printf ("\n%d", x) ;
 return 0 ;
}
```
- (c) 

```
#include <stdio.h>
int main()
{
 int x = 0, y = 5, z = 10, a ;
 a = x > 1 ? y > 1 : z > 1 ? 100 : 200 : 300 ;
 printf ("%d" , a) ;
 return 0 ;
}
```
- (d) 

```
#include <stdio.h>
int main()
{
 int x = 0, y = 5, z ;
 float a = 1.5, b = 2.2, c ;
 z = x || b ;
 c = a && b ;
 printf ("%d %f", z, c) ;
 return 0 ;
}
```

```
(e) #include <stdio.h>
int main()
{
 int a = 10, b = 5, c ;
 c += a *= b ;
 printf ("%d %d %d" , a, b, c) ;
}
```

**[E] Attempt the following:** [20 Marks, 5 Marks each]

- (1) Write a program to calculate the sum of the following series:  
 $1! 2! + 2! 3! + 3! 4! + 4! 5! + \dots + 9! 10!$
- (2) Write a program to enter the numbers till the user wants and at the end it should display the count of positive, negative and zeros entered.
- (3) Write a program to find the range of a set of numbers that are input through the keyboard. Range is the difference between the smallest and biggest number in the list.
- (4) If three integers are entered through the keyboard, write a program to determine whether they form a Pythagorean triplet or not.



**Periodic Test II**  
**(Based on Chapters 8 to 12)**

Time: 90 Minutes

Maximum Marks: 40

**[A] Fill in the blanks:****[5 Marks, 1 Mark each]**

- (1) \_\_\_\_\_ function is used to clear the screen.
- (2) \_\_\_\_\_ are variables, which hold addresses of other variables.
- (3) \_\_\_\_\_ is called an 'address of' operator.
- (4) The preprocessor directive that is used to give convenient names to difficult formulae is called \_\_\_\_\_.
- (5) For a call by reference we should pass \_\_\_\_\_ of variables to the called function.

**[B] State True or False:****[5 Marks, 1 Mark each]**

- (1) A function can return more than one value at a time.
- (2) A fresh set of variables are created every time a function gets called.
- (3) All types of pointers are 4 bytes long.
- (4) Any function can be made a recursive function.
- (5) The correct build order is Preprocessing – Compilation – Assembling – Linking.

**[C] Answer the following:****[10 Marks, 2 Marks each]**

- (1) Why are addresses of functions stored on the stack?
- (2) How do we decide whether a variable should be passed by value or by reference?
- (3) Size of a pointer is not dependent on whose address is stored in it. Justify.
- (4) At times there may be holes in a structure? Why do they exist? How can they be avoided?
- (5) A recursive call should always be subjected to an if. Why? Explain with an example.

[D] Attempt the following:

[20 Marks, 5 Marks each]

- (1) Define a function that receives 4 integers and returns sum, product and average of these integers.
- (2) Write a recursive function which prints the prime factors of the number that it receives when called from **main( )**.
- (3) Write macros for calculating area of circle, circumference of circle, volume of a cone and volume of sphere.
- (4) Write a program that prints sizes of all types of chars, ints and reals.

**Periodic Test III**  
**(Based on Chapters 13 to 17)**

Time: 90 Minutes

Maximum Marks: 40

**[A] Fill in the blanks:****[5 Marks, 1 Mark each]**

- (1) Mentioning name of the array yields \_\_\_\_\_ of the array.
- (2) C permits us to exceed \_\_\_\_\_ and \_\_\_\_\_ bounds of an array.
- (3) Size of an array is \_\_\_\_\_ of sizes of individual elements of an array.
- (4) Array elements are always counted from \_\_\_\_\_ onwards.
- (5) A structure is usually a collection of \_\_\_\_\_ elements.

**[B] State True or False:****[5 Marks, 1 Mark each]**

- (1) If the array is big its elements may get stored in non-adjacent locations.
- (2) All strings end with a '\0'.
- (3) Using **#pragma pack** you can control the layout of structure elements in memory.
- (4) Elements of 2D array are stored in the form of rows and columns in memory.
- (5) 3D array is a collection of several 1D arrays.

**[C] Answer the following:****[10 Marks, 2 Marks each]**

- (1) What is likely to happen if the bounds of an array are exceeded?
- (2) When you prefer a structure over an array to store similar elements? Explain with an example.
- (3) What is the limitation of an array of pointers to strings? How can it be overcome?
- (4) In a two-dimensional array **a[ 4 ][ 4 ]**, why do expressions **a** and **\*a** give base address?
- (5) How can we receive multi-word strings as input using **scanf( )** and using **gets( )**?

[D] Attempt the following:

[20 Marks, 5 Marks each]

- (1) Write a function that receives as parameters, a 1D array, its size and an integer and returns number of times the integer occurs in the array.
- (2) Create an array of pointers containing names of 10 cities. Write a program that sorts the cities in reverse alphabetical order and prints this reversed list.
- (3) Declare a structure called student containing his name, age and address. Create and initialize three structure variables. Define a function to which these variables are passed. The function should convert the names into uppercase. Print the resultant structure variables.
- (4) Write a program that checks and reports whether sum of elements in the  $i^{\text{th}}$  row of a 5 x 5 array is equal to sum of elements in  $i^{\text{th}}$  column.

**Periodic Test IV**  
**(Based on Chapters 18 to 23)**

Time: 90 Minutes

Maximum Marks: 40

**[A] Fill in the blanks:****[5 Marks, 1 Mark each]**

- (1) 0xAABB | 0xBBAA evaluates to \_\_\_\_\_.
- (2) The values of an **enum** are stored as \_\_\_\_\_.
- (3) Multiple signals can be registered at a time using a single call to \_\_\_\_\_ function.
- (4) The \_\_\_\_\_ function can block as well as unblock signals.
- (5) The \_\_\_\_\_ operator is used to invert the bits in a byte.

**[B] State True or False:****[5 Marks, 1 Mark each]**

- (1) To check whether a particular bit in a byte is on or off, the bitwise | operator is useful.
- (2) It is possible to create a union of structures.
- (3) The callback mechanism can be implemented using function pointers.
- (4) **fork( )** completely duplicates the code and data of the parent process into the child process.
- (5) If our signal handler gets called, the default signal handler still gets called.

**[C] Answer the following:****[10 Marks, 2 Marks each]**

- (1) What is the utility of <<, >>, & and | bitwise operators?
- (2) Define the **BV** macro. How would the following expressions involving the **BV** macro be expanded by the preprocessor?

```
int a = BV (5) ;
int b = ~ BV (5) ;
```

- (3) What does the following expression signify?

```
long (*p[3]) (int, float) ;
```

- (4) Suggest a suitable **printf( )** that can be used to print the grocery items and their prices in the following format:

|              |              |
|--------------|--------------|
| Tomato Sauce | : Rs. 225.50 |
| Liril Soap   | : Rs. 55.45  |
| Pen Refill   | : Rs. 8.95   |

- (5) When it is useful to make use of a union? What is the size of a union variable? How can the elements of a union variable be accessed?

**[D] Attempt the following:** [20 Marks, 5 Marks each]

- (1) Write a program to multiply two integers using bitwise operators.
- (2) Write a program to count number of words in a given text file.
- (3) Write a program that receives a set of numbers as command- line arguments and prints their average.
- (4) Write a program to check whether contents of the two files are same by comparing them on a byte-by-byte basis.



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# ***Index***

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*Dedicated to baba*  
*Who couldn't be here to see this day...*

# LET US JAVA

By: Yashavant Kanetkar ISBN: 978818333467 ₹ 297/-

**2<sup>nd</sup>  
Revised  
&  
Updated  
Edition**

“Today’s challenges” need “Today’s solutions”. The way this applies to read-word, it equally applies to programming world. To create solutions that meet today’s expectations, one needs to use a modern language like Java. Learning Java can be challenging for anyone. This is where you would find this book immensely useful. It follows simple and easy narration style. It doesn’t assume any programming background. It begins with the basics and steadily builds the pace so that the reader finds it easy to handle complex topics towards the end. Each chapter has been designed to create a deep and lasting impression on reader’s mind. OOP has been covered in detail to give a strong foundation for Java programming. Well thought out & fully working example programs and carefully crafted exercises of this book, cover every aspect of Java programming from the world famous author Yashavant Kanetkar.



## Contents

- An Overview of Java
- Getting Started
- More about Data Types
- Decision Control Instruction
- Loop Control Instruction
- Case Control Instruction
- Functions
- Introduction to OOP
- Classes and Objects
- Arrays
- Strings and Enums
- Inheritance
- Polymorphism
- Exception Handling
- Effective Input/Output
- Operations on Bits

## About the Author

Through his books and Quest Video Courseware DVDs on C, C++, Data Structures, VC++, .NET, Embedded Systems, etc. Yashavant Kanetkar has created, moulded and groomed lacs of IT careers in the last two decades. Yashavant's books and Quest DVDs have made a significant contribution in creating top-notch IT manpower in India and abroad.

Yashavant's books are globally recognized and millions of students / professionals have benefitted from them. Yashavant's books have been translated into Hindi, Gujarati, Japanese, Korean and Chinese languages. Many of his books are published in India, USA, Japan, Singapore, Korea and China.

Yashavant is a much sought after speaker in the IT field and has conducted seminars/workshops at TedEx, IITs, RECs and global software companies.

Yashavant has recently been honored with the prestigious "Distinguished Alumnus Award" by IIT Kanpur for his entrepreneurial, professional and academic excellence. This award was given to top 50 alumni of IIT Kanpur who have made significant contribution towards their profession and betterment of society in the last 50 years.

In recognition of his immense contribution to IT education in India, he has been awarded the "Best .NET Technical Contributor" and "Most Valuable Professional" awards by Microsoft for 5 successive years.

Yashavant holds a BE from VJTI Mumbai and M.Tech. from IIT Kanpur.

## Acknowledgments

Let Us C has become an important part of my life. I have created and nurtured it for last decade and half. While doing so, I have received, in addition to the compliments, a lot of suggestions from students, developers, professors, publishers and authors. So much have their inputs helped me in taking this book up to its fourteenth edition that ideally I should put their names too on the cover page.

In particular, I am indebted to Manish Jain who had a faith in this book idea, believed in my writing ability, whispered the words of encouragement and made helpful suggestions from time to time. I hope every author gets a publisher who is as cooperative, knowledgeable and supportive as Manish.

The Fourteen editions of this book saw several changes and facelifts. During this course many people helped in executing programs and spotting bugs. I trust that with their collective acumen, all the programs in this book would run correctly. I value the work that they did a lot. Any errors, omissions or inconsistencies that remain are, alas, my responsibility.

I thank Dada, Ammi—my parents, Seema—my wife, Aditya, Anuj—my sons for enduring the late nights, the clicking keyboard, and mostly for putting up with yet another marathon book effort.

Thinking of a book cover idea is one thing, putting it into action is a different cup of tea. This edition's cover idea has been implemented by Jayant. Many thanks to him.

And finally my heartfelt gratitude to the countless students who made me look into every nook and cranny of C. I want to remain in their debt. It is only because of them that Let Us C is now published from India, Singapore, USA, Japan, Korea and China in multiple languages.

## Preface

In this I have reorganized the contents of the book in a major way. After going through the thirteenth edition several times I decided to realign all the chapters in such a manner that if a C programming course is taught using Let Us C, it can roughly be finished in 23 lectures of one hour each, with one chapter's contents devoted to one lecture. I hope this would make the learning path trouble-free. Some end-of-chapter exercises in the book needed a second look to make them more practical. That also stands done now.

Many readers told me that they have immensely benefitted from the inclusion of the chapter on Interview FAQs. I have improved this chapter further. The rationale behind this chapter is simple—ultimately all the readers of Let Us C sooner or later end up in an interview room where they are required to take questions on C programming. I now have a proof that this chapter has helped to make that journey smooth and fruitful.

All the programs present in the book are available in source code form at [www.kicit.com/books/letusc/sourcecode](http://www.kicit.com/books/letusc/sourcecode). You are free to download them, improve them, change them, do whatever with them. If you wish to get solutions for the Exercises in the book they are available in another book titled 'Let Us C Solutions'. If you want some more problems for practice they are available in the book titled 'Let Us C Workbook'. As usual, new editions of these two books have also been launched along with 14<sup>th</sup> edition of Let Us C.

If you like 'Let Us C' and want to hear the complete video-recorded lectures created by me on C language (and other subjects like C++, VC++, C#, Java, .NET, Embedded Systems, etc.), then you can visit <http://quest.ksetindia.com> for more details.

'Let Us C' is as much your book as it is mine. So if you feel that I could have done certain job better than what I have, or you have any suggestions about what you would like to see in the next edition, please drop a line to **[bpbpublications@gmail.com](mailto:bpbpublications@gmail.com)**

Countless Indians have relentlessly worked for close to two decades to successfully establish "India" as a software brand. At times, I take secret pleasure in seeing that a Let Us C has contributed in its own little way in shaping so many careers that have made the "India" brand acceptable.

Recently I was presented with “Distinguished Alumnus Award” by IIT Kanpur. It was great to figure in a list that contained Narayan Murthy, Chief Mentor, Infosys, Dr. D. Subbarao, former Governor, Reserve Bank of India, Dr. Rajeev Motwani of Stanford University, Prof. H. C. Verma, Mr. Som Mittal President of NASSCOM, Prof. Minwalla of Harvard University, Dr. Sanjay Dhande former Director of IIT Kanpur, Prof. Arvind and Prof. Sur of MIT USA and Prof. Ashok Jhunjhunwala of IIT Chennai.

I think Let Us C amongst my other books has been primarily responsible for helping me get the “Distinguished Alumnus” award. What was a bit surprising was that almost all who were present knew about the book already and wanted to know from me what it takes to write a book that sells in millions of copies. My reply was—make an honest effort to make the reader understand what you have to say and keep it simple. I don’t know how convincing was this answer, but well, that is what I have been doing with this book in all its previous thirteen editions. I have followed the same principle with this edition too.

All the best and happy programming!

**Yashavant Kanetkar**



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